Antarctic Bottom Water in AWIESM-2.1 simulations with and without ice-shelf cavities

Historical and SSP1-2.6/2-4.5/3-7.0/5-8.5 (1851–2200): a multi-theme diagnostic report

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1 Executive summary

This report collects the diagnostics produced by ten analysis themes (setup and evaluation, atmosphere, sea ice and convection, ice-shelf melt and freshwater, bottom and shelf water, surface water-mass transformation, circulation and sections, overturning, volume census and deep change, abyssal signal propagation) for two AWIESM-2.1 simulation sets that differ only in whether the Antarctic ice-shelf cavities are resolved. Each theme section below starts with its findings and then shows all its figures with captions. The numbers quoted here come from those sections. All simulations are single members, and differences between the two sets that are smaller than the internal variability quoted in the theme sections should be read with care.

1.0.1 What the model can and cannot say about AABW

The model does not produce water in the observed AABW density class. South of 60S only 0.07 percent (cavity) and 0.04 percent (no cavity) of the volume has sigma-2 above 37.16, against 42 percent in WOA18. The abyss is filled with CDW-like water of 1.7 to 2.0 degC and sigma-2 near 36.9, so the deep Weddell and Ross basins are 2.4 to 2.6 degC too warm at the bottom. Dense shelf water exists on the western Weddell and Ross shelves but it is at most as dense as the deep water and no plume crosses the 1000 m isobath in either set. The shelves are about 0.2 psu too fresh, the winter sea-ice extent is 13.5 to 15 million km2 against about 18.5 observed, and February is almost ice free. Both pre-industrial controls drift, the deep ocean warming by about 0.08 degC per century and the densest 10 percent of the Southern Ocean lightening by 0.02 kg m-3 per century in every scenario, so deep "responses" must be drift corrected and the apparent decline of the model AABW class is to a large extent drift. AABW is therefore defined throughout with model-relative thresholds (sigma-2 above 36.90 to 36.92 from TEOS-10, or the densest 10 percent of the volume), and the report studies the precursors of AABW formation (shelf water properties, brine release, convection, surface transformation) and the abyssal overturning rather than a realistic AABW layer. Two further diagnostic facts matter for interpretation. Sea-surface salinity restoring is active in both sets (surf_relax_S in namelist.tra, 10 m per 60 days), and the model density field density_dMOC is quantised (1.2e-4 kg m-3) and narrower than TEOS-10 sigma-2 from the same temperature and salinity, so thresholds on it are noisy.

1.0.2 What the cavities change in the present-day state

The large-scale atmosphere is the same in both sets (jet strength and latitude, SAM, P-E over 50 to 70S all within noise), so every ocean difference below is oceanic in origin. The cavity set injects 2364 Gt per year of basal melt at depth under the ice shelves and has no Antarctic runoff, while the no-cavity set receives 2509 Gt per year of ice-sheet runoff at the surface in a 200 to 300 km coastal band. The amount of land-ice freshwater is thus nearly the same and only its route differs. This routing difference explains most of the present-day contrasts. With cavities the Weddell shelf bottom is saltier and denser (34.42 versus 34.33 psu, 37.00 versus 36.94 kg m-3), the Ross shelf is colder but fresher, Ice Shelf Water exists (282 thousand km3 inside the cavities, none at all without cavities), shelf brine release is larger on all shelves, and surface formation of water denser than sigma-2 37.0 is 5.1 versus 4.3 Sv (18 percent more, Ross plus 56 percent, Weddell plus 23 percent). The deep Weddell is 0.2 to 0.25 degC cooler and 0.03 psu fresher with cavities, the global abyssal cell is 0.5 to 0.9 Sv stronger (minus 10.4 versus minus 9.5 Sv), the model AABW class holds 6 to 10 percent more volume and its densest classes are 0.1 to 0.15 degC colder. The AMOC is about 1 Sv weaker with cavities. The no-cavity set convects in the open ocean near Maud Rise in 137 of 164 historical winters (September mixed layers deeper than 2000 m over 0.10 to 0.16 million km2) while the cavity set does so in 34 winters and over a tenth of the area. The cause is a 0.03 psu fresher and 0.01 kg m-3 more stratified upper ocean at Maud Rise in the cavity run, fed by the 316 Gt per year of Dronning Maud Land melt carried in the coastal current, whereas sea-ice insulation and heat loss are the same. The Maud Rise convection makes the density-space lower cell south of 50S look 9 Sv stronger in the no-cavity set (minus 30 versus minus 21 Sv) but this is a local recirculation and the export at 30S is the same (7 Sv). Outside the Weddell sector the cavity set has less winter sea ice (1.1 million km2), mostly in the Ross, Amundsen and Bellingshausen seas, where the atmosphere above is 2.7 K warmer in winter and loses 17 W m-2 more heat. The ACC transport differs by 14 Sv (93 versus 79 Sv) as an inheritance of the separate spin-ups, and the slope current is 35 percent weaker with cavities.

1.0.3 How the precursors of AABW respond to warming

Under SSP5-8.5 the Antarctic shelves warm by 1.5 to 2.2 degC and freshen by 0.2 to 0.6 psu at the bottom by 2171 to 2200, CDW at 400 m reaches every shelf after 2100 (plus 1.4 to 2.4 degC), and the shelf water becomes lighter than the deep ocean. The year in which the shelf minus deep density contrast first drops below zero is 2072 (cavity) and 2031 (no cavity) in the Weddell Sea and 2068 (cavity) and 2099 (no cavity) in the Ross Sea, so cavities bring the Ross collapse forward by three decades and delay the Weddell one. Surface formation of the dense class drops below a quarter of its present value in 2084 (cavity) and 2103 (no cavity) and is zero by

2171 to 2200 in SSP5-8.5 and SSP3-7.0, while the peak transformation migrates to classes 0.4 to 0.6 kg m-3 lighter because the shelves turn from net winter freezing (2.3 mm per day) to net melt (minus 1.4 mm per day). The Maud Rise convection of the no-cavity set stops between 2060 and 2120 in SSP2-4.5 and higher as the Weddell gyre stratification strengthens fourfold, and September sea-ice extent falls below 1 million km2 by 2146 (cavity) and 2156 (no cavity) in SSP5-8.5. Ice-shelf melt rises from 2364 to 12030 Gt per year (5.1 times) in SSP5-8.5, 10100 in SSP3-7.0, 4100 in SSP2-4.5 and 2500 in SSP1-2.6. The Filchner-Ronne cavity tips within a decade from about 200 to above 1500 Gt per year when the Weddell shelf bottom reaches minus 0.45 to minus 0.35 degC, which happens in 2078 (SSP5-8.5, 4.2 K global warming), 2093 (SSP3-7.0) and 2153 (SSP2-4.5) and never in SSP1-2.6, and warm water above 1 degC then fills the Filchner Trough to the grounding line. In the cavity set the total freshwater input south of 60S reaches 27100 Gt per year in 2171 to 2200 against 19450 in the no-cavity set, a 7600 Gt per year difference that only exists with cavities, and the cavity set loses shelf density about twice as fast. The global abyssal cell weakens by 25 percent (cavity) and 32 percent (no cavity), the density-space lower cell south of 50S by 69 and 75 percent, and the AMOC by 72 percent to about 4.5 Sv, both sets converging to the same weak state by 2200. South of 30S the abyss at 3000 m warms by about 0.3 degC, of which 0.06 to 0.17 degC survives drift correction, and the cavity set takes up 70 to 100 ZJ more heat below 2000 m and 90 ZJ less in the upper 1000 m. Westerlies strengthen by 0.02 to 0.04 Pa and shift 2.5 degrees poleward identically in both sets. South of 60S the cavity set warms 0.9 to 1.1 K less in the far period of SSP3-7.0 and SSP5-8.5 because the no-cavity set starts with more sea ice and loses more of its insulation.

In the abyss there is no propagating warming front from the Weddell Sea toward the equator. After drift correction the SSP5-8.5 warming of 0.3 to 0.6 degC by 2200 is confined to south of 40S and arrives at all latitude bands between 74S and 40S within the same two decades (2070 to 2095), while the Atlantic abyss north of 30S cools and freshens by 0.2 to 0.6 degC as the NADW weakens, spreading southward from 40N at 3 to 5 years per 1000 km. The deep Southern Ocean warms 0.28 (cavity) and 0.22 degC per century (no cavity) over 2015 to 2100 in every scenario, and the deep Weddell ends near 2.2 to 2.4 degC in 2171 to 2200 in all four scenarios. The model-relative dense layer (densest 10 percent of the present cavity volume) is 1211 m thick south of 60S with cavities and 175 m without, and it disappears from the Southern Ocean in both sets and in both SSP1-2.6 and SSP5-8.5 by 2171 to 2200, because the homogeneous abyssal mode lightens at the drift rate rather than because of the scenario.

Under SSP1-2.6 the precursors survive. Dense-class formation stays at 3.7 to 4.0 Sv, the global abyssal cell does not weaken, the density lower cell south of 50S recovers to 80 percent of its present value by 2200, the AMOC recovers to 13 to 14.5 Sv, September sea ice stays at 10 to 11 million km2, FRIS never tips and melt rises by 6 percent. The no-cavity set even convects more at Maud Rise in the far period (0.30 million km2, twice the present) and the cavity set begins to convect intermittently, so the open-ocean formation path survives only in the no-cavity set.

1.0.4 Story lines worth developing for a paper

The cleanest message is that the route of land-ice freshwater (basal melt at depth in the cavities versus runoff at the surface) controls whether the model forms dense water on the shelf or in the open ocean, and that this single difference produces the Weddell versus Ross asymmetry, the presence or absence of Maud Rise convection, and the differences in dense-water formation and abyssal overturning while the atmosphere is unchanged. A second line is the Filchner-Ronne tipping under SSP3-7.0 and higher and its 3200 Gt per year per degC sensitivity, with the resulting freshwater pulse doubling the loss of shelf density in the cavity set. A third line is that under strong warming both sets converge to the same state without dense-water formation, convection or winter sea ice, so the cavity matters for the timing (Ross earlier, Weddell later) and for the far-future Antarctic warming rather than for the end state, whereas under SSP1-2.6 the two sets keep different formation pathways. Any of these must be framed with the model caveats above (no observed AABW class, drift in both controls, single members, active SSS restoring, different coastal domains near Dronning Maud Land and Amery).

1.0.5 Cross-theme consistency notes

Sea-ice extent numbers differ between the overview (14.5 and 15.7 million km2, from the climatological concentration field) and the sea-ice theme (13.5 and 14.6, mean of yearly extents); the latter is the standard definition. The abyssal cell quoted in the guide as about minus 7 Sv was a two-year test; the 30-year present-day minimum is minus 10.4 and minus 9.5 Sv. The census theme uses TEOS-10 sigma-2 because density_dMOC is quantised; the bottom, circulation and propagation themes use density_dMOC for maps and sections, which is adequate for patterns but thresholds on it carry a 0.01 kg m-3 sawtooth. The gyre indices wg_psi, rg_psi and acc_psi in the time-series cache are sign-inverted and were replaced by the circulation theme. The FESOM density-class surface-flux diagnostics of cavity elements are binned into a stale class and are excluded in the transformation theme. The cavity run gains volume from basal melt (global mean sea level rises 0.77 m over the historical period), a setup artefact without physical consequence for the diagnostics here.

2 Data, methods and conventions

2.0.1 Simulations

Two sets of AWIESM-2.1-wiso simulations (ECHAM6.3 T63L47 coupled to FESOM2.1 on a CORE2-like unstructured mesh, run by Pengyang Song, AWI) are compared. The cavity set (PI_ICEv2 mesh, 132273 surface nodes, 47 levels, 6592 nodes under Antarctic ice shelves, `use_cavity` and `use_cavity_partial_cell` switched on) resolves the sub-ice-shelf cavities with a three-equation melt parameterisation. The no-cavity set (PI_CTRL mesh, 126858 nodes) has the same code, forcing and resolution but the ice-shelf cavities are closed and ice-shelf melt is not represented as basal melt. Both were spun up for 1000 years under pre-industrial conditions and branched at 1850 into a historical run (1851 to 2014) followed by SSP1-2.6, SSP2-4.5, SSP3-7.0 and SSP5-8.5 (2015 to 2200). The experiments are PI_ICEv2_hist and PI_ICEv2_ssp126/245/370/585 (cavity) and PI_CORE_hist and PI_CORE_ssp126/245/370/585 (no cavity). Data were retrieved from the DKRZ tape archive `/hsm/arch/ba0989/a270146/AWIESM2_cavity` and from `/work/ab0246/a270146/experiments` to `/work/ba1066/a270064/ice_cavity_data`.

2.0.2 Output used

FESOM2 monthly 2D fields (sea-ice concentration and thickness, SST, SSS, SSH, heat and freshwater fluxes, runoff, mixed-layer depths, wind stress) and annual-mean 3D fields (temperature, salinity, model sigma-2, velocities, vertical velocity, N2, diffusivities) plus the FESOM2 density-class diagnostics (density-space overturning transports U_rho_x_DZ and V_rho_x_DZ and surface buoyancy fluxes binned into 89 sigma-2 classes). ECHAM6 monthly GRIB output was extracted with cdo.

2.0.3 Conventions and periods

Analysis periods are early (1851 to 1880, a pre-industrial-like state), pd (1985 to 2014), mid (2041 to 2070), end (2071 to 2100) and far (2171 to 2200). In all line plots solid lines denote the cavity set and dashed lines the no-cavity set, with black for historical and blue, cyan, orange and red for SSP1-2.6, SSP2-4.5, SSP3-7.0 and SSP5-8.5. Differences between sets are computed after interpolating both meshes to a common regular grid. Ice-shelf melt is the freshwater flux under the cavity nodes (negative fw means freshwater into the ocean). Mixed-layer depths and surface fields under ice shelves are artefacts and are excluded from all open-ocean region means. Because the model does not produce water in the observed AABW density class (sigma-2 above 37.16), AABW is defined with model-relative thresholds (sigma-2 above 36.90 unless stated otherwise). Depth-space overturning is computed from the vertical velocity integrated from the north (NADW cell positive, abyssal cell negative). Density-space overturning uses the FESOM2 edge-transport diagnostics. Region masks, ice-shelf boxes and all diagnostics are implemented in the Python package `aabw` in `/work/ba1066/a270064/cc_projects/ice_cavity`.

3 Model setup, geometry and evaluation

Findings

3.1 Overview theme: findings (setup, geometry, evaluation)

3.1.1 A. Meshes and geometry

1. Mesh sizes. PI_ICEv2 (cavity mesh): 132273 surface nodes, 254926 triangles, 47 z-layers (48 interfaces, 5 m surface layer, 250 m layers below 2400 m, deepest interface 6250 m, deepest wet layer 6000 m), minimum depth 30 m, 6592 cavity nodes (12601 cavity elements). PI_CTRL (no-cavity mesh): 126858 nodes, 244659 triangles, same vertical grid, no cavities. Ocean area 365.2 (PI_ICEv2) versus 364.4 million km2 (PI_CTRL); ocean volume 1.3221 versus 1.3214 x 10^18 m3 (difference 0.65 x 10^15 m3, i.e. the cavity volume of 0.566 x 10^15 m3 plus a small shelf difference).

2. The two meshes are identical north of 60S (113584 identical node positions). South of 60S PI_ICEv2 has 18689 nodes and PI_CTRL 13274, so the cavity mesh also refines the open shelf and coastal zone (median spacing south of 60S 24 km versus 38 km; area-weighted resolution on the open shelf 29.9 versus 28.6 km, i.e. similar where both are wet, but PI_ICEv2 has many more small cells along the grounding line and in the cavities at 16 km).

3. Bathymetry. On a common 0.25 degree grid south of 55S the depths agree to 13 m on average (rms 86 m, 5 percent of common cells differ by more than 100 m, all near the coast and shelf break). Geometric differences that can matter for AABW are therefore (i) the cavities themselves, (ii) the coastline, and (iii) the different node layout on the shelf, not the deep bathymetry.

4. Cavity geometry. The cavities cover 1.425 million km2 (91 percent of the 1.56 million km2 of Rignot et al. 2013, approximate literature value), area-weighted mean draft 265 m (max 1040 m under Filchner-Ronne), mean sea-floor depth under ice 631 m, mean water-column thickness 366 m (min 25 m; 2 percent of the cavity area thinner than 100 m; max 2620 m where the ice front sits over a deep trough). Cavity volume 566 thousand km3.

5. Coastline difference. 1098 thousand km2 of cavity (77 percent, Filchner-Ronne, Ross, Getz, Pine Island/Thwaites, Abbot, George VI/Wilkins, half of Amery, two thirds of Larsen C) is land in PI_CTRL, whose coastline follows the ice front there. 327 thousand km2 (23 percent) of the PI_ICEv2 cavity area is open ocean in PI_CTRL (almost all Dronning Maud Land shelves 126 of 127 thousand km2, Shackleton/West 45 of 46, Totten/Moscow University 14 of 15, Amery 27 of 46, Larsen C 20 of 58). In these regions PI_CTRL exposes the sea surface to the atmosphere and sea ice where PI_ICEv2 has an ice shelf roof, so the no-cavity run can form sea ice and HSSW over areas that do not exist as open shelf in the cavity run. This is a second-order but real confound in shelf-region comparisons (dml_shelf, amery_shelf, adelie_shelf boxes). The open (non-cavity) shelf area south of 60S shallower than 1000 m is 2.38 million km2 in PI_ICEv2 and 2.92 million km2 in PI_CTRL; the difference of 0.55 million km2 is the open-in-CTRL cavity area plus coastal cells of different size.

6. Cavity areas per ice-shelf box compared with Rignot et al. (2013) areas (approximate, quoted from memory, several shelves grouped per box):

ice-shelf box	cavity nodes	model area (1e3 km2)	Rignot 2013 approx (1e3 km2)	ratio	mean draft (m)	mean water column (m)	volume (1e12 m3)
FRIS	1998	403.3	443.1	0.91	437	474	200.0
Ross	2113	470.7	500.8	0.94	263	356	170.9
Amery	187	46.1	60.7	0.76	249	570	30.8
LarsenC	253	58.3	46.5	1.25	120	265	16.9
Fimbul_DML	472	127.1	141.0	0.90	151	297	45.5
Shackleton_West	148	46.3	41.7	1.11	100	210	12.5
Totten_MU	63	14.7	11.8	1.25	147	327	5.9
Getz	168	33.5	34.0	0.98	167	338	14.0
PIG_Thwaites	145	23.5	20.8	1.13	145	328	9.2
Abbot_Venable	185	33.2	35.9	0.93	110	231	9.3
GeorgeVI_Wilkins	309	50.1	48.9	1.02	85	295	17.4

ice-shelf box	cavity nodes	model area (1e3 km2)	Rignot 2013 approx (1e3 km2)	ratio	mean draft (m)	mean water column (m)	volume (1e12 m3)
Ronne	1566	319.4	338.9	0.94	427	453	150.3
Filchner	417	80.4	104.3	0.77	491	546	46.3
all cavity nodes	6592	1425.0	1561	0.91	265	366	565.8

The two large cavities are 6 to 9 percent smaller than observed, Amery and Filchner are 23 to 24 percent too small, and the small East Antarctic and Peninsula shelves are 10 to 25 percent too large because 16 km resolution cannot resolve their outlines. The total is 9 percent low. Note that ice-shelf boxes are lon-lat boxes on cavity nodes, so neighbouring small shelves are grouped (see CAPTIONS.md for the grouping).

3.1.2 B. Key biases relevant for AABW (1985-2014 versus WOA18 2005-2017 and PHC3.0)

7. Deep warm bias. Both runs have no AABW in the observed sense. Bottom potential temperature in the deep Weddell box (>1000 m) is 1.74 degC (cavity) and 2.01 degC (no cavity) against -0.64 degC in WOA18, a bias of 2.4 and 2.6 degC; below 3000 m the values are 1.77 and 2.00 degC against -0.71 degC. Deep Ross box: 1.65 and 1.77 degC against -0.06 degC (bias 1.7 and 1.8 degC). Whole Southern Ocean south of 60S deeper than 1000 m: 1.69 and 1.90 degC against -0.25 degC. Zonal-mean bias below 2000 m between 80S and 30S is 1.36 (cavity) and 1.46 degC (no cavity), global deep ocean 1.06 and 1.10 degC. The warm bias starts below 150 m in the Weddell and Ross boxes (profiles), so the CDW layer is too warm as well (0.8 and 0.65 degC zonal-mean bias in 0 to 1000 m south of 30S).

8. Deep salinity is realistic. Bottom salinity biases in the deep basins are within 0.04 psu (Weddell +0.01 and +0.035 psu, Ross -0.04 and -0.03 psu; zonal mean below 2000 m -0.015 and -0.003 psu). The model abyss is therefore CDW-like (1.7 to 2.0 degC, 34.66 to 34.69 psu, sigma2 about 36.9) rather than AABW-like (-0.7 degC, 34.65 psu, sigma2 37.2). The density deficit is almost entirely thermal.

9. Density census. South of 60S, 42 percent of the WOA18 volume has sigma2 above 37.16 (AABW) and 88 percent above 37.0. In the cavity run the fractions are 0.07 and 0.6 percent, in the no-cavity run 0.04 and 0.2 percent (volumes of sigma2 > 37.16 water: 31 x 10^15 m3 observed, 0.05 and 0.03 x 10^15 m3 modelled). Any AABW census in this project must therefore use model-relative density classes (e.g. sigma2 above 36.9 or the densest few percent), not the observational 37.16 threshold.

10. Shelf water is too fresh and too warm. Bottom salinity on the Antarctic shelf (<1000 m, south of 60S) is 34.23 (cavity) and 34.29 psu (no cavity) against 34.55 psu in WOA18 (bias -0.20 and -0.18 psu). Weddell shelf: 34.30 and 34.30 against 34.60 psu (-0.21, -0.25); Ross shelf: 34.30 and 34.42 against 34.65 psu (-0.25, -0.17); Prydz Bay, Dronning Maud Land, Adelie shelves -0.16 to -0.27 psu. Bottom temperature on the shelf is 0.9 (cavity) and 1.05 degC (no cavity) too warm on average; only the Filchner-Ronne and Ross fronts in the cavity run stay below -1.5 degC. HSSW-like water (T < -1.7 degC, S > 34.5 psu) is 0.27 percent of the volume south of 60S in WOA18 and essentially absent in both runs (0.007 and 0.02 percent). The lack of HSSW (no dense shelf source) is the proximate cause of the missing AABW. Ice Shelf Water (T < -1.9 degC) exists only in the cavity run (270 thousand km3, WOA18 169 thousand km3 mostly under the Filchner-Ronne front).

11. Sea ice. SH extent 1985-2014: September 14.5 million km2 (cavity) and 15.7 million km2 (no cavity) against about 18.5 million km2 (NSIDC literature value), February 0.43 and 0.54 against about 3 million km2. Sea-ice area 10.5 and 11.8 million km2 in September. September volume 5.5 and 6.7 thousand km3. Winter ice is missing in the Amundsen-Ross sector north of 65S and in the eastern Weddell Sea; summer ice survives only in the western Weddell Sea. The too small and too seasonal ice cover means too little brine rejection on the shelves, consistent with the fresh shelves. The cavity run has 1.1 million km2 less September ice than the no-cavity run (Indian and Pacific sectors) and 0.5 degC cooler, 0.1 psu fresher surface water around the coast (zonal mean).

12. Open-ocean deep convection. The no-cavity run has a persistent winter convection site in the eastern Weddell gyre near Maud Rise (centroid 9W, 66.5S; MLD up to 5000 m). Per-year September area with MLD > 2000 m (open ocean south of 50S): mean 0.20, max 0.48 million km2, present in all 30 years; the 30-year climatological MLD exceeds 2000 m over 0.14 million km2. In the cavity run the same site is intermittent (15 of 30 years, mean 0.02, max 0.12 million km2) and the climatological MLD never exceeds 2000 m. This open-ocean convection is the likely source of whatever dense water the no-cavity

run produces and is not a realistic AABW pathway; it should be checked in the seaice/convection theme with the yearly ts cache (conv_area2000) and against the Weddell polynya literature. The cavity run is less prone to it, which is consistent with its fresher, more stratified surface layer.

13. Relative difference cavity minus no cavity (same forcing, same code). The cavity run is 0.25 degC (Weddell deep), 0.12 degC (Ross deep) and 0.2 degC (Southern Ocean below 1000 m) cooler at the bottom, 0.02 to 0.03 psu fresher in the deep Weddell and 0.1 psu fresher in the upper 500 m south of 65S, and has 0.12 degC colder bottom water on the Ross shelf. It has a slightly larger fraction of dense water (sigma2 > 37.0: 0.6 versus 0.2 percent). These differences are an order of magnitude smaller than the common bias against observations, so the paper story must be framed as “response of a CDW-dominated, AABW-poor state to cavities” rather than “realistic AABW with and without cavities”.

14. Climatology caveats. WOA18 A5B7 (2005-2017) is compared with 1985-2014 means; the time offset is negligible relative to the biases. PHC3.0 and WOA18 agree to 0.1 degC and 0.01 psu below 500 m in both boxes. Both datasets store in-situ temperature; the comparison converts them to potential temperature (gsw pt0_from_t). Note that FESOM2 is initialised from phc3.0 as if it were potential temperature, which makes the initial deep ocean about 0.25 to 0.3 degC too warm; this is only a small part of the 2 degC bias after 1000 years of spin-up. Model bottom values are taken at the deepest wet layer and WOA18 at the deepest valid standard level, so the two “bottoms” can differ by a few hundred metres on steep slopes; region means use the model bathymetry on the 0.25 degree grid for the shelf/deep split. Sea-ice reference values are literature numbers, no observational dataset is available locally.

3.1.3 C. Suggested story lines and issues

15. Story: the cavity run is systematically closer to observations in the Southern Ocean (cooler deep water, more ISW and slightly more dense water, less spurious open-ocean convection, colder Ross shelf) but the difference is small compared to the shared warm bias. Figures for a paper: eval_TS (shows the missing AABW class at a glance), eval_bottomT (d,e,f), eval_mld (convection difference), geom_domain_diff (what the cavity mesh changes geometrically).

16. Data and diagnostic issues. The open-in-CTRL cavity area (327 thousand km2, mostly Dronning Maud Land and Shackleton/West) means that shelf boxes dml_shelf, amery_shelf and adelie_shelf compare open water in PI_CTRL with sub-ice water in PI_ICEv2; interpret cavity minus no-cavity differences there with care. MLD is not meaningful on cavity nodes (excluded here; the precompute mld/conv_area diagnostics should do the same). The climatological MLD underestimates convection area in intermittent years; use yearly fields.

Scripts: analyses/overview/{geometry.py, prep_clim.py, evaluation.py, seaice_eval.py, conv_years.py, design_timeline.py}. Numbers: cache/overview/{geometry_stats.json, eval_stats.json, seaice_stats.json, conv_years.json, cavity_table.md}; climatologies on FESOM levels: cache/overview/woa18_fesomlev_0p5.nc, woa18_bottom_0p25.nc, phc3_fesomlev_1p0.nc (potential temperature).

Figures

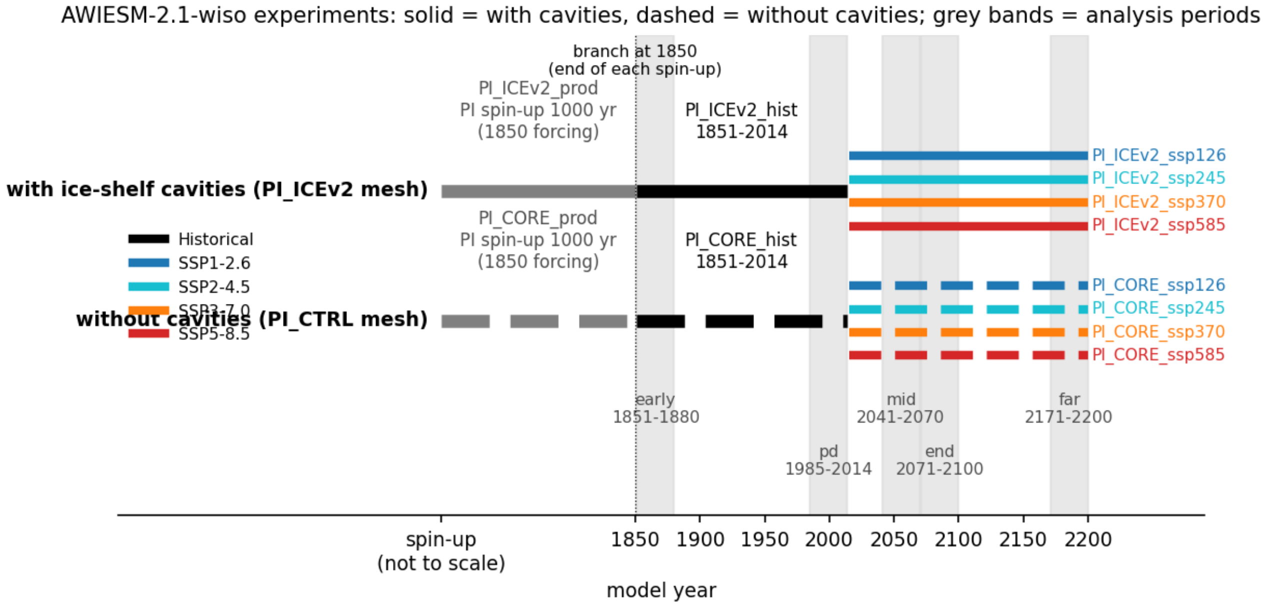

Figure 1: design_timeline (overview). Experiment design of the two AWIESM-2.1-wiso sets. Each set was spun up for 1000 years under 1850 forcing on its own FESOM2 mesh (PI_ICEv2 with ice-shelf cavities, PI_CTRL without), then branched at 1850 into a historical run (1851 to 2014) and four SSP continuations (2015 to 2200). Solid bars are the cavity set, dashed bars the no-cavity set. Grey bands mark the analysis periods early (1851 to 1880), pd (1985 to 2014), mid (2041 to 2070), end (2071 to 2100) and far (2171 to 2200). The spin-up axis is not to scale. Atmosphere (ECHAM6 T63L47), forcing and code are identical between the sets; only the ocean mesh differs.

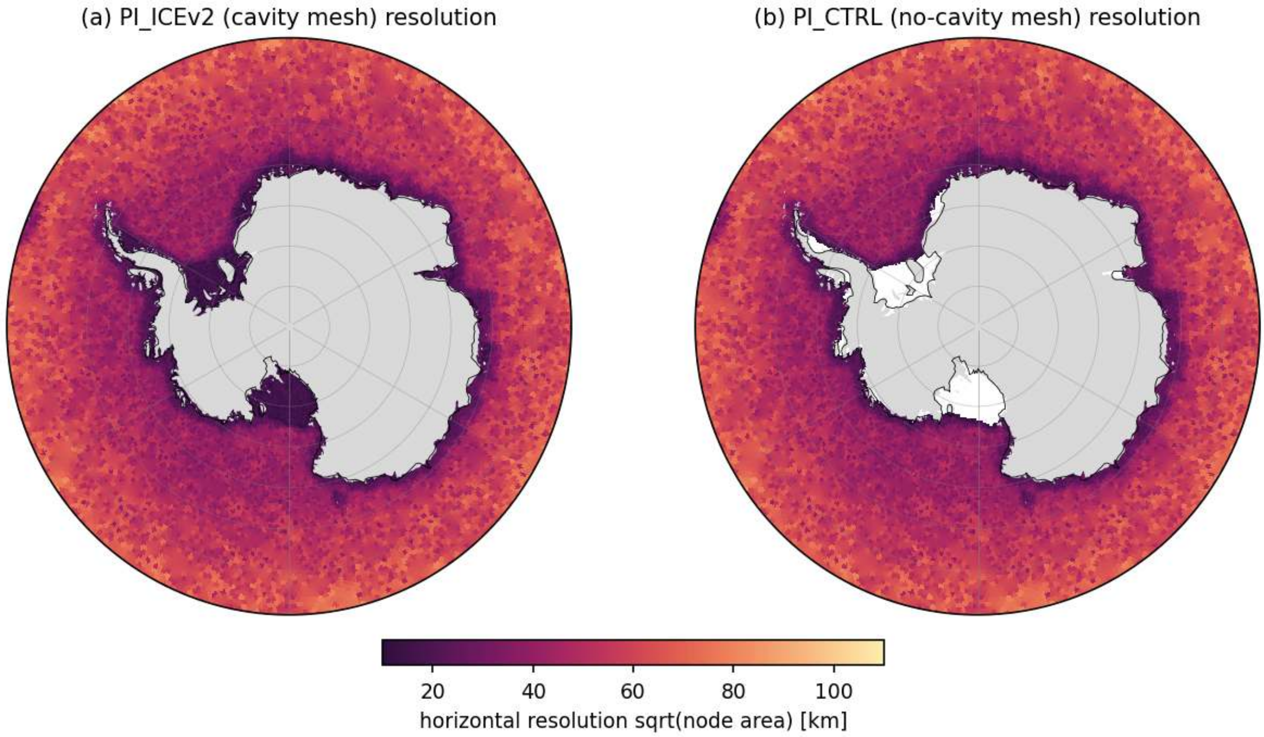

Figure 2: geom_resolution_so (overview). Horizontal resolution of the two FESOM2 meshes south of 55S, shown as the square root of the node area in km on a 0.25 degree grid. (a) PI_ICEv2 (cavity mesh, 132273 nodes). (b) PI_CTRL (no-cavity mesh, 126858 nodes). Land (grey) follows the Natural Earth coastline, so ice shelves appear white in (b) where the no-cavity mesh has no ocean. The meshes are identical north of 60S (113584 shared nodes). Area-weighted resolution is 16 km inside the cavities, 29 to 30 km on the open Antarctic shelf (depth below 1000 m, south of 60S), 49 km over the deep Southern Ocean south of 60S, 67 km between 50S and 60S and 76 km globally. South of 60S the median node spacing is 24 km in PI_ICEv2 and 38 km in PI_CTRL.

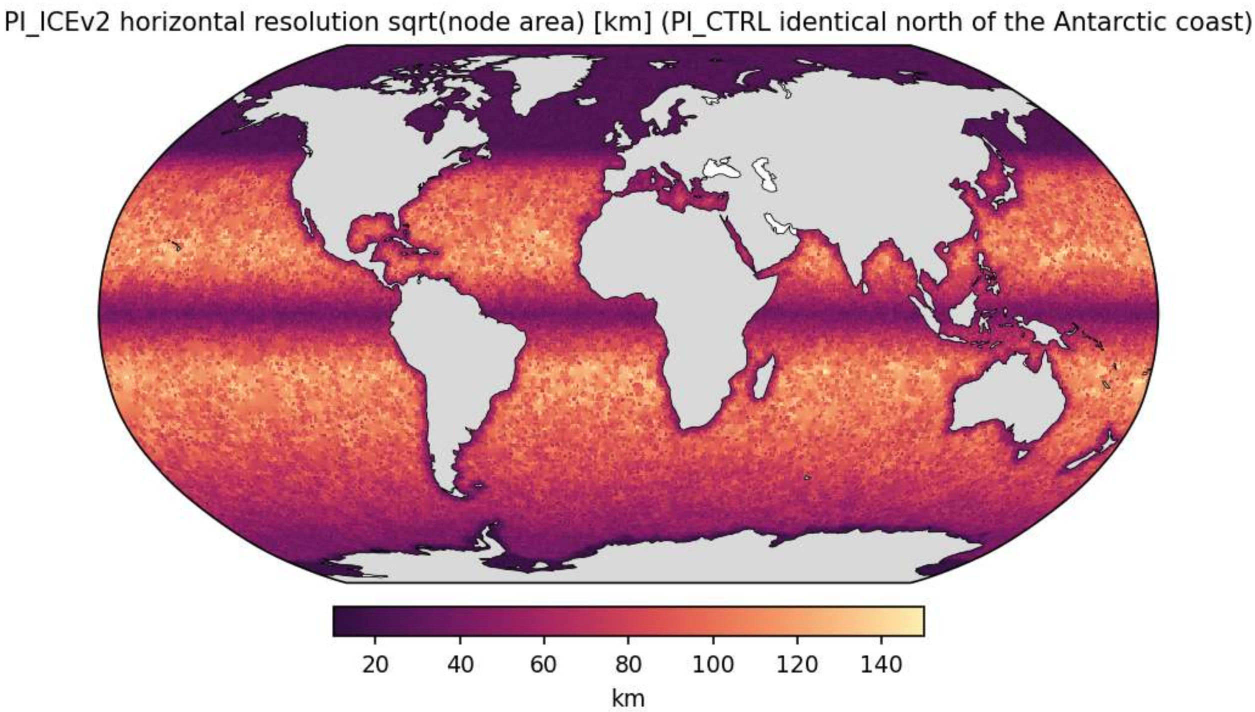

Figure 3: geom_resolution_global (overview). Global map of the PI_ICEv2 horizontal resolution (square root of node area, km). The CORE2-like mesh has 7 km minimum spacing, about 150 km in the subtropical gyres and refinement along coasts, the equator and the Arctic and Antarctic margins. PI_CTRL is identical outside the Antarctic margin.

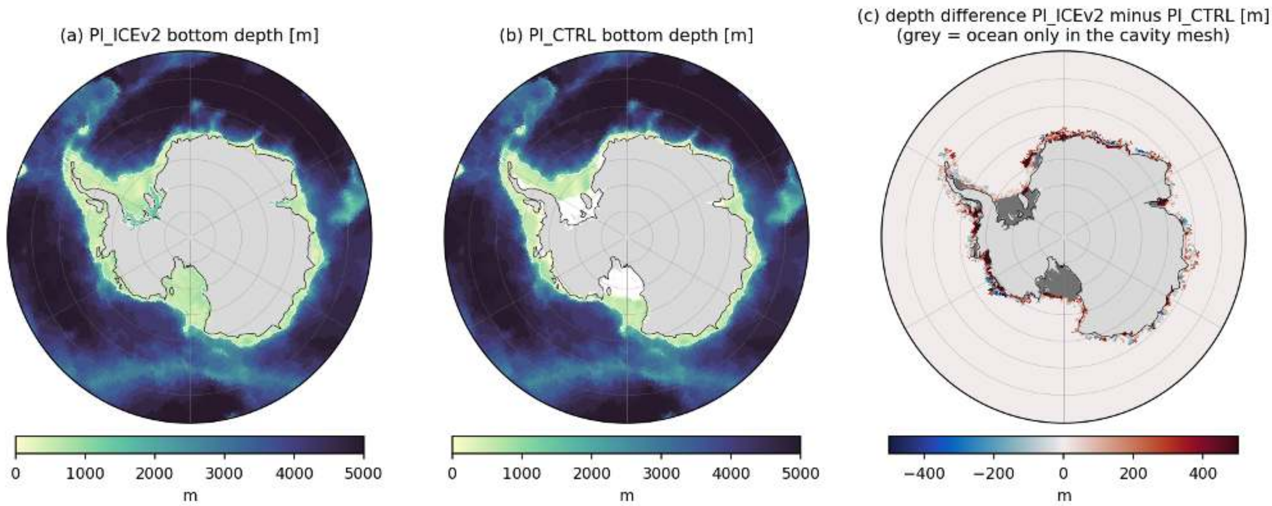

Figure 4: geom_bathymetry (overview). Bottom depth of the two meshes on a common 0.25 degree grid south of 55S. (a) PI_ICEv2 including the sea floor under ice shelves. (b) PI_CTRL. (c) Depth difference PI_ICEv2 minus PI_CTRL where both meshes have ocean; grey marks grid cells that are ocean only in the cavity mesh (mainly the Filchner-Ronne and Ross cavities, 8 percent of the cavity-mesh ocean cells on this grid south of 55S). White line in (a) and (b) is the 1000 m isobath. Where both meshes are wet the depths agree to 13 m on average with an rms difference of 86 m; only 5 percent of common cells differ by more than 100 m, all along the coast and shelf break. Both meshes use the same 47 z-levels from 5 m thickness at the surface to 250 m below 2400 m and a maximum depth of 6000 m.

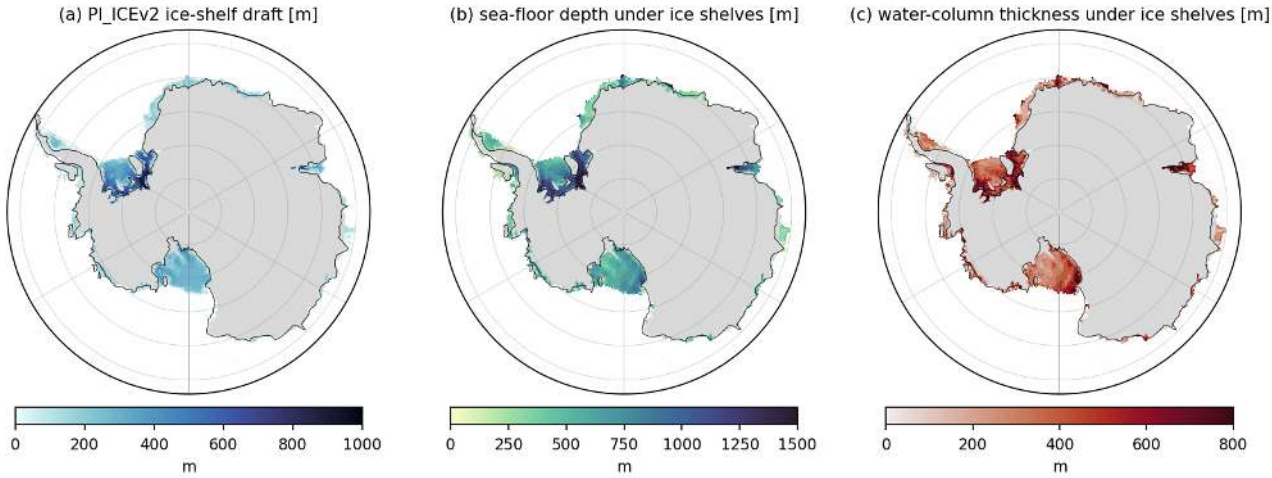

Figure 5: geom_draft_wct (overview). PI_ICEv2 cavity geometry on a 0.25 degree grid. (a) Ice-shelf draft (depth of the ice base, m). (b) Sea-floor depth under the ice shelves. (c) Water-column thickness, the difference of (b) and (a). The 6592 cavity nodes cover 1.425 million km2 with an area-weighted mean draft of 265 m (maximum 1040 m under the Filchner-Ronne ice shelf), a mean sea-floor depth of 631 m and a mean water-column thickness of 366 m (minimum 25 m, 2 percent of the cavity area thinner than 100 m). The cavity volume is 566 thousand km3, about 0.04 percent of the global ocean volume. With partial cells, every cavity column has at least 3 wet layers.

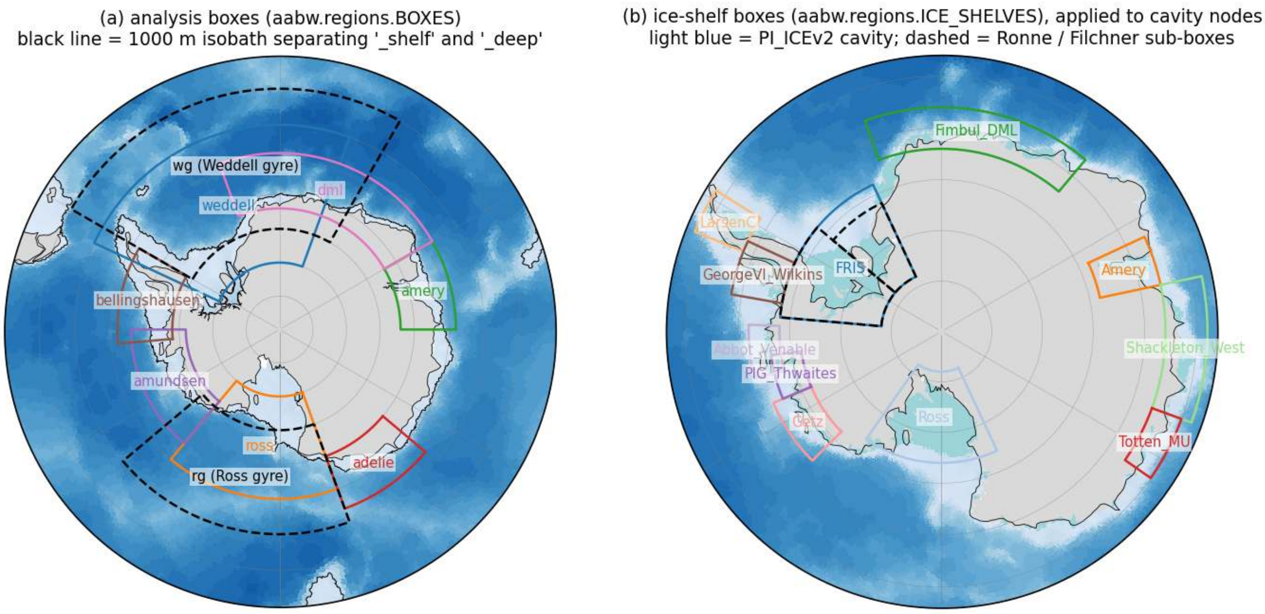

Figure 6: geom_regions (overview). Analysis regions used throughout the report. (a) Lon-lat boxes of aabw.regions.BOXES drawn over the PI_ICEv2 bathymetry; the black line is the 1000 m isobath, which separates the "_shelf" (shallower) and "_deep" regions of each box. Dashed boxes are the Weddell gyre (wg) and Ross gyre (rg) boxes; the so50, so30 and ssa (south of 60S) regions are latitude bands and are not drawn. (b) Ice-shelf boxes of aabw.regions.ICE_SHELVES, which are applied to cavity nodes only (light blue); Ronne and Filchner (dashed) are sub-boxes of FRIS.

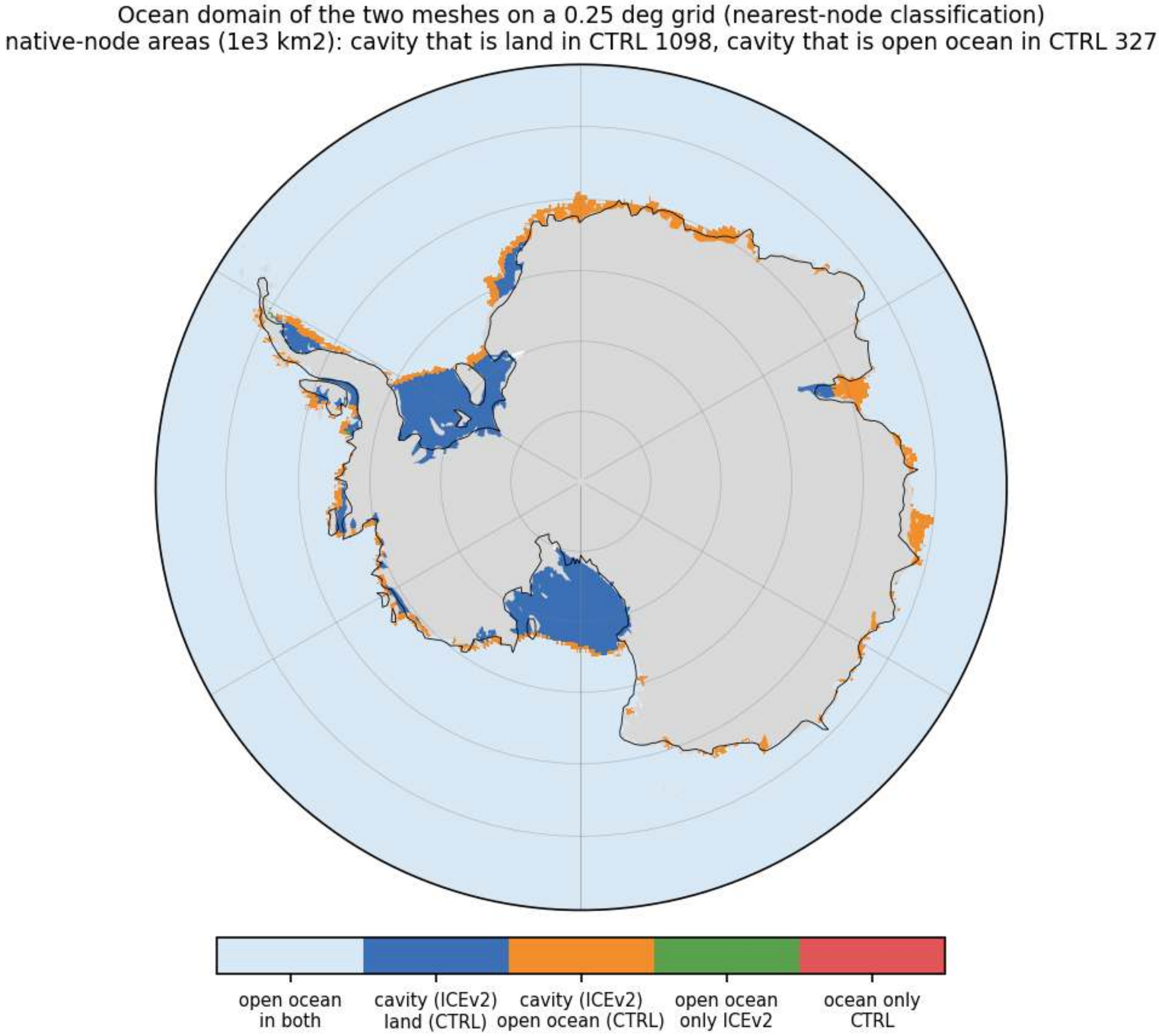

Figure 7: geom_domain_diff (overview). Ocean domain of the two meshes on a 0.25 degree grid south of 60S, classified by nearest node. Dark blue marks cavity area of PI_ICEv2 that is land in PI_CTRL (Filchner-Ronne, Ross, most of the Amundsen and Bellingshausen shelves, half of the Amery and two thirds of Larsen C). Orange marks cavity area that is open ocean in PI_CTRL (almost all of the Dronning Maud Land shelves, Shackleton and West, Totten and Moscow University, half of the Amery, one third of Larsen C). Using the native nodes, 1098 thousand km2 of the cavity area is land in PI_CTRL and 327 thousand km2 is open ocean there. Green and red (open ocean in one mesh only, outside cavities) are negligible (11 and 0 thousand km2). The grid-cell areas in the colourbar categories are inflated by the nearest-node tolerance and are not used for numbers.

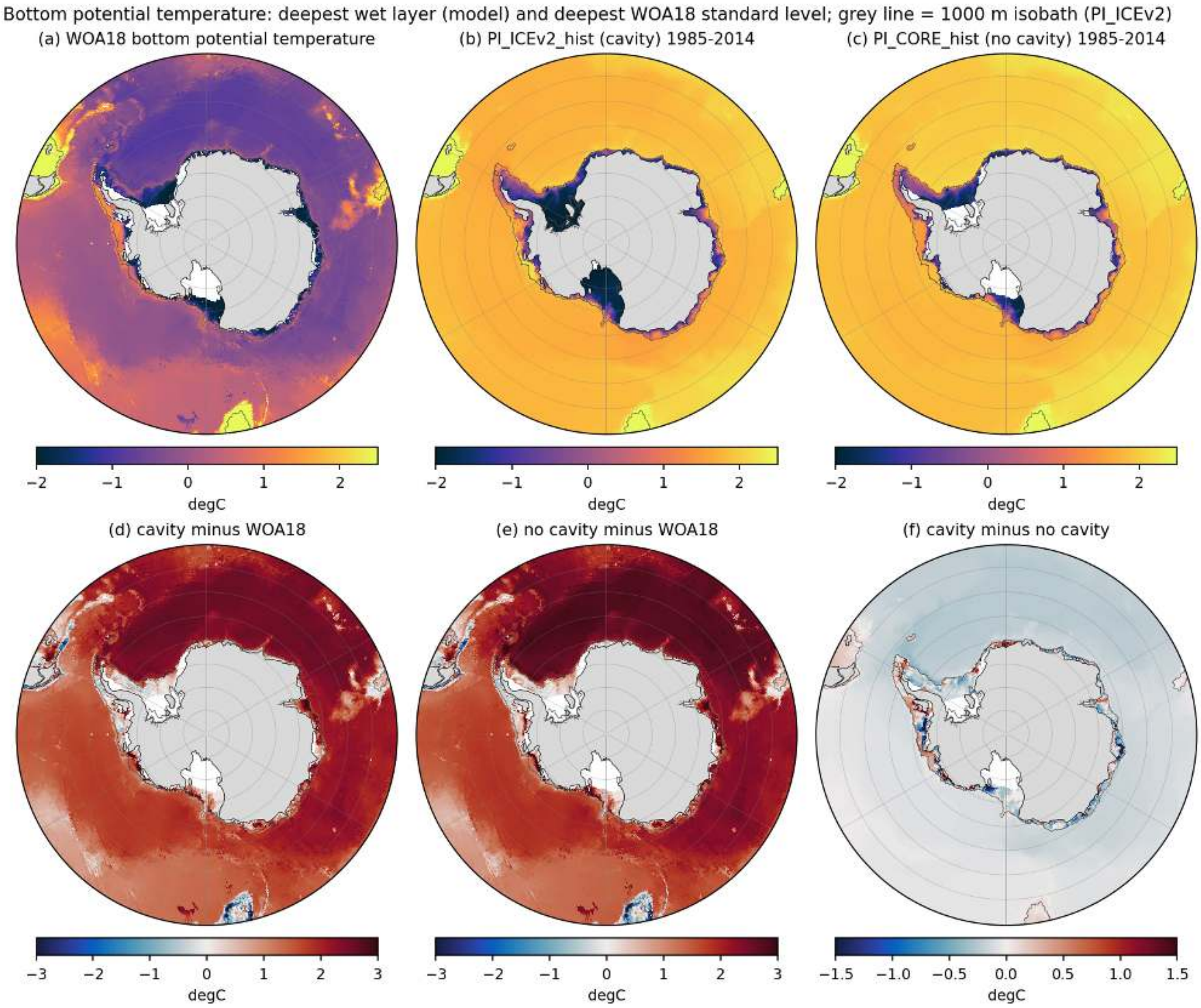

Figure 8: eval_bottomT (overview). Bottom potential temperature south of 50S, 1985-2014 mean, on a 0.25 degree grid. (a) WOA18 (2005-2017 objectively analysed annual mean, deepest valid standard level, converted from in-situ to potential temperature). (b) PI_ICEv2_hist (cavity), (c) PI_CORE_hist (no cavity), both taken at the deepest wet layer of the native mesh. (d, e) Model minus WOA18. (f) Cavity minus no cavity. Grey line is the 1000 m isobath of PI_ICEv2. Both runs fill the deep Southern Ocean with 1.6 to 2.0 degC water instead of AABW near or below 0 degC. The area-mean warm bias of the deep (>1000 m) Weddell box is 2.4 degC (cavity, bottom 1.74 degC versus WOA18 -0.64 degC) and 2.6 degC (no cavity, 2.01 degC). In the deep Ross box the bias is 1.7 and 1.8 degC (model 1.65 and 1.77 degC, WOA18 -0.06 degC). Over the whole Antarctic shelf (<1000 m, south of 60S) the bottom is 0.9 (cavity) and 1.1 degC (no cavity) too warm; cold shelf water colder than -1.5 degC survives only in the cavity run in front of the Filchner-Ronne and Ross ice shelves. The cavity run is 0.2 to 0.3 degC cooler than the no-cavity run in the deep Weddell and Ross basins and up to 1 degC cooler on the Ross and Weddell shelves.

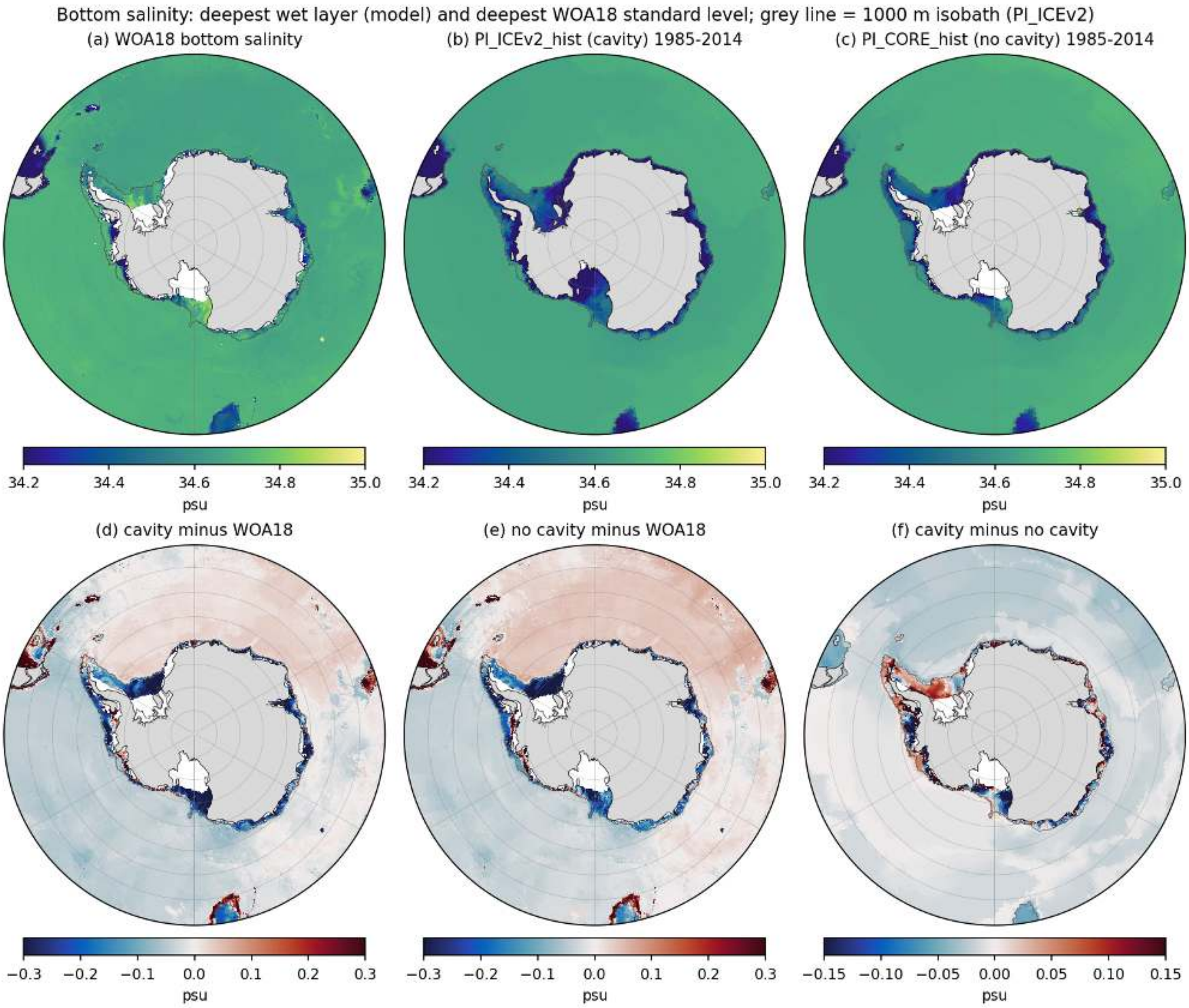

Figure 9: eval_bottomS (overview). As eval_bottomT.png for bottom salinity. Deep basin salinities are close to WOA18 (biases within 0.04 psu, the Weddell abyss is 0.01 to 0.04 psu too salty and the Ross abyss 0.03 to 0.04 psu too fresh). The shelves are too fresh everywhere: area-mean bottom salinity bias on the Antarctic shelf south of 60S is -0.20 psu (cavity) and -0.18 psu (no cavity), on the Weddell shelf -0.21 and -0.25 psu, on the Ross shelf -0.25 and -0.17 psu, and in Prydz Bay and Dronning Maud Land -0.2 to -0.27 psu. No shelf region reaches the 34.6 to 34.8 psu of observed HSSW. The cavity run is fresher than the no-cavity run by 0.1 to 0.15 psu on the Ross and Amundsen shelves and saltier in the southern Weddell Sea in front of Ronne (panel f).

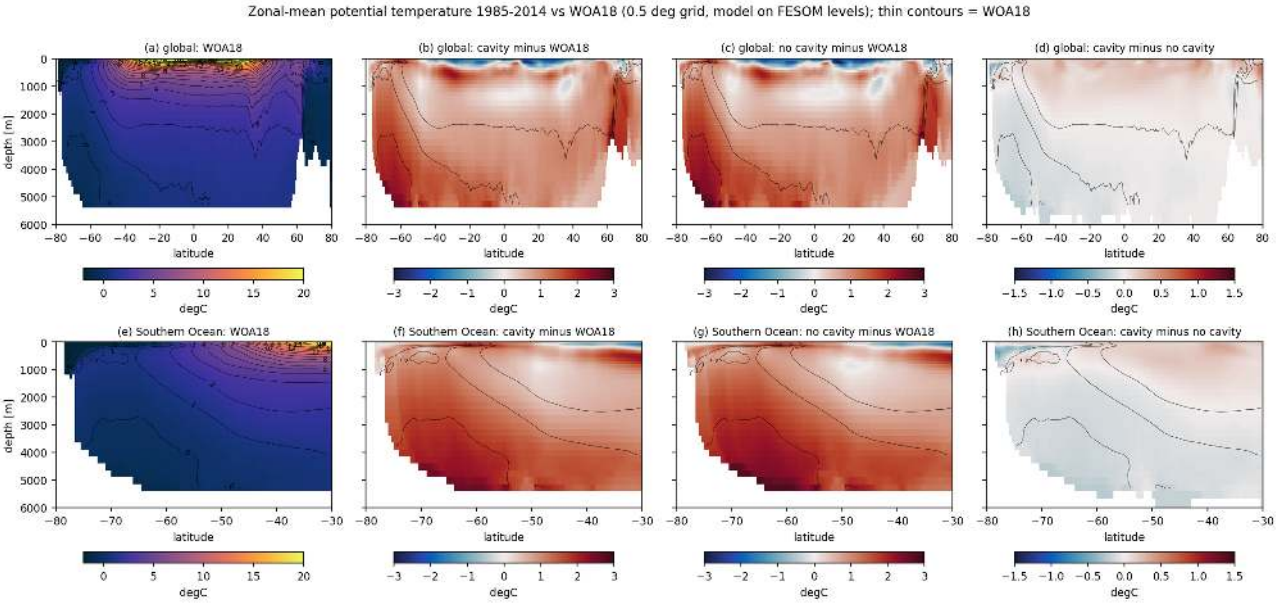

Figure 10: eval_zonal_T (overview). Zonal-mean potential temperature, 1985-2014 versus WOA18, both on the 0.5 degree grid and the 47 FESOM layers. Upper row global (80S to 80N), lower row Southern Ocean (80S to 30S). (a, e) WOA18 with contours. (b, f) Cavity minus WOA18, (c, g) no cavity minus WOA18, computed where both are valid. (d, h) Cavity minus no cavity. Thin contours are the WOA18 0, 1 and 2 degC isotherms. Below 2000 m the Southern Ocean is 1.4 (cavity) and 1.5 degC (no cavity) too warm, the global deep ocean 1.1 degC; the 0 to 1000 m layer south of 30S is 0.8 and 0.65 degC too warm with a cold bias in the surface 100 m near the coast. The cavity run is cooler than the no-cavity run by 0.1 to 0.3 degC everywhere south of 50S below 500 m and by up to 1 degC at 200 to 600 m south of 70S.

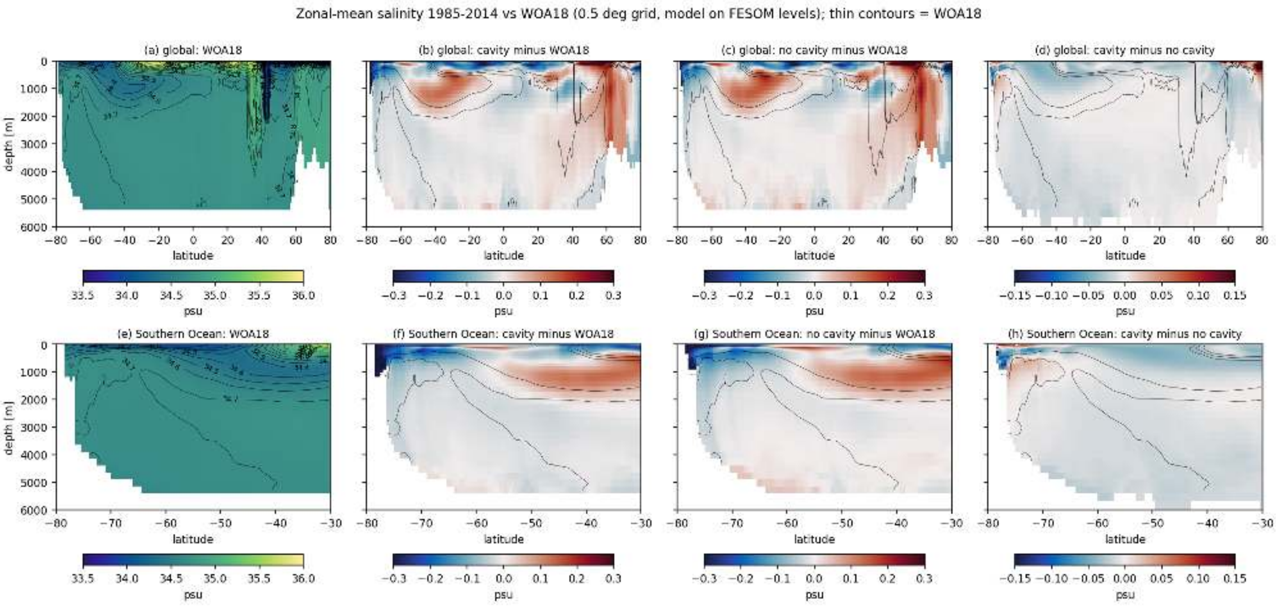

Figure 11: eval_zonal_S (overview). As eval_zonal_T.png for salinity (bias range 0.3 psu, difference range 0.15 psu; WOA18 34.6, 34.7, 34.8 contours). The deep ocean salinity bias is within 0.02 psu in both runs. The main biases are a fresh surface layer south of 60S (-0.2 to -0.5 psu in the upper 200 m near the coast) and a salty Circumpolar Deep Water core (+0.1 to 0.2 psu at 500 to 1500 m, 50S to 30S). The cavity run is fresher than the no-cavity run by up to 0.1 psu in the upper 500 m south of 65S and slightly saltier at 500 to 1500 m south of 75S.

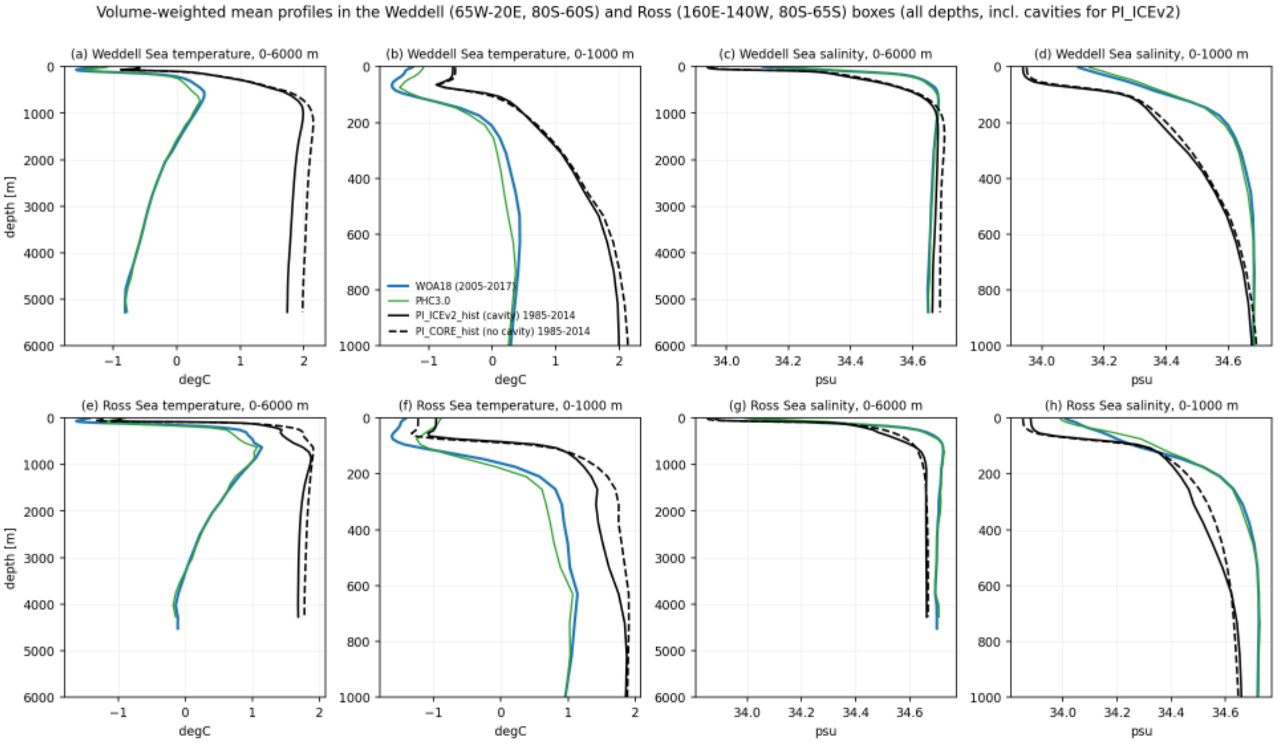

Figure 12: eval_profiles (overview). Volume-weighted mean profiles of potential temperature and salinity in the Weddell box (65W to 20E, 80S to 60S) and Ross box (160E to 140W, 80S to 65S) for WOA18 (blue), PHC3.0 (green, converted to potential temperature), PI_ICEv2_hist (black solid) and PI_CORE_hist (black dashed), 1985-2014. Left column full depth, right column upper 1000 m. Observations show the 0.3 to 1 degC CDW maximum at 500 to 1000 m and cooling to -0.65 degC (Weddell) and -0.15 degC (Ross) at 4000 m. Both runs instead have 1.7 to 2.1 degC from 1000 m to the bottom, i.e. 2.4 degC (Weddell) and 1.8 degC (Ross) too warm at 4000 m, and the warm bias already starts at 150 m. Salinity below 1000 m is within 0.01 (Weddell) to 0.04 psu (Ross) of WOA18, while the upper 300 m is 0.2 to 0.3 psu too fresh. The cavity run is 0.1 to 0.3 degC cooler than the no-cavity run throughout the water column in both basins.

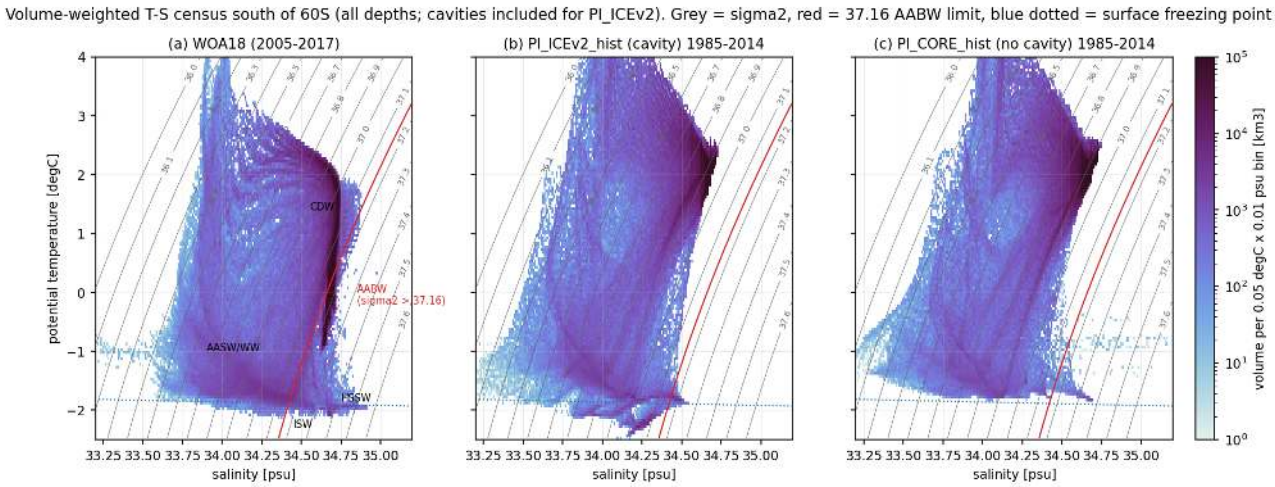

Figure 13: eval_TS (overview). Volume-weighted T-S census of all water south of 60S (0.05 degC x 0.01 psu bins, log colour scale, km3 per bin) for (a) WOA18 on the 0.5 degree grid and FESOM layers, (b) PI_ICEv2_hist and (c) PI_CORE_hist, 1985-2014, cavities included in (b). Grey lines are sigma2 (reference 2000 dbar), the red line sigma2 = 37.16 separates AABW, the blue dotted line is the surface freezing point. In WOA18 42 percent of the volume south of 60S has sigma2 above 37.16 and 88 percent above 37.0. In the models these fractions are 0.07 and 0.6 percent (cavity) and 0.04 and 0.2 percent (no cavity); the dense corner of the diagram is empty and the deep water sits at 1.7 to 2.0 degC and 34.65 to 34.7 psu (sigma2 about 36.9). Water colder than -1.9 degC (Ice Shelf Water) exists only in the cavity run (270 thousand km3 versus 169 thousand km3 in WOA18 and none in the no-cavity run). HSSW-like water (colder than -1.7 degC and saltier than 34.5 psu) is 0.27 percent of the volume in WOA18 and effectively absent in both runs.

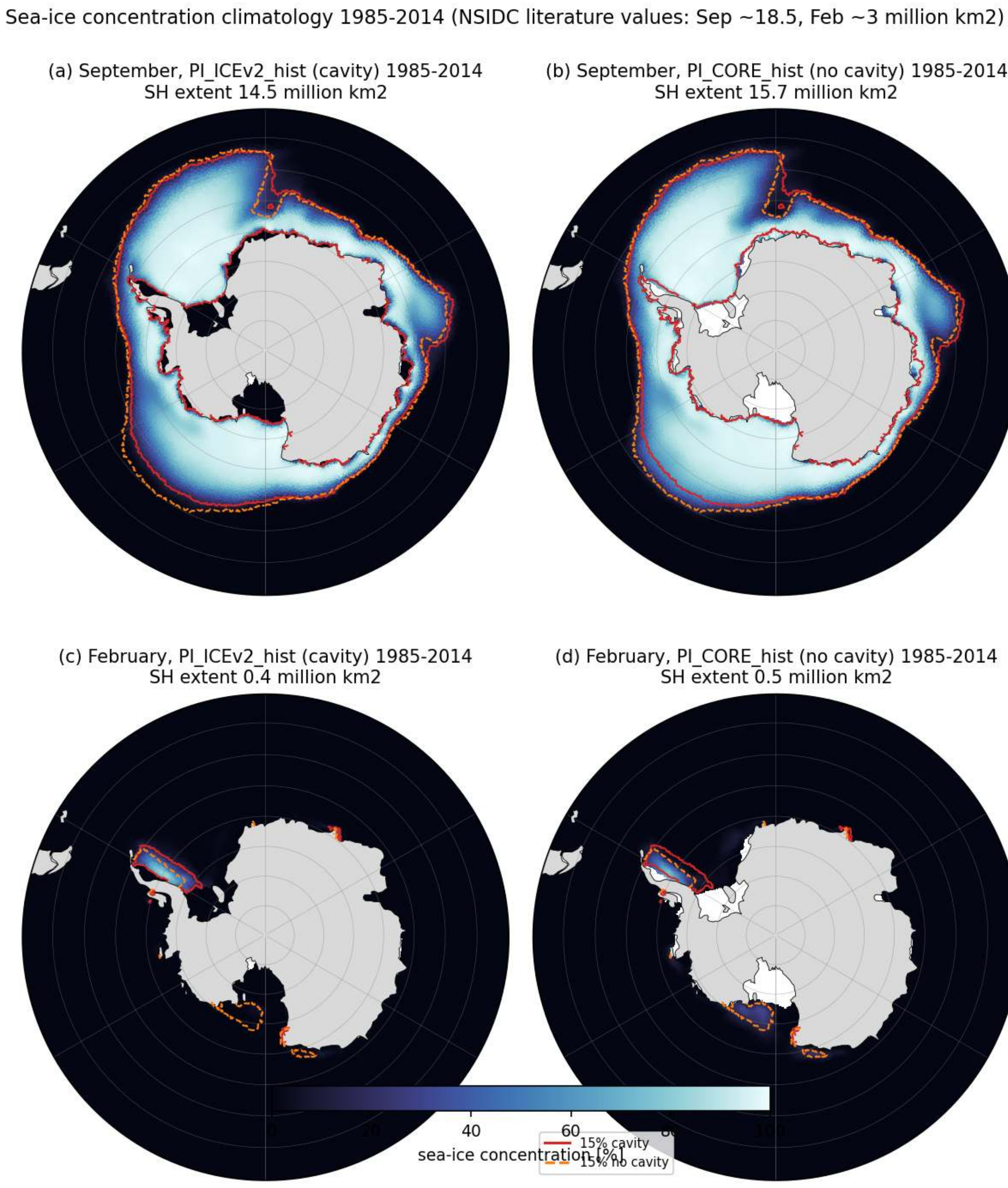

Figure 14: eval_seaice (overview). Sea-ice concentration climatology 1985-2014 in September (a, b) and February (c, d) for PI_ICEv2_hist (cavity) and PI_CORE_hist (no cavity). Red solid and orange dashed lines are the 15 percent contours of the cavity and no-cavity runs, drawn on all panels. September SH extent is 14.5 (cavity) and 15.7 million km2 (no cavity) against about 18.5 million km2 in the NSIDC satellite climatology (literature value); February extent is 0.4 and 0.5 million km2 against about 3 million km2. Winter ice is missing mainly in the Pacific sector (Amundsen and Ross seas north of 65S) and in the eastern Weddell Sea; summer ice survives only in the western Weddell Sea. The cavity run has less winter ice than the no-cavity run in the Indian and Pacific sectors.

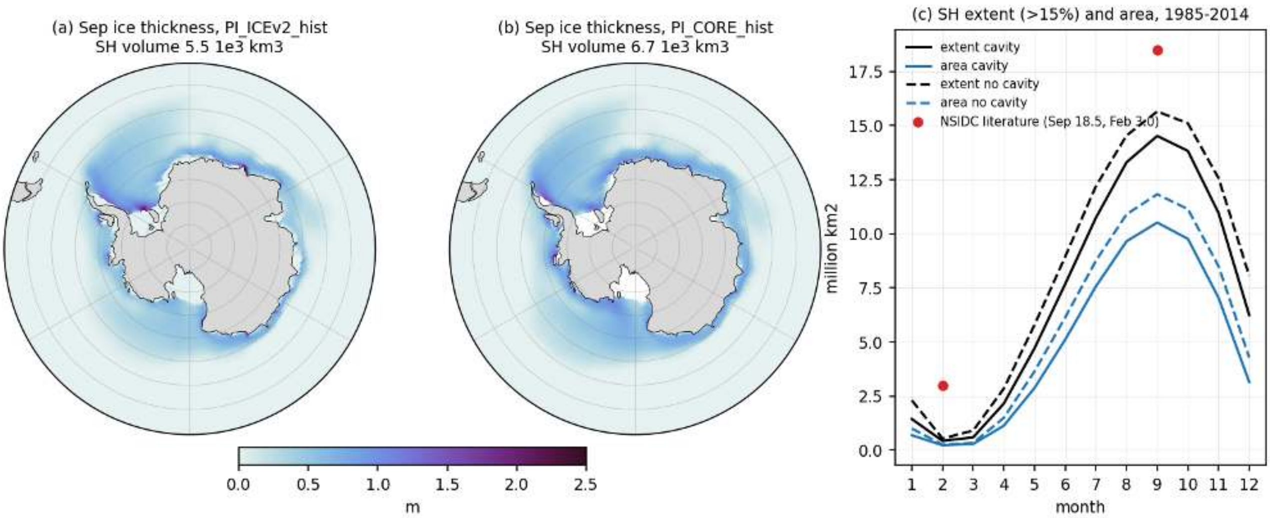

Figure 15: eval_seaice_thickness_cycle (overview). (a, b) September grid-mean sea-ice thickness 1985-2014 for the cavity and no-cavity runs; SH September volume 5.5 and 6.7 thousand km3. (c) Seasonal cycle of SH sea-ice extent (>15 percent, black) and area (blue) for both runs, with the NSIDC literature values for September (18.5) and February (3.0 million km2) as red dots. The model seasonal cycle is too large in amplitude: winter extent is 15 to 20 percent low and summer ice nearly vanishes.

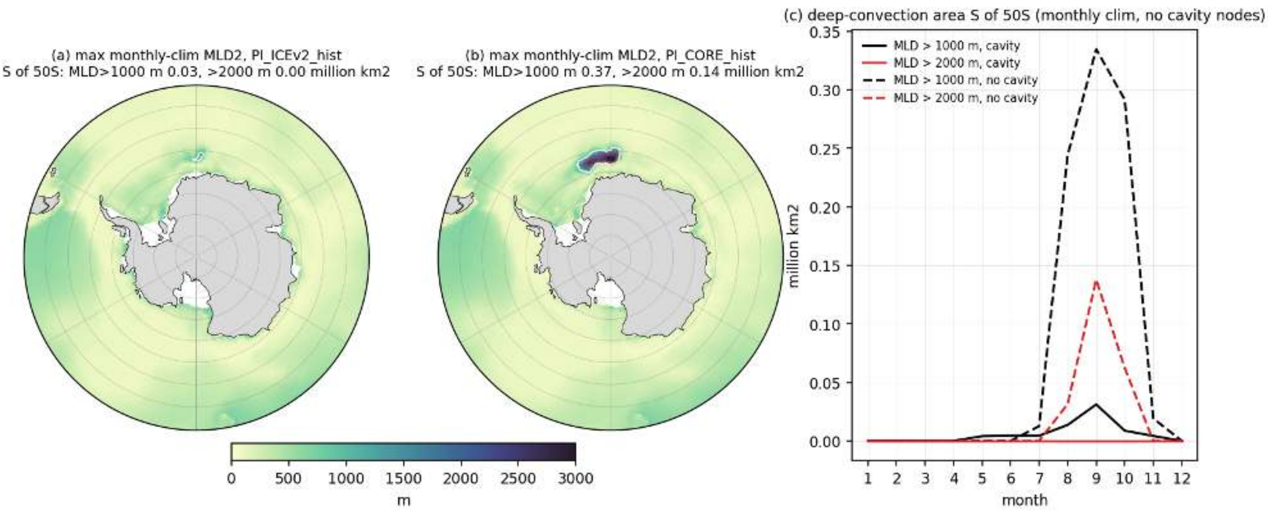

Figure 16: eval_mld (overview). Deep convection indicator. (a, b) Annual maximum of the monthly climatological mixed-layer depth MLD2 (1985-2014, cavity nodes excluded) with 1000 m (white) and 2000 m (red) contours. (c) Seasonal cycle of the open-ocean area south of 50S with climatological MLD deeper than 1000 m and 2000 m. The no-cavity run has a persistent winter open-ocean convection site in the eastern Weddell gyre near Maud Rise (centroid 9W, 66.5S) with climatological MLD above 2000 m over 0.14 million km2 (0.37 million km2 above 1000 m); per-year September areas with MLD above 2000 m are 0.2 million km2 on average, 0.48 million km2 at most, and occur in all 30 years. In the cavity run the same site convects only intermittently (15 of 30 years, mean 0.02, max 0.12 million km2) and the climatological MLD stays below 2000 m. Shelf convection does not reach the bottom of the deep basins in either run.

4 Atmospheric forcing and response (ECHAM6)

Findings

4.1 Findings, theme atmos (ECHAM6 atmosphere as driver of and response to Southern Ocean changes)

Data: monthly ECHAM6 `echam` stream of all 10 experiments extracted with cdo into `cache/echam/<EXP>_atmos_mon.nc` (codes 167 169 142 143 165 166 180 181 146 147 176 177 97 182 172 171 plus psl computed by cdo sealevel-pressure), region-mean monthly indices in `<EXP>_atmos_idx.nc`, seasonal period means on the T63 grid in `<EXP>_atmos_pmean.nc`. Scripts in `analyses/atmos/` (extract_echam.py, compute_indices.py, atmos_lib.py, plot_*.py). Periods: early 1851-1880, pd 1985-2014, mid 2041-2070, end 2071-2100, far 2171-2200.

4.1.1 Numbered findings

1. Mean-state differences between the cavity and no-cavity atmospheres are confined to the sea-ice zone and are largest in the Ross Sea. At pd the cavity set is warmer by 0.45 K (annual) over the ocean south of 60S and by 1.7 K (annual) and 2.7 K (JJA) over the Ross Sea, with local JJA maxima above 10 K at the Ross Ice Shelf front. Winter turbulent heat loss is 17 W/m2 larger in the Ross Sea and the net winter heat loss is 72 vs 55 W/m2 (cavity vs no cavity). The Weddell Sea shows no robust mean-state difference (-0.3 K JJA, -3 W/m2). These differences go together with a 1.3 million km2 smaller September sea-ice area in the cavity run (10.6 vs 11.9 million km2) and a 0.15 lower winter ice fraction in the Ross Sea. The atmospheric signal is therefore a response to the ocean-ice state (ice-shelf melt water and cavity circulation reduce Ross Sea ice) rather than an atmospheric cause.

2. Large-scale circulation is the same in both sets. The SH westerly jet (maximum zonal-mean zonal wind stress) is 0.200 Pa at 51.1S (cavity) and 0.198 Pa at 50.6S (no cavity) at pd. The difference never exceeds 0.01 Pa or 0.7 deg latitude in any 30-year period, below two standard deviations of the 11-year means in the historical period (0.004 Pa, 0.7 deg). The SAM-like index, normalised zonal-mean SLP 40S minus 65S, also agrees within noise. The 30-year pd SLP difference map shows a wave pattern of up to 3 hPa, which is internal variability of unpaired runs and should not be interpreted as a cavity effect.

3. Forced circulation changes are moderate and scenario-dependent. From 1851-1880 to pd the jet strengthened from 0.189 to 0.200 Pa and moved from 49.7S to 51.1S, with the SAM index rising by 1 standard deviation. Under SSP5-8.5 the jet reaches 0.222 Pa at 53.2S by 2071-2100 and 0.224 Pa at 53.6S by 2171-2200 (trend 2015-2100: +0.0027 Pa per decade and -0.26 deg per decade in the cavity set, +0.0020 Pa and -0.22 deg in the no-cavity set). SSP3-7.0 reaches similar values by 2200. In SSP1-2.6 the jet stays at pd strength and latitude (0.197 Pa, 50.9S at far) and the SAM index falls back to +0.5. The zonal wind stress over 50-60S increases by 0.02 to 0.04 Pa (10 to 20 percent) in the high scenarios, with weakening north of 45S (poleward shift). These changes strengthen Ekman upwelling and northward Ekman transport south of the jet and are identical in both sets.

4. Surface air temperature. Global warming relative to pd is 1.1/0.8 K (end/far) in SSP1-2.6, 2.1/2.6 K in SSP2-4.5, 3.2/6.2 K in SSP3-7.0 and 3.7/7.4 K in SSP5-8.5 for the cavity set; the no-cavity set warms 0.05 K more at end and 0.35 K more at far in the two strong scenarios (7.7 K global at far SSP5-8.5). South of 60S (land and ocean) the warming is 3.6/7.1 K (cavity) vs 4.0/8.2 K (no cavity) in SSP5-8.5 and 3.2/6.2 vs 3.5/7.0 K in SSP3-7.0, so the cavity set warms 0.3 K less at end and 0.9 to 1.1 K less at far. Over the 50-70S ocean the numbers are 4.9 vs 5.7 K at far SSP5-8.5. Antarctic amplification ratios (warming south of 60S divided by global) are 1.0 to 1.2 at end and 1.0 to 1.7 at far, weakest (0.97 to 1.07) in SSP5-8.5 and strongest (1.6 to 1.7) in SSP1-2.6 where the global signal is small and the delayed Southern Ocean response keeps rising. Antarctic land warms 4.3/9.1 K (cavity) vs 4.5/10.0 K (no cavity) in SSP5-8.5.

5. Why the cavity set warms less in the south. In the historical period the cavity set has less sea ice and is warmer (+0.44 K south of 60S). As warming removes the extra ice of the no-cavity set, the no-cavity set gains the ice-albedo and insulation feedback that the cavity set has partly already used, so the sign of the difference reverses around 2100 in SSP3-7.0 and SSP5-8.5 (cavity minus no cavity south of 60S: +0.30 K pd, -0.04 K end, -0.77 K far in SSP5-8.5; the winter ice-fraction difference goes from -0.05 to 0.00). Ice-shelf melt water increase in the cavity runs, which cools and freshens the surface and favours sea ice, can add to this but is not separable from the atmosphere alone. The difference at far (0.8 to 1.1 K) is five times the two-sigma noise of 11-year means (0.16 K) and is robust; the end-of-century difference (0.3 K) is marginal.

6. Surface heat loss relevant for dense-water formation. Winter (JJA) net heat loss of the ocean-ice column rises in the Weddell Sea from 63 W/m2 (1851-1880) to 73 W/m2 (pd) and 80 to 82 W/m2 by 2041-2100

in all scenarios (trend 1985-2100 about +1 W/m2 per decade), then stays at 82 to 88 W/m2 in SSP1-2.6 to SSP3-7.0 and falls back to 77 W/m2 by 2171-2200 in SSP5-8.5. The Ross Sea winter heat loss grows strongly under high forcing (72 to 98 W/m2 at end and 100 W/m2 at far in SSP5-8.5 cavity; 55 to 86 and 101 W/m2 no cavity). These increases are driven by sea-ice thinning and opening, not by colder air, since air temperature rises. Over 50-70S the winter heat loss decreases (92 to 73 W/m2 at far SSP5-8.5) and the annual mean flux turns into a net ocean heat gain after 2100 (-7 W/m2 at far SSP5-8.5), which means the Southern Ocean surface is heated from above and the buoyancy forcing for dense-water formation weakens north of 60S while the coastal seas still lose heat. The cavity effect on these series is a constant offset in the Ross Sea (+13 W/m2 historical) that disappears under SSP5-8.5 by 2150, and no offset in the Weddell Sea.

7. Freshwater input from the atmosphere. P minus E over the 50-70S ocean is 1.37 mm/day in 1851-1880, 1.48 mm/day at pd and reaches 1.90 mm/day at end and 2.32 mm/day at far under SSP5-8.5 (+57 percent, trend 0.048 mm/day per decade 1985-2100). SSP3-7.0 reaches 2.20 mm/day at far, SSP2-4.5 1.73 and SSP1-2.6 1.54. Almost all of this is precipitation (2.68 to 3.53 mm/day over 50-70S) while evaporation stays at 1.2 mm/day. South of 60S the increase is from 1.28 to 2.13 mm/day. Weddell and Ross P minus E rise from about 1.05 to 1.8 and 1.5 mm/day at far SSP5-8.5. Converted to a freshwater flux this is an extra 0.85 mm/day over 50-70S, equivalent to roughly 0.11 Sv over that belt, which is of the same order as the projected ice-shelf melt changes and acts to stratify the surface ocean in the same direction. Cavity minus no-cavity differences in P, E and P minus E are 0.01 to 0.04 mm/day and within noise.

8. Seasonal cycle of the cavity effect (pd). The Ross Sea cavity minus no-cavity net heat flux difference is -20 W/m2 in June (more loss with cavities), dominated by sensible heat (-12 W/m2) with latent -7 W/m2, and +20 W/m2 in December (more shortwave absorbed by open water). Around the whole continent the cavity set loses 3 to 4 W/m2 more in May to August and absorbs 6 W/m2 more shortwave in November and December. The Weddell Sea difference is within plus or minus 5 W/m2 and changes sign, so cavities do not change the Weddell Sea surface forcing seen by the atmosphere in this model.

9. Scenario response maps. Under SSP5-8.5 both sets show the same pattern: maximum warming over the retreating sea-ice zone, SLP decrease of 3 to 4 hPa around Antarctica and increase near 40S, westerly stress increase south of 50S, increased winter heat loss where ice retreats and more P minus E south of 50S. Response differences between the sets are small relative to the responses except for the Ross Sea (cavity warms 0.7 K less at end and 2.2 K less at far, and loses 18 W/m2 less extra winter heat at far because the no-cavity run has more ice to lose). Differences of the SLP and wind-stress responses are at most 0.9 hPa and 0.005 Pa in regional means.

10. Sea-ice consistency. ECHAM's sea-ice area (code 97 times ocean fraction) matches the FESOM cache to within 1 percent (September 10.6 vs 10.5 million km2 with cavities, 11.9 vs 11.8 without; February 0.3 million km2), so the atmospheric and oceanic diagnostics describe the same state. Note that the September extent from the cache is 13.5 to 14.7 million km2, lower than the 17 million km2 quoted in the guide.

4.1.2 Caveats

The two historical runs are unpaired single realisations; 30-year mean SLP and wind differences of 1 to 3 hPa and 0.5 m/s are internal variability. Only the Ross Sea and circum-Antarctic sea-ice related differences exceed the noise estimate (two sigma of 11-yr means) in the historical period, and the SAT difference after 2100 in SSP3-7.0 and SSP5-8.5. ECHAM heat fluxes are grid-box means over open water plus ice; the flux over the ice fraction includes conductive flux through ice, so "ocean heat loss" here is the loss of the ocean-ice column. Zonal wind stress is the all-surface value (land included) in the zonal means; over the Southern Ocean latitudes land is a small fraction south of 45S except Antarctica. SLP was derived, not output, and is sensitive over the high Antarctic plateau; the SAM index uses 40S and 65S which avoids the plateau. DJF means use December of the previous year so the first DJF of each chain segment (1851 and 2015) is dropped.

4.1.3 Suggested story lines

The atmosphere cannot be the cause of cavity versus no-cavity differences in AABW formation because the winds, SAM, heat loss and P minus E over the Weddell Sea and the Southern Ocean are the same within noise; any AABW differences come from the ocean-ice system (melt water, cavity circulation, sea-ice production). The one region where the atmosphere "sees" the cavities is the Ross Sea, where less winter sea ice with cavities gives 15 to 20 W/m2 more winter heat loss and 2 to 3 K warmer winter air, which should be related to Ross Sea shelf-water properties in the bottom and seaice themes. Under strong forcing the common atmospheric drivers move in a consistent direction for weaker dense-water formation north of 60S (net surface heat gain, +57 percent P minus E, stronger and poleward westerlies) while coastal winter heat loss increases as ice retreats; the cavity set warms about 1 K less south of 60S by 2200 because it starts with less ice, which delays but does not change the response.

Figures

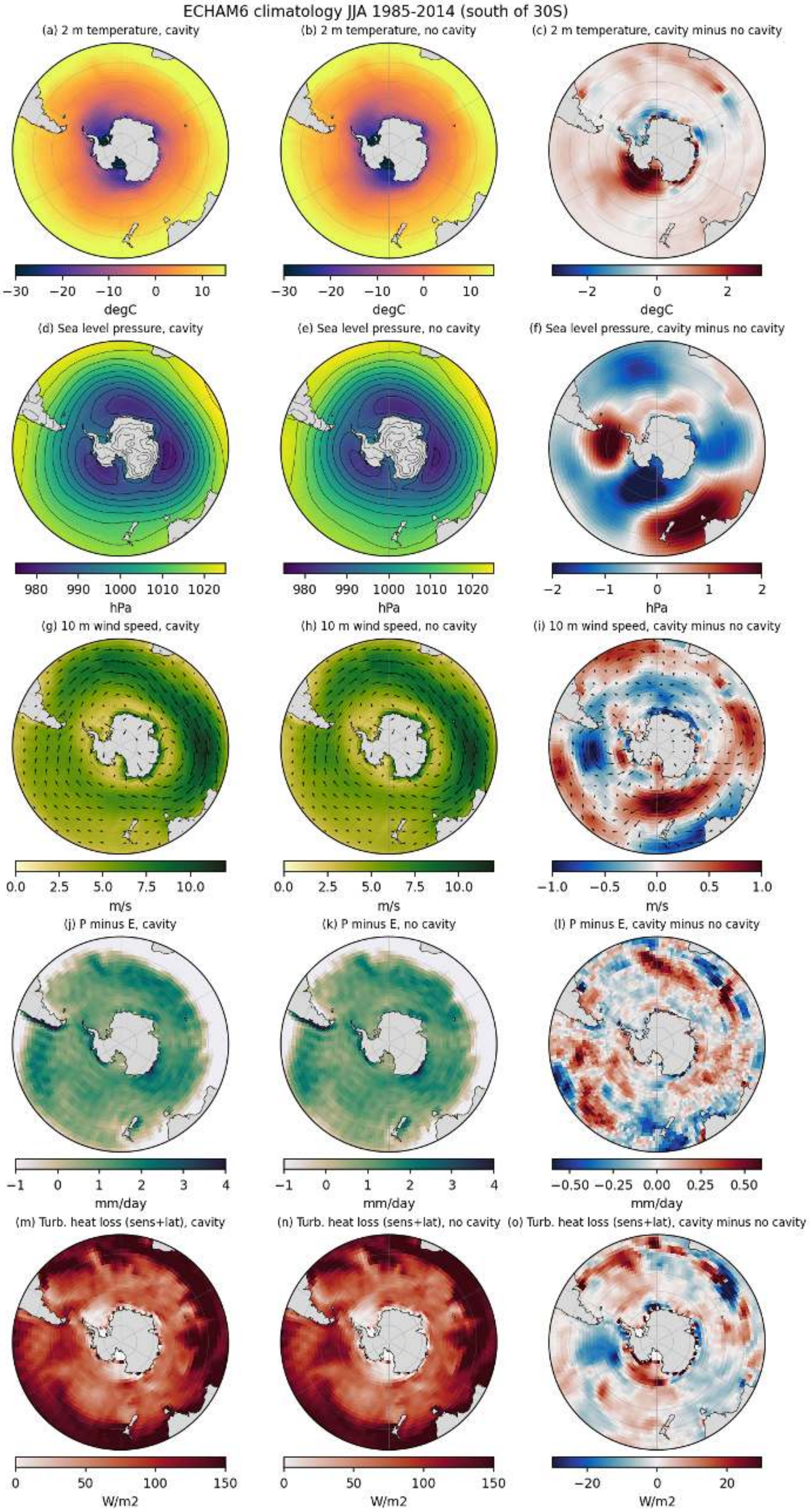

Figure 17: atmos_clim_pd_JJA (atmos). Winter (JJA) climatology 1985-2014 south of 30S for the cavity historical run (left), the no-cavity run (middle) and cavity minus no cavity (right). Rows: 2 m temperature (degC), sea level pressure (hPa, contours every 5 hPa), 10 m wind speed with vectors (m/s), P minus E (mm/day) and turbulent ocean heat loss (sensible plus latent, W/m2, ocean points only). The two runs share the same large-scale climate. The cavity run is warmer over the Ross Sea (region mean +2.7 K in JJA, local maximum 12 K near the Ross Ice Shelf front) and has 17 W/m2 more winter turbulent heat loss there. Over the Weddell Sea the differences are small (-0.3 K, -3 W/m2). The SLP difference pattern (up to 3 hPa) is a wave-like pattern that is of the size of internal 30-year variability rather than a forced signal.

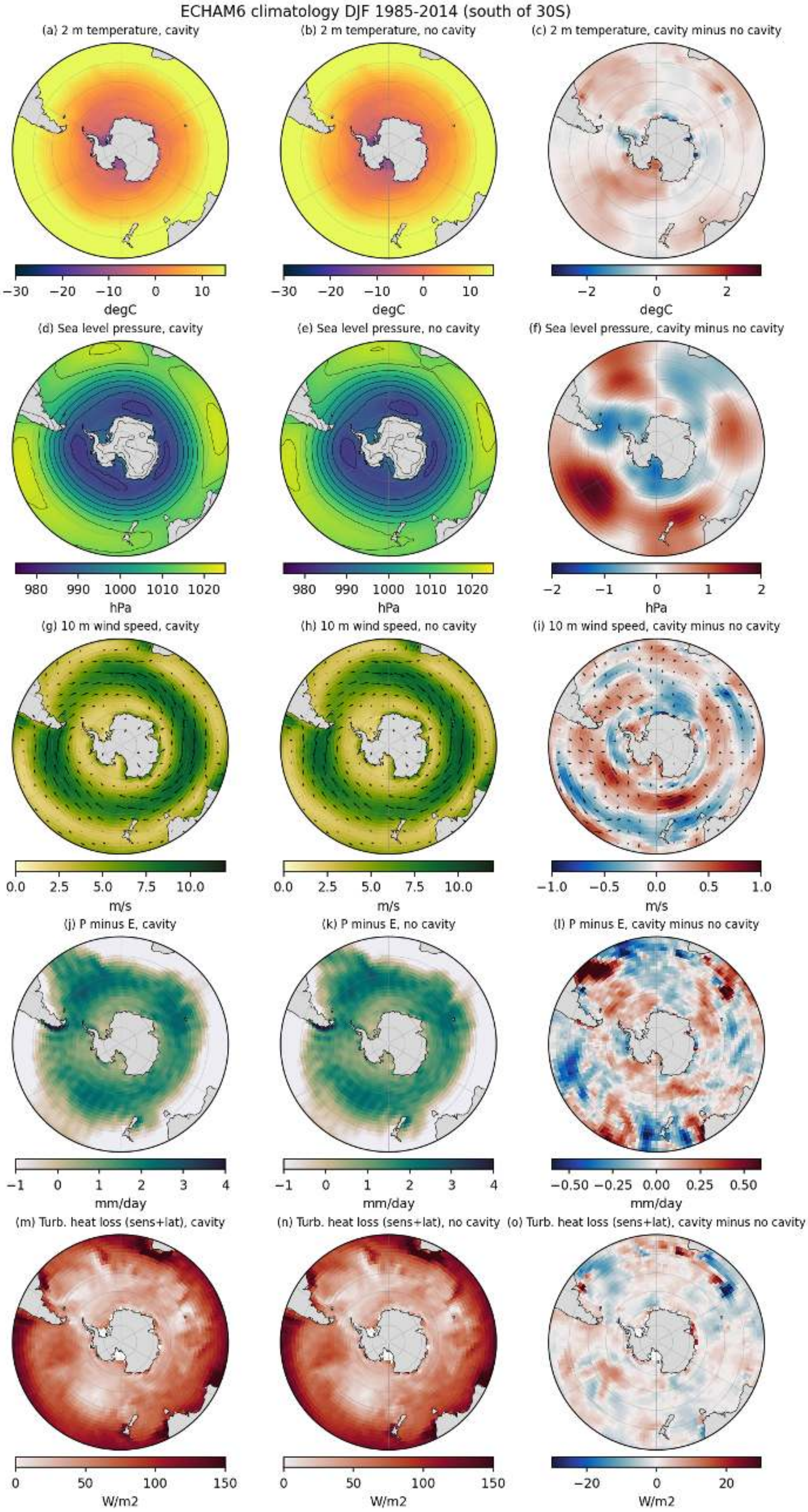

Figure 18: atmos_clim_pd_DJF (atmos). Same as the JJA figure but for summer (DJF, December of the previous year with January and February). Cavity minus no-cavity differences are much smaller in summer (Ross +0.7 K, 50-70S ocean +0.1 K, turbulent flux differences below 1 W/m2 in the regional mean), which shows that the winter signal comes through sea-ice differences in the Ross sector.

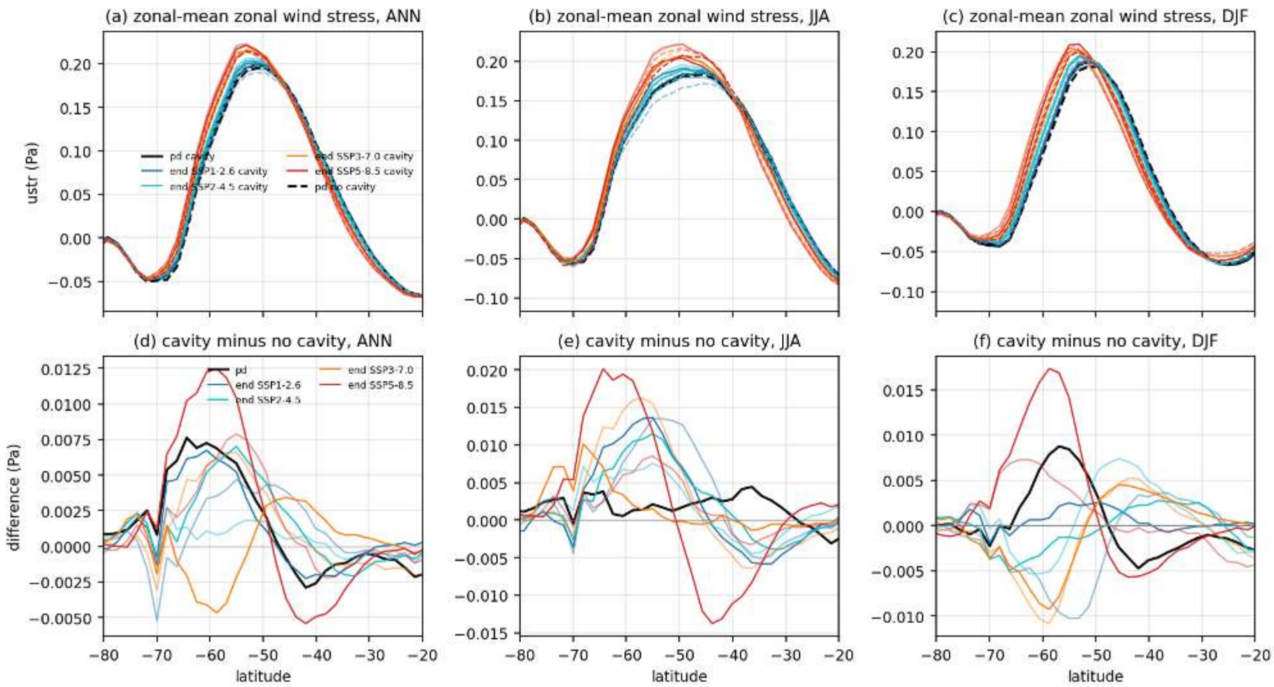

Figure 19: atmos_westerlies_profile (atmos). Zonal-mean zonal wind stress over all surfaces as a function of latitude for annual mean (a), JJA (b) and DJF (c). Black is pd, scenario colours give end (2071-2100, full colour) and far (2171-2200, light) means, solid cavity, dashed no cavity. Panels d to f show the cavity minus no-cavity profiles for the same periods. The pd westerly maximum is 0.20 Pa at 51S in both sets. Under SSP5-8.5 the maximum strengthens to 0.22 Pa and moves to 53.5S; in SSP1-2.6 it stays near pd. Cavity minus no-cavity differences are below 0.02 Pa everywhere, i.e. below 10 percent of the jet strength, and have no consistent sign across scenarios.

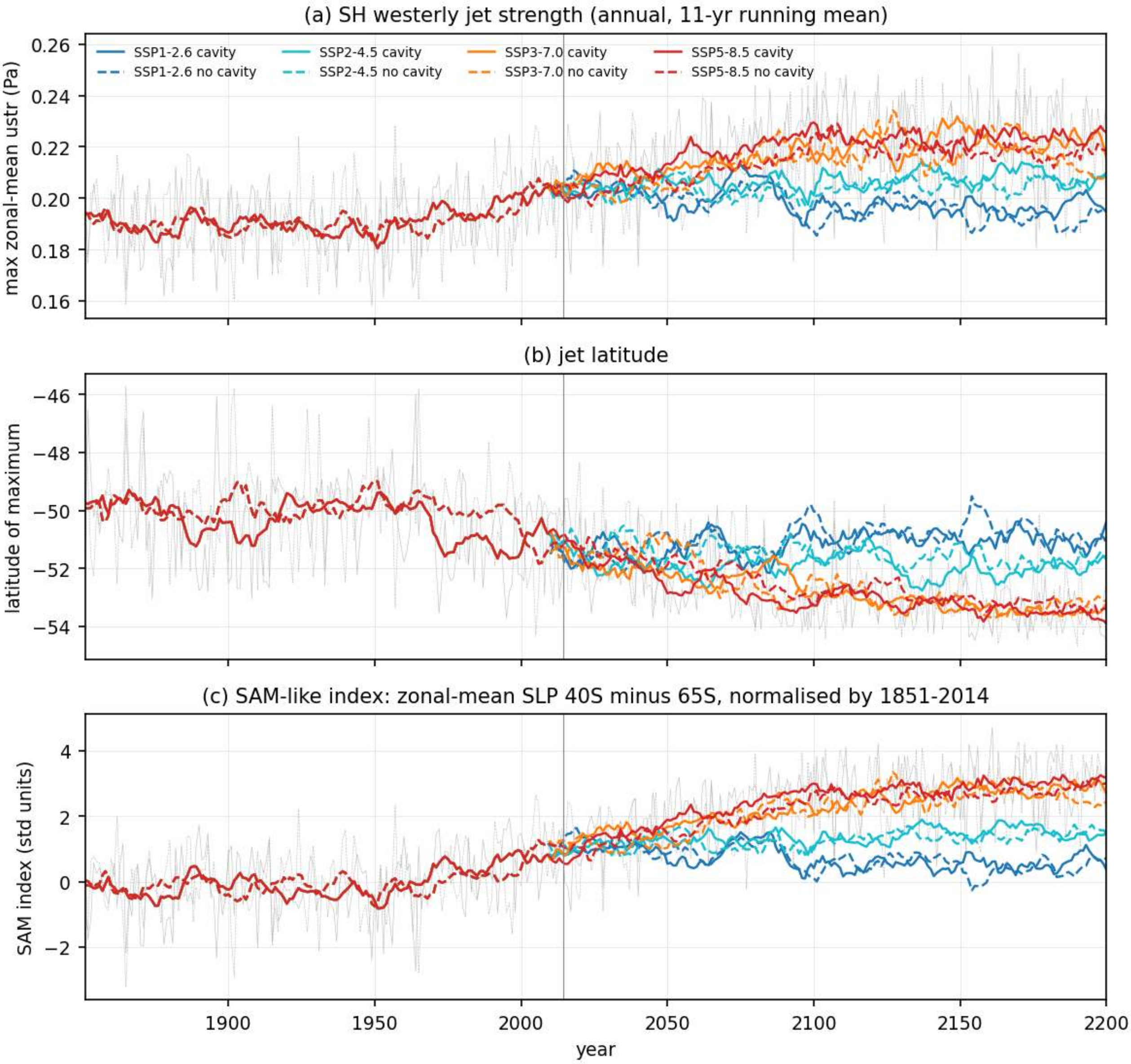

Figure 20: atmos_jet_sam_timeseries (atmos). Annual time series 1851-2200 (11-yr running mean, thin grey lines are annual values of the SSP5-8.5 chains) of (a) the SH westerly jet strength (maximum of the zonal-mean zonal wind stress between 70S and 30S, parabolic fit), (b) the latitude of that maximum and (c) a SAM-like index, zonal-mean SLP at 40S minus 65S normalised by the 1851-2014 mean and standard deviation of each set. All four scenarios and both sets. Jet strength rises from 0.19 Pa (1851-1880) to 0.20 Pa (pd) and 0.22 Pa (SSP5-8.5 far), the jet shifts poleward from 49.7S to 51.1S (pd) and 53.6S (SSP5-8.5 far), and the SAM index increases by 3 standard deviations by 2200 under SSP5-8.5 and SSP3-7.0 while it stabilises near +0.5 to +1.5 in SSP1-2.6 and SSP2-4.5. Cavity and no-cavity chains are indistinguishable within interannual variability.

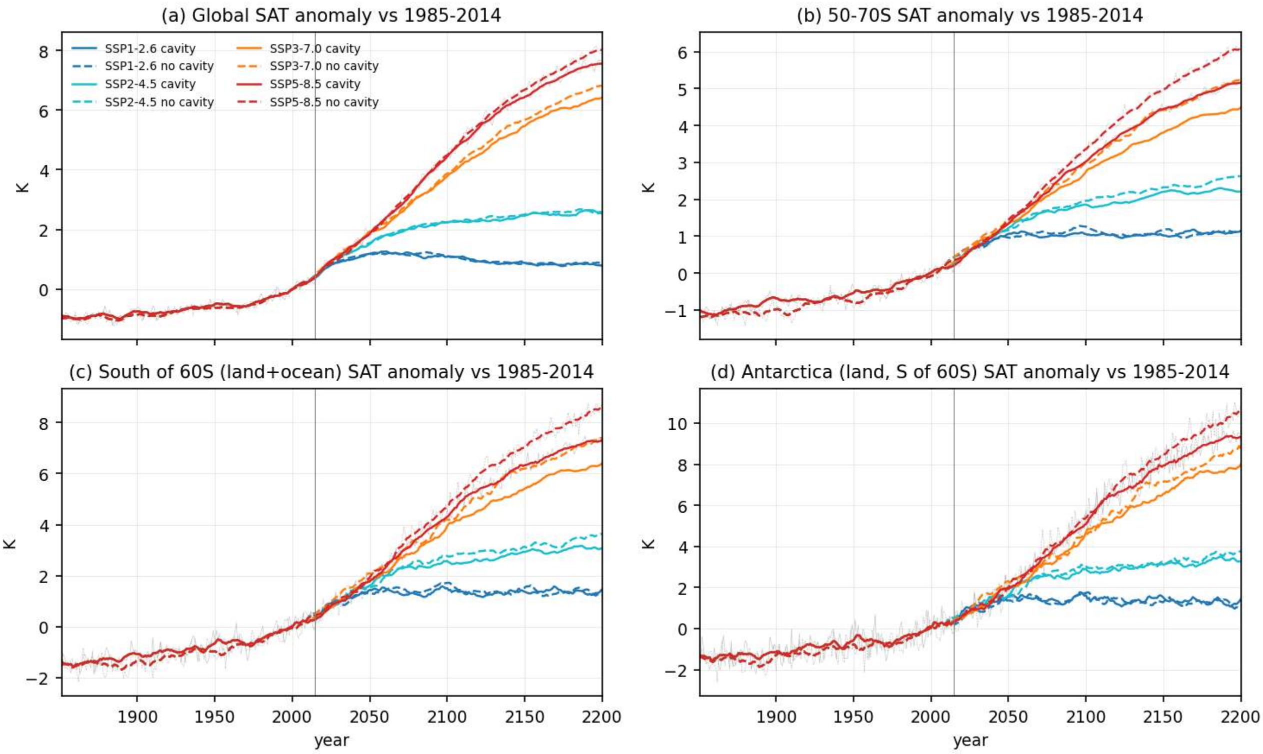

Figure 21: atmos__sat__timeseries (atmos). Annual-mean 2 m air temperature anomalies relative to 1985-2014 for (a) the globe, (b) 50-70S (land and ocean), (c) south of 60S (land and ocean) and (d) Antarctic land south of 60S, 11-yr running means, both sets, four scenarios. Global warming by 2171-2200 reaches 7.4 K (cavity) and 7.7 K (no cavity) in SSP5-8.5 and 0.8 K in SSP1-2.6. South of 60S the no-cavity set warms more than the cavity set in the high scenarios (8.2 K vs 7.1 K far SSP5-8.5, 7.0 vs 6.2 K SSP3-7.0). In the historical period the cavity set is 0.3 to 0.4 K warmer south of 60S than the no-cavity set.

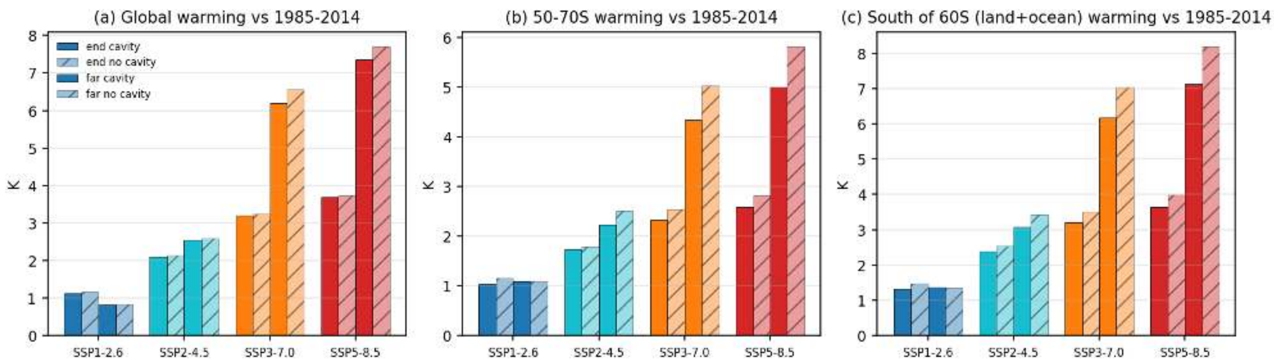

Figure 22: atmos__warming__bars (atmos). Warming (period mean minus 1985-2014) of annual 2 m temperature for end (2071-2100, left pair of bars in each scenario group) and far (2171-2200, right pair) for (a) global, (b) 50-70S and (c) south of 60S, full bars cavity, hatched bars no cavity. The cavity set warms less than the no-cavity set south of 60S by 0.3 K at end and 0.9 to 1.1 K at far in SSP3-7.0 and SSP5-8.5, while the global difference is 0.05 K at end and 0.35 K at far. Antarctic amplification (warming south of 60S divided by global warming) is 1.0 to 1.2 at end and 1.0 to 1.7 at far; it is weakest in the strong scenarios and in the cavity set.

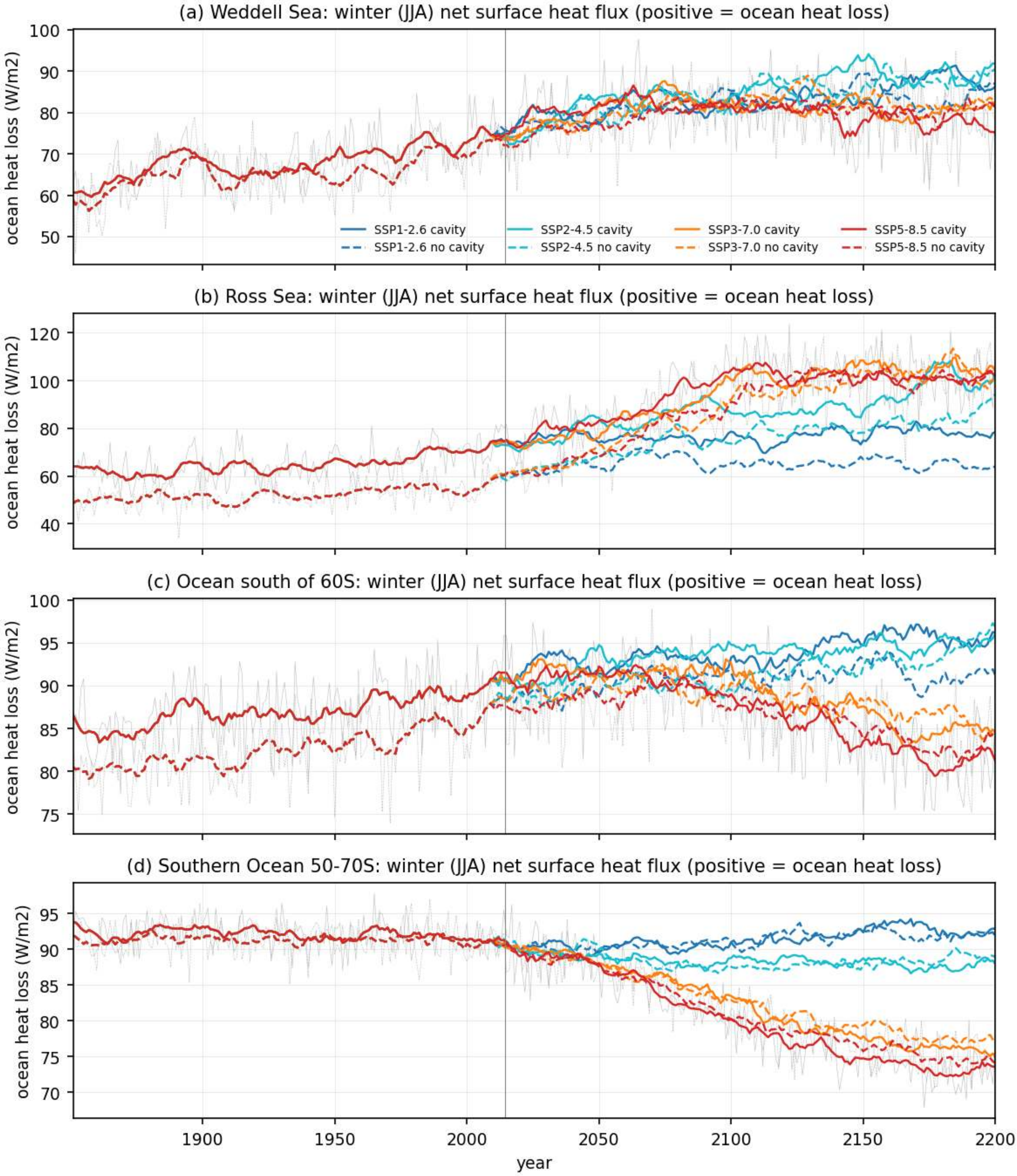

Figure 23: atmos_heatloss_winter_timeseries (atmos). Winter (JJA) net surface heat loss of the ocean-ice column (minus the ECHAM net surface heat flux, sensible plus latent plus net shortwave plus net longwave, grid-box mean) for (a) Weddell Sea, (b) Ross Sea, (c) ocean south of 60S and (d) 50-70S, 1851-2200, 11-yr running means, both sets, four scenarios. Winter heat loss in the Weddell Sea rises from 63 W/m2 (1851-1880) to 73 W/m2 (pd) and 82 W/m2 (end, all scenarios) because sea ice thins and opens. In the Ross Sea the cavity run loses 12 to 17 W/m2 more heat than the no-cavity run throughout the historical period (72 vs 55 W/m2 at pd); the difference closes only after 2100 under SSP5-8.5 when both reach about 100 W/m2. Over 50-70S winter heat loss decreases under strong warming (92 to 73 W/m2 by 2171-2200 in SSP5-8.5) because the ocean surface warms less than the air.

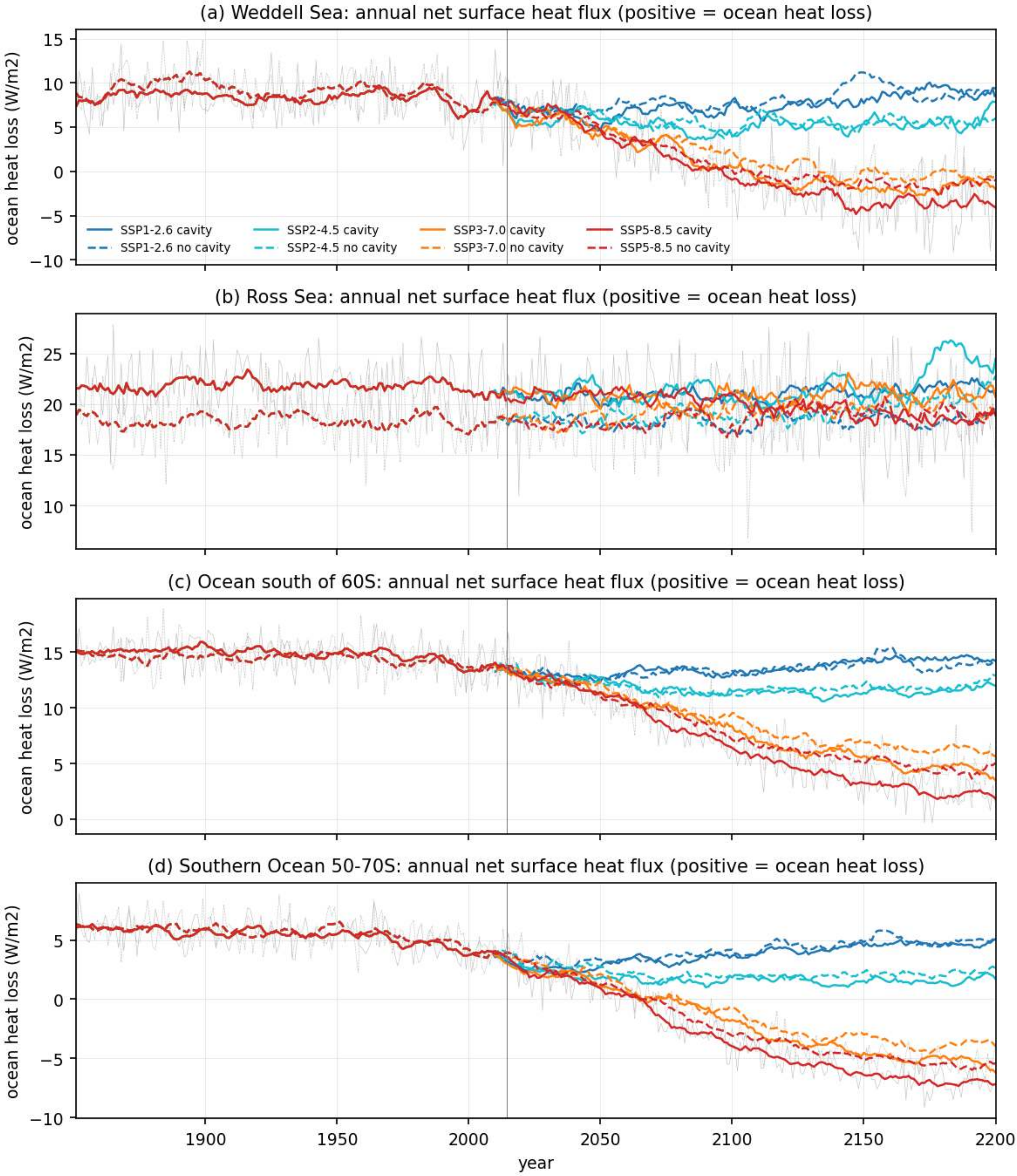

Figure 24: atmos_heatloss_annual_timeseries (atmos). Same as the winter figure but annual means. Annual heat loss is small (4 to 22 W/m2 at pd) and becomes negative (net heat gain) in SSP5-8.5 after 2100 for 50-70S and the Weddell Sea. The Ross Sea keeps a net annual heat loss of about 20 W/m2 in all scenarios, 3 W/m2 higher with cavities.

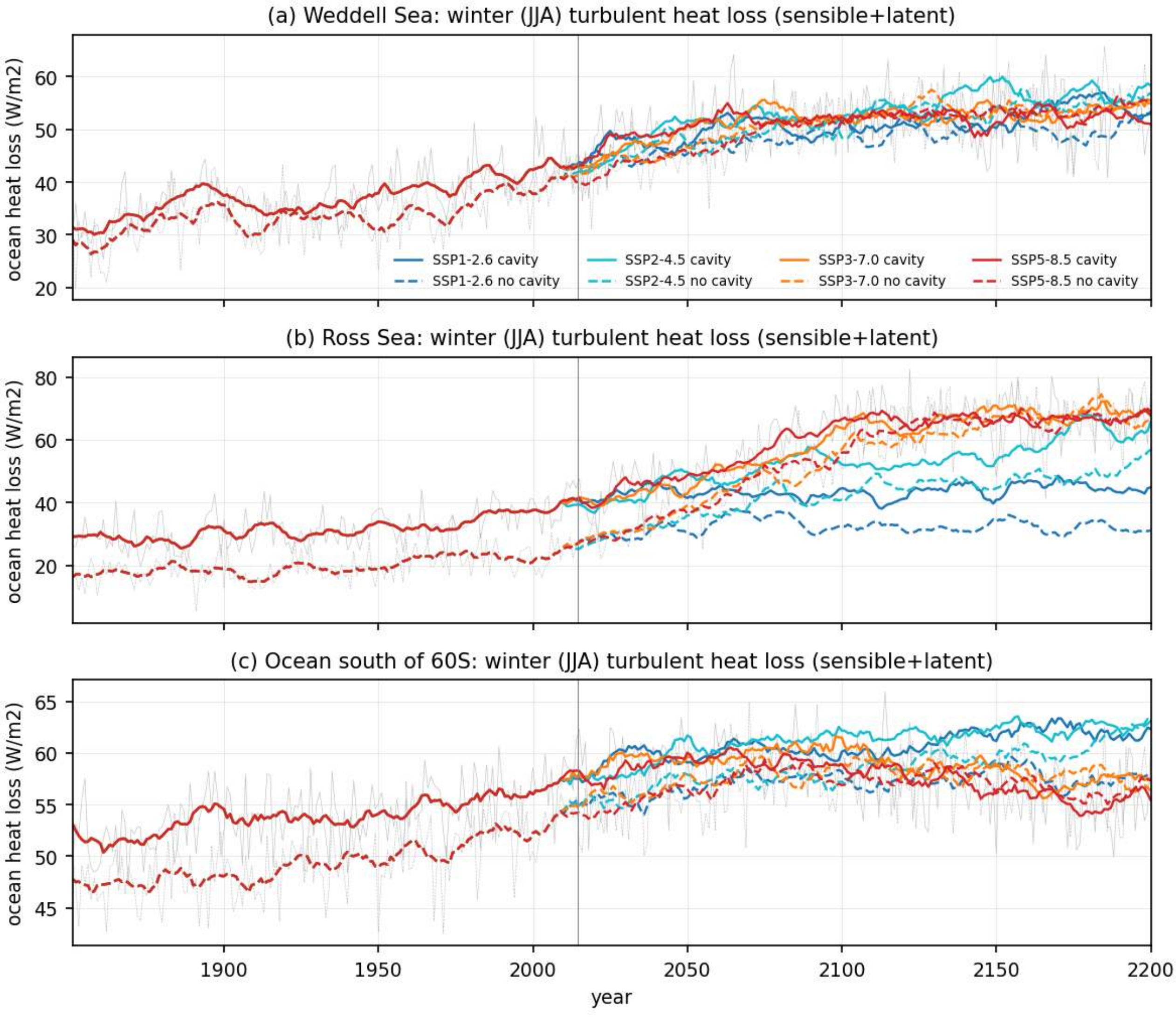

Figure 25: atmos_turbflux_winter_timeseries (atmos). Winter (JJA) turbulent heat loss (sensible plus latent) for Weddell, Ross and ocean south of 60S, 1851-2200. It grows in all scenarios as sea ice retreats (Weddell 32 to 52 W/m2 from 1851-1880 to 2071-2100) and is 15 W/m2 larger in the Ross Sea with cavities at pd (38 vs 23 W/m2).

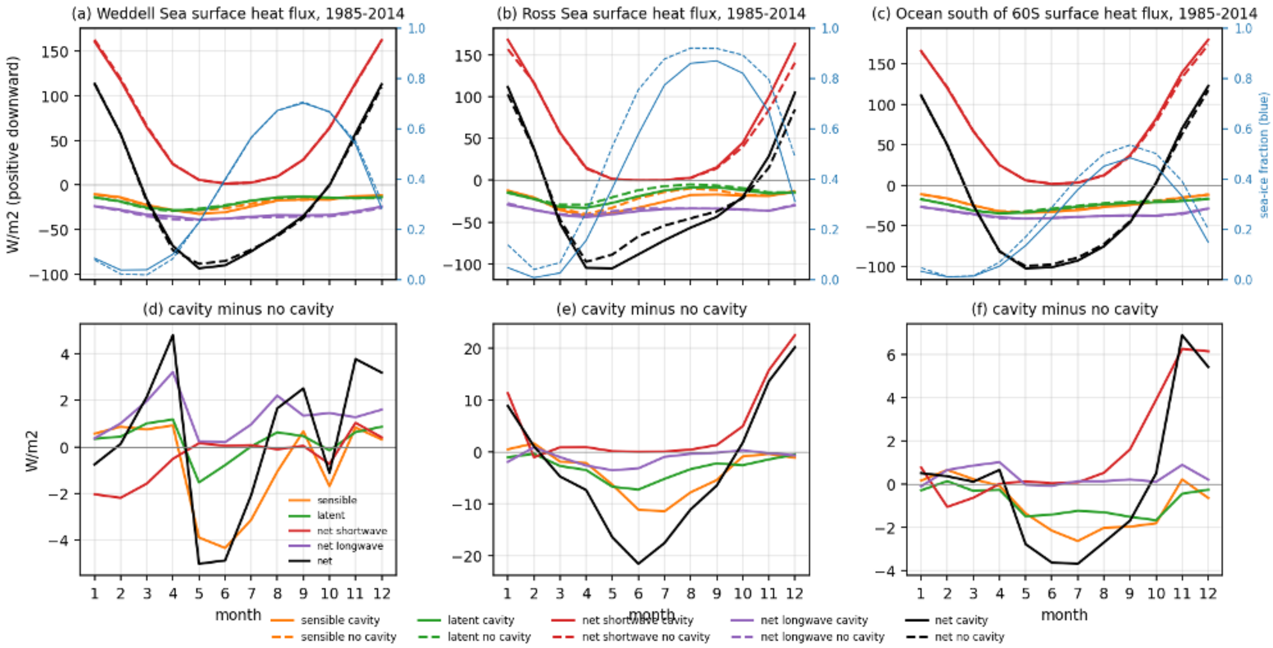

Figure 26: atmos_heatflux_seasonal_cycle_pd (atmos). Seasonal cycle 1985-2014 of the surface heat flux components (positive downward) for Weddell Sea (a), Ross Sea (b) and ocean south of 60S (c), solid cavity, dashed no cavity, with the ECHAM sea-ice fraction on the right axis (blue). Panels d to f give cavity minus no cavity. The Ross Sea shows the clearest cavity signal: winter sea-ice fraction is 0.15 lower and the net winter heat loss is 20 W/m2 larger with cavities, mostly from sensible heat (-12 W/m2 in June). Around the whole continent the cavity set has 3 to 4 W/m2 more net heat loss in May to August and 6 W/m2 more shortwave absorption in November and December (less ice, lower albedo). The Weddell Sea difference is within plus or minus 5 W/m2 and changes sign through the year.

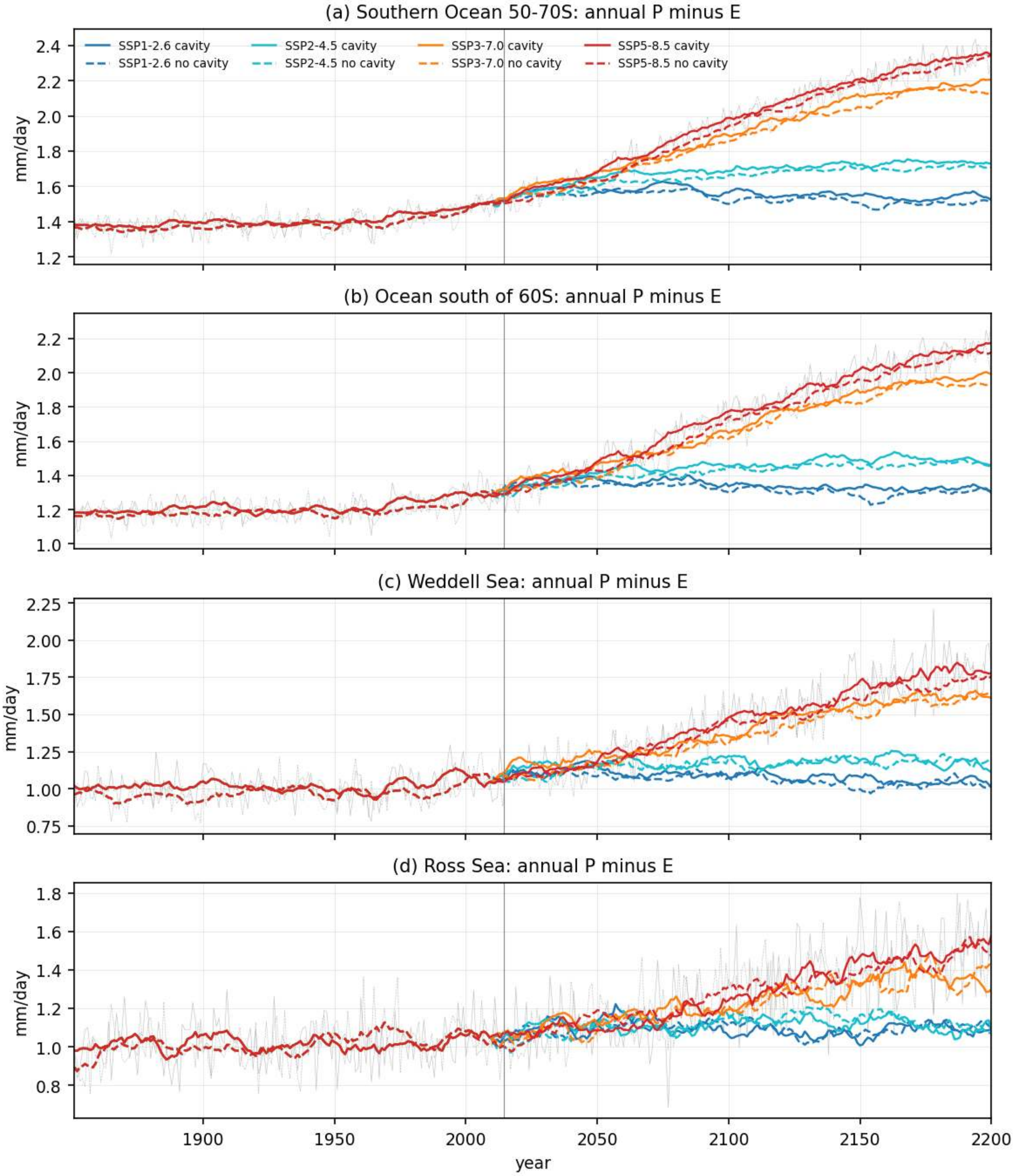

Figure 27: atmos_pme_timeseries (atmos). Annual-mean precipitation minus evaporation over the ocean for 50-70S (a), south of 60S (b), Weddell Sea (c) and Ross Sea (d), 1851-2200. P minus E over 50-70S increases from 1.37 mm/day (1851-1880) to 1.48 (pd), 1.90 (end) and 2.32 mm/day (far) in SSP5-8.5, i.e. +57 percent, while SSP1-2.6 levels off at 1.55 mm/day. Cavity minus no-cavity differences are 0.01 to 0.04 mm/day, within the grey noise band of the cavity-difference figure.

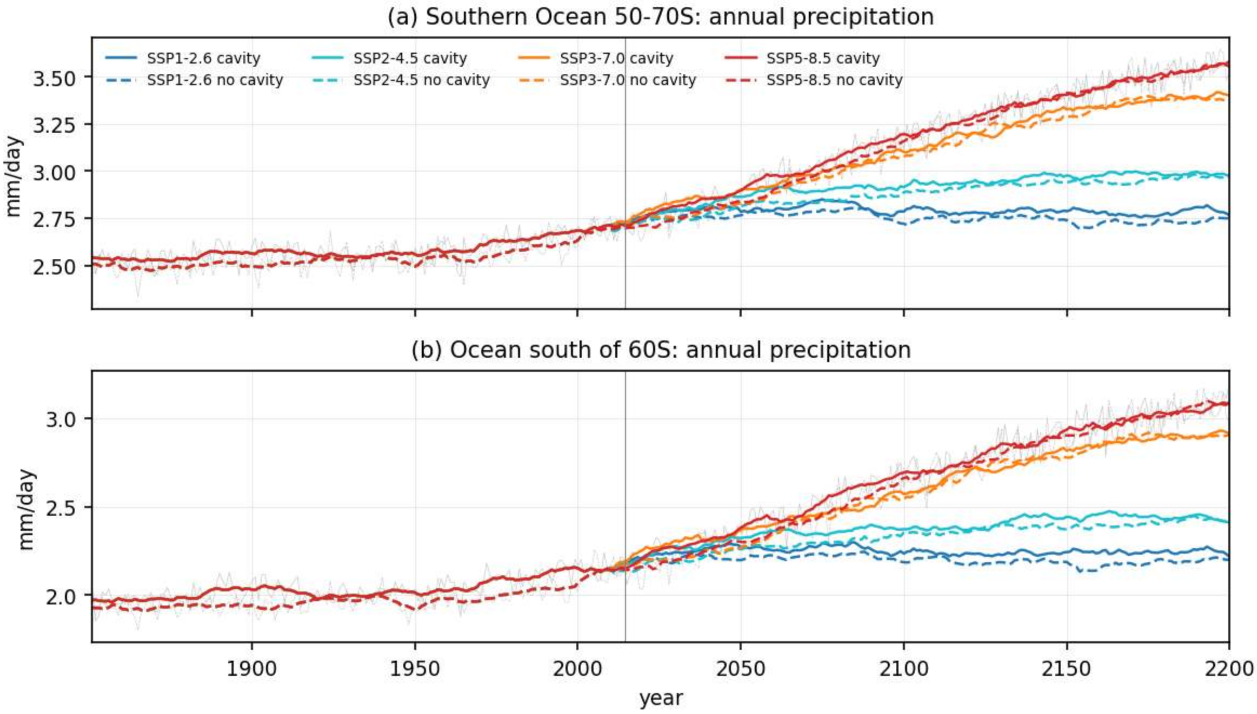

Figure 28: atmos__precip__timeseries (atmos). Annual precipitation over the ocean 50-70S and south of 60S. Precipitation rises from 2.68 mm/day (pd) to 3.53 mm/day (far SSP5-8.5) over 50-70S. The two sets agree within 0.05 mm/day.

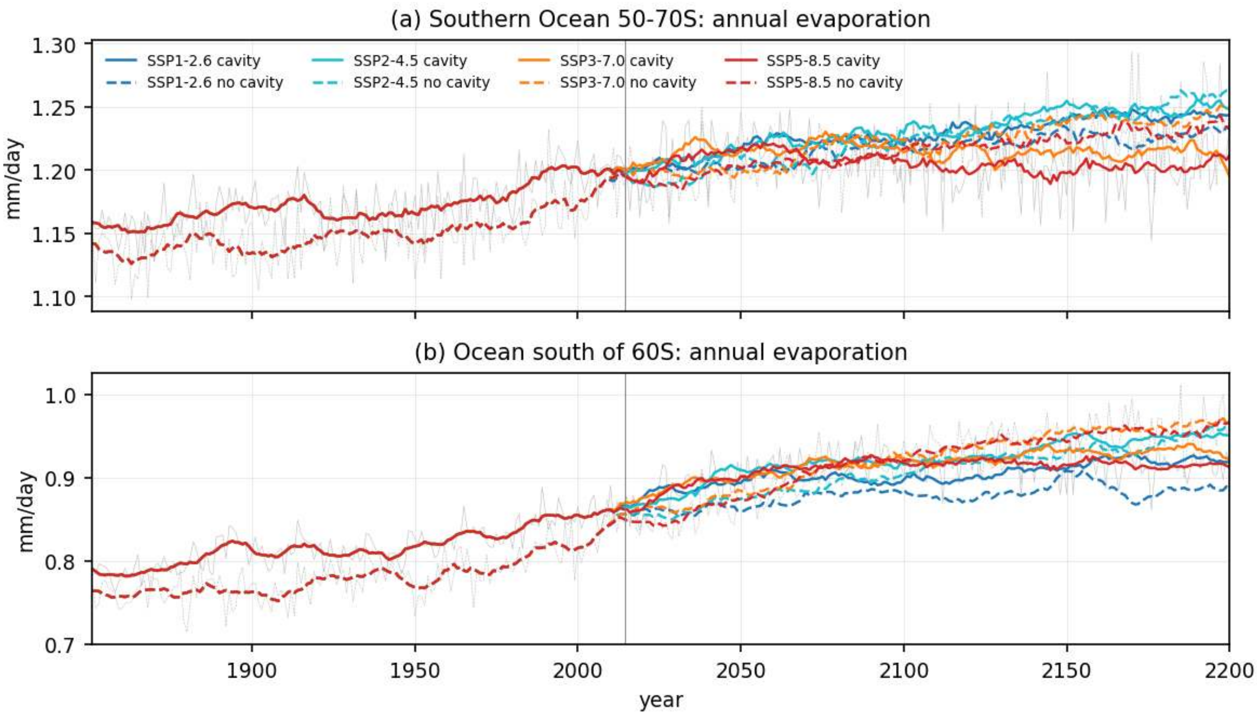

Figure 29: atmos__evap__timeseries (atmos). Annual evaporation over the ocean 50-70S and south of 60S. Evaporation changes little (1.20 to 1.24 mm/day over 50-70S) so the P minus E increase is almost entirely a precipitation increase. South of 60S the no-cavity set evaporates slightly more at far SSP5-8.5 (0.96 vs 0.91 mm/day) because its sea ice retreats further.

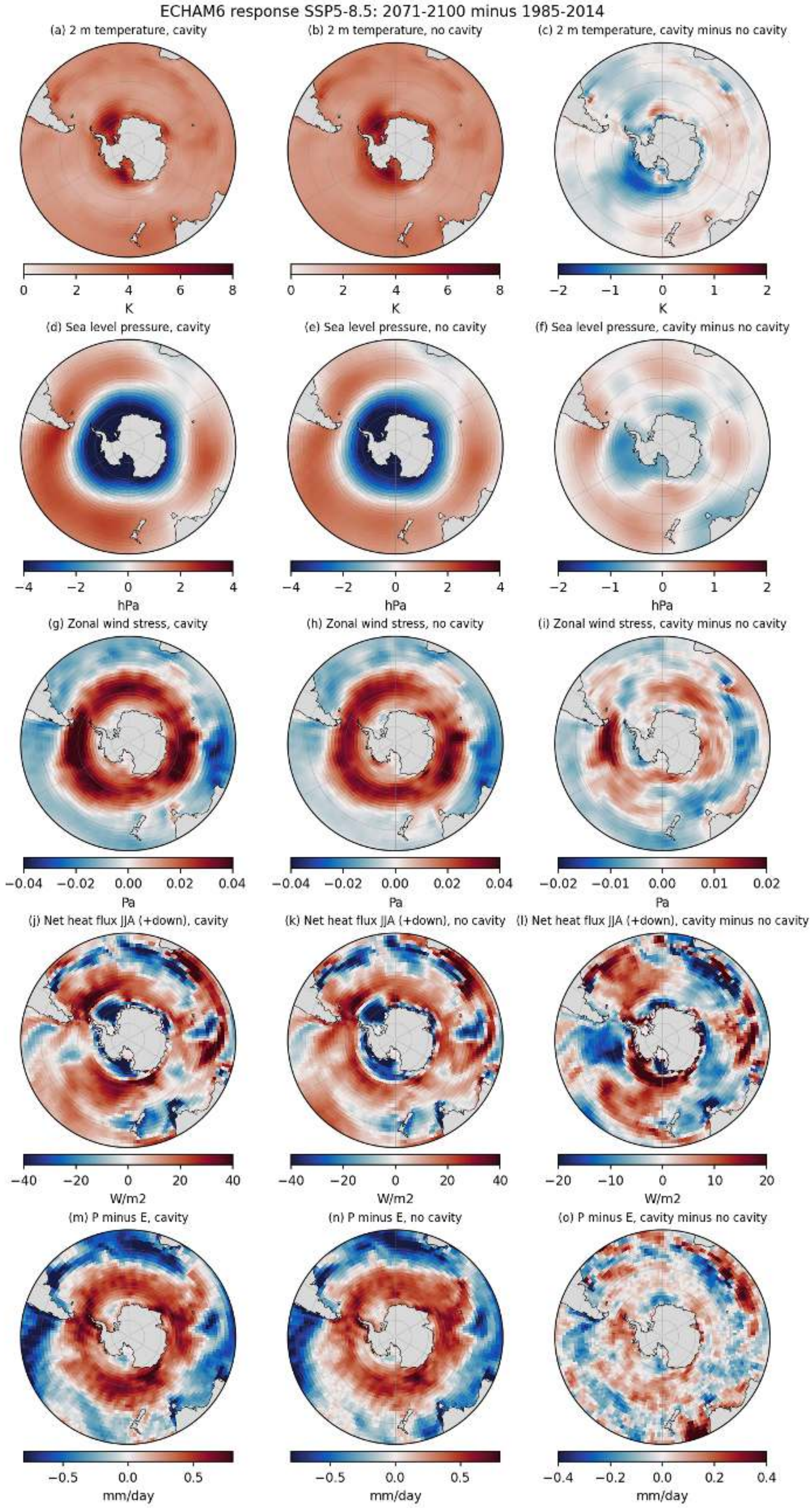

Figure 30: atmos_response_ssp585_end (atmos). Response maps for SSP5-8.5, 2071-2100 minus 1985-2014, south of 30S: annual 2 m temperature (K), annual SLP (hPa), annual zonal wind stress (Pa), JJA net surface heat flux (W/m2, positive downward) and annual P minus E (mm/day). Columns: cavity response, no-cavity response, difference of the responses. Both sets show 3 to 6 K warming with a maximum over the sea-ice zone, a 3 to 4 hPa SLP drop around Antarctica with rises at 40S (SAM-like), a westerly increase of 0.02 to 0.04 Pa at 50-60S with weakening north of 45S, increased winter heat loss over the former ice zone and increased P minus E south of 50S. The difference of the responses is small compared with the response except over the Ross Sea (cavity warms 0.7 K less, loses 5 W/m2 less extra winter heat).

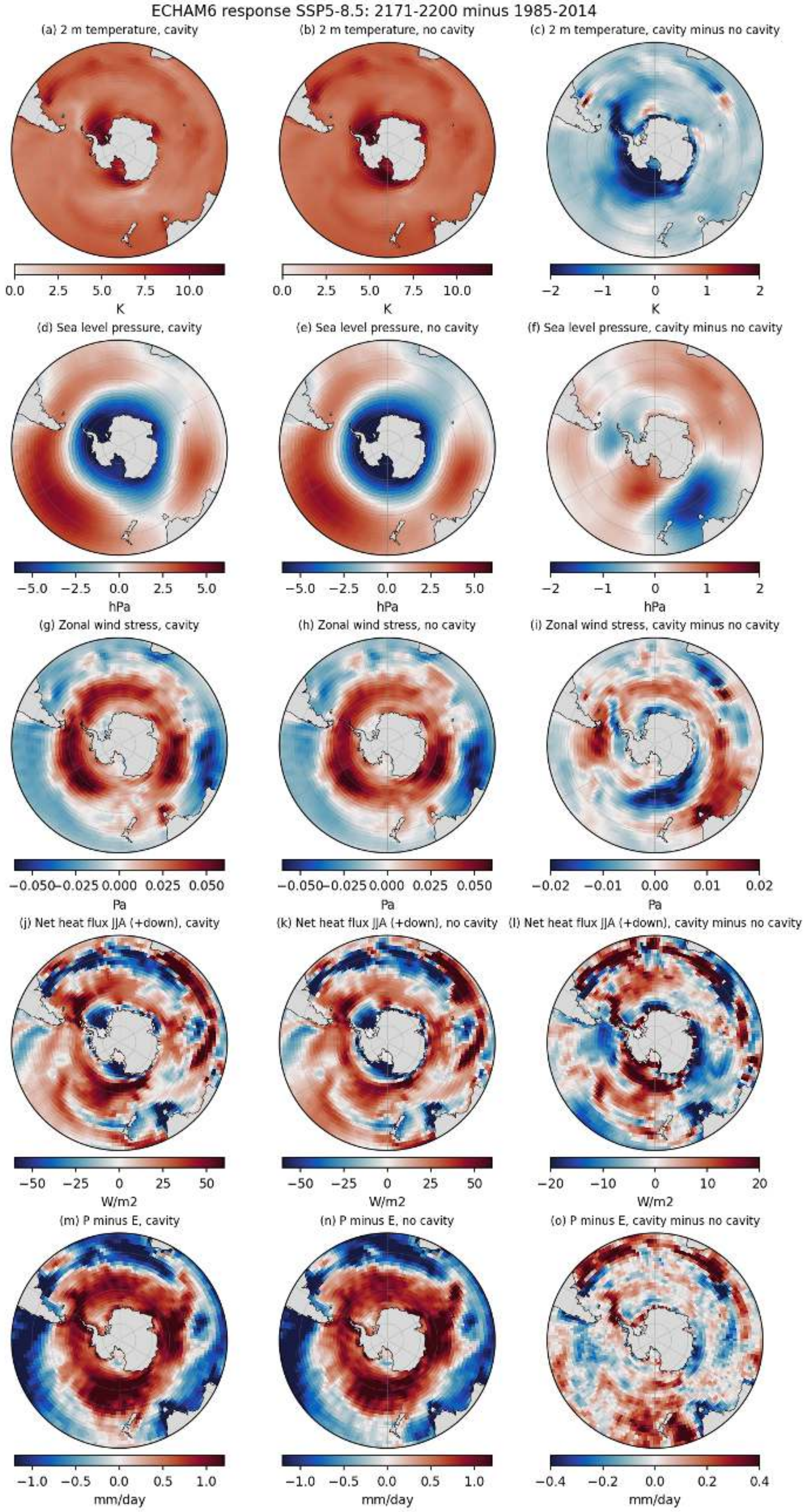

Figure 31: atmos_response_ssp585_far (atmos). Same as above for 2171-2200 minus 1985-2014 (colour ranges widened). Warming is 6 to 10 K over the Southern Ocean; the cavity set warms 1.2 K less over the ocean south of 60S and 2.2 K less over the Ross Sea than the no-cavity set. The Ross Sea winter heat loss response differs by 18 W/m2 (cavity 28 W/m2 vs no cavity 46 W/m2 extra heat loss) because the no-cavity Ross sea ice starts from a higher pd cover and loses more. SLP, wind-stress and P minus E responses agree between sets within 0.9 hPa, 0.005 Pa and 0.04 mm/day in the regional means.

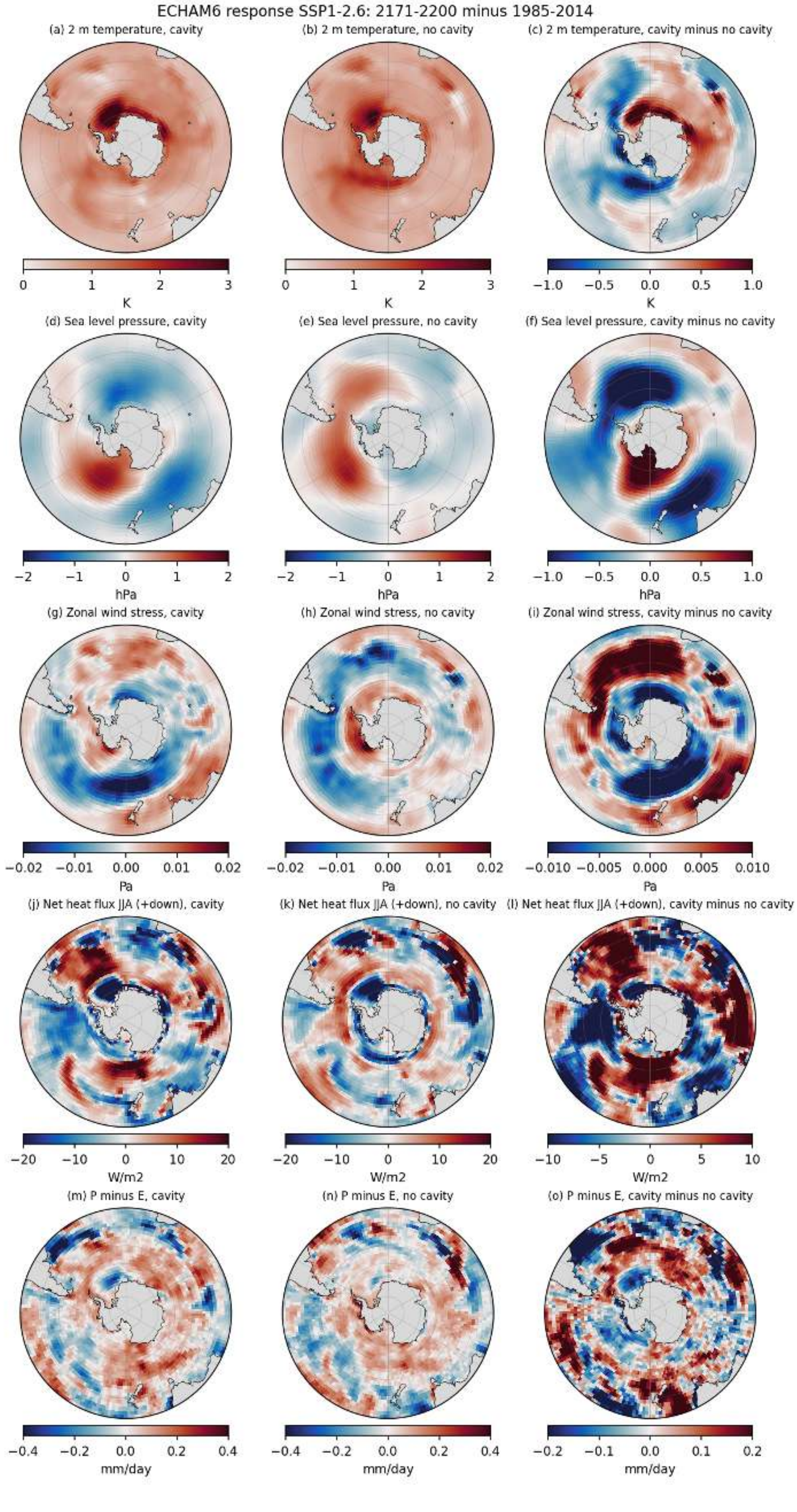

Figure 32: atmos_response_ssp126_far (atmos). Same for SSP1-2.6, 2171-2200 minus 1985-2014. Warming is 1 to 2 K, the circulation response is weak (SLP within plus or minus 1 hPa, wind stress within 0.005 Pa) and the two sets differ by up to 0.6 K in the Ross Sea and 0.4 K in the Weddell Sea, which is of the size of 30-year internal variability.

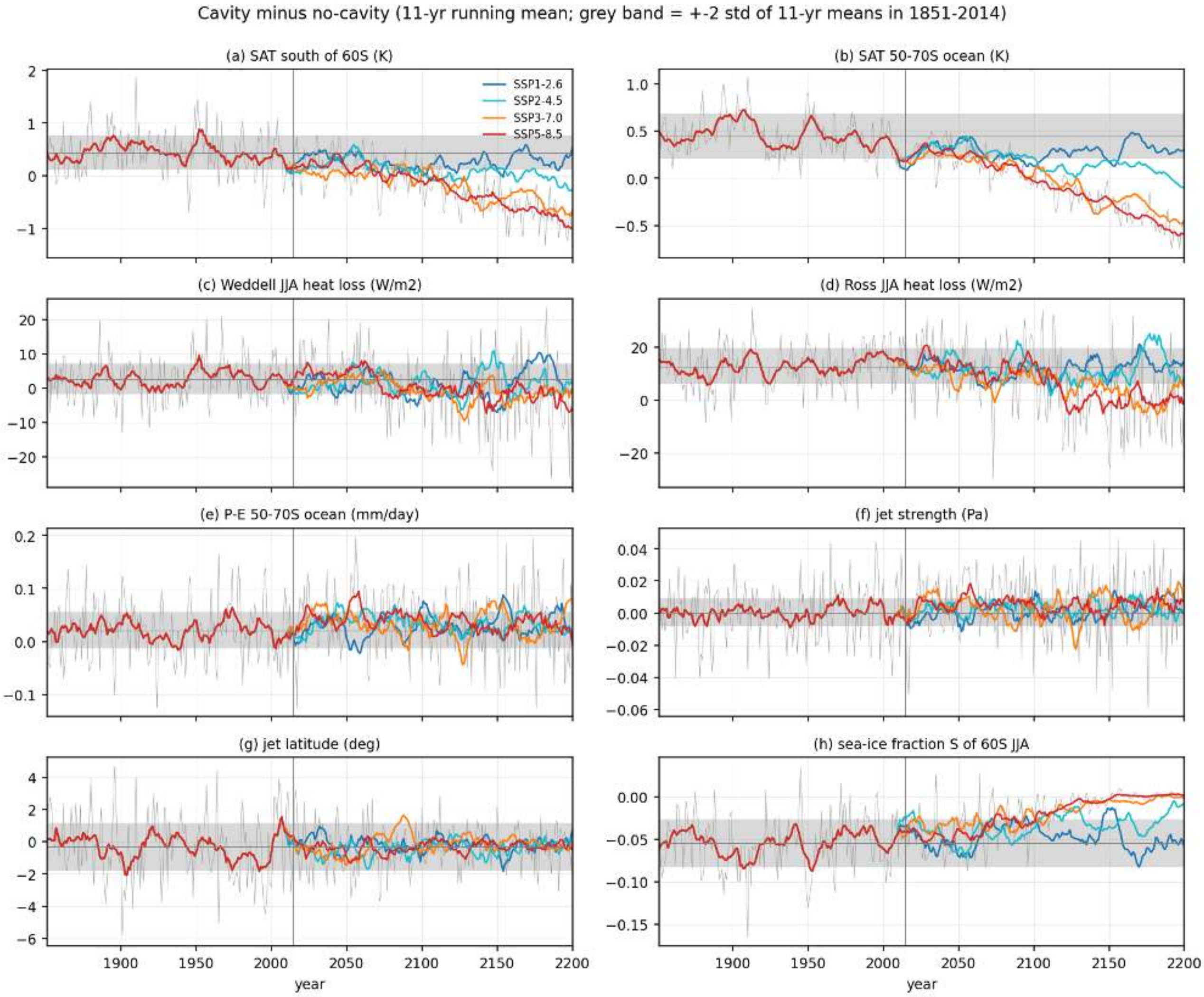

Figure 33: atmos_cavity_minus_nocavity_timeseries (atmos). Cavity minus no-cavity differences (11-yr running means) of (a) SAT south of 60S, (b) SAT over the 50-70S ocean, (c) Weddell and (d) Ross winter heat loss, (e) P minus E over 50-70S, (f) jet strength, (g) jet latitude and (h) winter sea-ice fraction south of 60S, for all four scenario chains. The grey band is plus or minus two standard deviations of the 11-yr means of the historical difference around its mean. In the historical period the cavity set is 0.44 K warmer south of 60S, has a 0.05 lower winter ice fraction and 13 W/m2 more Ross winter heat loss. After 2100 the SAT difference reverses to -0.8 K (SSP5-8.5) and -0.7 K (SSP3-7.0) south of 60S and the sea-ice and Ross heat-loss differences vanish as the no-cavity set loses its extra ice. Jet strength, jet latitude and P minus E differences never leave the noise band.

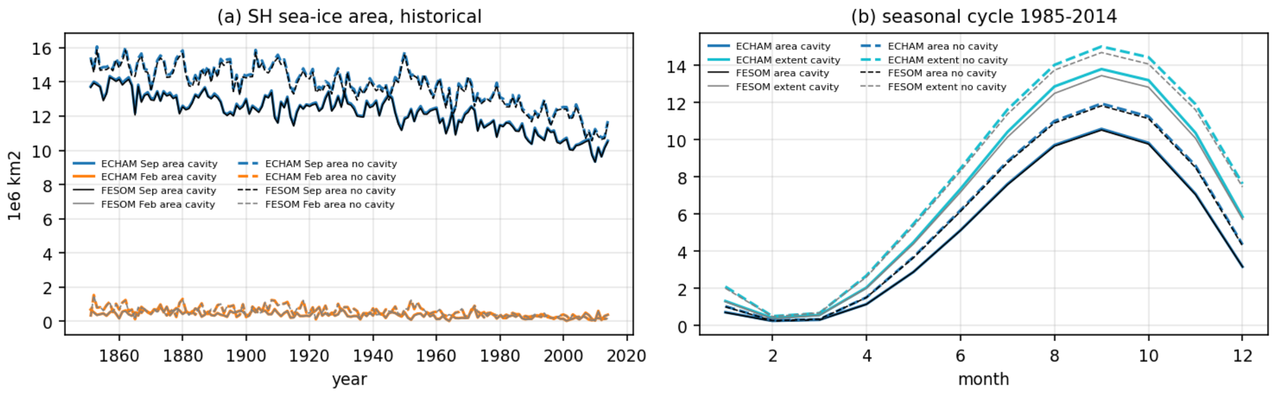

Figure 34: atmos__seaice__echam__vs__fesom (atmos). Consistency check of the Southern Hemisphere sea-ice area seen by ECHAM (code 97 times ocean fraction) against the FESOM cache ('cache/ts'): (a) September and February area 1851-2014, (b) mean seasonal cycle 1985-2014 of area and extent. ECHAM and FESOM agree within 1 percent (September area 10.6 vs 10.5 million km2 with cavities, 11.9 vs 11.8 without), so the atmospheric diagnostics are consistent with the ocean-ice state. The cavity run has 1.3 million km2 less September sea-ice area than the no-cavity run.

5 Sea ice, polynyas, convection and mixed layers

Findings

5.1 seaice theme: findings (sea ice, polynyas, open-ocean convection, mixed layers as AABW precursors)

Sets: cavity = PI_ICEv2 (hist + SSP), no cavity = PI_CORE. Periods early 1851-1880, pd 1985-2014, mid 2041-2070, end 2071-2100, far 2171-2200. Numbers are period means unless stated. Region means exclude cavity nodes. Supporting numbers are in figures/seaice/si_*_stats.txt and si_thresholds.txt.

5.1.1 Sea-ice state and bias

1. Both runs have too little Antarctic sea ice. September extent at pd is 13.5e6 km2 (cavity) and 14.6e6 km2 (no cavity), against about 18.5e6 km2 observed; February extent 0.36 and 0.49e6 km2 against about 3e6 km2. September volume is 5.5 and 6.7e3 km3. The early-historical state (1851-1880) had 16.6 and 18.3e6 km2 in September and 7.6 and 9.1e3 km3, so half of the present-day shortfall is a model bias already in the PI state and the rest is the historical decline (about 3e6 km2 and 2e3 km3 over 1880-2014 in both sets, larger than observed). The overview note of 14.5 and 15.7e6 km2 was computed from the climatological mean concentration field; the mean of yearly extents used here is 1 to 1.1e6 km2 smaller. An ice-free February (extent below 1e6 km2) occurs in nearly every year of the historical period (cavity from 1852, no cavity persistently from 1935), so it cannot be used as a projected threshold.

2. Cavity minus no cavity at pd: 1.1e6 km2 less September extent, 1.3e6 km2 less September area, 1.2e3 km3 less September volume, and the same offsets hold in 1851-1880. The loss is concentrated in the Pacific sector (Ross, Amundsen, Bellingshausen, concentration lower by 0.2 to 0.5 and thickness by 0.2 to 0.5 m) and along the outer Indian Ocean ice edge. The cavity run has more ice (by up to 0.4 in concentration) over the eastern Weddell gyre near Maud Rise and in parts of the East Antarctic coastal zone, and thicker summer ice in the western Weddell Sea. The Ross Sea difference matches the atmos finding of a 2.7 K warmer winter Ross Sea in the cavity run (less ice, more heat loss); since the cavity run loses this ice already in the PI state, the Ross cavity (melt 257 Gt/yr, ice-shelf water flux and circulation change under the Ross Ice Shelf) is the likely origin.

3. Scenario response. September extent falls below 10e6 km2 in SSP3-7.0 and SSP5-8.5 around 2040 (cavity 2038 and 2041) to 2060 (no cavity 2059 for SSP3-7.0), below 5e6 km2 around 2095-2115 (SSP5-8.5: cavity 2094, no cavity 2102), and below 1e6 km2 in 2146 (SSP5-8.5) and 2174 (SSP3-7.0) in the cavity run and 2156 and 2194 in the no-cavity run. By 2171-2200 the maximum monthly extent is 10.0/11.3e6 km2 (SSP1-2.6 cavity/no cavity), 6.4/7.1 (SSP2-4.5), 1.1/1.5 (SSP3-7.0) and 0.6 (SSP5-8.5 cavity), with maximum volume 3.7/4.6, 2.1/2.4, 0.3/0.4 and 0.1e3 km3. SSP1-2.6 stabilises at about 75 percent of the pd September extent after 2050. The no-cavity run keeps 1 to 1.5e6 km2 more ice in all scenarios until the pack is gone.

5.1.2 Sea-ice production (brine) on the shelves and coastal polynyas

4. Winter (Apr-Oct) net freezing, measured by the positive surface freshwater flux, is larger in the cavity run on all shelves at pd: Weddell shelf 5.5 vs 3.6 mm/day, Ross shelf 8.8 vs 7.1, Amundsen shelf 2.1 vs 2.9 (here the no-cavity run is larger), whole Antarctic shelf 6.2 vs 4.7 mm/day. The difference maps show the extra freezing of the cavity run right at the ice-shelf fronts and along the coast, consistent with colder ice-shelf-water outflow and a narrower, colder coastal shelf zone. The Amundsen shelf freezing has already dropped from 5.1 (early) to 2.1 mm/day (pd) in the cavity run, the largest historical change of any shelf.

5. Future loss of sea-ice production. Shelf-mean winter freezing decreases under all scenarios: Antarctic shelf (cavity) 6.2 (pd) to 5.0 (SSP1-2.6 far), 3.9 (SSP2-4.5 far), -0.5 (SSP3-7.0 far) and -1.1 mm/day (SSP5-8.5 far). The no-cavity run ends even lower (Antarctic shelf SSP5-8.5 far -4.8 mm/day, Weddell shelf -3.3, Ross shelf -2.3). Negative values mean the shelf receives net freshwater in winter, i.e. the brine source for dense shelf water is gone. The Ross shelf keeps positive winter freezing longest (cavity SSP5-8.5 far 1.3 mm/day, no cavity SSP3-7.0 far -0.9). The no-cavity run loses its shelf freezing earlier and more completely (Weddell shelf SSP3-7.0 far -3.0 vs -0.4 mm/day cavity). Integrated winter freezing (sum of positive fw over the shelf, Apr-Oct rate) is 1714 (cavity) and 1585 Gt/yr (no cavity) on the Weddell shelf and 1526 and 1363 Gt/yr on the Ross shelf at pd. By 2171-2200 it falls to 1587/1343 (SSP1-2.6), 1622/971 (SSP2-4.5), 521/169 (SSP3-7.0) and 338/70 Gt/yr (SSP5-8.5) on the Weddell shelf and to 1394/1260, 1207/1116, 595/376 and 475/206 Gt/yr on the Ross shelf; the Amundsen shelf goes from 516 (early) to 319 (pd) to 10 Gt/yr (SSP5-8.5 far) in the cavity run. Coastal polynya activity (winter open water with concentration

below 0.7 within 150 km of the coast or ice front) is larger in the cavity run at pd (Weddell 0.33 vs 0.29, Ross 0.14 vs 0.10, Amundsen 0.17 vs 0.11e6 km2) and increases in all scenarios, reaching 0.75 to 0.80e6 km2 of the 0.82e6 km2 Weddell band under SSP3-7.0 and SSP5-8.5, i.e. the coastal zone becomes open water in winter while freezing stops, so open water no longer means ice production.

5.1.3 Open-ocean convection and Maud Rise polynya

6. Deep convection (September MLD2 > 2000 m) occurs essentially only north-east of Maud Rise (0-10E, 63-66S). The no-cavity run convects in 137 of 164 historical years (84 percent) with a mean convective area of 0.10e6 km2 (1851-1880) to 0.16e6 km2 (1985-2014) and up to 0.4e6 km2 in single years; the cavity run convects in 34 years (21 percent) with a mean area of 0.01 to 0.02e6 km2 and a maximum of 0.2e6 km2. Deep mixing below 1000 m occurs in more than half of the years over 0.094e6 km2 in the no-cavity run and over 0.004e6 km2 in the cavity run. Convective winters in both runs have the same signature over a 300 by 600 km patch east of Maud Rise: concentration lower by up to 0.4, SST 0.5 K warmer, SSS 0.2 psu saltier (box means 0.09 to 0.13 lower ice, 0.1 to 0.17 K, 0.015 psu).

7. Why the no-cavity run convects every winter and the cavity run does not (tested hypotheses with numbers). Stratification: the basin-scale Weddell gyre stratification is almost identical (0-200 m minus 200-1000 m sigma2 -0.22 cavity vs -0.19 kg m-3 no cavity at pd) and has no predictive skill for convection (correlation -0.1). Over Maud Rise the cavity run is 0.03 psu fresher in the upper 200 m in every decade of the historical period (34.204 vs 34.233 psu; 0-50 m 33.94 vs 33.97) and 0.01 kg m-3 more stratified (-0.123 vs -0.112 kg m-3). This is small but systematic, and within each run the upper-ocean salinity is the best predictor of convection (correlation with the convective area 0.70 in the no-cavity run and 0.38 in the cavity run; lag-1 0.44 and 0.25), whereas the density contrast itself is not (correlation -0.06 to -0.08). The no-cavity run sits just on the convective side of the salinity threshold, the cavity run just below it. Freshwater: the most plausible source of the extra 0.03 psu freshwater is ice-shelf melt upstream. The Fimbul and other Dronning Maud Land shelves melt 316 Gt/yr in the cavity run (the largest single contributor after the total of 2364 Gt/yr), and this water enters the coastal current that feeds the Maud Rise region. The DML coastal box (20W-30E, 71-67S) is 0.04 psu fresher in the upper 200 m (34.10 vs 34.14 psu) and 0.2 K colder in the cavity run. Freshening the upper 200 m of the 3.2e6 km2 Maud Rise box by 0.03 psu needs 570 Gt of freshwater, about two years of DML melt, so the melt flux is ample. In the no-cavity mesh the DML cavity area is open ocean that forms sea ice and rejects brine (annual net freshwater loss over the dml_shelf region 370 Gt/yr vs 203 Gt/yr in the cavity run, different domains). Sea ice: September ice in the Weddell gyre box is slightly thinner in the cavity run (0.26 vs 0.28 m) and the concentration is the same (0.48 vs 0.49), so ice does not insulate the cavity run more; convective winters have thinner ice in both runs (0.30 vs 0.33 m cavity, 0.35 vs 0.40 m no cavity) as a consequence of the heat released by convection, not a cause. Heat: winter ocean heat loss over the Weddell gyre box is the same (82 vs 81 W m-2) and the 200-1000 m layer is 0.05 K warmer in the no-cavity run (1.97 vs 1.92 degC at Maud Rise), a consequence of the different convective history rather than a driver. Conclusion: the cavity run lacks the Maud Rise polynya because basal melt of the DML ice shelves freshens the surface layer of the eastern Weddell gyre by a few hundredths of a psu, enough to keep the column below the convection threshold in most winters. Because both runs are single members the exact frequency (21 vs 84 percent) carries sampling uncertainty, but the systematic salinity offset over 164 years makes the sign robust.

8. Future convection. Under SSP2-4.5, SSP3-7.0 and SSP5-8.5 open-ocean convection stops in the no-cavity run between about 2060 and 2120 (Maud box area with MLD2 > 2000 m, no cavity: 0.14, 0.07 and 0.09e6 km2 in 2041-2070, 0.03, 0.04 and 0.00 in 2071-2100, 0.01, 0.00 and 0.00 in 2171-2200 for SSP2-4.5, SSP3-7.0 and SSP5-8.5) and never recovers in the cavity run, because the upper ocean freshens and the stratification (0-200 minus 200-1000 m sigma2 in the Weddell gyre) strengthens from -0.2 to -0.53 (SSP3-7.0) and -0.63 kg m-3 (SSP5-8.5) by 2171-2200 while the surface salinity contrast grows from -0.44 to -0.61 psu. Under SSP1-2.6 the no-cavity run convects more in the far period (0.30e6 km2 with MLD2 > 2000 m, double the pd value) as the thinning winter ice reduces the freshwater barrier while the deep heat reservoir remains, and the cavity run begins to convect intermittently (0.02e6 km2). Open-ocean convection therefore survives only in the low-emission world, and only in the set without cavities.

5.1.4 Mixed layers on the shelves

9. September shelf mixed layers are deep (300 to 700 m, frequently reaching the bottom) and deeper in the cavity run (Weddell shelf 324 vs 292 m, Ross shelf 383 vs 274 m at pd), consistent with its stronger winter freezing. They do not shoal with warming until after 2150; the Ross shelf MLD2 in the cavity run even deepens to 400-420 m in 2071-2100 under SSP2-4.5 to SSP5-8.5 before falling to 339 m by 2171-2200 under SSP5-8.5. The no-cavity shelf mixed layers stay near 270-300 m in all scenarios. Since the shelf water is

0.2 psu too fresh and HSSW is absent (overview theme), deep shelf mixing here homogenises a light water column and does not by itself make AABW.

5.1.5 Implications for AABW formation

10. The precursors of AABW formation in the model are weak at present day and vanish in the future. At pd the shelves do release brine each winter (5 to 9 mm/day freshwater loss) and mix to the bottom, but the resulting shelf water is not dense enough to reach the abyss (overview theme). In the no-cavity run the dominant deep-water formation pathway is the Maud Rise open-ocean polynya, a mode that is rare in observations and intermittent in the cavity run. Under SSP3-7.0 and SSP5-8.5 both pathways close: shelf freezing turns to net melt by 2171-2200, the winter pack disappears (September extent below 1e6 km2 after 2146-2174) and open-ocean convection ceases before 2100. Under SSP1-2.6 shelf freezing keeps about 80 percent of its present strength and the no-cavity run intensifies its Maud Rise convection, so the two sets diverge most in the low-emission scenario.

5.1.6 Caveats

11. Single members: hist and SSP are one member per set, so the 1 to 1.1e6 km2 sea-ice offset is well established over 164 years but the convective frequencies have sampling uncertainty. Region boxes near the coast compare different domains (327e3 km2 of PI_ICEv2 cavity is open ocean in PI_CTRL), which affects the dml and amery shelf diagnostics and the coastal-band polynya areas. fw is the total surface freshwater flux (ice growth and melt, precipitation, evaporation, runoff on some nodes), so the winter positive values are a net-freezing proxy and not ice production alone. The observed extent reference is an approximate NSIDC climatology quoted from memory, not a processed data set. MLD2 on cavity nodes is an artefact and has been excluded everywhere.

Figures

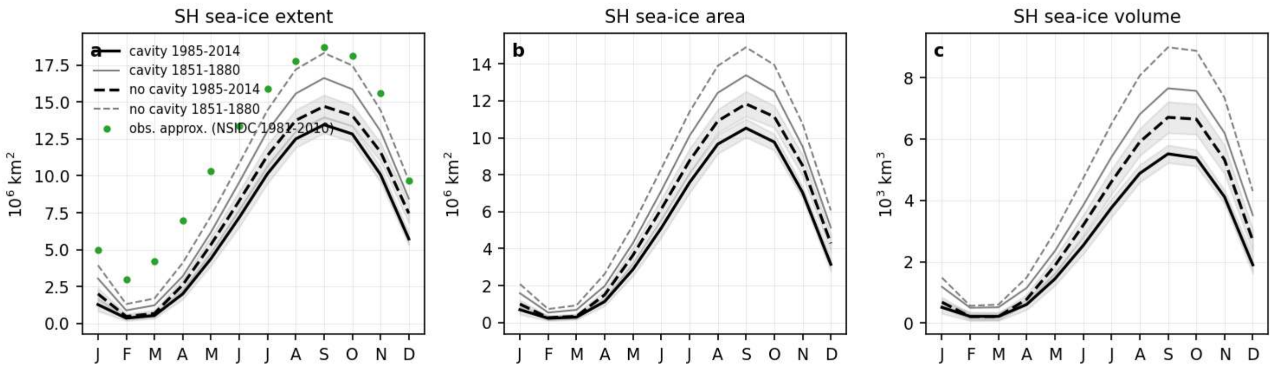

Figure 35: si_seasonal_cycle (seaice). Monthly climatology of Southern Hemisphere sea-ice extent (a, concentration above 15 percent), sea-ice area (b) and sea-ice volume (c) for 1985-2014 (thick, shading one standard deviation of the monthly values) and for 1851-1880 (thin grey) in the cavity (solid) and no-cavity (dashed) runs. Green dots in (a) give an approximate observed NSIDC 1981-2010 climatology for reference. Units 1e6 km2 and 1e3 km3. Take-away. September extent at present day is 13.5e6 km2 (cavity) and 14.6e6 km2 (no cavity) against about 18.5e6 km2 observed, and the February minimum is 0.36 and 0.49e6 km2 against about 3e6 km2 observed, so the model has too little ice in all seasons and almost no summer ice. In 1851-1880 the September extent was 16.6 and 18.3e6 km2 and the ice volume 7.6 and 9.1e3 km3, so the cavity run carries about 1.1e6 km2 less winter extent and 1.2e3 km3 less September volume throughout the historical period.

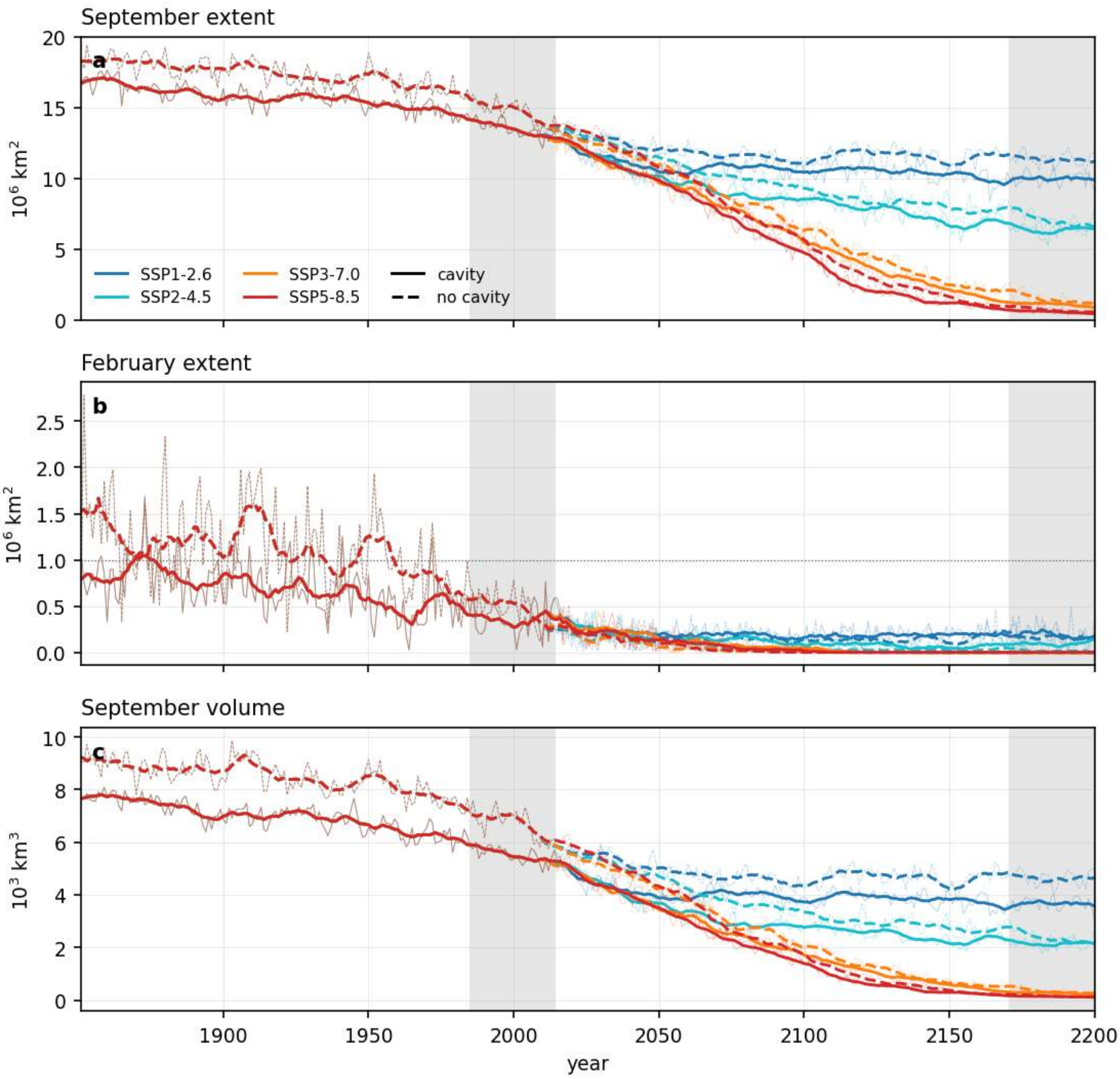

Figure 36: si_timeseries (seaice). Annual September extent (a), February extent (b) and September volume (c) of Southern Hemisphere sea ice from 1851 to 2200 for all four scenarios in both sets. Thin lines are annual values, thick lines 11-year running means. Grey bands mark the pd and far periods. The dotted line in (b) marks 1e6 km2 (ice-free summer). Take-away. February extent is below 1e6 km2 in almost all years already in the historical period (first five consecutive years 1852 cavity, 1935 no cavity), so an ice-free summer is a pre-existing model bias and not a projected transition. September extent falls below 10e6 km2 around 2040-2050 in SSP3-7.0 and SSP5-8.5 (cavity 2038/2041, no cavity 2059 for SSP3-7.0) and the September winter pack is essentially gone (below 1e6 km2) by 2146 in SSP5-8.5 and 2174 in SSP3-7.0 in the cavity run; SSP1-2.6 stabilises near 10e6 km2 (cavity) and 11e6 km2 (no cavity) and SSP2-4.5 near 6.5 and 7e6 km2.

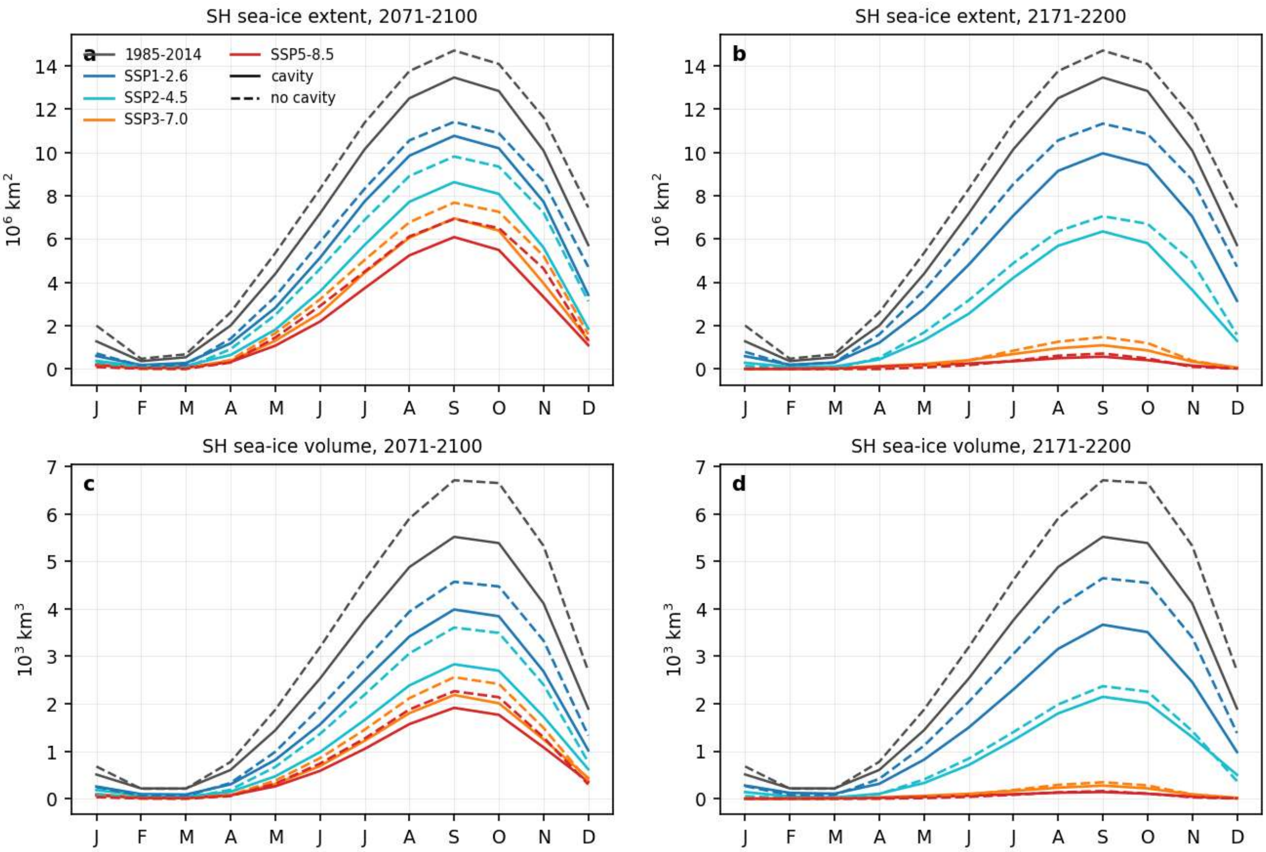

Figure 37: si_scen_cycle (seaice). Monthly climatology of Southern Hemisphere sea-ice extent (a, b) and volume (c, d) in 2071-2100 (left) and 2171-2200 (right) for all scenarios and both sets, with the 1985-2014 climatology in grey. Take-away. By 2171-2200 the seasonal cycle survives only in SSP1-2.6 (max extent 10.0 cavity and 11.3e6 km2 no cavity) and SSP2-4.5 (6.4 and 7.1e6 km2); under SSP3-7.0 and SSP5-8.5 the ocean south of 60S is essentially ice free all year with a September maximum below 1.5e6 km2 and a maximum volume below 0.4e3 km3.

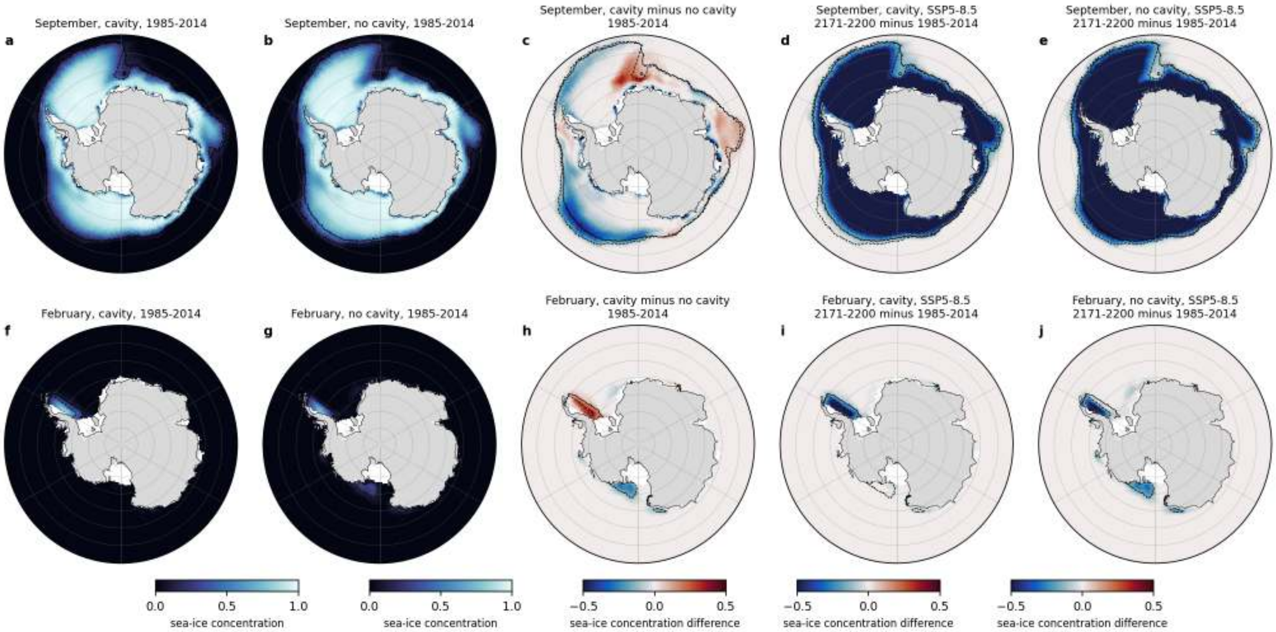

Figure 38: si_maps_conc (seaice). September (top) and February (bottom) sea-ice concentration: 1985-2014 climatology of the cavity run (a, f) and no-cavity run (b, g), cavity minus no cavity (c, h), and the SSP5-8.5 2171-2200 minus 1985-2014 change for the cavity run (d, i) and the no-cavity run (e, j). Black contours mark the 15 percent concentration edge of the pd climatologies (solid cavity, dashed no cavity). Take-away. The cavity run has less winter ice in the Amundsen, Bellingshausen and Ross sectors and along the outer Indian Ocean edge, and more ice over the eastern Weddell gyre near Maud Rise where the no-cavity run opens a recurrent polynya. In February only the western Weddell retains ice; the cavity run has more of it. Under SSP5-8.5 the entire September pack is lost by 2171-2200.

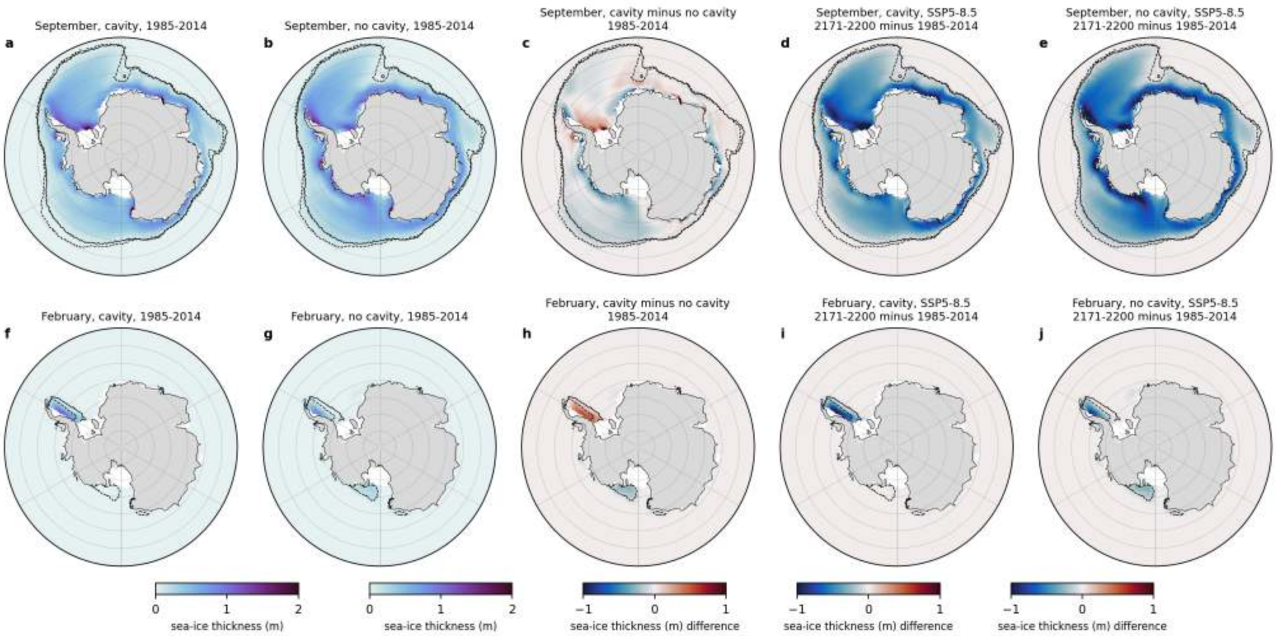

Figure 39: si_maps_thick (seaice). As si_maps_conc.png but for grid-mean sea-ice thickness (m). Take-away. Thickest ice (above 1.5 m) sits in the western Weddell Sea and along the coast; the cavity run is thinner by 0.2 to 0.5 m in the Amundsen and Ross sectors and at the outer edge, and thicker along parts of the East Antarctic coast where the PI_ICEv2 mesh has an ice-shelf front instead of open ocean.

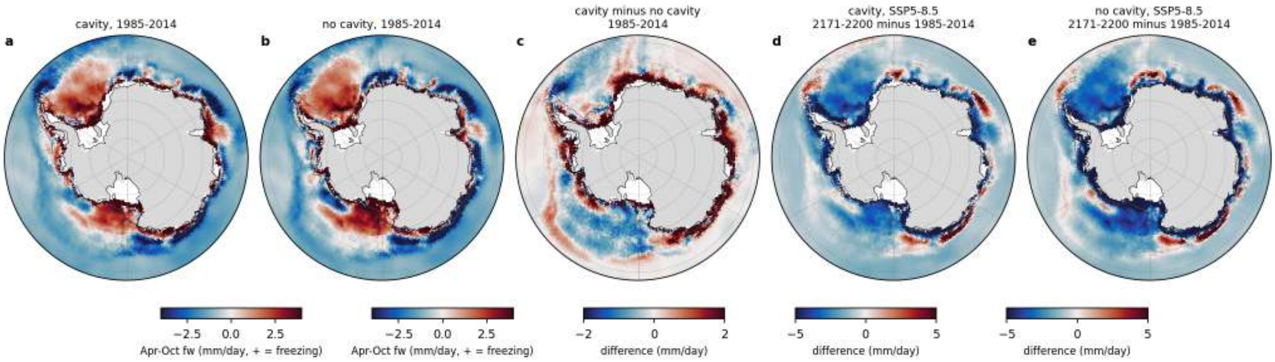

Figure 40: si_brine_maps (seaice). Winter (April to October) mean surface freshwater flux fw in mm/day (positive = freshwater loss, i.e. net sea-ice formation and brine release) for 1985-2014 in the cavity run (a) and no-cavity run (b), cavity minus no cavity (c), and SSP5-8.5 2171-2200 minus 1985-2014 for the cavity run (d) and the no-cavity run (e). The grey line is the 1000 m isobath. Take-away. Net freezing is concentrated on the continental shelves and in coastal polynyas (2 to more than 4 mm/day), with net melt north of the ice edge. The cavity run freezes more strongly along most of the coast (shelf means 5.5 vs 3.6 mm/day Weddell shelf and 8.8 vs 7.1 mm/day Ross shelf), consistent with the colder ice-shelf-water outflow at the ice fronts, and freezes less over the eastern Weddell gyre. By 2171-2200 under SSP5-8.5 the shelf freezing signal has vanished and the whole Southern Ocean receives net freshwater (negative values).

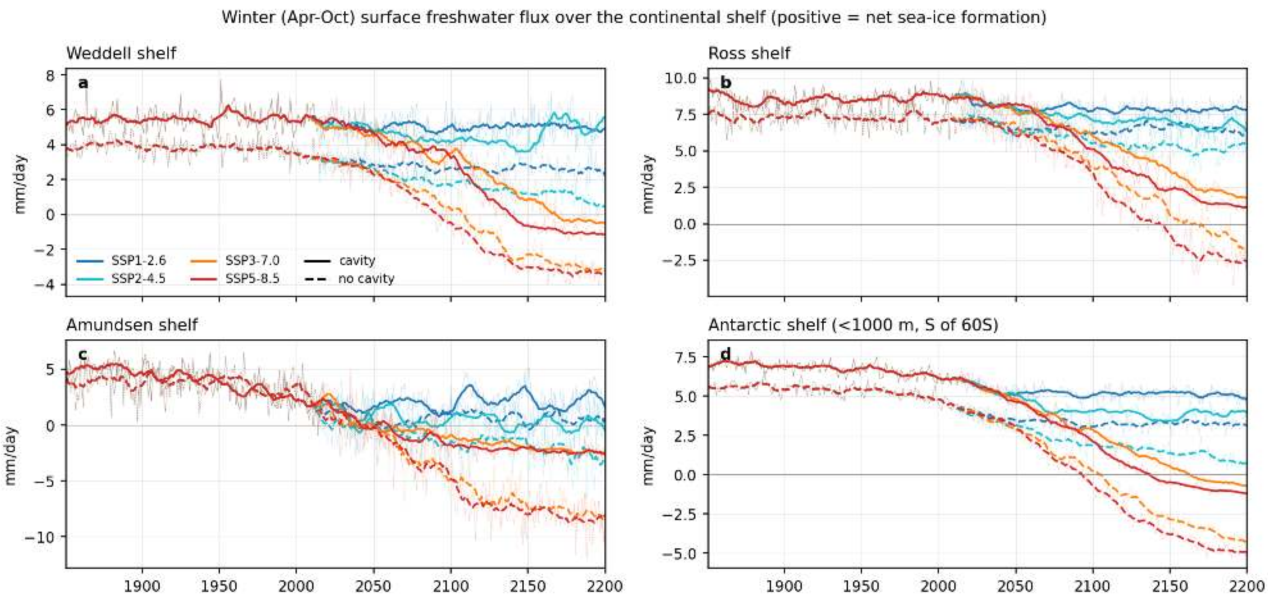

Figure 41: si_shelf_fw_ts (seaice). Time series 1851-2200 of the winter (April to October) shelf-mean surface freshwater flux (mm/day, positive = net freezing) for the Weddell shelf (a), Ross shelf (b), Amundsen shelf (c) and the whole Antarctic shelf south of 60S (d), all scenarios, both sets. Thin lines annual, thick lines 11-year running means. Take-away. Present-day net winter freezing is larger in the cavity run on every shelf (Antarctic shelf 6.2 vs 4.7 mm/day). Freezing declines under all scenarios and turns to net melt (negative) on the Weddell, Amundsen and whole Antarctic shelves by 2171-2200 under SSP3-7.0 and SSP5-8.5 (Antarctic shelf cavity -0.5 and -1.1 mm/day), i.e. the brine source for dense shelf water disappears. SSP1-2.6 keeps about 80 percent of the present-day freezing.

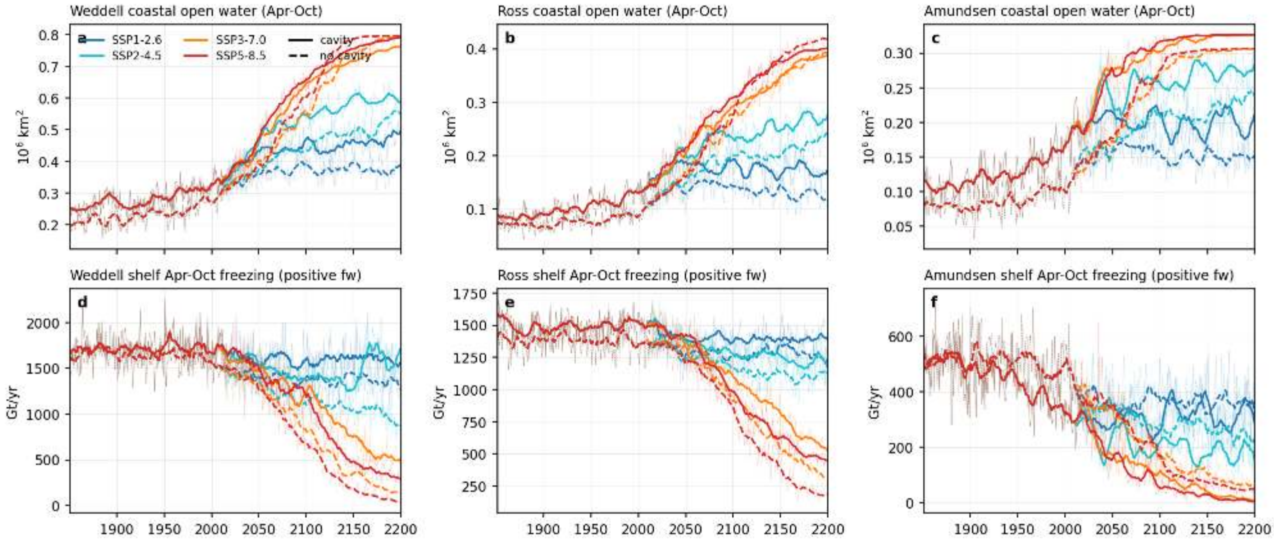

Figure 42: si_coastal_polynya (seaice). Coastal polynya activity and sea-ice production. Top row: April to October mean open-water area (concentration below 0.7) within a 150 km band along the coast and ice-shelf fronts in the Weddell (a), Ross (b) and Amundsen (c) sectors, 1e6 km2. Bottom row: April to October mean area-integrated freezing (sum of positive fw over the shelf region, expressed as Gt/yr rate) for the Weddell shelf (d), Ross shelf (e) and Amundsen shelf (f). All scenarios, both sets. Take-away. The cavity run has more winter coastal open water than the no-cavity run at pd (Weddell 0.33 vs 0.29, Ross 0.14 vs 0.10, Amundsen 0.17 vs 0.11e6 km2 out of band areas of 0.82, 0.45 and 0.33e6 km2) and also more integrated shelf freezing (Weddell shelf 1714 vs 1585 Gt/yr, Ross shelf 1526 vs 1363 Gt/yr), so its coastal polynyas are more active. Coastal open water roughly doubles by 2171-2200 under SSP3-7.0 and SSP5-8.5 (Weddell 0.75 to 0.80e6 km2, nearly the whole band) while the shelf freezing collapses to 340 (cavity) and 70 Gt/yr (no cavity) on the Weddell shelf and 475 and 206 Gt/yr on the Ross shelf; under SSP1-2.6 freezing stays within 10 percent of present day. The Amundsen shelf freezing already fell from 516 to 319 Gt/yr between 1851-1880 and 1985-2014 in the cavity run.

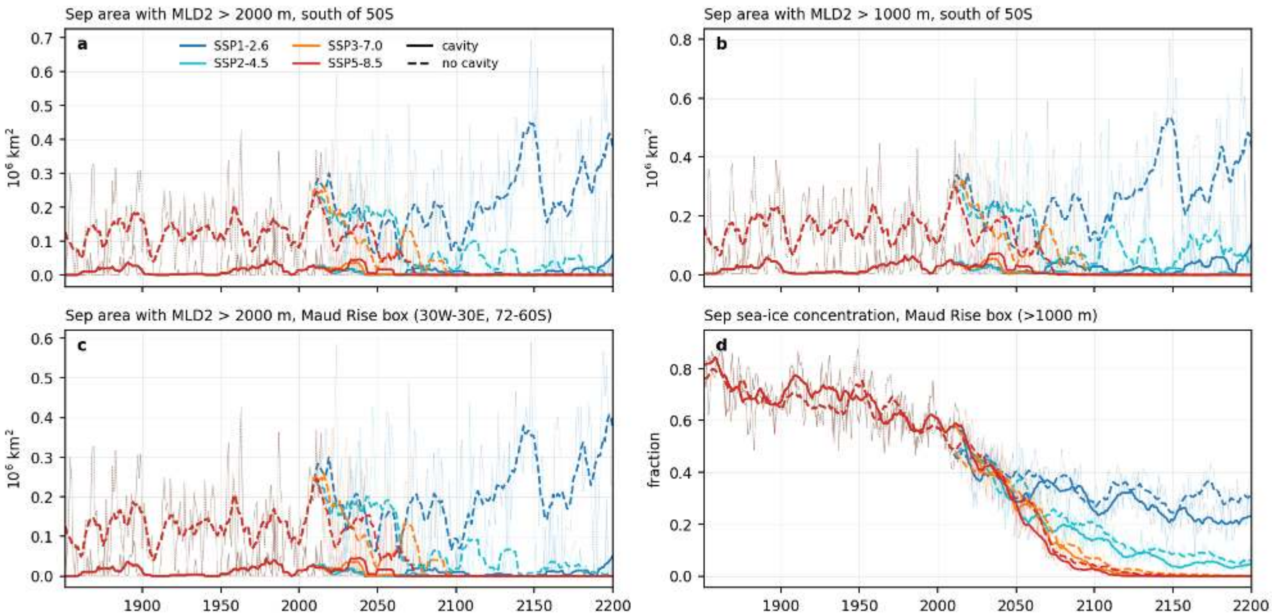

Figure 43: si_convection_ts (seaice). Open-ocean convection 1851-2200. September area with MLD2 deeper than 2000 m (a) and 1000 m (b) south of 50S, the same for 2000 m inside the Maud Rise box 30W-30E, 72-60S (c), and the September sea-ice concentration averaged over the deep (>1000 m) part of the Maud Rise box (d). All scenarios, both sets, thin annual and thick 11-year means. Take-away. Deep convection occurs almost only at Maud Rise. The no-cavity run convects in 84 percent of the historical years (mean September area with MLD2 > 2000 m 0.10e6 km2 in 1851-1880 and 0.16e6 km2 in 1985-2014), the cavity run in 21 percent (0.01 to 0.02e6 km2). Convection collapses under SSP2-4.5 and stronger forcing after about 2050 to 2100 in the no-cavity run, but intensifies under SSP1-2.6 (0.30e6 km2 in 2171-2200) as the winter pack thins while the deep heat reservoir remains.

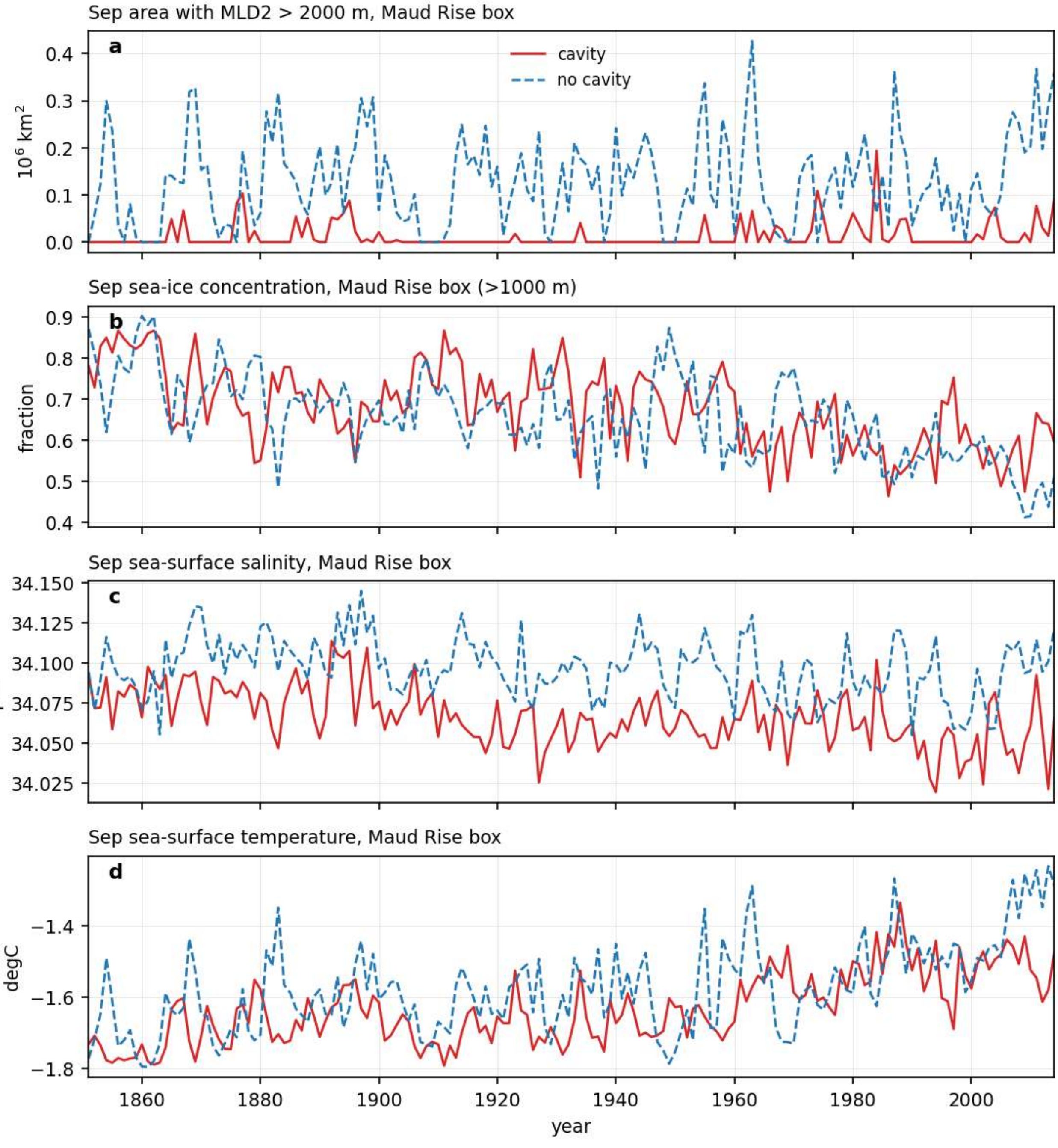

Figure 44: si_maud_hist (seaice). Historical (1851-2014) Maud Rise detail for the cavity (red) and no-cavity (blue) runs: September area with MLD2 > 2000 m in the Maud Rise box (a), September sea-ice concentration (b), sea-surface salinity (c) and sea-surface temperature (d) averaged over the deep part of the box. Take-away. The no-cavity run opens the Maud Rise polynya almost every winter (137 of 164 years), the cavity run in 34 years, mostly during 1880-1900 and after 1960. Convective winters show 0.09 to 0.13 lower ice concentration, 0.1 to 0.17 K warmer and 0.015 psu saltier surface water over the box.

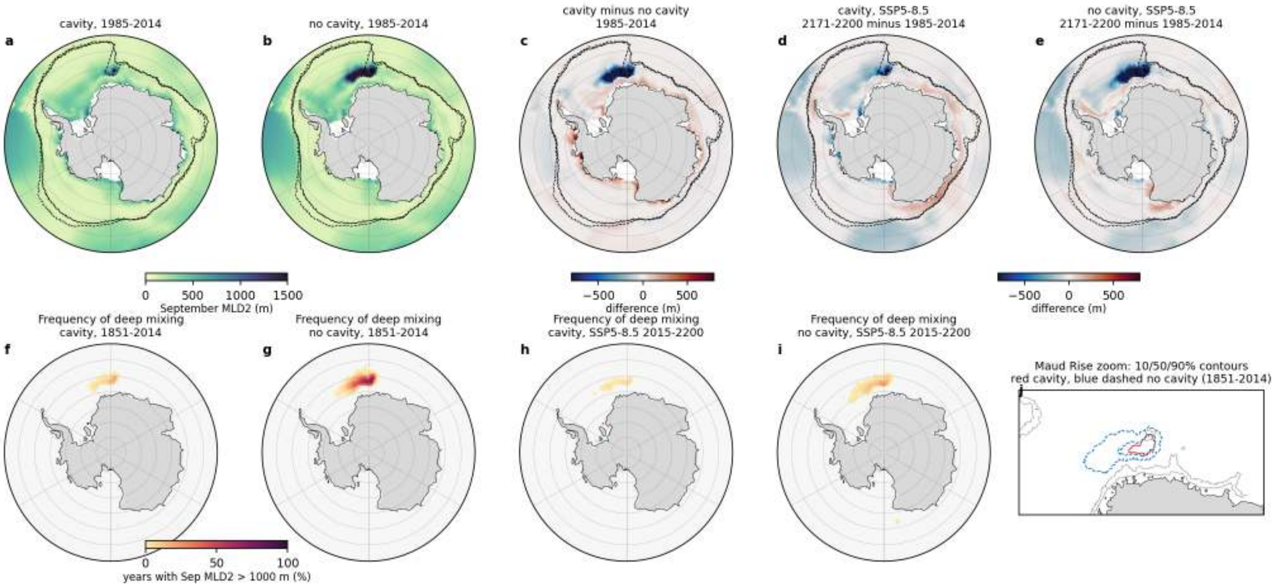

Figure 45: si_mld_maps (seaice). Top: September MLD2 climatology 1985-2014 for the cavity run (a) and no-cavity run (b), cavity minus no cavity (c), and SSP5-8.5 2171-2200 minus 1985-2014 for the cavity (d) and no-cavity (e) runs; black contours are the pd September ice edges. Bottom: percentage of years with September MLD2 deeper than 1000 m per node for 1851-2014 (f cavity, g no cavity) and for SSP5-8.5 2015-2200 (h, i), with a Maud Rise zoom (j) showing the 10, 50 and 90 percent contours of both historical runs (red cavity, blue dashed no cavity) and the 1000 and 3000 m isobaths. Take-away. Outside Maud Rise, September mixed layers are 100 to 500 m everywhere south of the ice edge and on the shelves. Deep mixing below 1000 m happens in more than 50 percent of years over 0.09e6 km2 north-east of Maud Rise in the no-cavity run but only over 0.004e6 km2 in the cavity run; the Maud Rise box-mean frequency is 4.8 percent (no cavity) against 0.6 percent (cavity). Under SSP5-8.5 deep mixing disappears in both sets.

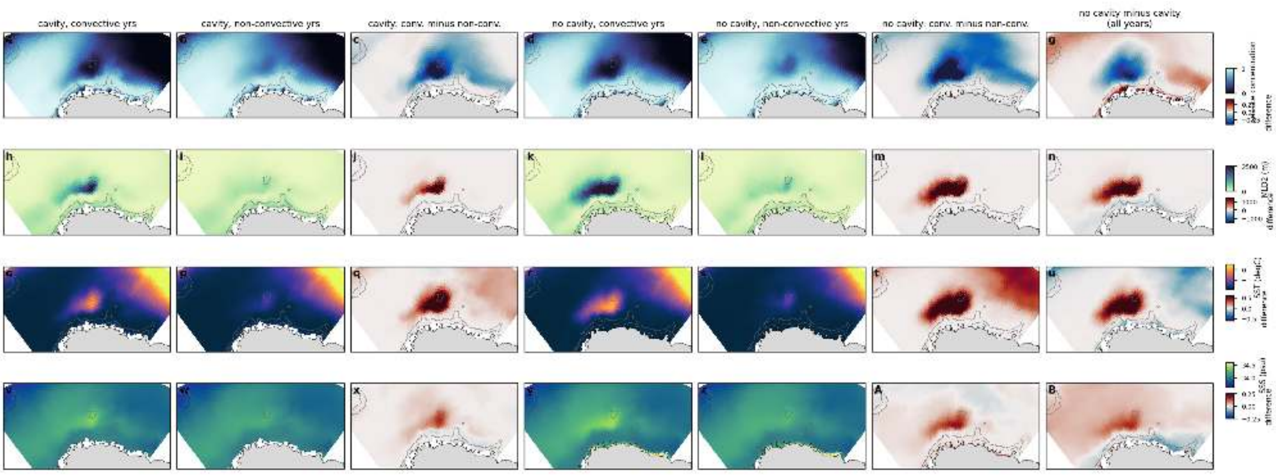

Figure 46: si_maud_composite (seaice). September composites over the Maud Rise region (stereographic view, 32W-32E, 73-59S; grey contours 1000 and 3000 m isobaths) of sea-ice concentration (row 1), MLD2 (row 2), SST (row 3) and SSS (row 4) for 1851-2014. Columns: cavity run convective years (34) and non-convective years (130) and their difference; no-cavity run convective (137) and non-convective (27) years and their difference; and the all-year no-cavity minus cavity difference. A convective year has more than 0.02e6 km2 with MLD2 > 2000 m inside the box. Take-away. Convective winters in both runs show the same signature, a polynya with concentration reduced by up to 0.4, MLD2 deeper by 1500 m, SST 0.5 K warmer and SSS 0.2 psu saltier over a 300 by 600 km patch east of Maud Rise. The all-year difference between the runs (last column) is this same pattern, so the no-cavity run differs from the cavity run mainly by having the polynya in most years.

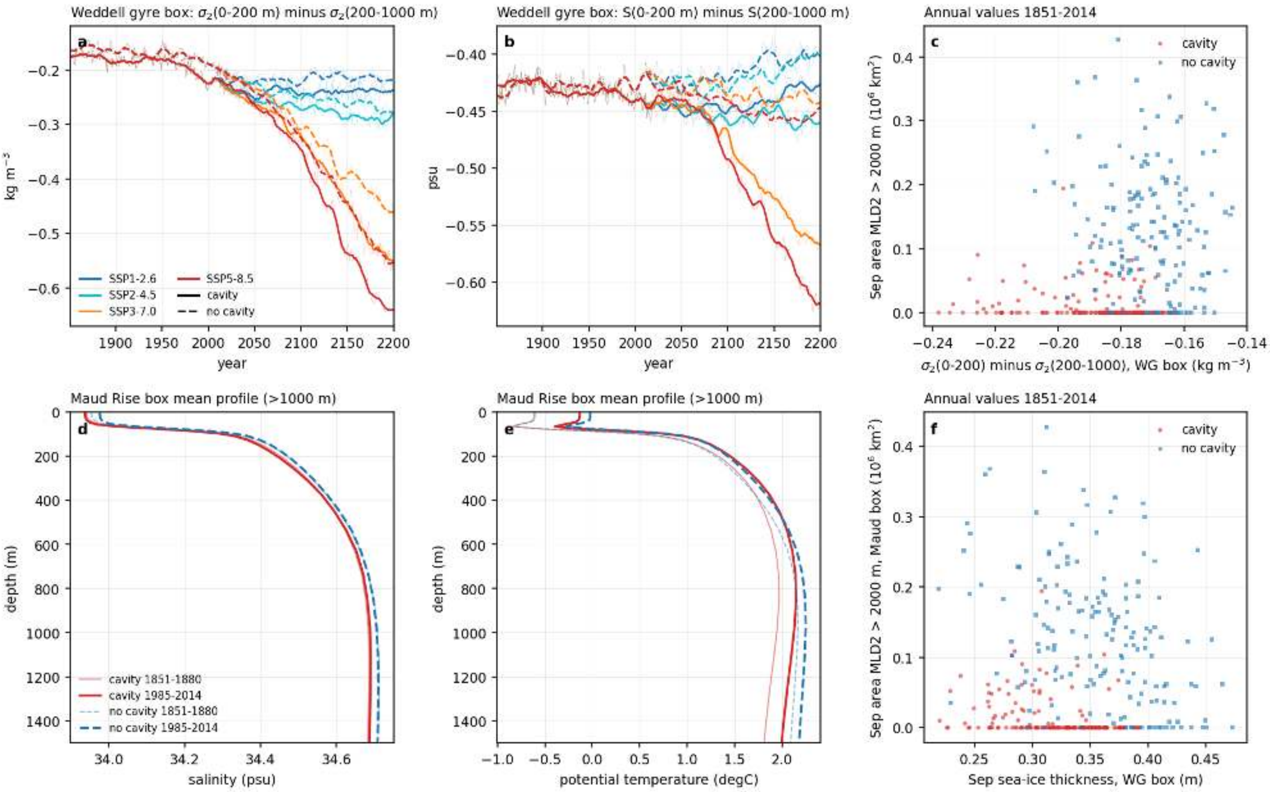

Figure 47: si_stratification (seaice). Stratification control. (a) Weddell gyre box (60W-30E, 75-55S) sigma2 of 0-200 m minus 200-1000 m and (b) the same for salinity, 1851-2200, all scenarios, both sets. (c) Annual September area with MLD2 > 2000 m south of 50S against the Weddell gyre stratification, 1851-2014. (d, e) Maud Rise box (>1000 m) mean salinity and potential temperature profiles for 1851-1880 (thin) and 1985-2014 (thick). (f) September convective area in the Maud Rise box against the September sea-ice thickness in the Weddell gyre box, 1851-2014. Take-away. The basin-scale stratification is nearly identical in both sets (pd density contrast -0.22 cavity vs -0.19 kg m-3 no cavity) and is not a good predictor of convection (correlation -0.1). The Maud Rise column is 0.03 to 0.05 psu fresher in the upper 200 m in the cavity run (34.20 vs 34.24 psu) and slightly more stratified (-0.138 vs -0.124 kg m-3). Convective winters have thinner ice (0.30 vs 0.33 m cavity, 0.35 vs 0.40 m no cavity). Stratification strengthens strongly after 2050 under SSP3-7.0 and SSP5-8.5 (to -0.53 and -0.63 kg m-3 by 2171-2200 in the cavity run) as the surface freshens by 0.15 to 0.2 psu relative to 200-1000 m.

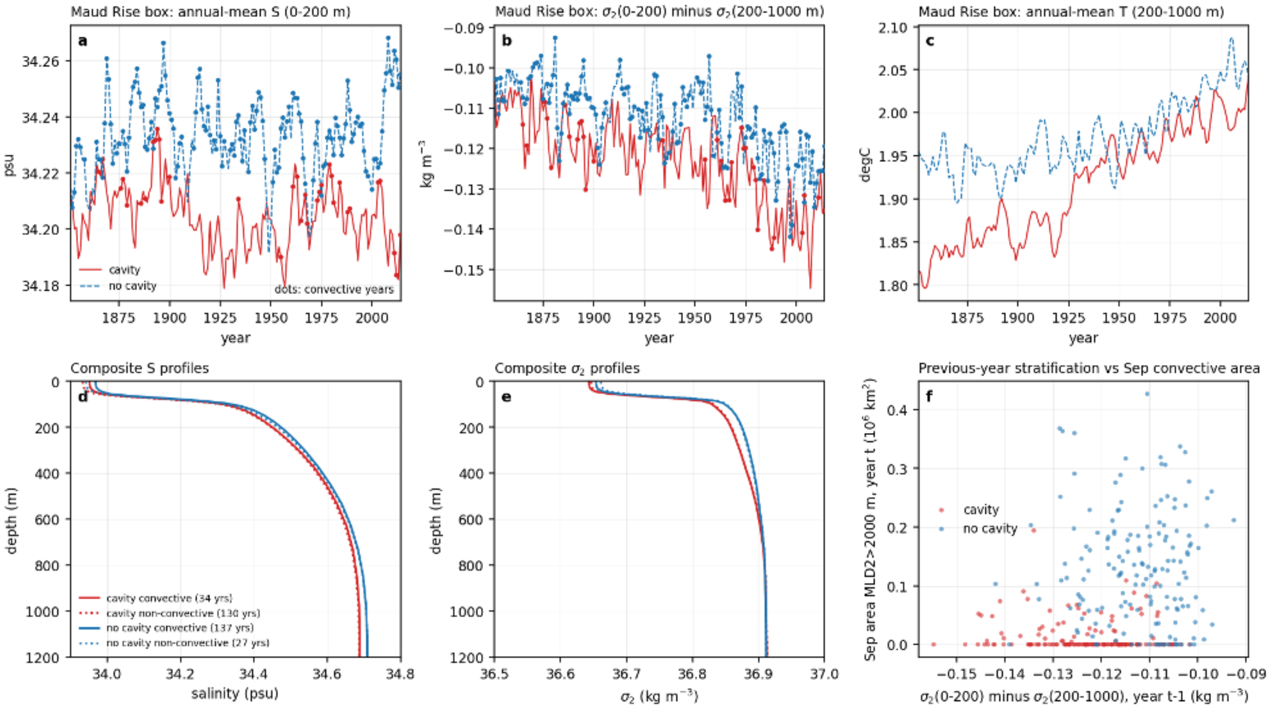

Figure 48: si__maud__strat (seaice). Maud Rise box (30W-30E, 72-60S, >1000 m, cavity excluded) annual-mean upper-ocean salinity 0-200 m (a), sigma2 difference 0-200 m minus 200-1000 m (b) and temperature 200-1000 m (c) for 1851-2014 with convective years dotted; composite salinity (d) and sigma2 (e) profiles for convective and non-convective years; (f) previous-year stratification against the September convective area. Take-away. The cavity run is 0.03 psu fresher in the upper 200 m over the whole historical period (34.204 vs 34.233 psu) and 0.01 kg m-3 more stratified. Within each run convection follows the upper-ocean salinity (correlation 0.70 no cavity, 0.38 cavity, lag-1 0.44 and 0.25), while the density contrast itself is a poor predictor. The deep layer warms by 0.2 K over the historical period in both runs.

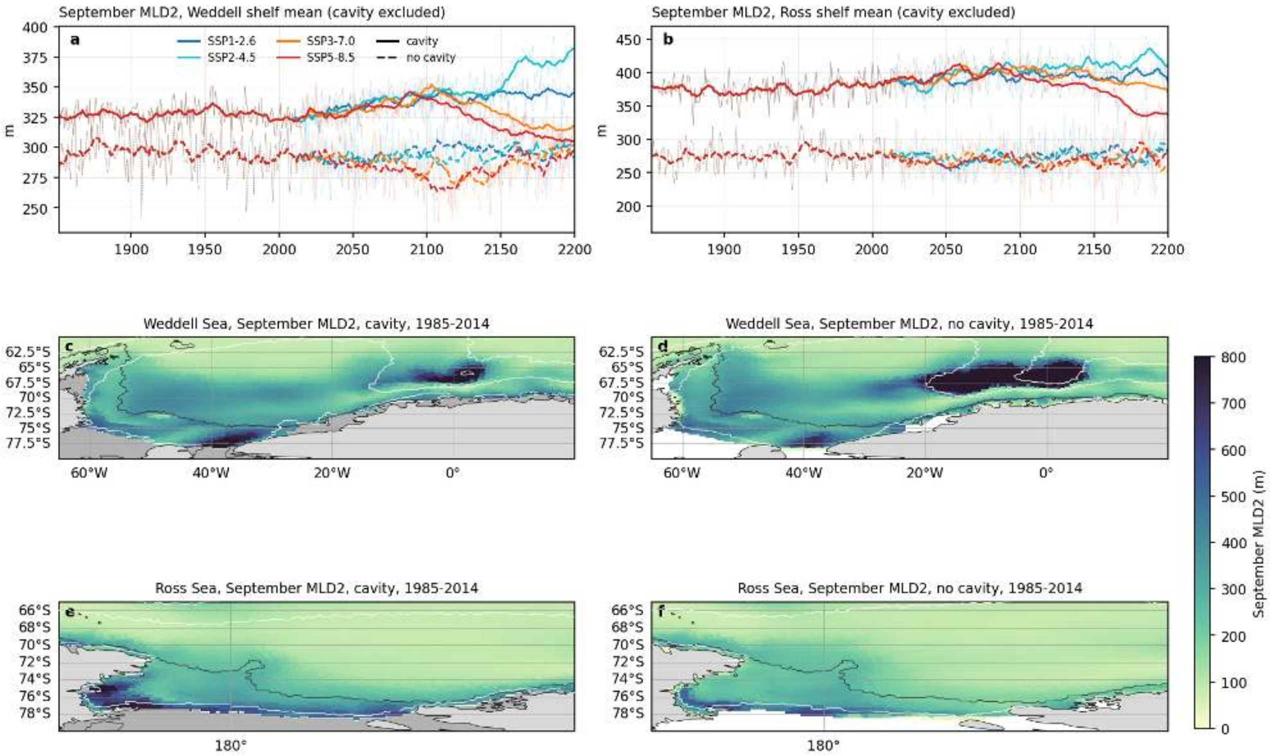

Figure 49: si_shelf_mld (seaice). September MLD2 averaged over the Weddell shelf (a) and Ross shelf (b) 1851-2200 (all scenarios, both sets), and maps of the September MLD2 climatology 1985-2014 for the Weddell Sea (c cavity, d no cavity) and Ross Sea (e, f); dark line 1000 m isobath, white contours the 15 and 80 percent September ice concentration, light grey the ice-shelf cavities of PI_ICEv2. Take-away. Shelf mixed layers reach 300 to 700 m and often the sea floor; they are deeper in the cavity run (Weddell 324 vs 292 m, Ross 383 vs 274 m) and remain so or deepen slightly until 2100. Only the strong scenarios reduce them after 2150 (Ross shelf SSP5-8.5 cavity 339 m by 2171-2200). Deep open-ocean mixing east of Maud Rise appears also in the cavity pd climatology but over a much smaller area.

6 Ice-shelf basal melt and freshwater budget

Findings

6.1 Findings, theme melt (ice-shelf basal melt, cavity hydrography, freshwater budget)

Scripts: analyses/melt/ (fig01__melt__clim.py, fig02__melt__ts.py, fig04__drivers.py, fig06__cavity__hydro.py, fig08__fw__budget.py, fig09__isw.py, cav__ts.py (SLURM, yearly cavity hydrography and ISW volumes into cache/melt/__cav__ts.nc), check__runoff.py, melt__lib.py). Tables: melt__table.md, melt__change__table.md, melt__sensitivity__table.md, fw__budget__table.md.

1. Present-day melt is too large by a factor 1.6 to 2.1 and wrongly distributed. PI__ICEv2__hist 1985-2014 gives 2364 plus or minus 143 Gt/yr (1.66 m/yr over 1.42e6 km2 of cavity) against 1100 to 1500 Gt/yr from satellites; in 1851-1880 it was 1930 Gt/yr. Filchner-Ronne (191 Gt/yr, 0.47 m/yr), Shackleton/West (118), Totten/Moscow University (81) and Getz (226) are within a factor 1.6 of the observations. The cold-water cavities are far too productive: Ross 257 Gt/yr (observed about 48, 5.4x), Amery 252 (36, 7x, 5.5 m/yr), Larsen C 122 (21, 5.8x), Dronning Maud Land 316 (about 100, 3.2x), George VI/Wilkins 305 (107, 2.9x). Pine Island and Thwaites are too low (88 vs about 199 Gt/yr, 3.8 m/yr instead of 17) although the Amundsen shelf bottom water is warm (1.35 degC); the small deep cavities are coarsely resolved and their melt is limited by the resolved exchange with the shelf. The latent heat of the melt is 25 TW; the diagnosed heat flux into the ice is 28.5 TW.

2. Thermal forcing at the ice base is small at present (cavity mean 0.17 K, ice-base T -1.89 degC, S 33.89) with ISW under Filchner-Ronne, Ross and Amery (ISW volume 282e3 km3, half of the cavity volume; cavity-mean temperature -1.62 degC, Filchner-Ronne -2.12, Ross -1.86; basal thermal forcing 0.04 K under Filchner-Ronne, 0.05 K under Ross, 0.35 K under Amery). The excessive melt of Ross and Amery is therefore produced by a small thermal forcing acting on large areas plus high melt rates on the thin shelf-front nodes, consistent with the generally too warm and too fresh continental shelf (bottom theme: shelf bottom salinity 0.2 psu too fresh, no HSSW). Melt refreezing patches exist under Ronne and central Ross.

3. Scenario response is strongly nonlinear and dominated by Filchner-Ronne. Total melt in 2171-2200 is 2500 (SSP1-2.6, +6 percent), 4100 (SSP2-4.5), 10100 (SSP3-7.0) and 12030 Gt/yr (SSP5-8.5, 5.1x). Filchner-Ronne switches within one decade from 200 to more than 1500 Gt/yr when the 11-year mean Weddell-shelf bottom temperature (outside the cavity) reaches about -0.45 to -0.35 degC: in 2078 under SSP5-8.5 (global warming 4.2 K above 1851-1880), 2093 under SSP3-7.0 (4.4 K) and 2153 under SSP2-4.5 (3.4 K), never under SSP1-2.6; it then rises to 4924 Gt/yr (26x) by 2171-2200 in SSP5-8.5 (Filchner 2518, Ronne 2380), i.e. warm water above 1 degC fills the Filchner Trough and reaches the grounding line at 1200 m. Ross (2016 Gt/yr, 7.8x), Amery (1104, 4.4x), Dronning Maud Land (1264, 4x) and Shackleton (433, 3.7x) increase gradually and accelerate after 2100. The Amundsen and Bellingshausen shelves change least (Getz 2.1x, PIG/Thwaites 3.2x, George VI 1.0x, Abbot 1.3x) and Larsen C decreases in all scenarios (0.5 to 0.9x); melt under the western Peninsula shelves decreases even in SSP1-2.6 because the Bellingshausen shelf cools and freshens.

4. Sensitivity to the adjacent shelf bottom temperature: Filchner-Ronne 3200 Gt/yr per K once tipped (r = 0.98), Ross 600 (steepening above 1 degC), Amery 430, Getz 190, PIG/Thwaites 150, and the Antarctic total 5900 Gt/yr per K of mean Antarctic shelf bottom warming (about 250 percent per K of the present value). Against the in-situ thermal forcing at the ice base (cavity-area mean) the melt follows a near-linear power law (exponent 1.1 for Filchner-Ronne and Ross, 1.4 for the total, 1.4 to 1.5 for Amery, Dronning Maud Land and PIG/Thwaites) with no threshold; the Filchner-Ronne step is a jump of the basal thermal forcing from 0.04 to 0.2 K (0.59 K by 2171-2200) when warm water crosses the sill, so the tipping sits in the cavity access, not in the melt law. Ocean heat loss to the ice grows from 28 TW (pd) to 154 TW (SSP5-8.5 far). The seasonal cycle of melt is weak (plus or minus 10 percent, maximum February to April).

5. The two sets differ in how, not how much, land-ice freshwater reaches the ocean. South of 60S the total freshwater input at present day is 11100 Gt/yr (cavity) vs 11200 Gt/yr (no cavity): P-E over the open ocean 9870 vs 9830 Gt/yr, sea ice (net freezing and export) -1140 vs -1110 Gt/yr. The land-ice term is 2364 Gt/yr of basal melt injected at depth inside the cavities in the cavity set (Antarctic runoff is exactly zero there), and 2509 Gt/yr of runoff in the no-cavity set (the ECHAM/JSBACH P-E of the Antarctic ice sheet routed to the coast, also 2290 Gt/yr in 1851-1880 vs 1930 Gt/yr of melt) spread at the surface over a 200 to 300 km wide coastal band at 0.3 to 3 m/yr with a winter maximum. Under SSP5-8.5 the cavity

set total reaches 27100 Gt/yr in 2171-2200 (melt 12030, P-E 16400) whereas the no-cavity total reaches 19450 Gt/yr (runoff 4280, P-E 16390, sea ice -1220) because its land-ice term only follows precipitation; so the sets diverge by 7600 Gt/yr of freshwater in the far future (end: 16520 vs 14930). Global open-ocean fw integrates to zero in both sets (FESOM balances the coupled water flux), hence the cavity-run melt is a net volume gain: global mean SSH rises 0.77 m between 1851-1880 and 1985-2014 in PI_ICEv2_hist and not at all in PI_CORE_hist (about 6 mm/yr, consistent with 2360 Gt/yr). This is a diagnostic artefact of the setup to be mentioned, not a physical signal.

6. Consequence for shelf stratification and AABW. In the no-cavity set 2500 Gt/yr of fresh water are placed at the surface of the coastal ocean where sea ice forms, directly opposing brine rejection; in the cavity set the same amount enters at 200 to 1000 m under the ice shelves, is mixed into cold cavity water and leaves as ISW or fresh shelf water at mid depth. Both routes freshen the shelf (bottom theme: shelves 0.2 psu too fresh in both sets), which helps explain the missing HSSW in both. The cavity run additionally holds 282e3 km3 of ISW inside the cavities at present day (50 percent of the cavity volume, Filchner-Ronne 189e3 and Ross 92e3 km3), but in annual-mean fields only 2 to 5e3 km3 of ISW are found on the open shelf (0.01 to 0.02e6 km2 of columns, mostly in front of Filchner-Ronne and Ross) and none in the no-cavity runs in any year; shelf water in front of the large cavities is 0.1 to 0.2 K above freezing in the cavity run against 0.2 to 0.5 K in the no-cavity run. Under SSP5-8.5 the cavity ISW shrinks to 184e3 km3 (2071-2100) and 35e3 km3 (2171-2200) and a transient 8 to 15e3 km3 ISW outflow pulse follows each Filchner-Ronne regime change (2085, 2100 and 2165 for SSP5-8.5, SSP3-7.0, SSP2-4.5); under SSP1-2.6 the cavity ISW stays at 278e3 km3. So cavity-driven cold fresh outflow is a distinctive feature only of the cavity set, so cavity-driven cold fresh outflow is a distinctive feature only of the cavity set, and the far-future 12000 Gt/yr meltwater pulse under SSP5-8.5 is a strong stratifying agent that does not exist in the no-cavity set.

Caveats and issues. Satellite comparison values are quoted from memory (approximate) and the model shelf groups follow lon/lat boxes, so the Dronning Maud Land, Shackleton/West and Peninsula groups are sums of several shelves. The sea-ice freshwater term is a residual (open-ocean -fw minus runoff minus ECHAM P-E) and absorbs small differences between the ECHAM ocean mask and FESOM open-ocean nodes. Melt per area is taken on cavity nodes inclusive of partially cavity-covered front nodes. The ISW diagnostics use the surface freezing point of the 30-year or annual mean fields (annual means underestimate seasonal ISW presence on the open shelf). PI_CORE_ssp585 panels are included only if its cache existed at run time (see the tables).

Story lines. (i) A simple cavity parameterisation at CORE2 resolution gives the right order of magnitude of total melt but a wrong pattern (cold cavities too active, warm cavities too weak), which is the signature of a too warm too fresh shelf without HSSW. (ii) Filchner-Ronne has a sharp threshold at Weddell shelf bottom T near -0.4 degC reached at about 3.5 to 4.5 K of global warming; this is the tipping element of the Antarctic freshwater budget. (iii) Cavities change the depth and timing of the land-ice freshwater injection rather than its present-day amount; only under strong warming does the melt feedback add 8000 Gt/yr that the no-cavity set cannot produce.

Figures

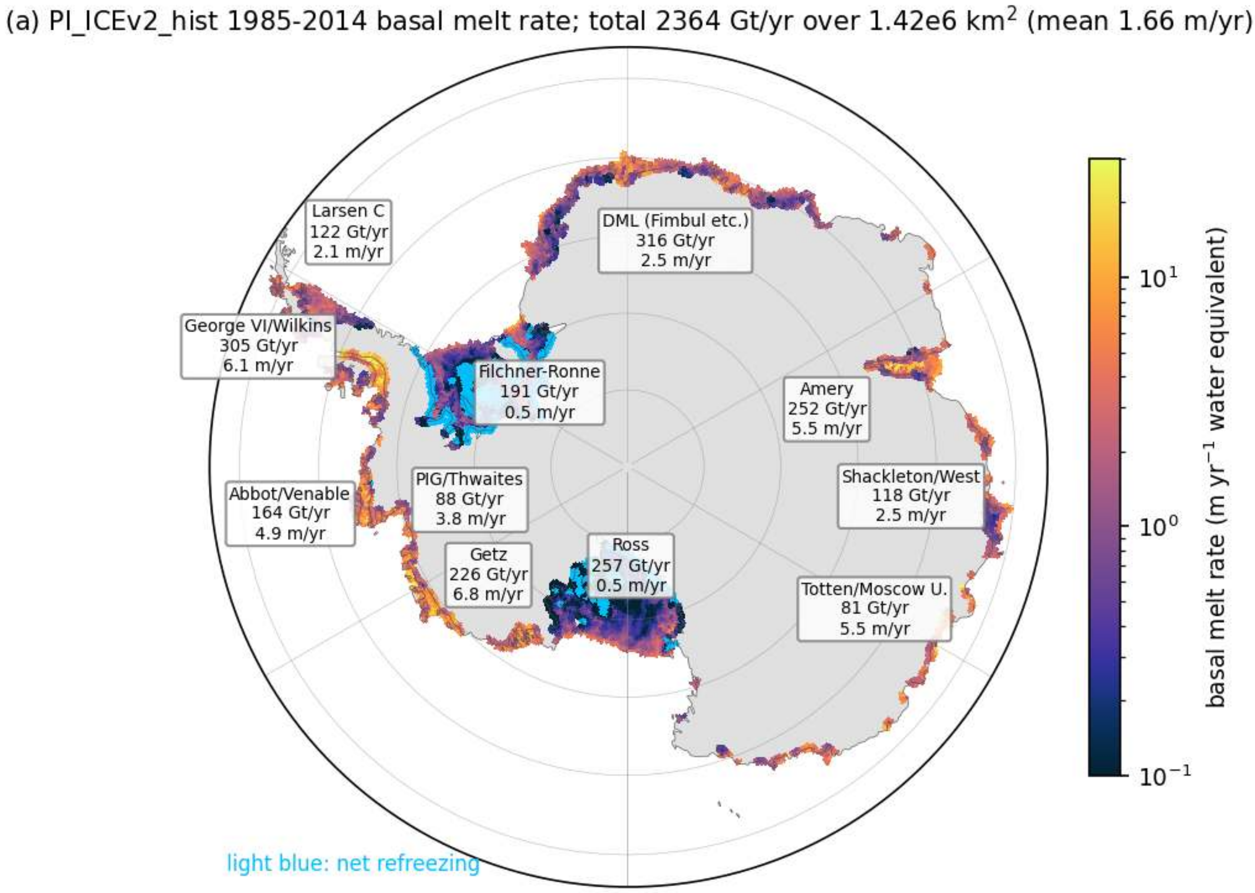

Figure 50: melt_clim_map (melt). Present-day (1985-2014) basal melt rate (m/yr water equivalent, log colour scale) of PI_ICEv2_hist on the cavity nodes, regridded to 0.1 degree, South Polar view south of 63S. Light blue marks net refreezing. Labels give the mass loss per ice-shelf group in Gt/yr and the area-mean melt rate in m/yr. The total is 2364 Gt/yr over 1.42e6 km2 of cavity (mean 1.66 m/yr). Melt is concentrated along the deep grounding lines of the small warm-water shelves (Getz, George VI, Abbot, Amery, Larsen C at 4 to 7 m/yr area mean, locally above 10 m/yr) while the two large cold cavities melt at about 0.5 m/yr with refreezing patches under Ronne and central Ross. The Filchner-Ronne value (191 Gt/yr) is close to observations, the Ross (257) and Amery (252) values are 5 to 7 times too high.

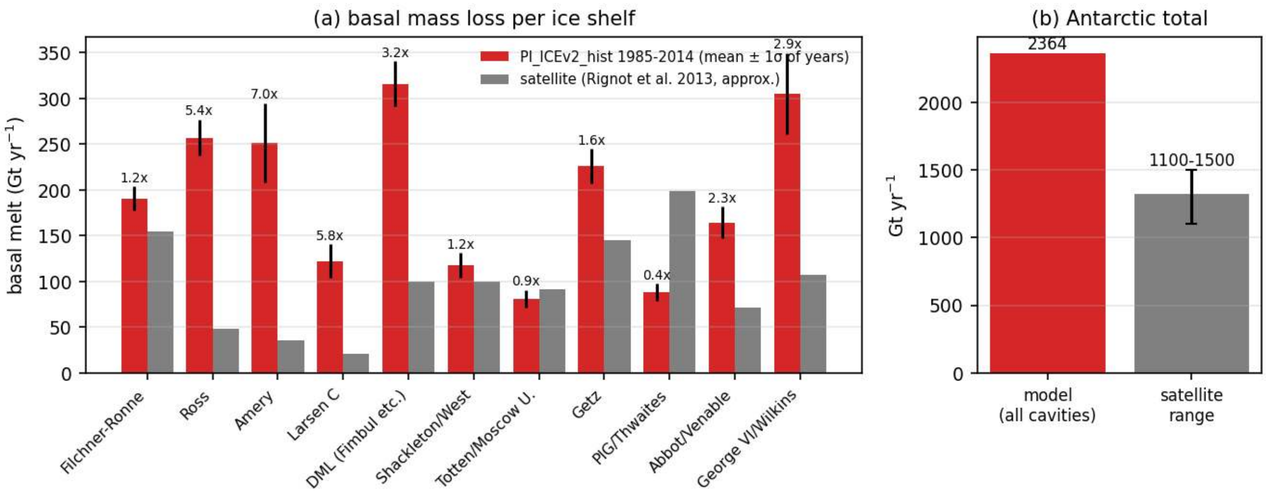

Figure 51: melt_vs_obs (melt). (a) Present-day basal mass loss per ice-shelf group in PI_ICEv2_hist (red, mean of 1985-2014 with one standard deviation of annual values) compared with approximate satellite estimates after Rignot et al. (2013) (grey); numbers give the model-to-observation ratio. (b) Antarctic total against the range of satellite estimates (1100 to 1500 Gt/yr). The model total of 2364 Gt/yr is 1.6 to 2.1 times the observed total. The excess comes from the cold-water shelves (Ross 5.4x, Amery 7x, Larsen C 5.8x, Dronning Maud Land 3.2x, George VI 2.9x) while Filchner-Ronne (1.2x), Shackleton (1.2x), Totten (0.9x) and Getz (1.6x) are within a factor 1.6 and Pine Island plus Thwaites is too low (0.4x).

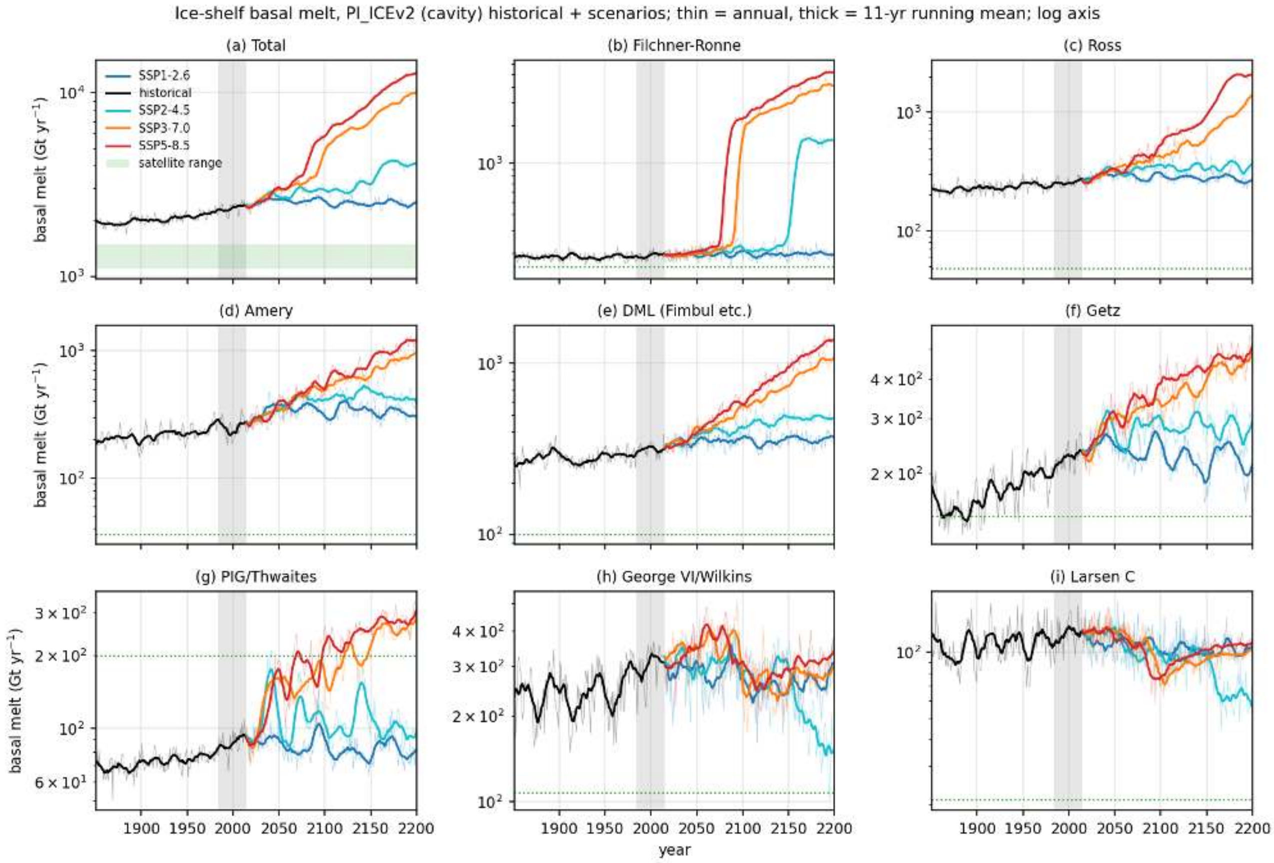

Figure 52: melt_ts_shelves (melt). Annual (thin) and 11-year running mean (thick) basal melt 1851-2200 for the Antarctic total and eight ice-shelf groups, PI_ICEv2 historical (black) followed by SSP1-2.6, SSP2-4.5, SSP3-7.0 and SSP5-8.5 (colours), logarithmic axis. Green shading or dotted line gives the satellite value. The total rises from 1930 Gt/yr in 1851-1880 to 2364 in 1985-2014 (+22 percent) and to 2500 (SSP1-2.6), 4100 (SSP2-4.5), 10100 (SSP3-7.0) and 12030 Gt/yr (SSP5-8.5) in 2171-2200. Filchner-Ronne shows an abrupt regime change (from about 200 to more than 1500 Gt/yr within a decade) in 2078 (SSP5-8.5), 2093 (SSP3-7.0) and 2153 (SSP2-4.5) and none in SSP1-2.6. Ross, Amery and Dronning Maud Land increase gradually by factors 4 to 8 in SSP5-8.5. The Amundsen and Bellingshausen shelves (Getz, PIG/Thwaites, George VI) change least (factor 1 to 3) and Larsen C even decreases.

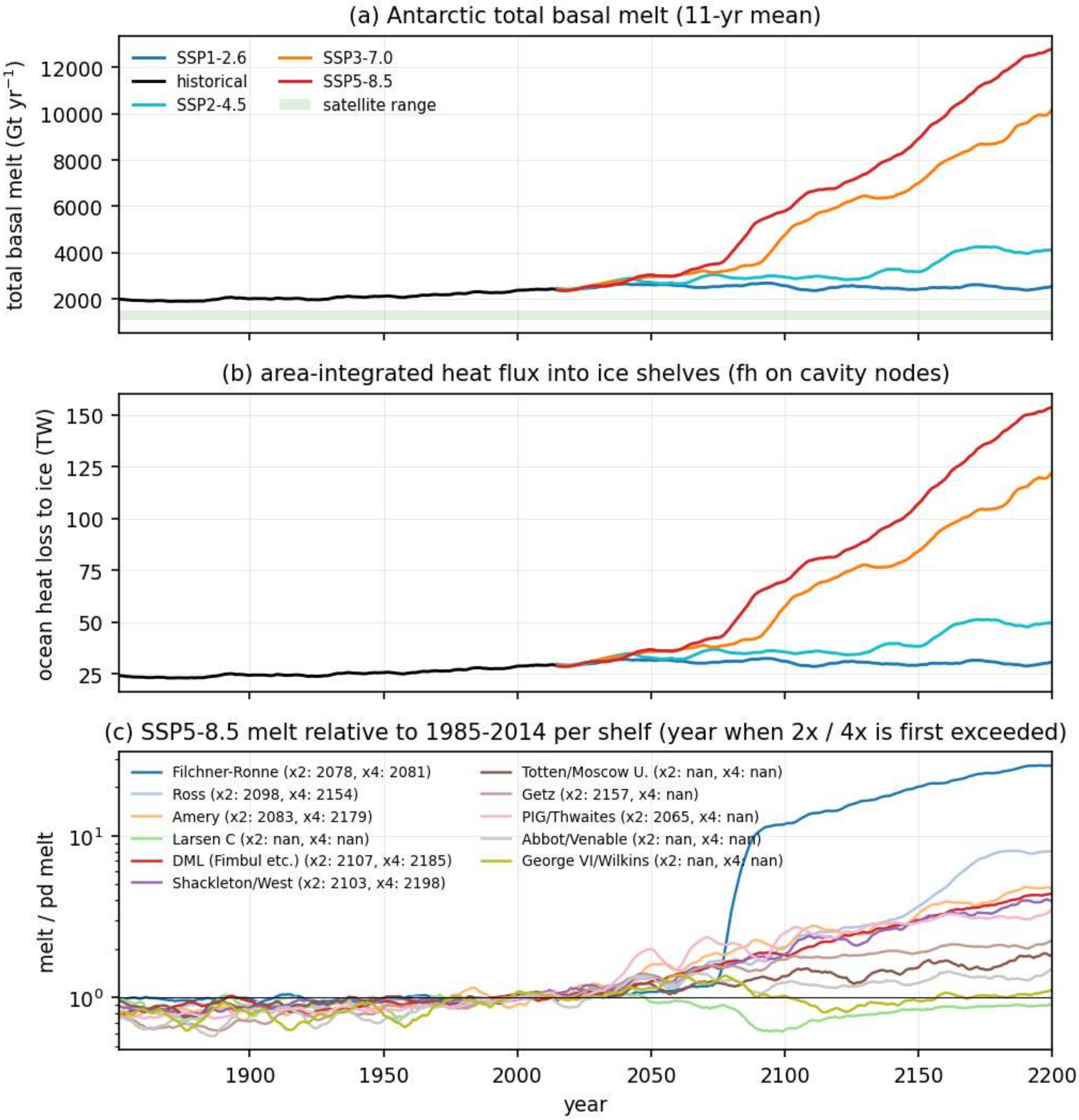

Figure 53: melt_total_heat_ts (melt). (a) Antarctic total basal melt (11-year mean) for all scenario chains, cavity runs. (b) Area-integrated ocean heat loss to the ice shelves (fh on cavity nodes, TW); 28.5 TW at present day (2364 Gt/yr of melt corresponds to 25 TW of latent heat). (c) SSP5-8.5 melt of each ice-shelf group divided by its 1985-2014 value (log axis) with the first year in which the 11-year mean exceeds two and four times the present-day value. Under SSP5-8.5 the total melt doubles by 2086 and quadruples by 2155; Filchner-Ronne doubles in 2078 and quadruples three years later, Pine Island/Thwaites doubles in 2065, Ross in 2098, Amery in 2083, Dronning Maud Land in 2107; Larsen C, Totten, Abbot and George VI never double.

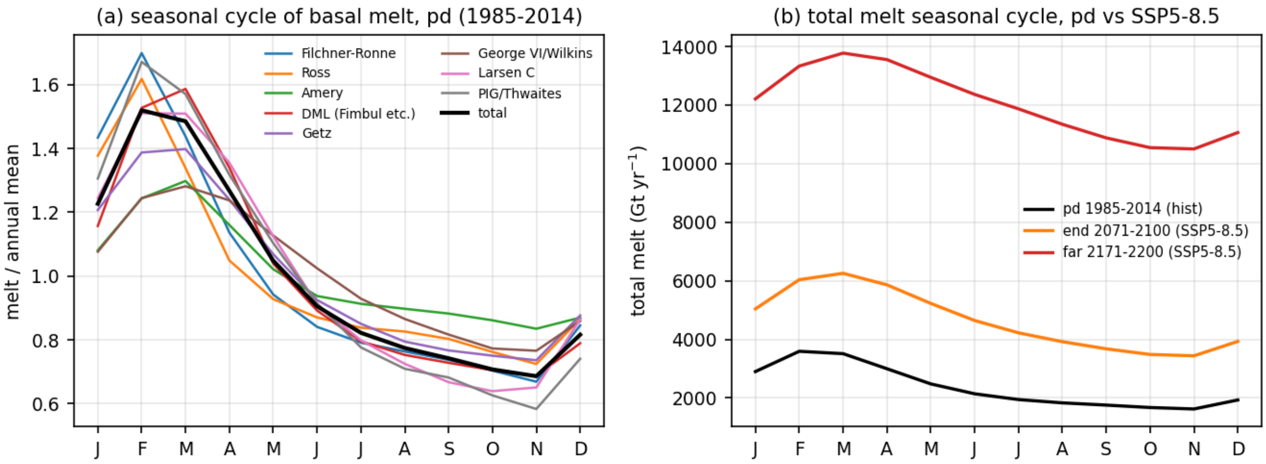

Figure 54: melt_seasonal (melt). (a) Mean seasonal cycle (1985-2014) of basal melt of selected ice shelves divided by their annual mean, and of the total (black). (b) Seasonal cycle of the total melt in pd (black), end (orange) and far (red) of SSP5-8.5. The present-day seasonal cycle is weak (total within plus or minus 10 percent, maximum in February to April when the summer-warmed surface layer reaches the shallow shelf fronts; Larsen C and George VI vary most, about 30 percent). In the far SSP5-8.5 period the total melt is 10000 to 14000 Gt/yr throughout the year with a late-summer maximum.

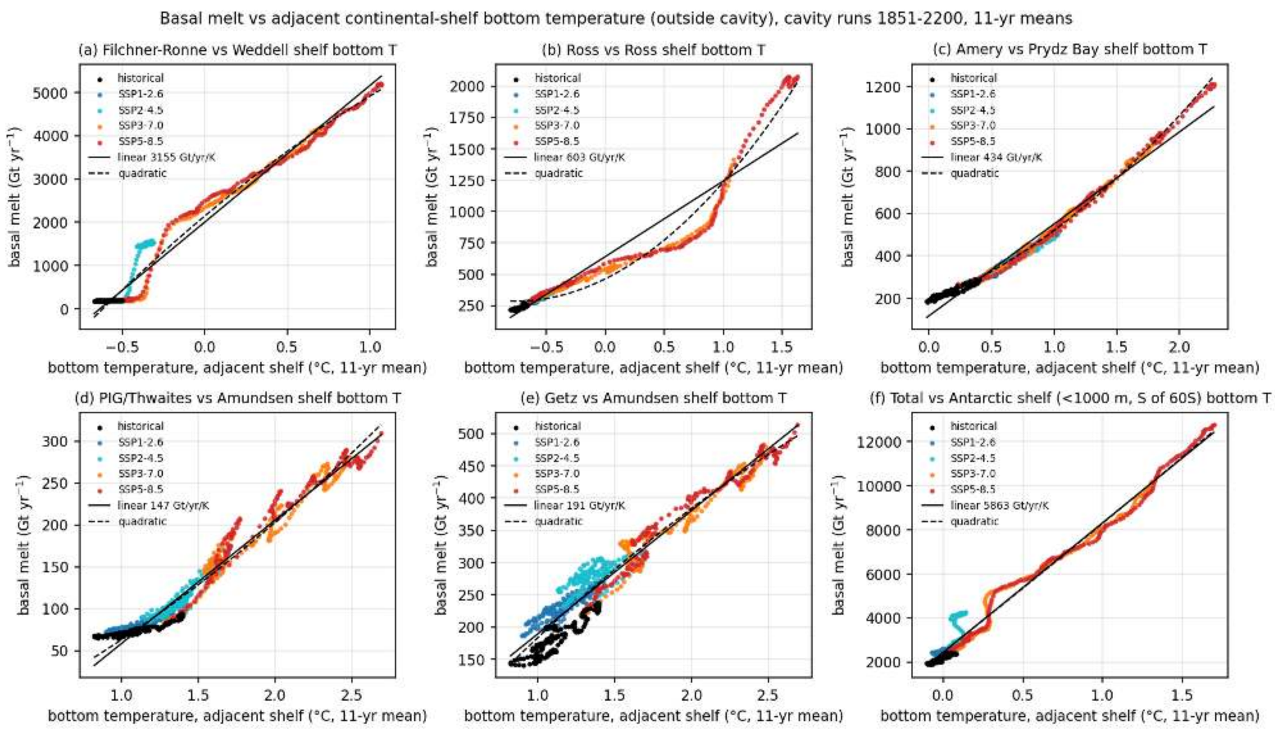

Figure 55: melt_vs_shelfT (melt). Scatter of 11-year mean basal melt against the 11-year mean bottom temperature of the adjacent continental-shelf region outside the cavity (cache bT, cavity nodes excluded) for all years 1851-2200 and all scenario chains (black historical, colours scenarios) with linear and quadratic fits. Filchner-Ronne vs Weddell shelf, Ross vs Ross shelf, Amery vs Prydz Bay shelf, PIG/Thwaites and Getz vs Amundsen shelf, total vs all Antarctic shelf (less than 1000 m, south of 60S). The Filchner-Ronne melt is a step function of Weddell shelf bottom temperature: below about -0.45 degC it stays near 200 Gt/yr, between -0.45 and -0.3 degC it jumps to 1500 to 2000 Gt/yr and above that it increases linearly at 3200 Gt/yr per K. Ross (600 Gt/yr per K, accelerating above 1 degC), Amery (430 Gt/yr per K), Getz (190) and PIG/Thwaites (150) respond quasi-linearly; the total responds at 5900 Gt/yr per K of Antarctic shelf bottom warming (250 percent per K relative to present day).

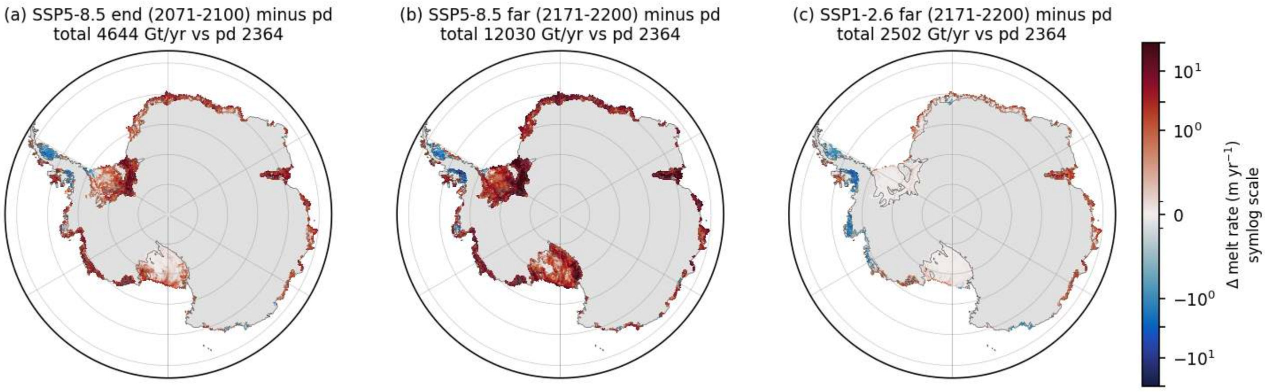

Figure 56: melt__change__maps (melt). Change of the basal melt rate (m/yr, symmetric log colour scale) relative to pd for (a) SSP5-8.5 end, (b) SSP5-8.5 far and (c) SSP1-2.6 far, cavity runs, with the total melt in the title. The largest increases (more than 10 m/yr) occur deep inside the Filchner-Ronne cavity (far SSP5-8.5), under Ross, Amery and the Dronning Maud Land shelves. Melt decreases under the western Antarctic Peninsula shelves (George VI, Wilkins, Abbot, Larsen C) in all cases including SSP1-2.6 where freshening and cooling of the Bellingshausen shelf reduce the melt by 1 to 3 m/yr.

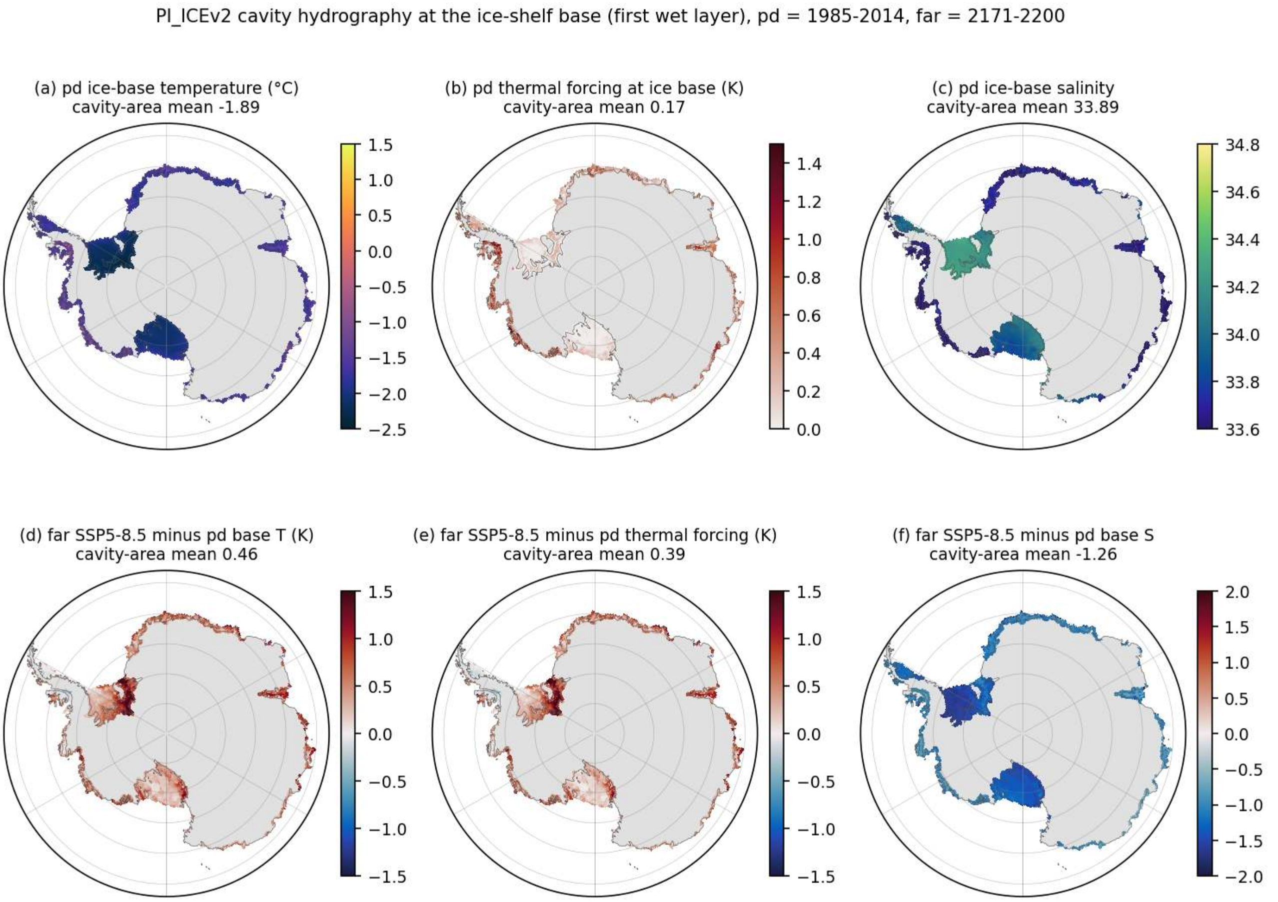

Figure 57: cavity__base__TS (melt). Ice-base hydrography of the cavity run (first wet layer under the ice shelf): (a) temperature, (b) thermal forcing T minus the in-situ freezing point at the draft depth, (c) salinity for pd, and (d-f) the far SSP5-8.5 minus pd differences. Cavity-area means are given in the titles. Present-day basal thermal forcing is only 0.17 K on average (0.05 to 0.1 K under Filchner-Ronne and Ross, 0.5 to 1 K under the small warm shelves); the ice base is at -1.89 degC and 33.89 psu. By 2171-2200 under SSP5-8.5 the base warms by 0.46 K (thermal forcing +0.39 K, up to +1.5 K in the Filchner-Ronne interior) and freshens by 1.26 psu because of the meltwater.

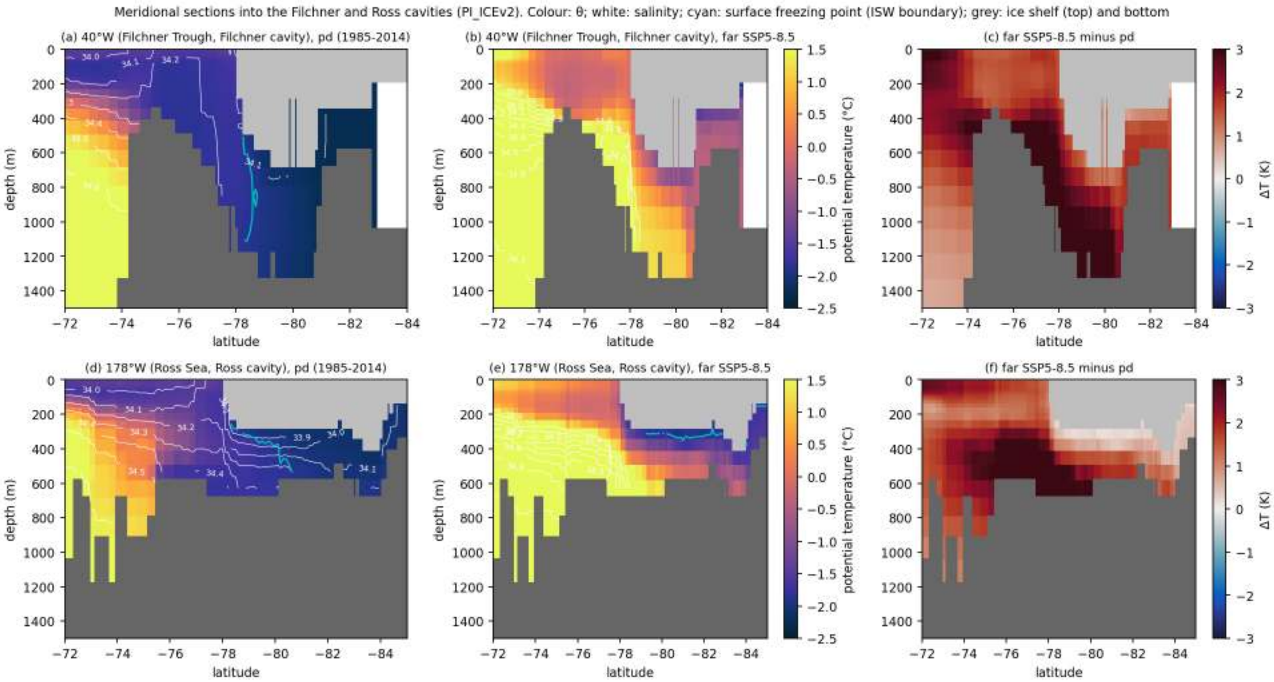

Figure 58: cavity_sections (melt). Meridional sections of potential temperature (colour), salinity (white contours) and the surface freezing point (cyan, ISW boundary) along 40W through the Filchner Trough into the Filchner cavity (top) and along 178W into the Ross cavity (bottom) for pd (left) and far SSP5-8.5 (middle) with the difference (right). Grey areas are the ice shelf and the sea floor along the nearest-node path. At present day both cavities are filled with water near -2 degC below about 34.1 (Filchner) and 34.4 psu (Ross), with ISW (below the surface freezing point) at mid depth under the ice; warm water above 0.5 degC stays north of the shelf break (Filchner) or on the outer shelf at 300 to 600 m (Ross). In far SSP5-8.5 water warmer than 1 degC fills the Filchner Trough and reaches the grounding line at 1200 m (warming of 2 to 3 K inside the cavity), and the Ross cavity warms by 1 to 2 K at depth while a thin fresh cold layer remains under the ice.

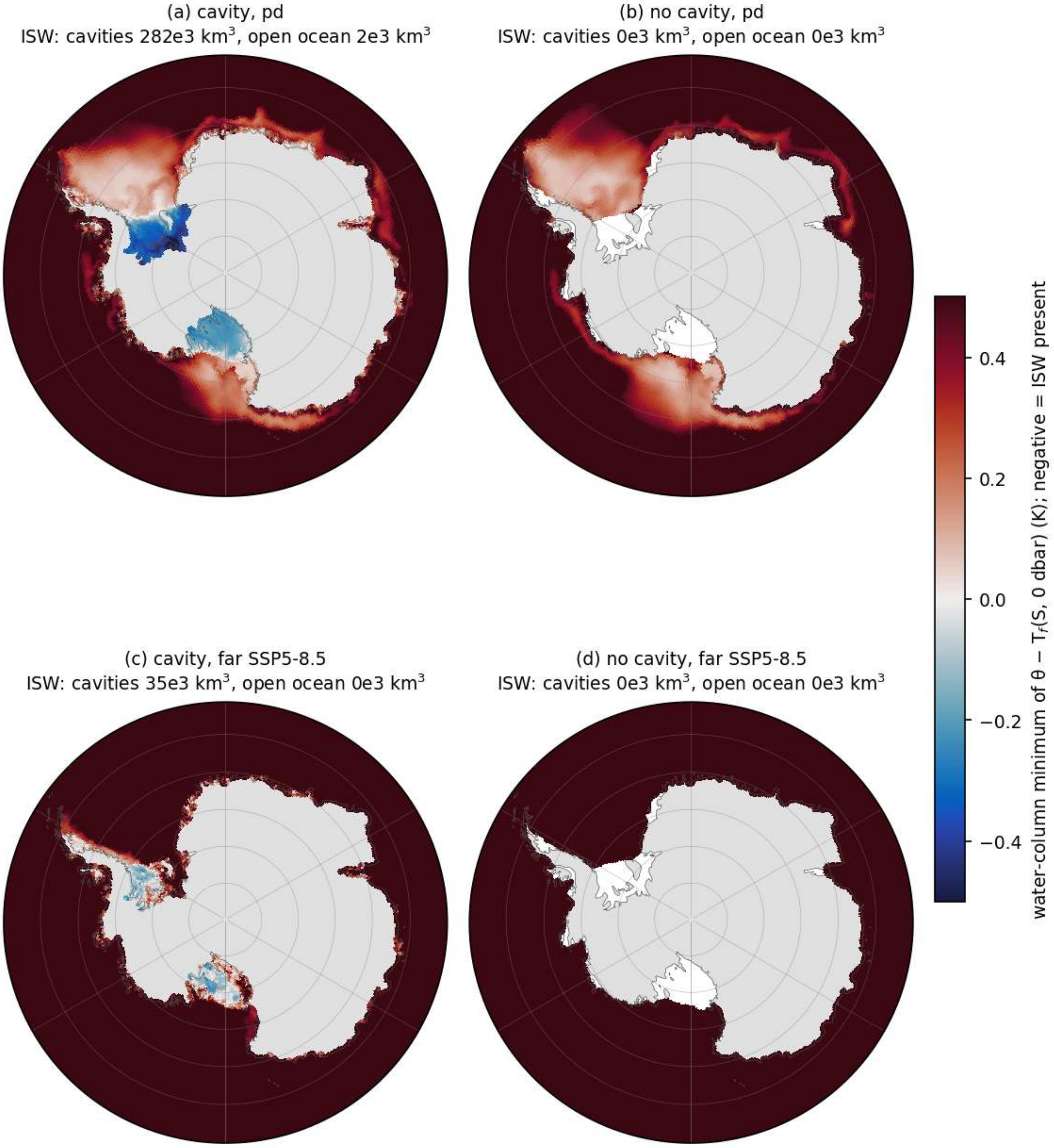

Figure 59: isw_maps (melt). Water-column minimum of theta minus the surface freezing point Tf(S, 0 dbar) south of 60S (K, negative = Ice Shelf Water present) for the cavity run (a) pd and (c) far SSP5-8.5 and the no-cavity run (b) pd and (d) far SSP5-8.5 (30-year mean fields). ISW volumes inside the cavities and in the open ocean are given in the titles. ISW exists only inside the Filchner-Ronne, Ross and Amery cavities of the cavity run (282e3 km3 at pd, 35e3 km3 in far SSP5-8.5) and essentially nowhere in the open ocean in the 30-year mean fields; the no-cavity run has no water below the surface freezing point at all. Open-shelf water in front of the big cavities is within 0.1 to 0.2 K of the freezing point in the cavity run but 0.2 to 0.5 K above it in the no-cavity run.

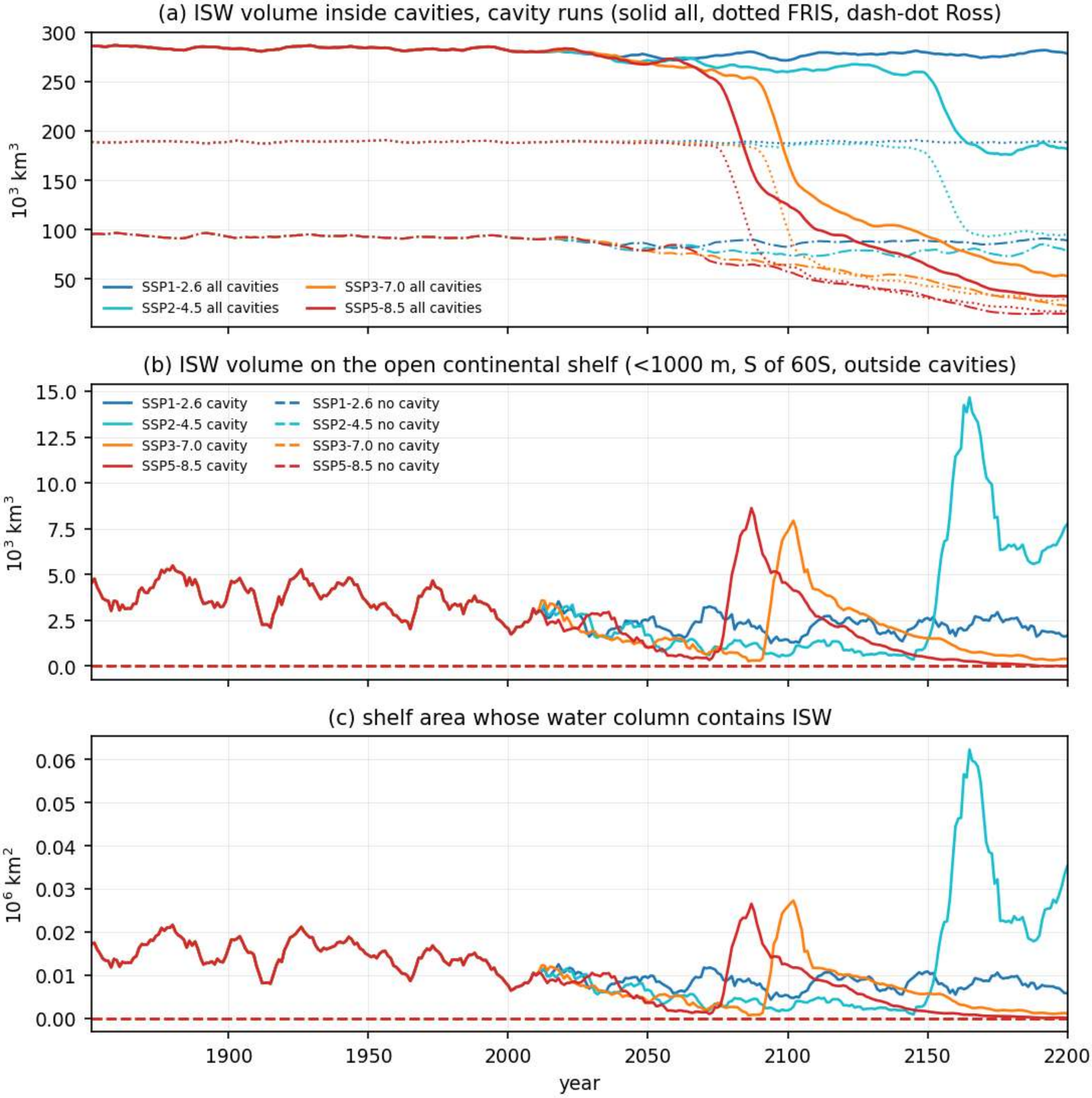

Figure 60: isw_ts (melt). 11-year running means 1851-2200 of (a) the ISW volume (theta below the surface freezing point of the local salinity, annual mean fields) inside all cavities (solid), Filchner-Ronne (dotted) and Ross (dash-dot) of the cavity runs for the four scenarios, (b) the ISW volume on the open continental shelf (less than 1000 m, south of 60S, outside the cavities) for cavity (solid) and no-cavity (dashed) runs, and (c) the shelf area whose water column contains ISW. The cavities hold 282e3 km3 of ISW at present day (50 percent of the cavity volume; Filchner-Ronne 189e3, Ross 92e3), unchanged under SSP1-2.6 (278e3 in 2171-2200) but reduced to 184e3 km3 (SSP5-8.5 end) and 35e3 km3 (SSP5-8.5 far, 6 percent of the cavity volume), with Filchner-Ronne losing its ISW within two decades after the melt regime change (2078 SSP5-8.5, 2093 SSP3-7.0, 2153 SSP2-4.5). On the open shelf only 2 to 5e3 km3 (0.01 to 0.02e6 km2 of columns) contain ISW in the annual mean of the cavity runs, declining through the 21st century, and a transient ISW outflow pulse of 8 to 15e3 km3 leaves the Filchner-Ronne cavity in the decade after each regime change (2085, 2100, 2165). The no-cavity runs never contain ISW (exactly zero in all years).

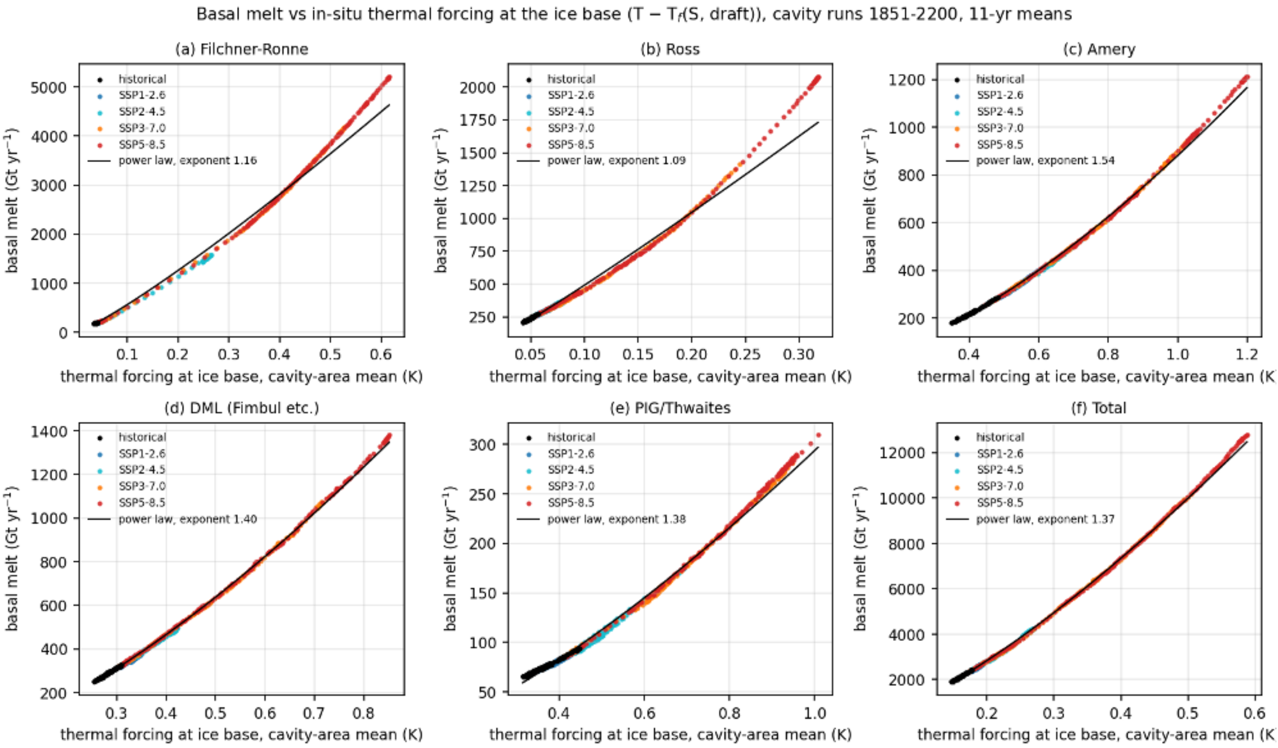

Figure 61: melt_vs_thermal_forcing (melt). Scatter of 11-year mean basal melt against the cavity-area mean in-situ thermal forcing at the ice base (T minus the freezing point at the draft depth, from cache/melt cav_ts) for Filchner-Ronne, Ross, Amery, Dronning Maud Land, PIG/Thwaites and the total, all years and scenario chains, with power-law fits. Melt scales nearly linearly with the basal thermal forcing (exponents 1.1 for Filchner-Ronne and Ross, 1.4 for the total, 1.4 to 1.5 for Amery, Dronning Maud Land and PIG/Thwaites) without the threshold seen against the outside shelf temperature; the Filchner-Ronne regime change is a jump of the cavity thermal forcing from 0.04 to 0.2 K when warm water crosses the sill, not a change of the melt law. Present-day basal thermal forcing is 0.04 K (Filchner-Ronne), 0.05 K (Ross), 0.35 K (Amery) and 0.17 K (all cavities); in far SSP5-8.5 it reaches 0.59, 0.31, 1.2 and 0.56 K.

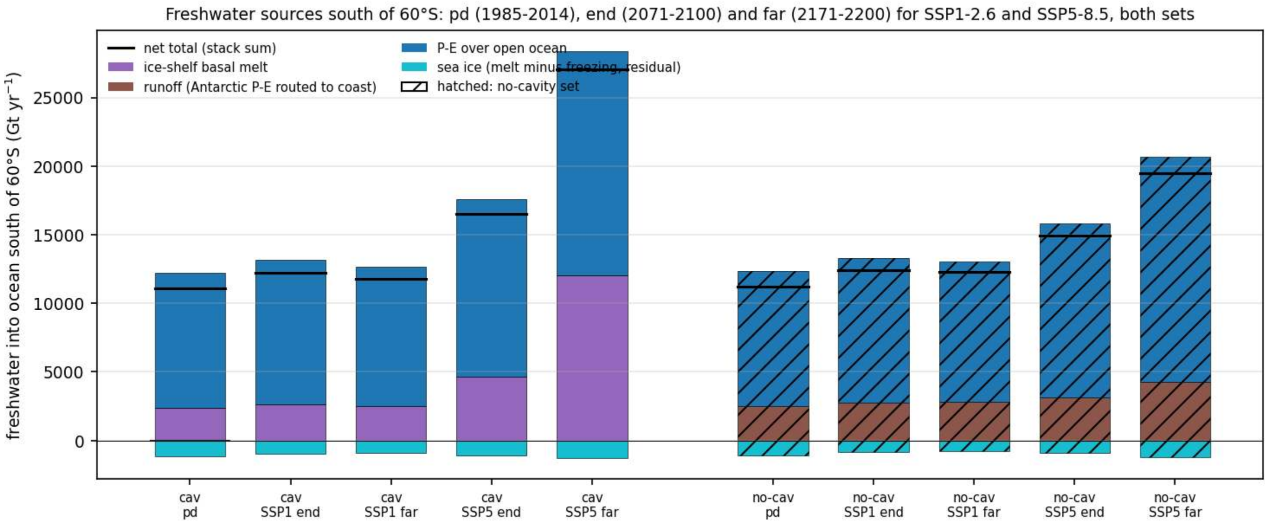

Figure 62: fw__budget__bars (melt). Freshwater sources of the ocean south of 60S (Gt/yr, positive into the ocean) for pd, SSP1-2.6 end and far and SSP5-8.5 end and far, cavity set (left, plain) and no-cavity set (right, hatched): ice-shelf basal melt (cavity nodes), runoff (FESOM runoff field, open-ocean nodes), precipitation minus evaporation over the open ocean (ECHAM, ocean cells south of 60S) and the sea-ice contribution (melt minus freezing, obtained as residual of the total FESOM water flux on open-ocean nodes). Black line is the net total. At present day both sets receive about 11100 to 11200 Gt/yr in total: the cavity set gets 2364 Gt/yr as basal melt at depth inside the cavities and zero Antarctic runoff, the no-cavity set gets 2509 Gt/yr of Antarctic runoff (the ECHAM/JSBACH P-E over the ice sheet routed to the coast) at the surface of a 200 to 300 km wide coastal band and no melt. P-E (9800 to 9900 Gt/yr) and the sea-ice sink (about -1100 Gt/yr net freezing and export) are the same in both sets. Under SSP5-8.5 the cavity set total grows to 16500 (end) and 27100 Gt/yr (far), 12000 of which is melt, while the no-cavity set reaches 14900 (end) and 19450 Gt/yr (far) with runoff growing only to 3140 and 4280 Gt/yr.

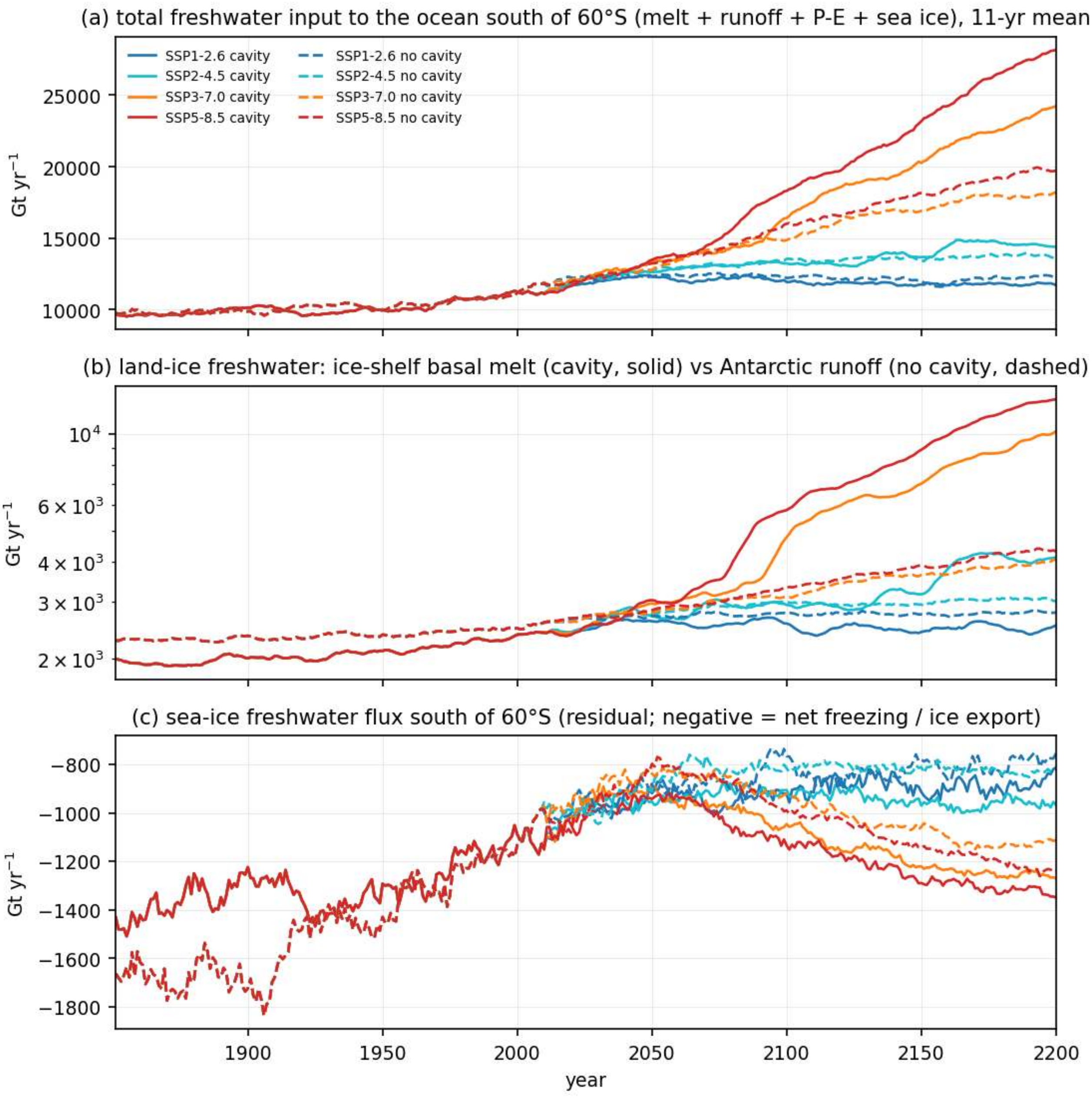

Figure 63: fw_budget_ts (melt). 11-year running means 1851-2200 of (a) the total freshwater input to the ocean south of 60S, (b) the land-ice component (ice-shelf basal melt in the cavity runs, solid; Antarctic runoff in the no-cavity runs, dashed; log axis) and (c) the sea-ice component (residual, negative = net freezing and ice export), for all scenarios (colours), cavity solid and no cavity dashed. The land-ice input is 15 percent smaller in the cavity set before 1950 (1930 vs 2290 Gt/yr) and equal around 2030; afterwards the melt grows exponentially under SSP3-7.0 and SSP5-8.5 (10000 to 12000 Gt/yr by 2200) while the runoff grows linearly with precipitation to 3000 to 4000 Gt/yr. The sea-ice sink weakens from -1400 to -1700 Gt/yr in the 19th century to about -900 Gt/yr in 2050 and then strengthens again under strong warming (more freezing of the ice-free winter ocean and export).

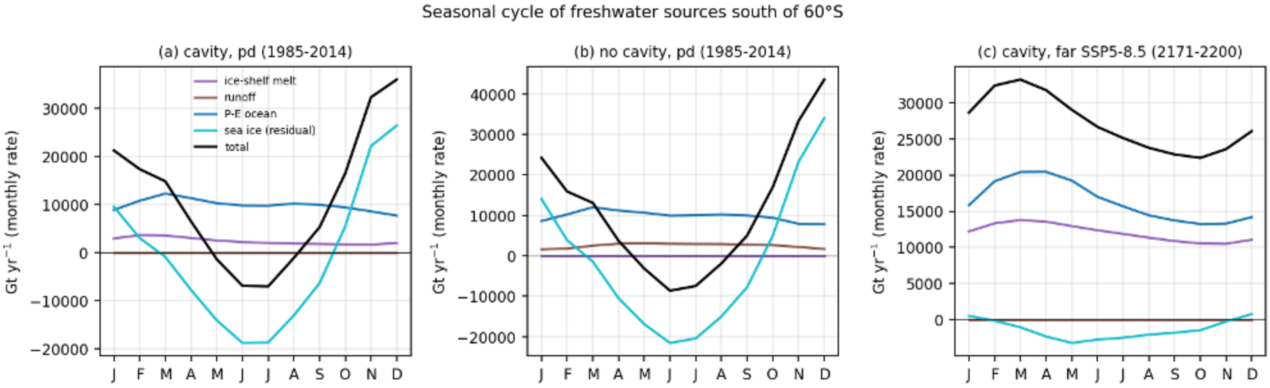

Figure 64: fw_budget_seasonal (melt). Seasonal cycle of the freshwater components south of 60S (monthly rates in Gt/yr) for (a) cavity pd, (b) no cavity pd and (c) cavity far SSP5-8.5. Ice-shelf melt has almost no seasonal cycle (2000 to 3500 Gt/yr). The Antarctic runoff in the no-cavity run peaks in winter (3000 Gt/yr in May to September, 1500 in January) following the accumulation season. Sea ice dominates the seasonal cycle (about -20000 Gt/yr freezing in June and July, +25000 to +35000 Gt/yr melt in November and December). In far SSP5-8.5 the sea-ice cycle nearly disappears and melt plus P-E supply 22000 to 33000 Gt/yr all year.

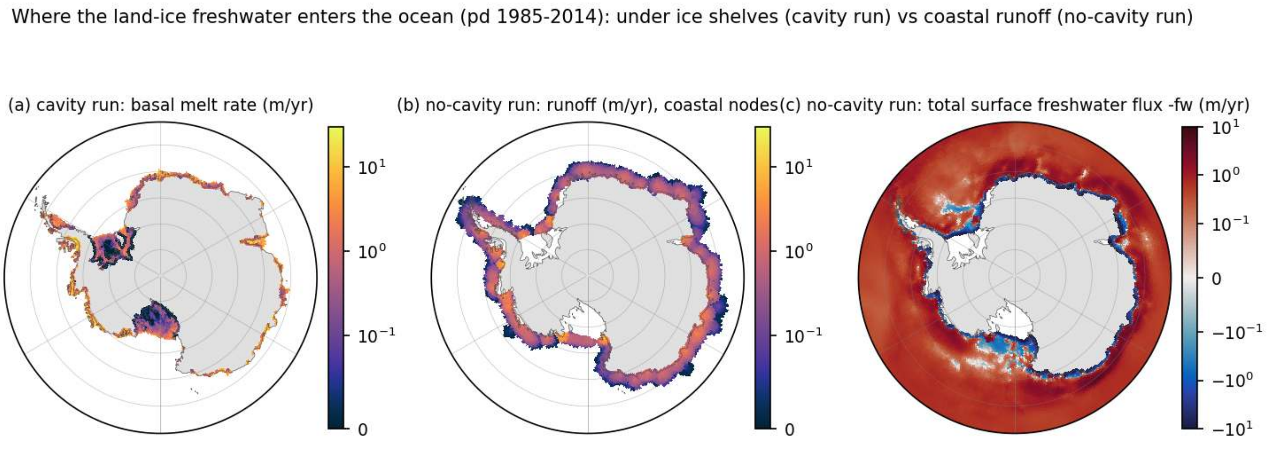

Figure 65: fw_runoff_vs_melt_map (melt). Where the land-ice freshwater enters the ocean at present day: (a) basal melt rate under the ice shelves in the cavity run, (b) runoff in the no-cavity run (m/yr, only nodes above 0.01 m/yr, log colour), (c) total surface freshwater flux -fw in the no-cavity run (m/yr, symmetric log). In the no-cavity run the 2500 Gt/yr of Antarctic runoff are spread by the coupler over a 200 to 300 km wide coastal band at rates of 0.3 to 3 m/yr, which is of the same magnitude as the basal melt of the small warm cavities but is applied at the surface. On the continental shelf the net surface flux of the no-cavity run is negative (brine rejection) only in the Ross and western Weddell polynya areas.

7 Bottom water and shelf water properties

Findings

7.1 Bottom theme: bottom water and shelf water properties (source and product of AABW)

Data: cache/pmean period means (maps, T-S histograms), cache/ts bottom-layer region means and basin profiles (time series, profiles), all with cavity nodes excluded unless stated. Scripts in analyses/bottom (prep_bottom.py, plot_maps.py, plot_shelf_ts.py, plot_timeseries.py, plot_profiles.py, summarize.py). Numbers tables: cache/bottom_theme ts_numbers.json, summary.txt. "cav / nocav" below means cavity run / no-cavity run. Periods: early 1851-1880, pd 1985-2014, end 2071-2100, far 2171-2200.

7.1.1 A. Present-day state

1. No AABW layer at the sea floor. The area-mean bottom potential temperature of the deep Southern Ocean south of 60S (>1000 m) is 1.75 degC (cav) and 1.90 degC (nocav), with 34.66 to 34.68 psu and sigma2 36.92 kg m-3 in both runs (bottom_pd_maps, bottom_ts_T). Deep Weddell 1.76 / 2.01 degC, deep Ross 1.66 / 1.77 degC. The cavity run is 0.25 (Weddell), 0.11 (Ross) and 0.15 degC (whole deep SO) cooler at the bottom and the difference is not a steady state (see finding 10).

2. Cold dense bottom water exists only on the Weddell and Ross shelves. On the 0.25 degree grid the open shelf south of 60S (<1000 m) has mean bottom values of -0.07 degC, 34.36 psu, 36.92 kg m-3 (cav) and -0.21 degC, 34.29 psu, 36.87 kg m-3 (nocav). Weddell shelf bottom means (region mean from the ts cache): -0.56 degC, 34.42 psu, 37.00 kg m-3 (cav) against -0.56 degC, 34.33 psu, 36.94 kg m-3 (nocav). Ross shelf: -0.68 degC, 34.40 psu, 37.00 kg m-3 (cav) against -0.48 degC, 34.47 psu, 37.03 kg m-3 (nocav). The cavity run therefore has a saltier and denser Weddell shelf (+0.09 psu, +0.07 kg m-3) and a colder but fresher and slightly lighter Ross shelf (-0.2 degC, -0.07 psu, -0.03 kg m-3). This is the Ross versus Weddell asymmetry of the cavity effect at present day. The Amundsen shelf bottom is warm (1.35 / 1.12 degC, 34.40 psu, 36.81 / 36.85 kg m-3) and always lighter than the deep ocean.

3. Dense shelf water defined relative to the model deep ocean. Using the area-mean bottom sigma2 of the adjacent deep basin (36.92 kg m-3) as threshold, 56 percent of the open Weddell shelf volume (195 thousand km3, mean -1.05 degC, 34.37 psu) and 38 percent of the Ross shelf volume (89 thousand km3, -1.43 degC, 34.39 psu) is denser than the deep bottom water in the cavity run; nocav 47 percent (176 thousand km3) and 35 percent (75 thousand km3). The densest 10 percent of the shelf volume has sigma2 37.11 / 37.08 (Weddell cav / nocav) and 37.13 / 37.23 kg m-3 (Ross cav / nocav), temperature -1.55 / -1.48 and -1.80 / -1.73 degC, salinity 34.45 / 34.36 and 34.47 / 34.63 psu (shelf_TS_pd, shelf_dense_water, table dense_shelf_water.md). Over all shelves south of 60S, 26 / 25 percent of the open-shelf volume (303 / 278 thousand km3) is denser than the deep bottom water. So the model does produce shelf water that is 0.1 to 0.3 kg m-3 denser than its own abyss, but this water is 0.2 to 0.3 psu too fresh and the density excess is small, and the bottom_pathways maps show no coherent dense layer below the 1000 m isobath: the abyss is filled from elsewhere (open-ocean convection, see seaice theme) and the shelf export is too weak to cool it.

4. Distance from freezing and ISW. Mean bottom temperature above the in-situ freezing point is 1.7 K on the Weddell shelf (both runs), 1.6 / 1.8 K on the Ross shelf and 2.3 / 2.2 K over all shelves. Bottom water within 0.5 K of the in-situ freezing point covers 19 percent of the Weddell shelf area in the cavity run against 9 percent without cavities, and 5 / 6 percent of the Ross shelf. Water below the surface freezing point (Ice Shelf Water) sits on the bottom of 0.8 percent of the open Weddell shelf (in front of Filchner-Ronne) in the cavity run and nowhere in the no-cavity run (bottom_freezing). HSSW-like water (T < -1.7 degC, S > 34.5 psu) occurs only as the coldest few percent of the Ross and Weddell shelf volume.

5. Spurious salty cells in the no-cavity run. The no-cavity run has a tail of near-freezing water with salinity up to 35.1 (Ross), 35.4 (Prydz Bay) and 35.8 psu (Amundsen, all shelf maximum 35.76) in isolated coastal cells (shelf_TS_pd bottom row, dark spots in bottom_pathways_regional e-h). The cavity run maximum is 34.66 psu. These cells are small, contribute below 0.5 percent of the shelf volume and do not change the region means, but they explain the Ross no-cavity densest-10-percent value of 37.23 kg m-3 and the isolated 37.2+ patch near Terra Nova Bay; they are probably brine accumulation in closed coastal cells where the PI_CTRL coastline follows the ice front. Treat Ross no-cavity dense-water maxima with care.

7.1.2 B. Scenario response

6. Shelf warming and freshening. Cavity run, bottom-layer region means, change relative to 1985-2014 (end / far): Weddell shelf SSP1-2.6 +0.02 / -0.04 degC and -0.10 / -0.12 psu, SSP5-8.5 +0.29 / +1.54 degC and -0.24 / -0.64 psu; Ross shelf SSP1-2.6 +0.04 / +0.02 degC and -0.08 / -0.08 psu, SSP5-8.5 +0.39 / +2.21 degC and -0.19 / -0.54 psu; Amundsen shelf SSP1-2.6 -0.12 / -0.22 degC, SSP5-8.5 +0.38 / +1.20 degC and -0.08 / -0.32 psu. No-cavity run: Weddell shelf SSP1-2.6 +0.06 / +0.05 degC and -0.04 / -0.02 psu, SSP5-8.5 +0.42 / +2.08 degC and -0.11 / -0.18 psu; Ross shelf SSP1-2.6 +0.12 / +0.10 degC and -0.05 / -0.03 psu, SSP5-8.5 +0.35 / +1.94 degC and -0.11 / -0.24 psu. Under SSP1-2.6 the cavity run freshens the shelves two to four times more than the no-cavity run (more ice-shelf melt water) while temperatures hardly change; under SSP5-8.5 both runs warm the shelves by 1.5 to 2.1 degC by 2171-2200, but the cavity run freshens them by 0.5 to 0.6 psu against only 0.2 psu without cavities, so the shelf density loss is twice as large with cavities (Weddell shelf bottom sigma2 -0.73 against -0.33 kg m-3, Ross -0.78 against -0.35 kg m-3). The no-cavity run warms the Weddell shelf more (+2.1 against +1.5 degC), the Ross shelf warms by 2 degC in both. By 2171-2200 in SSP5-8.5 the whole shelf is 3 to 4 K above the in-situ freezing point (Weddell 3.2, Ross 3.8 K) and no bottom water is within 0.5 K of freezing (bottom_freezing c, f).

7. Shelf density loss and the shelf-deep contrast. Weddell shelf bottom sigma2 falls from 37.00 (pd) to 36.94 (SSP1-2.6 far) and 36.27 kg m-3 (SSP5-8.5 far) in the cavity run, Ross shelf from 37.00 to 36.96 and 36.22 kg m-3; deep-basin bottom density is constant within 0.01 kg m-3 (warming and salinification compensate). The shelf minus deep bottom density contrast (11-yr smoothed, bottom_ts_contrast) is: Weddell 0.11 (early), 0.09 (pd), 0.03 (SSP1-2.6 end and far), -0.08 (SSP5-8.5 end), -0.63 kg m-3 (SSP5-8.5 far) in the cavity run; 0.06, 0.02, -0.005, 0.01 (SSP1-2.6 end, far) and -0.07 (SSP5-8.5 end), -0.30 kg m-3 (SSP5-8.5 far) in the no-cavity run. Ross: 0.13, 0.08, 0.04, 0.05, -0.07, -0.69 (cav) and 0.13, 0.11, 0.07, 0.09, 0.01, -0.23 (nocav). First year (after 2014) when the smoothed contrast drops below zero: Weddell cav SSP5-8.5 2072, SSP3-7.0 2094 (SSP2-4.5 and SSP1-2.6 never, SSP1-2.6 levels at 0.03); Weddell nocav SSP1-2.6 2046 (noise around zero), SSP5-8.5 2031 (the no-cavity Weddell contrast is already near zero at present day); Ross cav SSP5-8.5 2068 (SSP1-2.6 never), Ross nocav SSP5-8.5 2099 (its contrast is still 0.01 kg m-3 in 2071-2100). The contrast halves relative to pd already by 2035 to 2045 in all scenarios of the cavity run. So the ability of shelf water to sink collapses around 2070 under SSP5-8.5 and by 2100 under SSP3-7.0, but survives at a reduced level under SSP1-2.6 and SSP2-4.5. The dense shelf volume fraction (denser than the deep bottom water) goes from 56 / 38 percent (Weddell / Ross, pd) to 30 / 33 percent (SSP1-2.6 far), 0 / 10 percent (SSP5-8.5 end) and 0 / 0 percent (SSP5-8.5 far) in the cavity run, and stays at 47 / 34 percent under SSP1-2.6 in the no-cavity run (shelf_dense_water).

8. Deep-ocean response and the cavity drift. The deep Southern Ocean bottom (>1000 m, S of 60S) warms by +0.23 / +0.38 degC (SSP1-2.6 end / far) and +0.24 / +0.50 degC (SSP5-8.5) in the cavity run but only +0.15 / +0.21 degC (SSP1-2.6) and +0.19 / +0.41 degC (SSP5-8.5) in the no-cavity run, with +0.02 to +0.04 psu salinification and a density change below 0.01 kg m-3. The deep warming is nearly scenario independent in the cavity run (bottom_ts_T d-f), which together with the 0.16 degC warming already during 1851-2014 (deep Weddell 1.60 to 1.76 degC, deep SO 1.59 to 1.75 degC, global 2.88 to 3.01 degC) shows that most of it is a spin-up drift of the cavity run's deep ocean (the no-cavity run warms only 0.08 degC in the deep Weddell over the historical period). Consequence: the cavity minus no-cavity deep temperature difference (-0.3 degC at 2000 to 4000 m in 1851-1880, -0.22 degC in pd) vanishes by 2171-2200 under SSP1-2.6 (bottom_profiles_diff). Deep bottom temperature differences between the two sets should not be interpreted as a cavity effect without subtracting the drift (use early minus pd trends, or the PI control if retrieved). The response difference maps (bottom_resp_T right column) are dominated by this drift in the deep basins; on the shelves the cavity run response is 0.1 degC cooler and 0.05 psu fresher than the no-cavity response under SSP1-2.6.

9. Vertical structure (bottom_profiles). Weddell box mean profiles: surface layer -0.75 degC, 33.95 psu; temperature maximum 2.0 degC at 1000 m (cav, 2.1 nocav), 1.75 degC at 4000 m (2.0 nocav); salinity 34.67 below 1000 m. Under SSP5-8.5 2171-2200 the upper 100 m warms by 3.2 degC and freshens by 0.25 psu, the 200 to 500 m layer warms 1.4 to 1.7 degC, and sigma2 falls by 0.7 kg m-3 at the surface, 0.5 at 200 m, 0.18 at 500 m and less than 0.04 kg m-3 below 1000 m. The Ross box behaves the same with a 1.3 degC colder surface layer. Under SSP1-2.6 the 200 m layer cools slightly (-0.18 degC) and freshens (-0.10 psu) in the cavity run, a signature of increased melt water, while it warms in the no-cavity run. The density stratification between 100 and 1000 m therefore strengthens strongly in all scenarios, which is the same information as the collapsing shelf-deep contrast.

7.1.3 C. Cavity versus no cavity, summary

10. Where the cavities matter. (i) Weddell shelf: denser, saltier, colder near-freezing water with cavities (ISW present, 19 against 9 percent of the shelf bottom within 0.5 K of freezing). (ii) Ross shelf: colder but fresher and lighter with cavities, and the densest Ross water of the no-cavity run is partly an artefact of closed coastal cells. (iii) Under SSP1-2.6 the cavity run keeps freshening the shelves (-0.1 psu by 2100) and loses dense shelf volume (56 to 30 percent on the Weddell shelf) while the no-cavity run keeps its pd dense volume (47 percent); the cavity run also loses its shelf-deep contrast earlier (halved by 2041 against 2029 in Weddell nocav, but from a higher starting value). (iv) Under SSP5-8.5 the Weddell contrast reverses in 2072 (cav) and 2031 (nocav, from a lower start), the Ross contrast in 2068 (cav) and 2099 (nocav): the cavity run loses its Ross dense-water source three decades earlier because its shelf freshens by 0.54 psu instead of 0.24 psu by 2171-2200, while its Weddell shelf starts from a denser state and reverses later than the no-cavity Weddell. The Ross shelf is where cavities accelerate the loss, the Weddell shelf is where they delay it. (v) Deep-ocean differences between the sets are dominated by the cavity run's drift and should not be read as cavity effects.

7.1.4 D. Caveats

11. Shelf water is 0.2 to 0.3 psu too fresh and there is no observed-type AABW or HSSW; all "dense" classes are model-relative (denser than the model's own 36.92 kg m-3 abyss or the densest 10 percent of shelf volume). Region boxes dml_shelf, amery_shelf and adelie_shelf compare different domains (327 thousand km2 of the cavity area is open ocean in PI_CTRL), so the Prydz Bay and Adelie values in the table should not be compared between runs. The Weddell box used for profiles includes the shelf. Historical runs are single members, differences below 0.05 degC or 0.02 psu between sets are within internal variability. The cavity run's deep ocean drifts by about 0.1 degC per century. The no-cavity run has spurious hypersaline coastal cells (finding 5). The bottom_freezing maps use gsw freezing temperatures with the model's practical salinity; the model's internal freezing formula may differ by a few hundredths of a degree.

7.1.5 E. Suggested story lines

12. Figure set for a paper: shelf_TS_pd (what dense water exists and how the two runs differ), bottom_ts_contrast (when the shelf loses the ability to sink, per scenario), bottom_resp_T/S (SSP5-8.5 rows, where the shelves warm and freshen), bottom_freezing (a,d: cavities create near-freezing and ISW bottom water), bottom_profiles (stratification change). Story: with or without cavities the model has CDW-like abyssal water and a weak dense-shelf-water source confined to the western Weddell and Ross shelves; cavities add Ice Shelf Water and a saltier Weddell shelf but a fresher Ross shelf; under strong forcing the shelf-deep density contrast reverses around 2070 in both runs, under SSP1-2.6 it survives at a third of its pre-industrial value, and the cavity run loses dense shelf water faster under low forcing because of extra melt water.

Figures

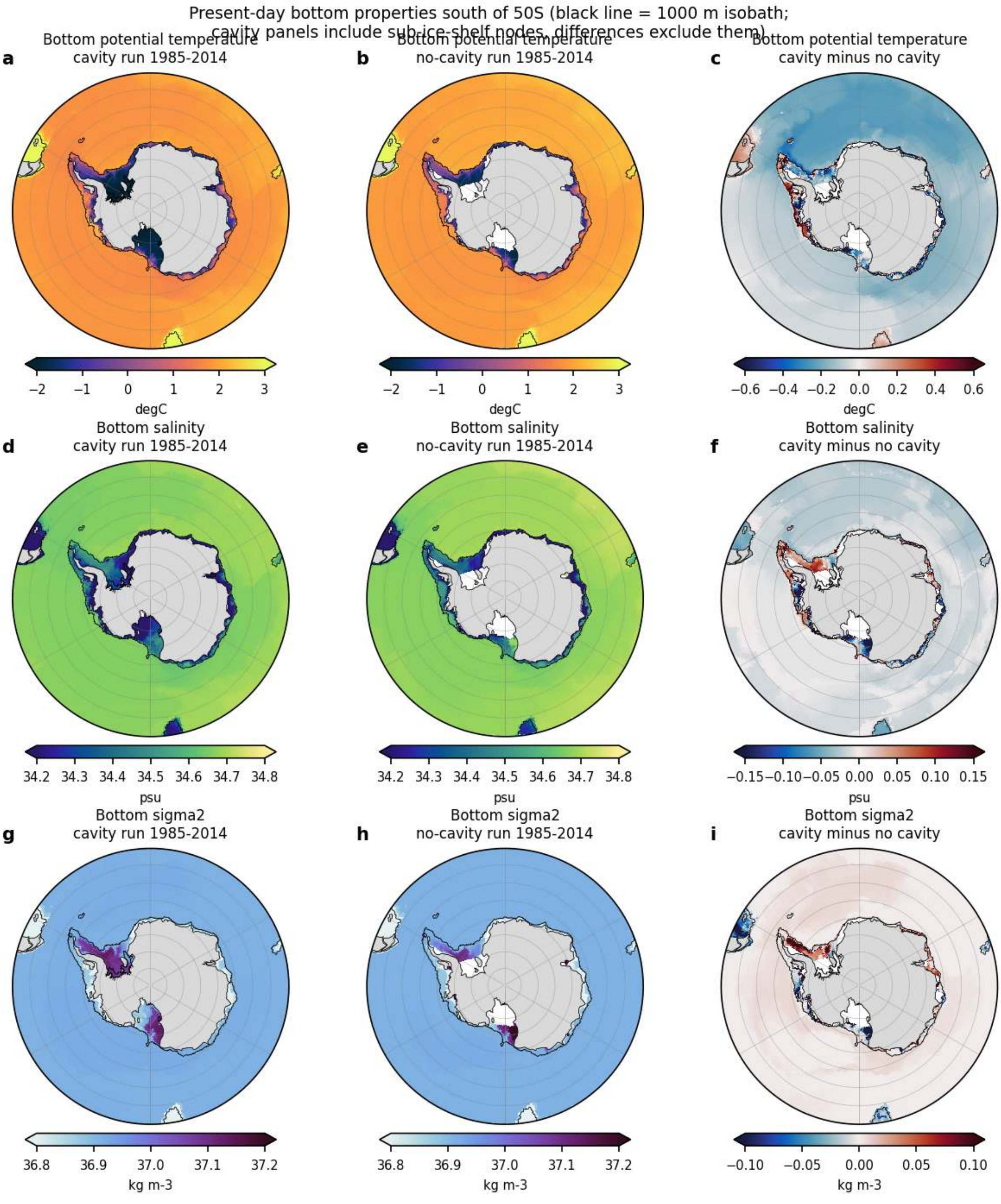

Figure 66: bottom_pd_maps (bottom). Present-day (1985-2014) bottom potential temperature (a-c), salinity (d-f) and sigma2 (g-i) south of 50S on a 0.25 degree grid. Left column cavity run including the sub-ice-shelf nodes, middle column no-cavity run, right column cavity minus no cavity on the common open-ocean grid. Black line is the 1000 m isobath. Units degC, psu, kg m-3. The abyss is 1.7 to 2.0 degC and 36.9 kg m-3 in both runs, so there is no cold AABW layer. Cold (below -1.5 degC) and dense (above 37.0 kg m-3) bottom water exists only on the Weddell and Ross shelves and under the Filchner-Ronne and Ross ice shelves. The cavity run is 0.1 to 0.3 degC cooler over the whole deep Southern Ocean, and its western Weddell shelf is 0.05 to 0.1 psu saltier and 0.05 to 0.1 kg m-3 denser, whereas its Ross shelf is fresher and lighter by a similar amount.

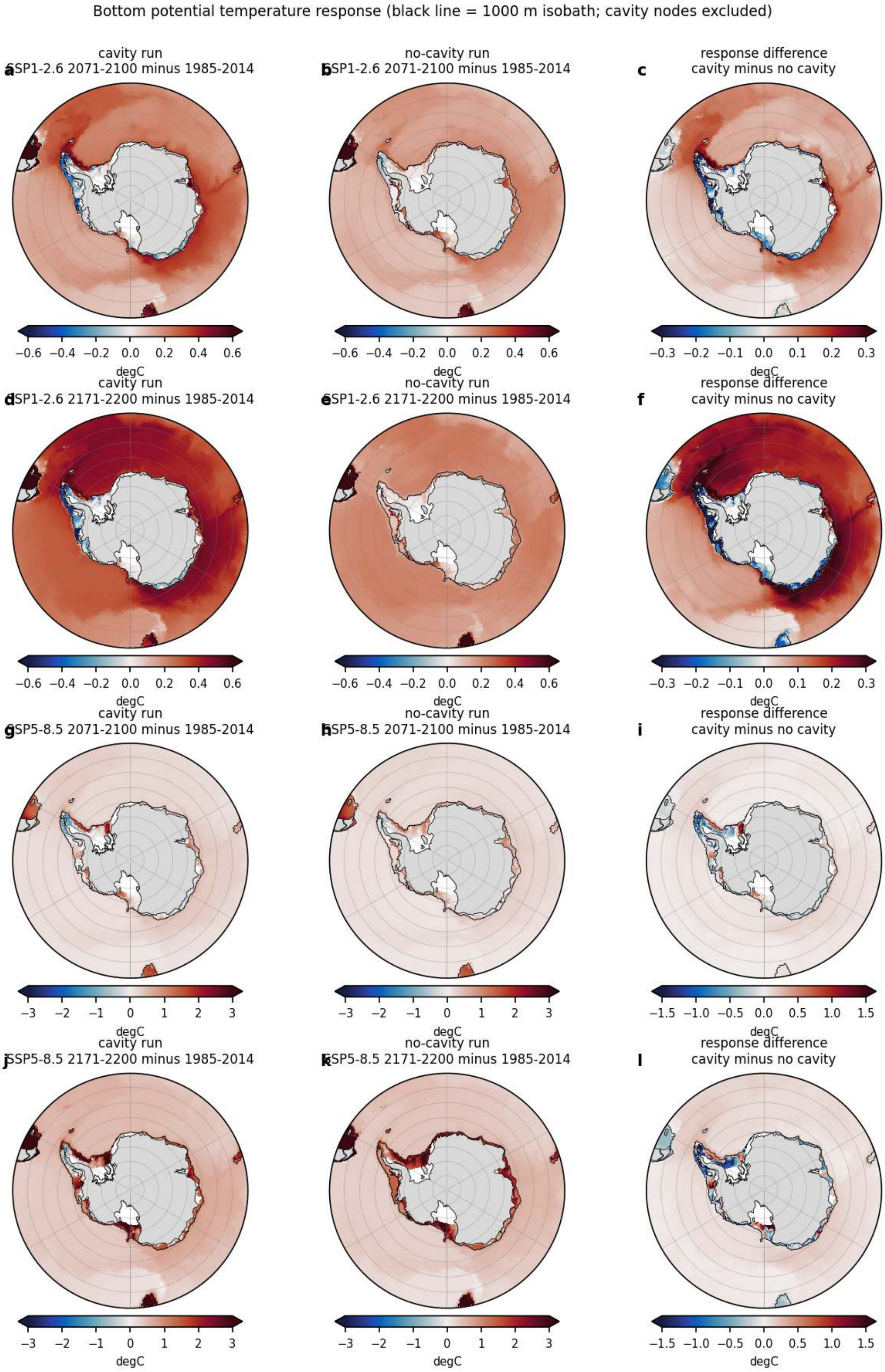

Figure 67: bottom_resp_T (bottom). Change of bottom potential temperature (degC) relative to 1985-2014 for SSP1-2.6 (rows 1 and 2) and SSP5-8.5 (rows 3 and 4) at 2071-2100 and 2171-2200. Left cavity run, middle no-cavity run, right the difference of the two responses (cavity response minus no-cavity response, colour range halved). Cavity nodes excluded. Colour range 0.6 degC for SSP1-2.6 and 3 degC for SSP5-8.5. In SSP1-2.6 the deep Southern Ocean bottom warms 0.2 to 0.4 degC, twice as much in the cavity run because of a persisting spin-up drift of its deep water, while the Weddell shelf stays near its present temperature. In SSP5-8.5 the shelves warm by 1.5 to 2.5 degC by 2171-2200 and the warming is largest where the shelf water was coldest.

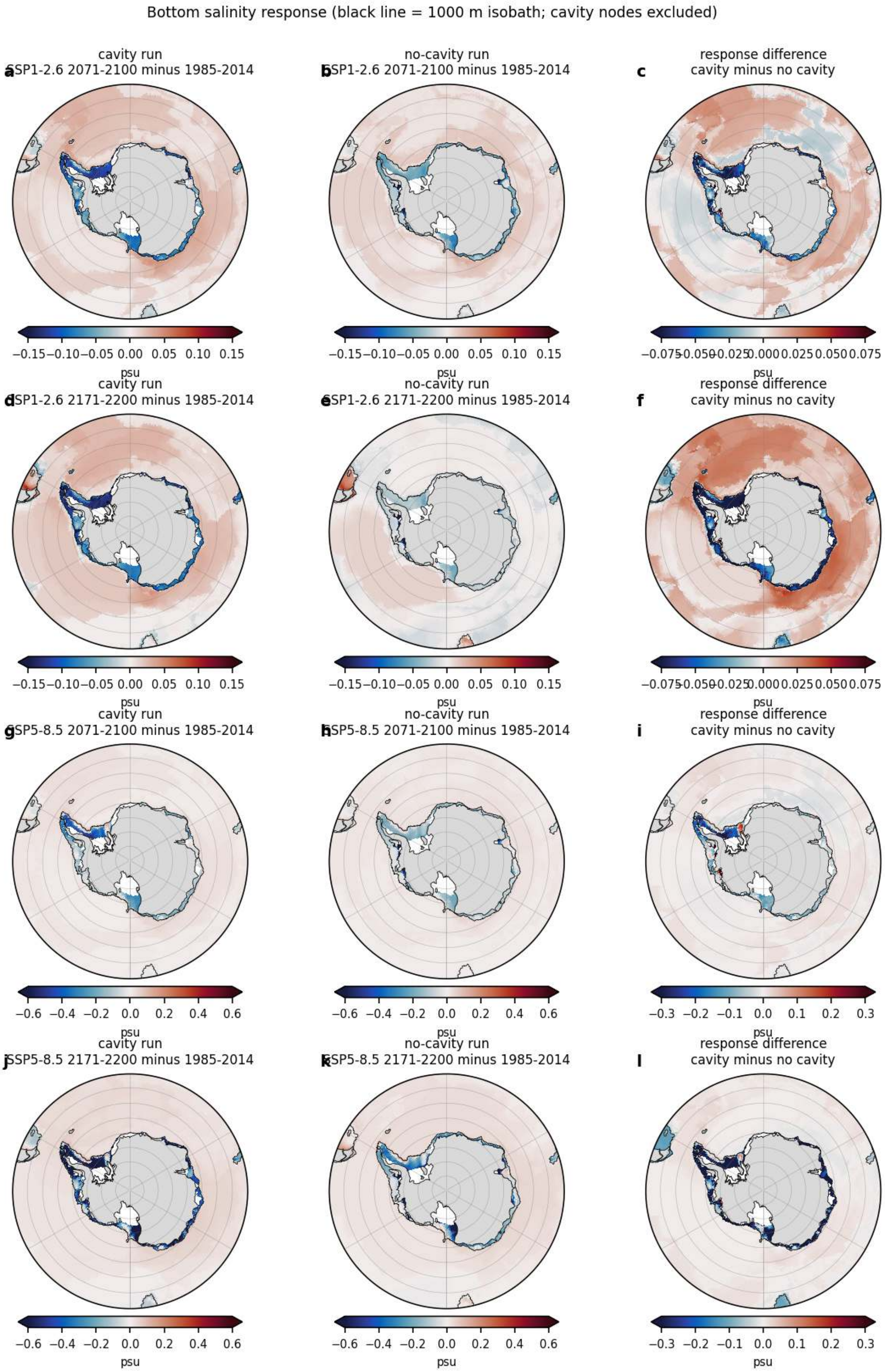

Figure 68: bottom_resp_S (bottom). Same as bottom_resp_T for bottom salinity (psu). Colour range 0.15 psu for SSP1-2.6 and 0.6 psu for SSP5-8.5. Shelf bottom water freshens everywhere, most on the Weddell and Ross shelves (0.1 psu in SSP1-2.6, 0.5 to 0.7 psu in SSP5-8.5 by 2171-2200), while the deep basins become slightly saltier (0.02 to 0.04 psu) as dense shelf water is no longer exported.

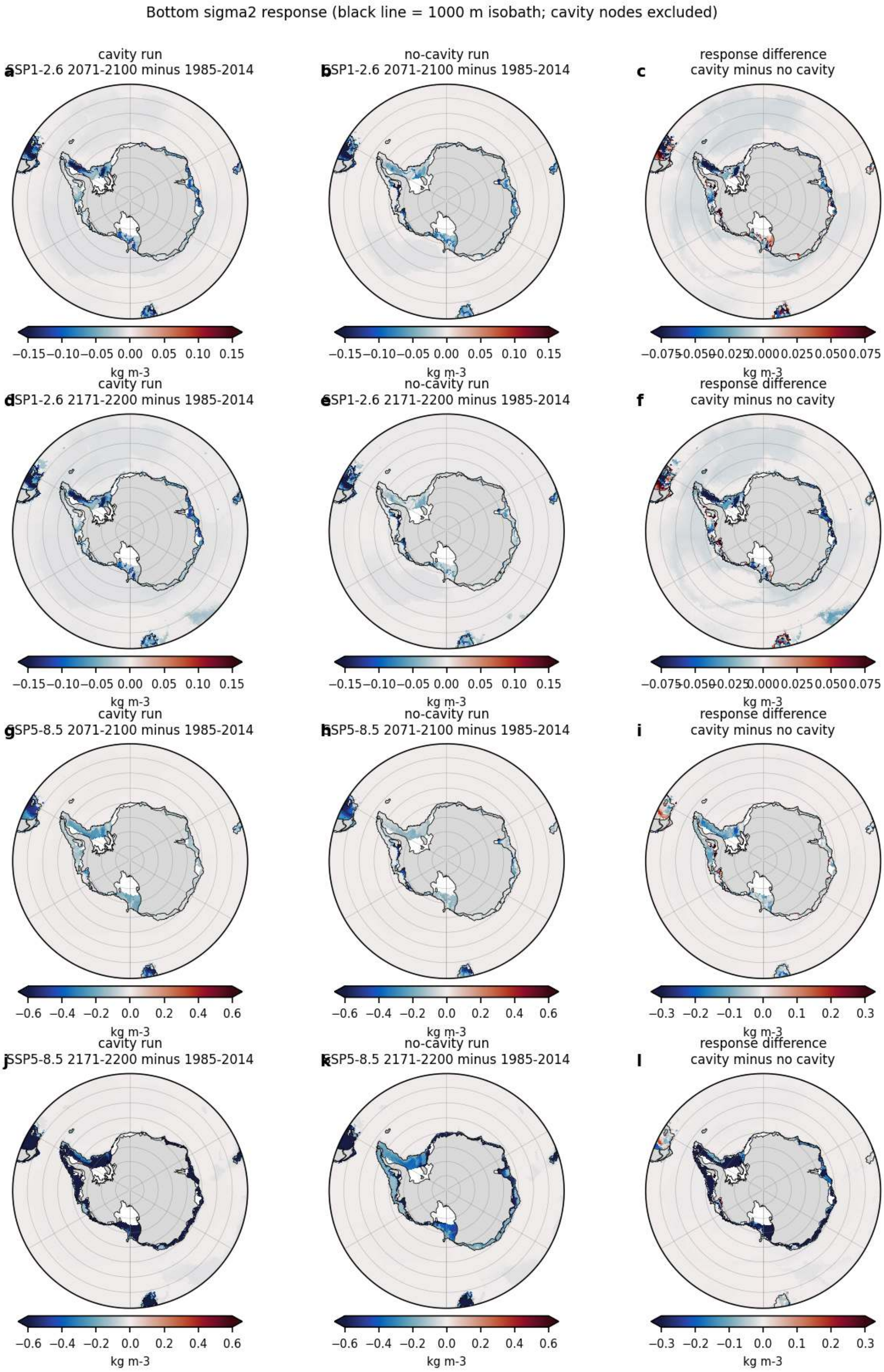

Figure 69: bottom_resp_Sig (bottom). Same as bottom_resp_T for bottom sigma2 (kg m-3). Colour range 0.15 kg m-3 for SSP1-2.6 and 0.6 kg m-3 for SSP5-8.5. The density loss is confined to the shelves and slope. The densest present-day shelf water (western Weddell and Ross) loses 0.6 to 0.9 kg m-3 by 2171-2200 in SSP5-8.5 and becomes lighter than the deep basin water.

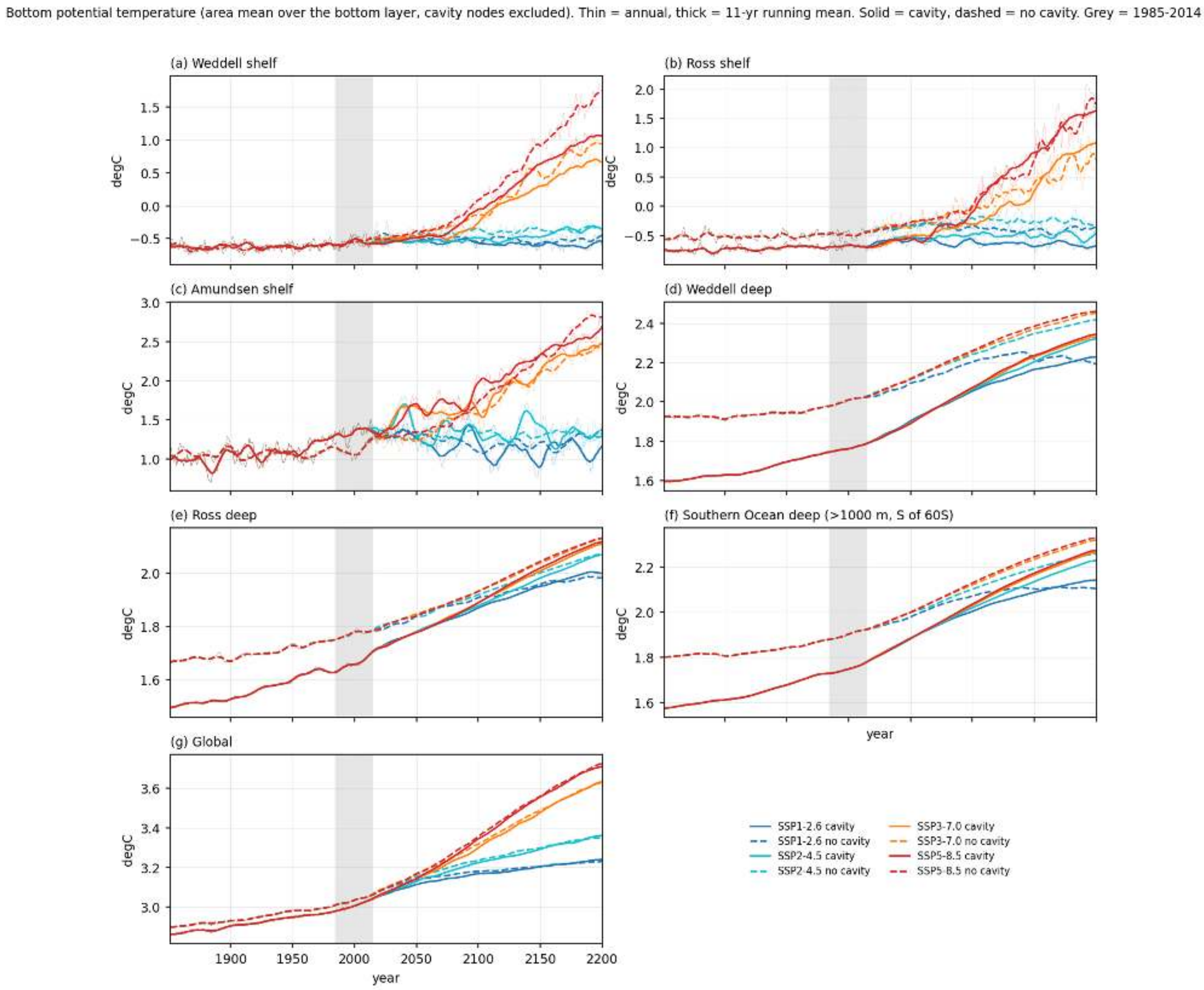

Figure 70: bottom_ts_T (bottom). Time series 1851-2200 of area-mean bottom potential temperature for the Weddell, Ross and Amundsen shelves (<1000 m), the deep Weddell and Ross basins, the deep Southern Ocean south of 60S and the global ocean. Thin lines annual means, thick lines 11-yr running means. Historical part in the scenario colours. Grey band 1985-2014. Region means exclude cavity nodes. The deep basins of the cavity run warm by 0.16 degC over the historical period (drift inherited from the spin-up) and by a further 0.25 (end) and 0.45 to 0.55 degC (far) in all scenarios, while the no-cavity run drifts little and warms 0.16 to 0.2 degC by 2200. Shelf bottom temperatures stay within 0.1 degC of present day under SSP1-2.6 and rise by 1.5 (Weddell) and 2.2 degC (Ross) under SSP5-8.5 by 2171-2200.

Bottom salinity (area mean over the bottom layer, cavity nodes excluded). Thin = annual, thick = 11-yr running mean. Solid = cavity, dashed = no cavity. Grey = 1985-2014

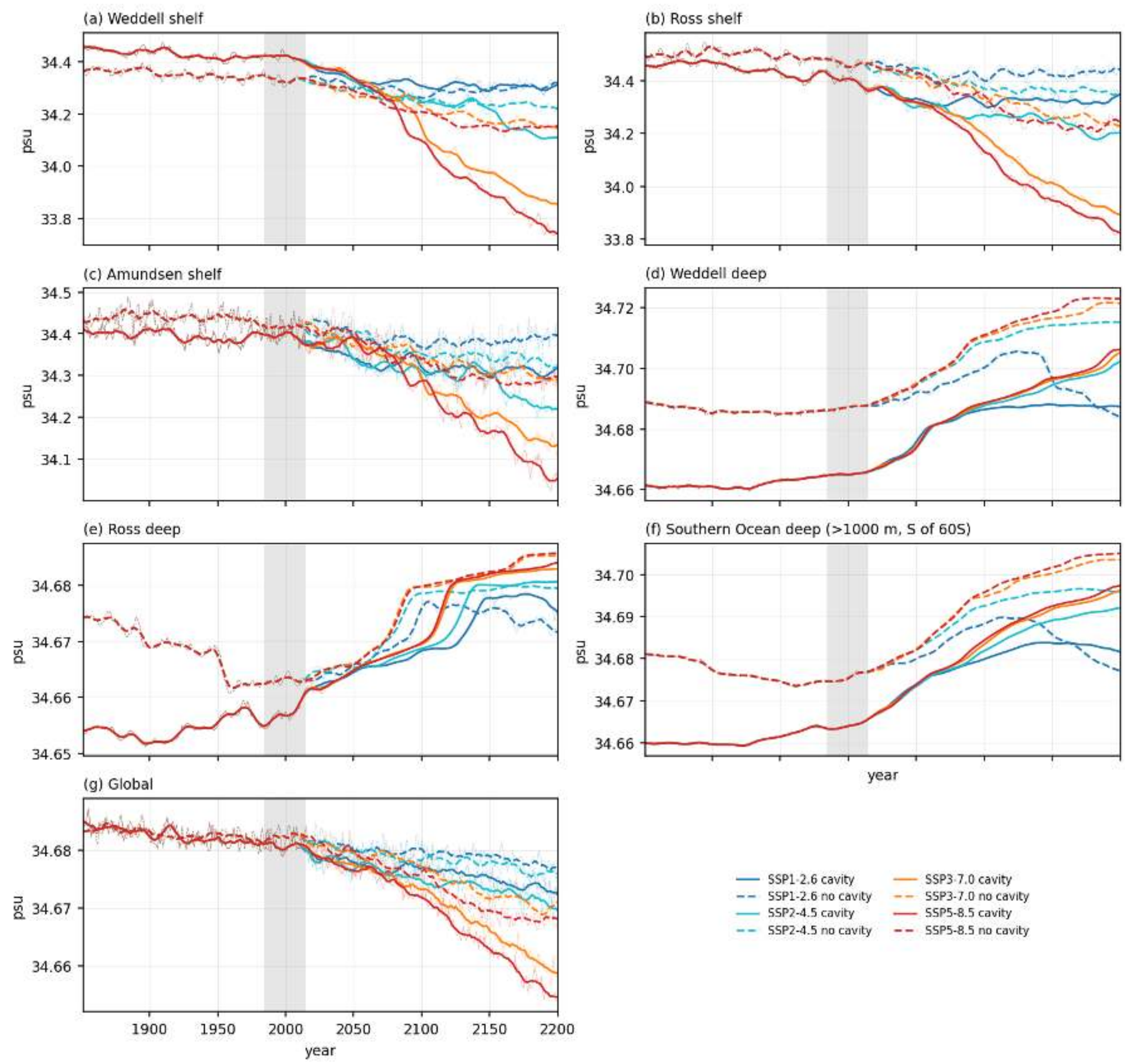

Figure 71: bottom_ts_S (bottom). Same as bottom_ts_T for bottom salinity (psu). Weddell and Ross shelf bottom salinities fall by 0.1 psu (SSP1-2.6) and 0.5 to 0.6 psu (SSP5-8.5) in the cavity run, while the deep basins become 0.02 to 0.04 psu saltier.

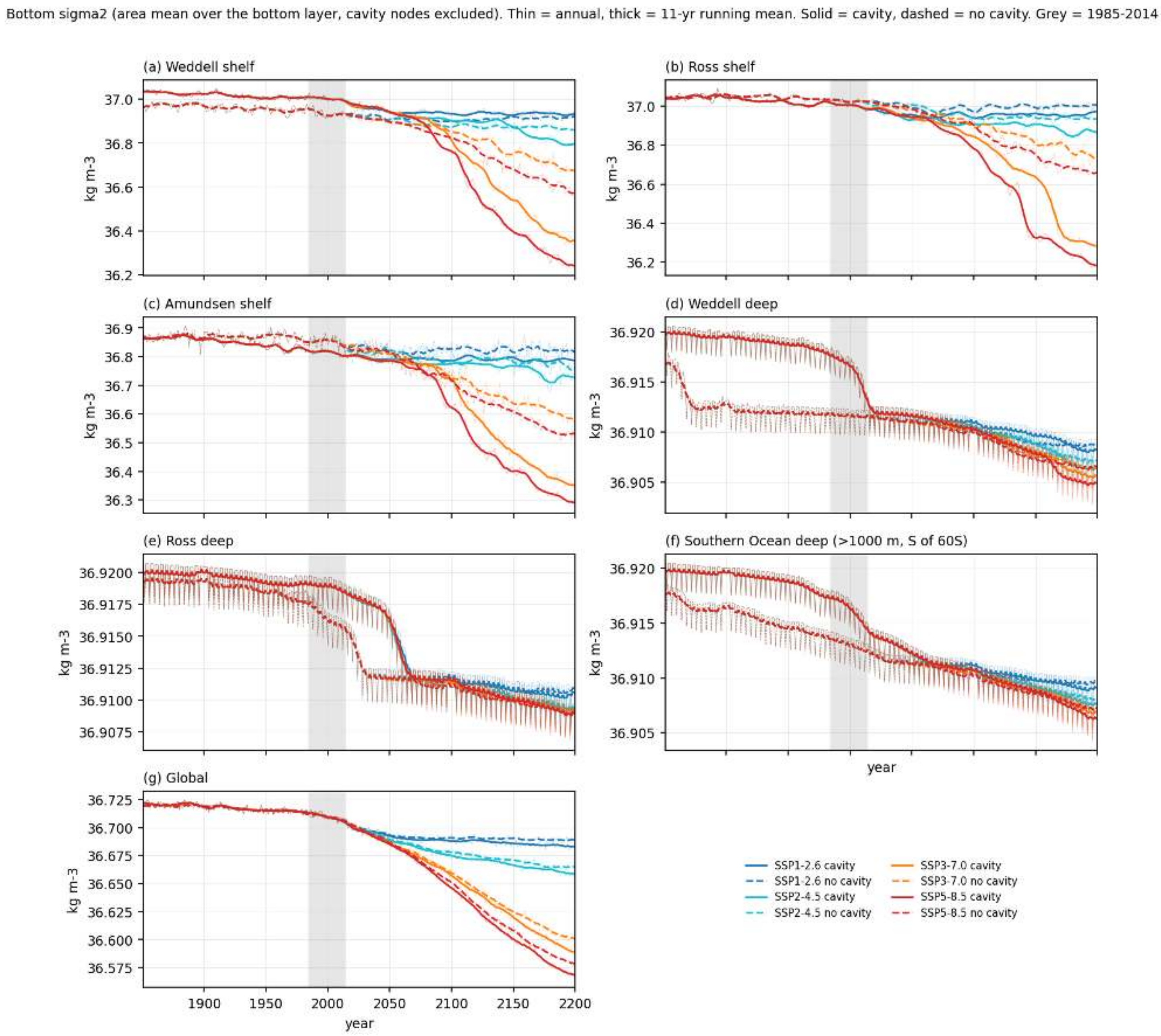

Figure 72: bottom_ts_Sig (bottom). Same as bottom_ts_T for bottom sigma2 (kg m-3). Shelf bottom density decreases in all scenarios, deep-basin density is almost constant (changes below 0.01 kg m-3) because the deep warming and salinification compensate.

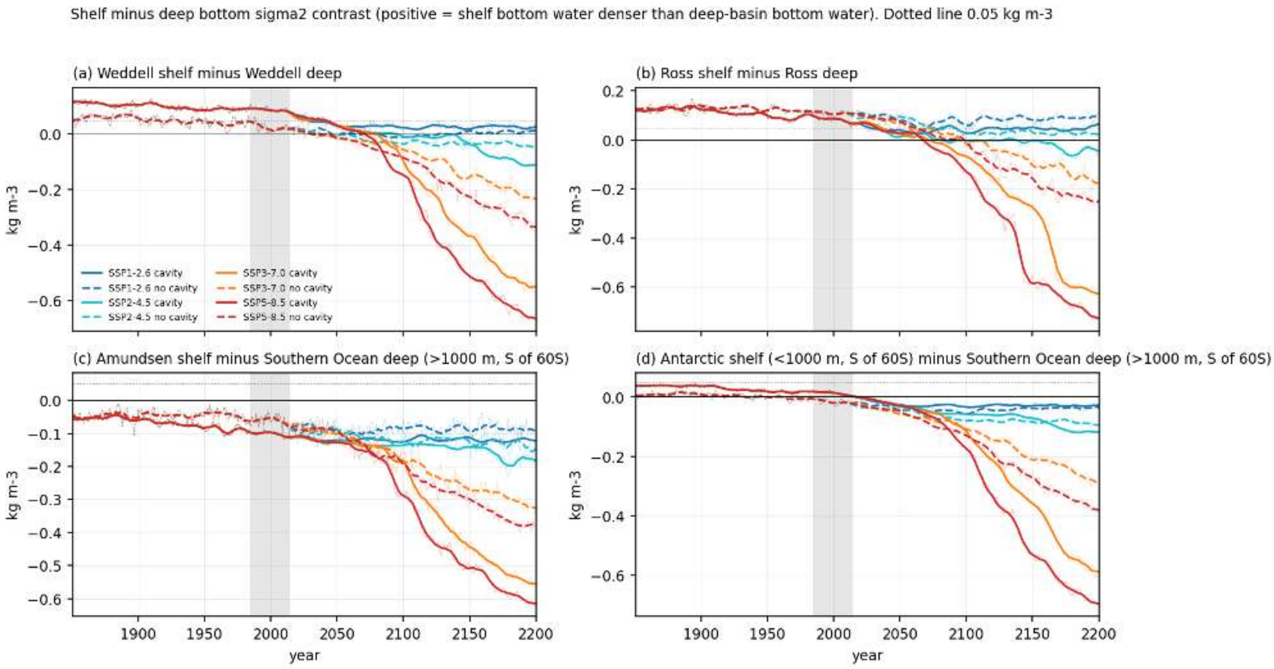

Figure 73: bottom_ts_contrast (bottom). Shelf minus deep bottom sigma2 contrast (kg m-3): Weddell shelf minus deep Weddell (a), Ross shelf minus deep Ross (b), Amundsen shelf minus deep Southern Ocean (c) and all Antarctic shelf minus deep Southern Ocean (d), 1851-2200, 11-yr running means over annual values. Positive values mean the shelf bottom water is on average denser than the deep bottom water and can in principle sink. Dotted line 0.05 kg m-3. The Weddell and Ross contrasts are 0.08 to 0.12 kg m-3 in 1851-1880, have already halved by 2014 in the no-cavity Weddell, and reach zero around 2070 (Weddell cavity run SSP5-8.5 in 2072, Ross in 2068). Under SSP1-2.6 they level off at 0.02 to 0.05 kg m-3 and never vanish. The Amundsen shelf is always lighter than the deep ocean.

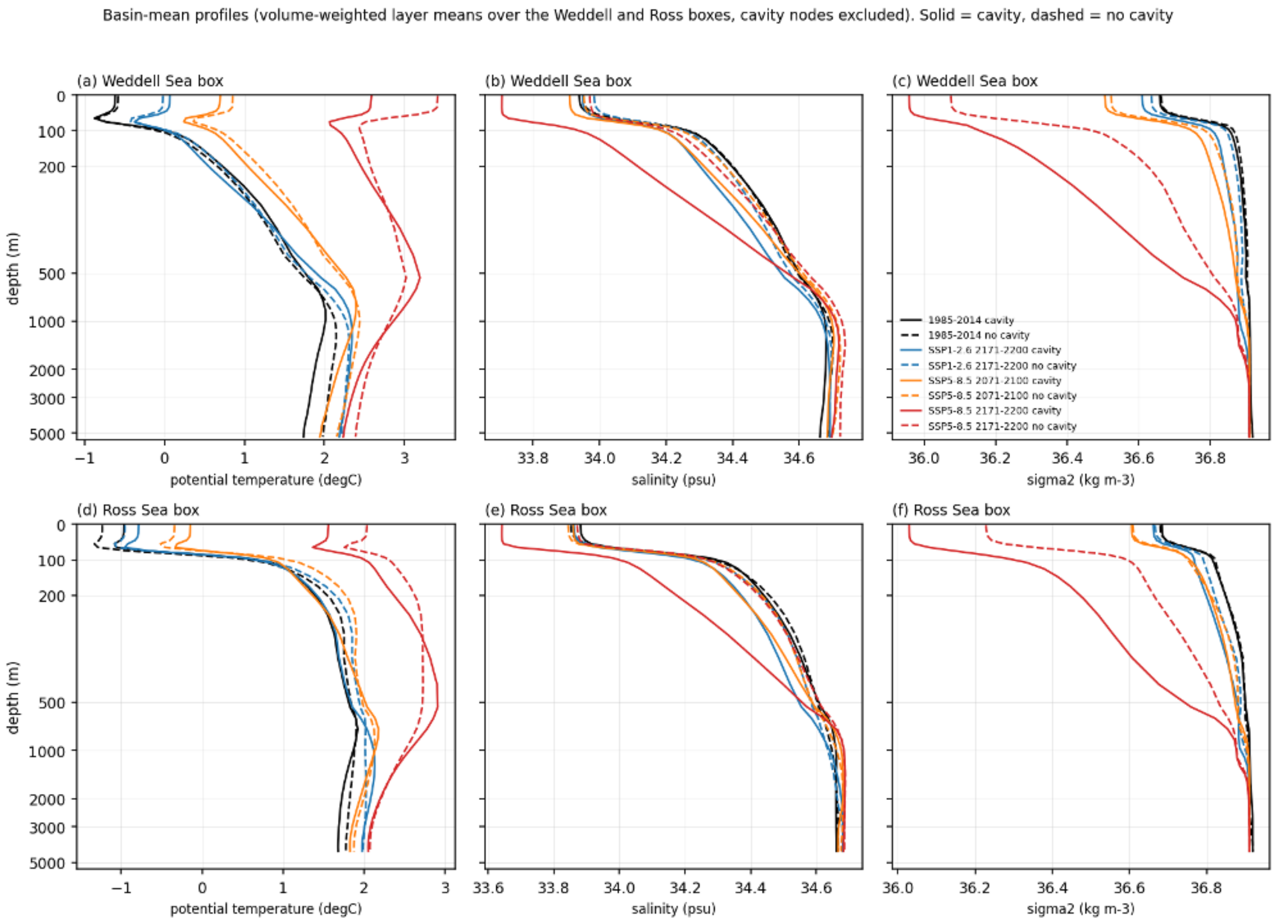

Figure 74: bottom_profiles (bottom). Volume-weighted mean profiles of potential temperature, salinity and sigma2 in the Weddell box (65W-20E, 80S-60S) and the Ross box (160E-140W, 80S-65S), cavity nodes excluded, for 1985-2014 (black), SSP1-2.6 2171-2200 (blue), SSP5-8.5 2071-2100 (orange) and SSP5-8.5 2171-2200 (red). Depth axis symlog. The cold surface layer is 0.8 to 1.3 degC below zero in the Weddell and Ross boxes at present; under SSP5-8.5 the upper 500 m warms by 1.5 to 3 degC and freshens by 0.25 psu, and the sigma2 profile loses 0.5 to 0.7 kg m-3 in the upper 200 m, which removes the weak present-day density connection between shelf and deep water.

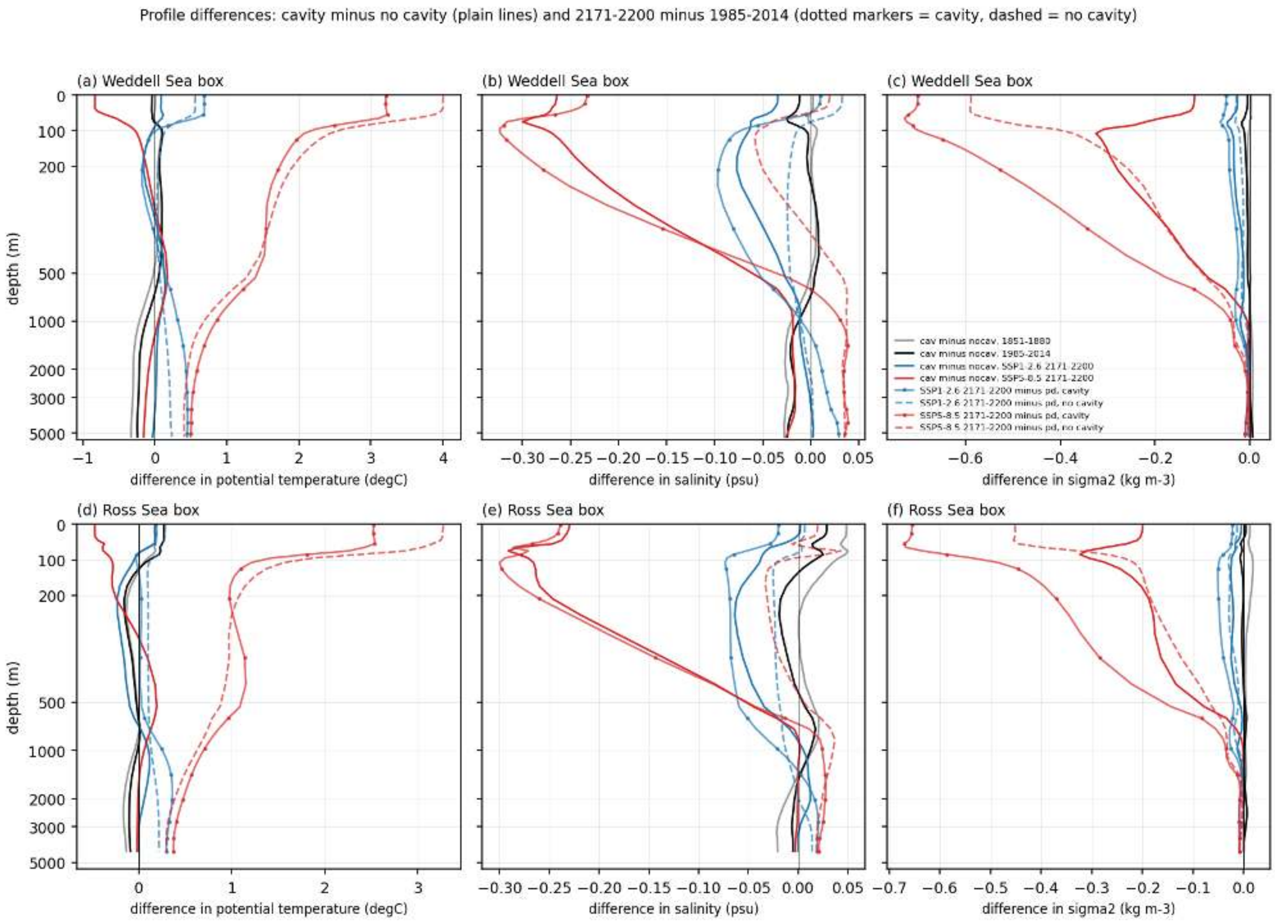

Figure 75: bottom_profiles_diff (bottom). Profile differences for the same boxes: cavity minus no cavity (plain lines for 1851-1880 grey, 1985-2014 black, SSP1-2.6 2171-2200 blue, SSP5-8.5 2171-2200 red) and 2171-2200 minus 1985-2014 (dotted markers cavity, dashed no cavity; blue SSP1-2.6, red SSP5-8.5). The cavity run is 0.2 to 0.3 degC cooler than the no-cavity run below 1000 m in the Weddell box and 0.15 degC cooler in the Ross box in 1851-1880, and the difference shrinks to 0.1 to 0.2 degC by 1985-2014 and to zero by 2200 under SSP1-2.6, which is the signature of the cavity run's deep drift. Density differences between the two runs are below 0.01 kg m-3 below 500 m at all times.

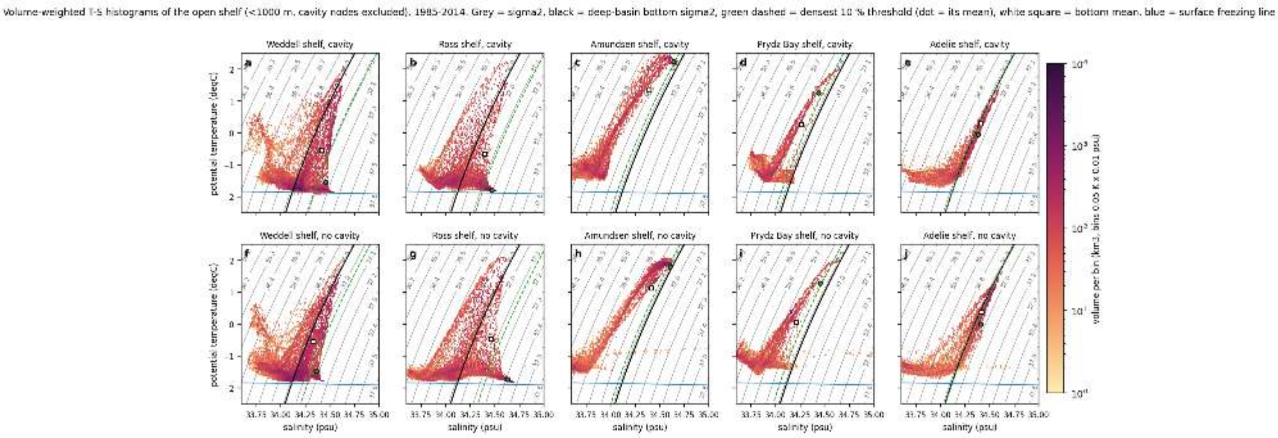

Figure 76: shelf_TS_pd (bottom). Volume-weighted T-S histograms (log colour, km3 per 0.05 K by 0.01 psu bin) of the open shelf (<1000 m, cavity nodes excluded) for the Weddell, Ross, Amundsen, Prydz Bay and Adelie shelf boxes, 1985-2014, cavity run (top) and no-cavity run (bottom). Grey contours sigma2 (TEOS-10 at the mean shelf pressure), black contour the area-mean bottom sigma2 of the corresponding deep basin, green dashed contour the sigma2 threshold of the densest 10 percent of the shelf volume with its mean as green dot, white square the area-mean bottom T-S, blue line the surface freezing point. The densest 10 percent of the Weddell and Ross shelf volume sits at -1.5 to -1.8 degC and 34.45 to 34.5 psu (37.1 kg m-3) in the cavity run, which is 0.2 kg m-3 denser than the deep basin bottom water. The no-cavity run has the same or slightly denser classes on the Ross shelf (34.6 psu) but fresher, lighter water on the Weddell shelf (34.36 psu). Amundsen and Prydz Bay shelf water is never denser than the deep basin. The no-cavity run shows a tail of spurious very salty (above 34.7 psu) near-freezing water in small coastal cells.

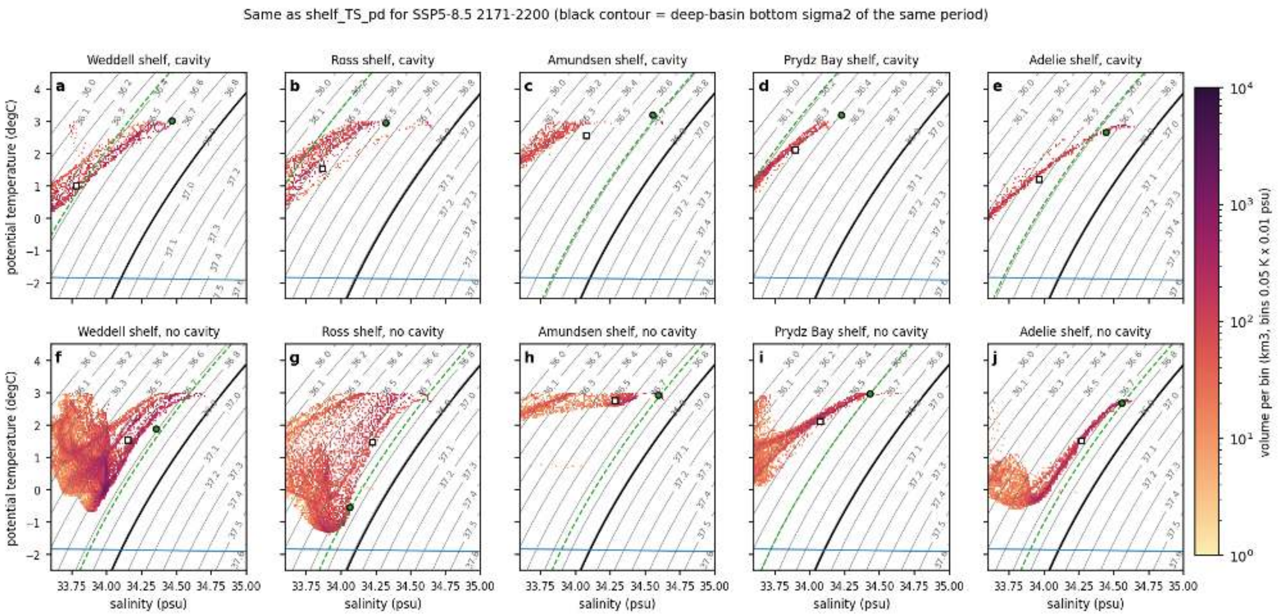

Figure 77: shelf_TS_far585 (bottom). Same as shelf_TS_pd for SSP5-8.5 2171-2200. The whole shelf volume has moved to warmer and fresher classes. The densest 10 percent is 36.4 to 36.7 kg m-3 on the Weddell and Ross shelves in the cavity run and 36.7 to 36.8 kg m-3 without cavities, in both cases 0.1 to 0.5 kg m-3 lighter than the deep basin bottom water, and no shelf water remains denser than the deep ocean. The no-cavity shelves are 0.3 to 0.4 psu saltier than the cavity shelves in this period.

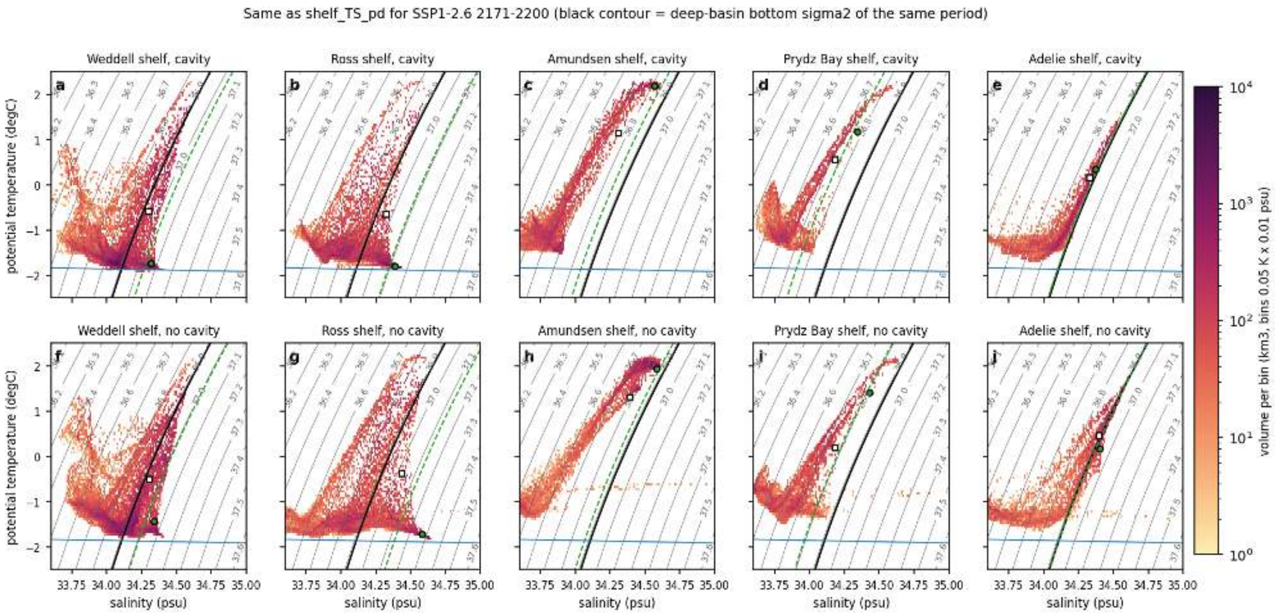

Figure 78: shelf_TS_far126 (bottom). Same as shelf_TS_pd for SSP1-2.6 2171-2200. The densest classes survive at about 37.0 kg m-3 but their volume shrinks, and the Ross shelf in the no-cavity run keeps more dense water than the cavity run.

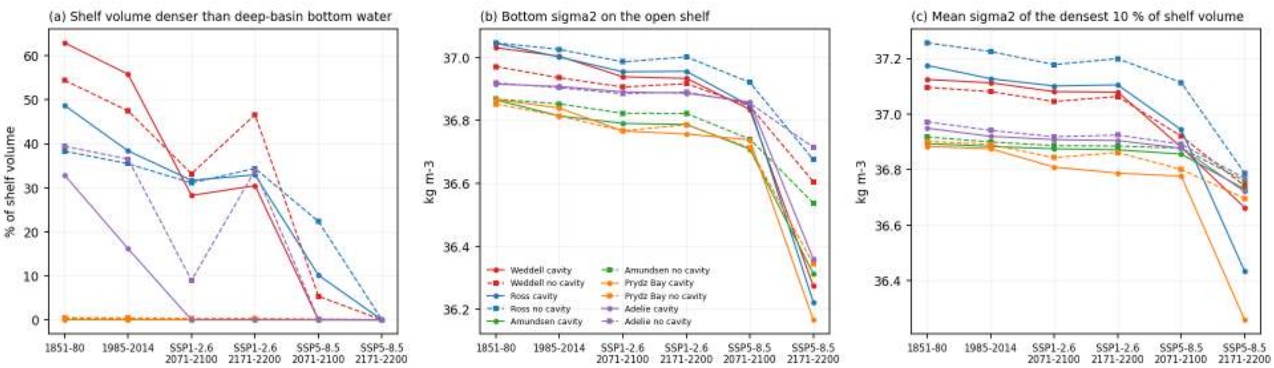

Figure 79: shelf_dense_water (bottom). Dense shelf water summary per shelf region (colours) for the periods 1851-1880, 1985-2014 and the SSP1-2.6 and SSP5-8.5 end and far periods: (a) percent of the open-shelf volume denser than the area-mean bottom sigma2 of the adjacent deep basin, (b) area-mean bottom sigma2 on the shelf, (c) mean sigma2 of the densest 10 percent of the shelf volume. Circles cavity, squares no cavity. Half of the Weddell shelf volume and 35 to 40 percent of the Ross shelf volume is denser than the deep bottom water at present; these fractions drop to 20 to 30 percent under SSP1-2.6 and to zero under SSP5-8.5 by 2100 to 2200.

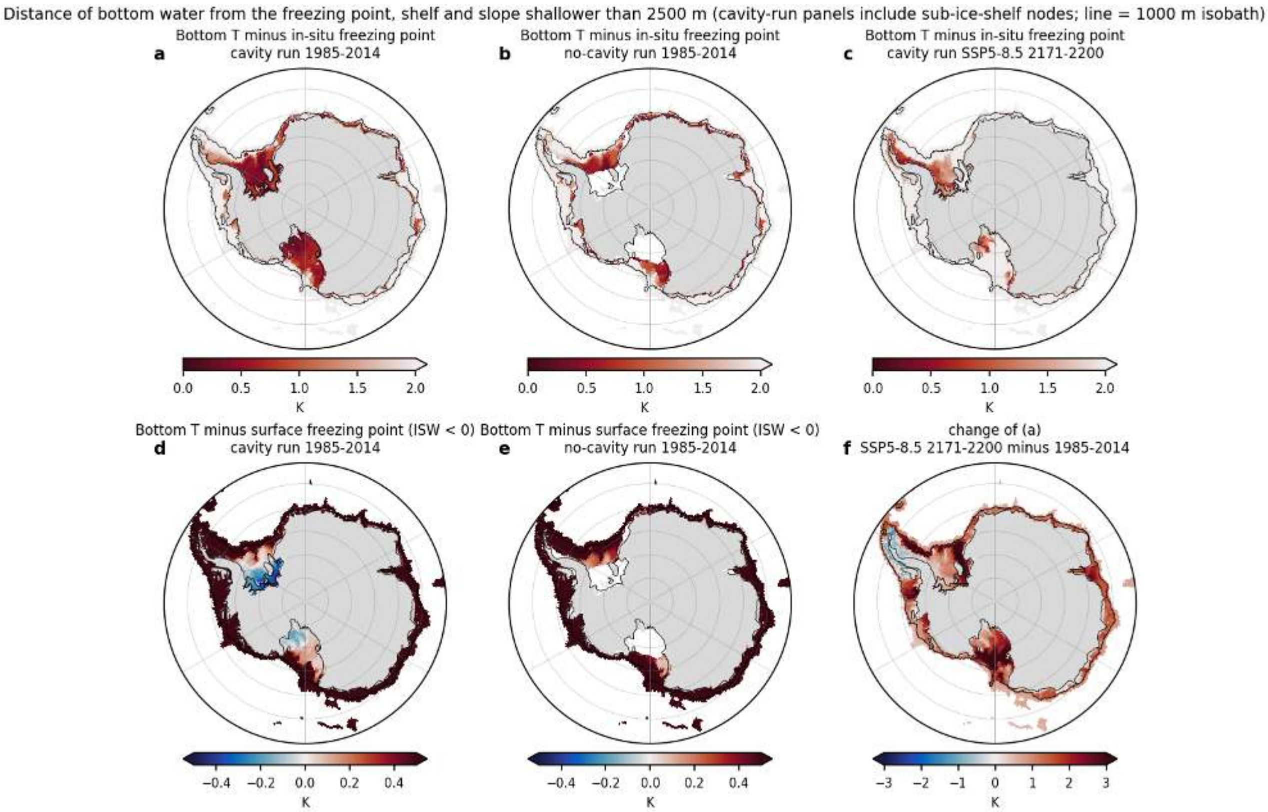

Figure 80: bottom_freezing (bottom). Bottom potential temperature minus the in-situ freezing point (a-c, K, gsw, using bottom salinity and depth) and minus the surface freezing point (d,e, negative = Ice Shelf Water) for the shelf and slope shallower than 2500 m, 1985-2014 cavity run (a,d, cavity nodes included), no-cavity run (b,e) and cavity run SSP5-8.5 2171-2200 (c), with the change (f). Black line 1000 m isobath. Bottom water within 0.5 K of the in-situ freezing point covers 19 percent of the Weddell shelf in the cavity run (9 percent without cavities) and 5 to 6 percent of the Ross shelf. Water below the surface freezing point exists only under and in front of the Filchner-Ronne and Ross ice shelves in the cavity run. By 2171-2200 under SSP5-8.5 the shelf bottom is 1.5 to 3 K further from freezing, except under the Filchner-Ronne cavity where the water cools slightly.

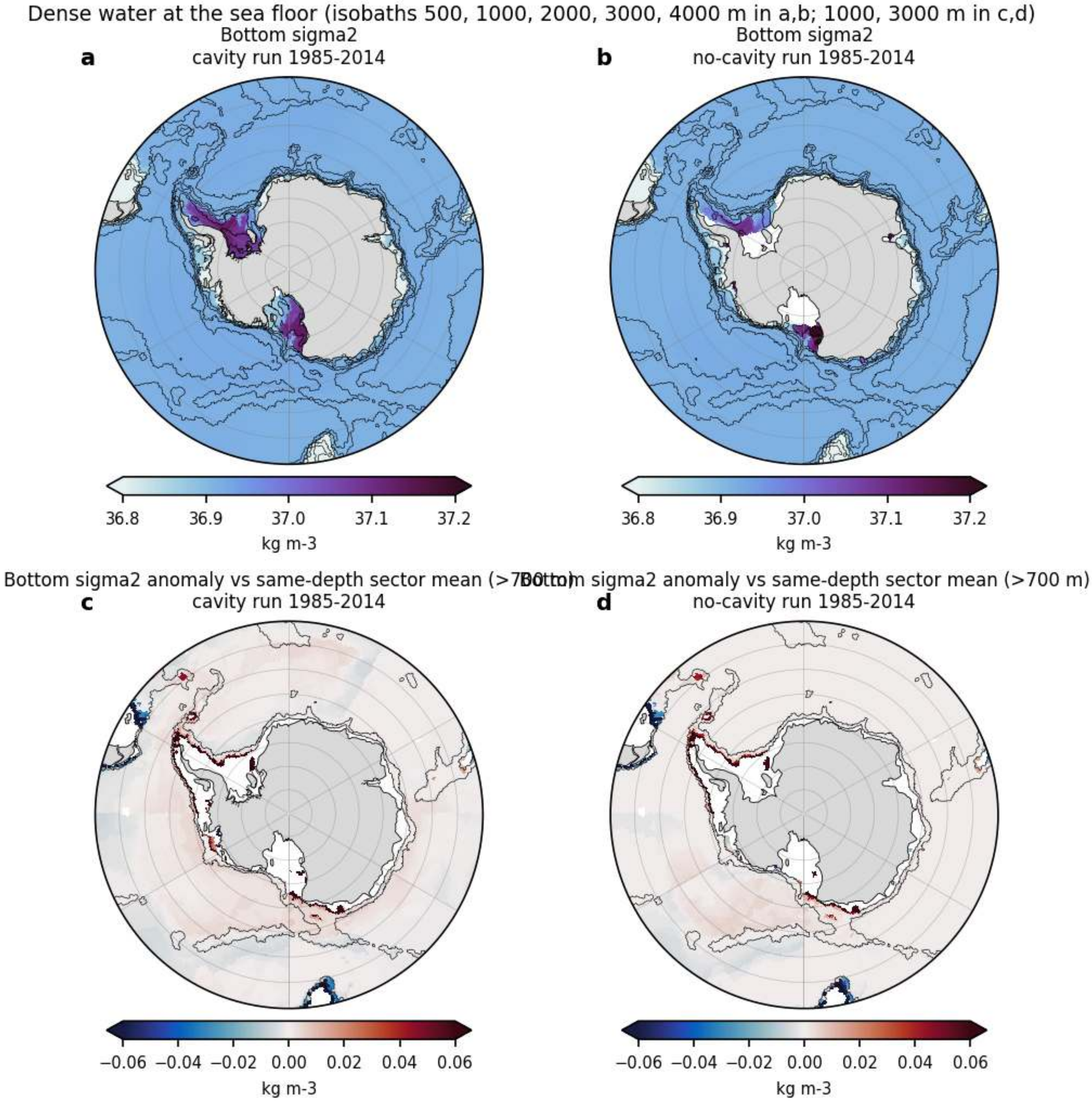

Figure 81: bottom_pathways (bottom). Bottom sigma2 (a,b; isobaths 500, 1000, 2000, 3000, 4000 m) and bottom sigma2 anomaly relative to the area mean of all nodes in the same 30 degree longitude sector and 250 m bottom-depth bin, shown for depths greater than 700 m (c,d; isobaths 1000 and 3000 m), 1985-2014, cavity run (a,c) and no-cavity run (b,d). Dense shelf water (above 37.0 kg m-3) is confined to the western Weddell and Ross shelves and does not reach below the 1000 m isobath as a coherent dense layer in either run. Positive anomalies on the slope mark the Filchner trough mouth, the western Ross slope and the Adelie and Prydz Bay shelf breaks, with amplitudes of only 0.02 to 0.05 kg m-3.

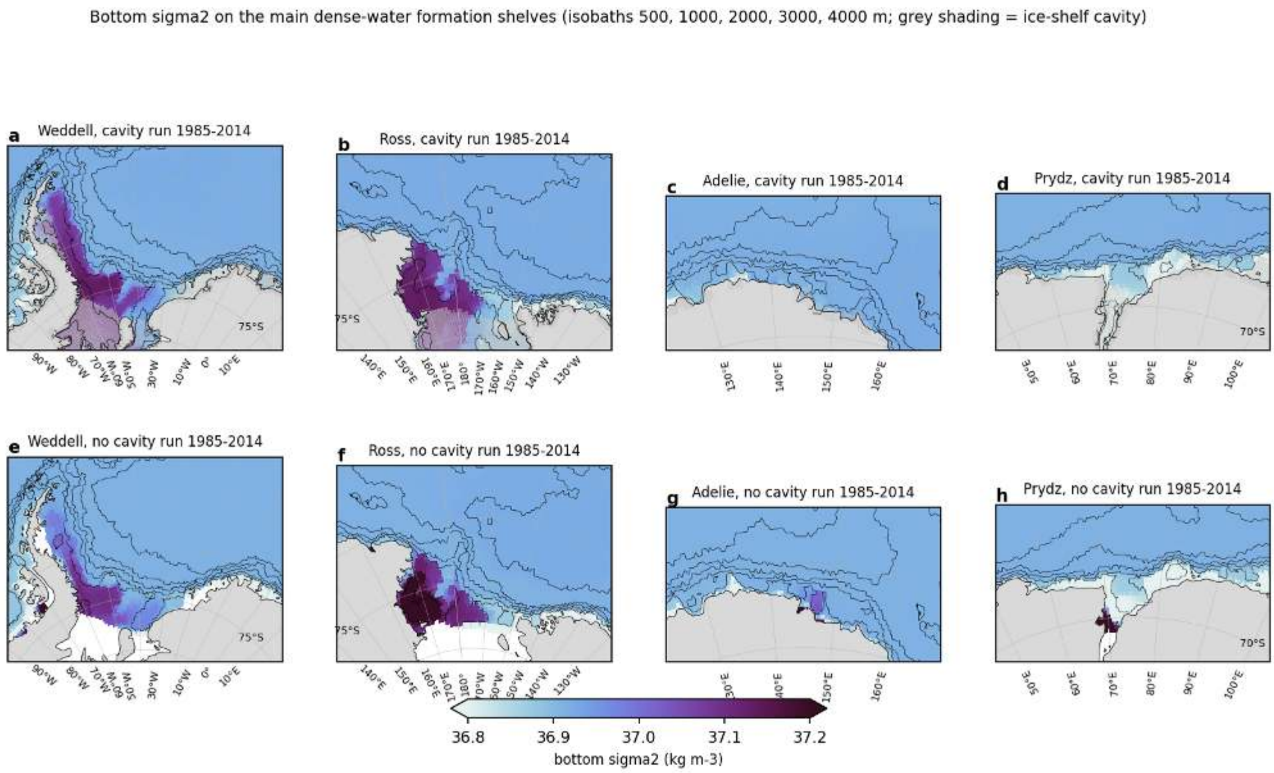

Figure 82: bottom_pathways_regional (bottom). Bottom sigma2 on the Weddell, Ross, Adelie and Prydz Bay shelves (isobaths 500 to 4000 m; grey shading = ice-shelf cavity), 1985-2014, cavity run (a-d) and no-cavity run (e-h). Dense water in the cavity run fills the whole western Weddell shelf and the Ross shelf down to the 1000 m isobath; in the no-cavity run the densest Ross water (above 37.2 kg m-3) is trapped in the south-western corner near Terra Nova Bay and on the Adelie and Prydz shelves dense water appears only in isolated coastal cells.

8 Surface water-mass transformation and buoyancy forcing

Findings

8.1 wmt theme: surface water-mass transformation and buoyancy forcing (Walin framework)

8.1.1 How the diagnostic was derived and validated

FESOM2 (gen_modules_diag.F90, routine compute_diagnostics, lines 503-541 of the awiesm-2.1-wiso source) computes for every surface element the density flux split into three parts and bins it into 89 sigma2 classes. The class index is the std_dens value nearest to the surface sigma2 of the element (class centres, not edges, so the class width is half the distance to each neighbour). The stored quantities are std_heat_flux = elem_area * mean(sw_alpha * heat_flux_in) / vcpw, std_frwt_flux = elem_area * mean(sw_beta * water_flux * S_surface) and std_rest_flux = elem_area * mean(sw_beta * relax_salt), with sw_alpha in 1/K, sw_beta in 1/psu, vcpw = rho*cp = 4.2e6 J m-3 K-1, heat_flux_in positive for ocean heat loss and water_flux positive for water leaving the ocean. The stored units are therefore m3/s (area-integrated density flux divided by a reference density), not the kg*m/s written in io_meandata.F90. The transformation rate across isopycnal sigma_i is F(sigma_i) = rho0 * sum_elements(std_flux_i) / dsigma_i / 1e6 Sv with rho0 = 1030 kg m-3 (FESOM density_0) and dsigma_i the nearest-node class width. Positive F means transformation towards denser water. Formation between two isopycnals is F(lighter) minus F(denser); the formation rate of water denser than a threshold is F(threshold). Because class spacing is uneven (several classes are only 0.004 kg m-3 apart), derivatives are taken on a uniform 0.025 kg m-3 grid after linear interpolation.

Validation (wmt_fig04): south of 50S (open ocean, 1985-2014) the freshwater term summed over classes is -2.05e7 kg/s (cavity run) versus -2.04e7 kg/s from an independent node-based estimate using gsw beta, the monthly fw and SSS climatologies. The heat term is 0.93e6 kg/s versus 1.2e6 kg/s from the monthly gsw estimate; the annual-mean based estimate (5.6e6 kg/s) is wrong by a factor 6 because heat loss occurs in winter at low alpha and heat gain in summer at high alpha. The seasonal covariance must be resolved, which the model diagnostic does at every time step. The sum of the three class-binned terms (-2.2e7 kg/s) agrees with the separately stored density_flux_e (-2.3e7 kg/s) to 5 percent (the latter uses alpha and beta at the first wet layer of every node, the class-binned fields use layer 1 values).

Two caveats were found in the diagnostic. 1. The restoring term is NOT zero. namelist.oce contains surf_relax_s = 0 but inside the namelist group oce_tra, which the code never reads. The value actually used comes from namelist.tra, group tracer_phys: surf_relax_S = 1.929e-6 m/s (10 m in 60 days) in all ten experiments, both sets. Surface salinity is relaxed towards the PHC3.0 climatology, the global mean of the relaxation flux is removed every step (ice_oce_coupling.F90 lines 475-500) and the relaxation is switched off under ice shelves. South of 50S this term removes salt (freshening, density flux -2.2e6 kg/s cavity, -1.5e6 kg/s no cavity, about 10 percent of the freshwater term) but in the dense classes it reaches several Sv (see findings). Restoring damps salinity anomalies and therefore weakens the freshwater forcing differences between the two sets. 2. For elements under ice shelves (cavity run) the class index is taken from dens(1), which is never set for elements whose first wet layer is below 1. The value is left over from the previously processed element, so the cavity fluxes are binned at essentially random densities. The cavity-only class sums are also inconsistent with the fluxes: the freshwater term summed over the cavity elements is positive (+4.4e5 kg/s, densification) while the meltwater input of 2360 Gt/yr implies -1.8e6 kg/s. All transformation numbers in this theme therefore EXCLUDE cavity elements (recomputed from the raw yearly std_* files, cache/wmt/*wmt.nc, script analyses/wmt/compute_wmt_series.py). The cache/ts wmt** sums include the cavity elements and should not be used for the cavity runs. The buoyancy input by ice-shelf melt is a real forcing of the cavity run but it acts at the ice-shelf base, not at the sea surface, and is treated by the melt theme.

8.1.2 Numbered findings

1. Where and how much dense water is formed at the surface. Present-day (1985-2014) surface transformation south of 50S peaks at 6.7 Sv at sigma2 36.94 in the cavity run and 6.0 Sv at 36.92 in the no-cavity run. The formation rate of water denser than the model-relative threshold sigma2 37.0 is 5.1 Sv (cavity) and 4.3 Sv (no cavity); for the threshold 36.9 it is 6.4 and 5.4 Sv, for 37.1 it is 3.5 and 2.8 Sv. All of this happens south of 60S (the curves for south of 50S and south of 60S coincide above 36.9) in the coastal band, mainly in the Weddell Sea (2.3 and 1.8 Sv) and the Ross Sea (1.8 and 1.2 Sv) boxes, which together account for 80 and 70 percent. The freshwater term (brine rejection in coastal polynyas) carries 85 percent of this formation (4.3 and 3.7 Sv), the heat term 0.6 and 1.0 Sv, the restoring +0.2 and -0.4 Sv. The densest surface class that is still transformed is 37.17 to 37.18 in both sets, so no surface water reaches the observed AABW density (sigma2 above 37.16 is the 2-class tail). (wmt_fig01, fig02, fig13, fig14)

2. The surface formation has no interior counterpart. The volume of sigma2 above 37.0 water south of 60S is 0.38 (cavity) and 0.12 times 1e15 m3 (0.5 and 0.2 percent of the volume south of 60S) and the densest overturning cell (sigma2 above 36.95, 70-55S) is 3.1 and 1.8 Sv. With 5.1 and 4.3 Sv surface formation the residence time of the dense surface water is 2.4 and 0.9 years, so the brine-formed dense shelf water is mixed into the 36.9 to 37.0 interior mode rather than sinking to the abyss. This is the surface-forcing view of the weak AABW cell found by the overturning and census themes: the forcing exists (5 Sv, the right order of magnitude) but the density reached (37.0 to 37.18) is 0.2 kg m-3 short of real AABW, consistent with the 0.2 psu fresh shelf bias. The 11-yr smoothed dense cell follows the surface formation along the scenario chains (correlation 0.86 to 0.96). (wmt_fig08)

3. Cavity effect. With cavities the dense-water formation is 0.8 Sv (18 percent) larger at present (5.1 versus 4.3 Sv at 37.0), the difference being 0.7 Sv in the Ross Sea (+56 percent) and 0.4 Sv in the Weddell Sea (+23 percent). The peak transformation is 0.7 Sv larger and 0.02 kg m-3 denser. This goes with the fact that the cavity run has more coastal densification (coastal band difference up to +1.5e-6 kg m-2 s-1, winter shelf brine flux 2.5 versus 2.0e-6 kg m-2 s-1, winter shelf fw 7.8 versus 6.3 mm/day). The main reason is the different freshwater routing, not sea ice alone: in the no-cavity run 1.1 to 1.8 mm/day of ice-sheet runoff enters at the coast and lightens the shelf water, in the cavity run the same water enters as 2360 Gt/yr of basal melt under the ice shelves (excluded here, it acts as a subsurface buoyancy source) and coastal runoff south of 60S is zero. The Ross Sea, where the model has a large Ross Ice Shelf cavity and the largest winter open-water difference (atmos theme), shows the largest effect. The cavity run also shows a pre-industrial to present decline of the dense formation (5.7 to 5.1 Sv) that is absent in the no-cavity run (4.6 to 4.3 Sv). The densest cell in density space is stronger with cavities (3.1 versus 1.8 Sv) and the dense volume three times larger. (wmt_fig01, fig05, fig08, fig13, fig14)

4. Scenario evolution. Under SSP5-8.5 the dense formation (F at 37.0) falls to 1.4 Sv (cavity, -73 percent) and 1.8 Sv (no cavity, -58 percent) in 2071-2100 and to 0.0 and 0.1 Sv in 2171-2200. It drops below 25 percent of present in 2084 (cavity) and 2103 (no cavity) and below 10 percent in 2089 and 2148. The heat contribution vanishes first (0.02 and 0.1 Sv by 2071-2100), the freshwater term follows. The Weddell formation vanishes in 2084/2086 (cavity, 25/10 percent) and 2091/2106 (no cavity), the Ross formation in 2089/2095 and 2121/2152. Under SSP3-7.0 formation ends around 2100 (cavity) and reaches 0.4 Sv by 2171-2200 (no cavity). Under SSP2-4.5 the cavity run loses 74 percent by 2171-2200 (1.3 Sv) including the whole Weddell formation (after 2157), the no-cavity run keeps 2.7 Sv. Under SSP1-2.6 the formation stabilises at 3.7 (cavity, -28 percent) and 4.0 Sv (no cavity, -7 percent). The cavity run is therefore more sensitive to warming in every scenario. The peak transformation itself does not weaken (5.7 to 8 Sv until 2100) but migrates to lighter classes (36.92 to 36.8 by 2071-2100 and to 36.3 to 36.5 by 2171-2200 under SSP5-8.5), and the zero crossing of F moves from 36.7 to 36.0. The loss of dense formation is a freshening effect: shelf fw changes from +2.3 to -1.4 mm/day (cavity) and +0.6 to -4.5 (no cavity), whereas the shelf heat loss increases (Ross shelf 32 to 58 and 30 to 70 W m-2) as sea ice retreats. (wmt_fig03, fig05, fig06, fig07, fig09, fig10, fig12, fig15)

5. Surface buoyancy fluxes. Present-day winter (JJA) heat loss is 88 and 86 W m-2 south of 60S, 65 and 48 W m-2 on the Antarctic shelf, 62 and 55 on the Ross shelf and 41 and 39 on the Weddell shelf (cavity, no cavity). The annual density flux south of 50S is -0.48e-6 kg m-2 s-1 (buoyancy gain) with heat +0.02 and freshwater -0.45; the shelf is the only densifying region (+0.82 and +0.46). In winter the density flux south of 50S is +1.7e-6 (heat driven), on the shelf +3.0 and +2.4e-6 (brine driven). Under SSP5-8.5 the mean density flux south of 50S changes by -0.20 (cavity) and -0.28e-6 kg m-2 s-1 (no cavity) by 2171-2200, with the coastal band losing 1.0 to 1.5e-6 of its freshwater-driven densification. (wmt_fig11 to fig15)

6. SSS restoring. Both sets restore SSS to PHC3.0 with 1.929e-6 m/s (10 m per 60 days), globally balanced and off under ice shelves. Its density flux adds salt in the Weddell Sea (+0.12 and +0.14e-6 kg m-2 s-1), on the shelf (+0.08 and +0.22) and in the Ross Sea of the no-cavity run (+0.10), and removes salt over the ACC. In single dense classes it reaches 3.9 Sv (cavity) and 2.9 Sv (no cavity), and it contributes +0.2 and -0.4 Sv to F at 37.0. The sign pattern (more salt added where the model is fresher) means the restoring counteracts the freshening by melt and runoff and therefore damps the cavity versus no-cavity difference in the shelf salinity, the density reached by the shelf water and the dense-water formation. This should be stated in any paper and is a reason to treat the shelf salinity bias (0.2 psu fresh despite the restoring) as robust. (wmt_fig16, fig01)

8.1.3 Caveats and data issues

Cavity elements are excluded from all F values (diagnostic binning bug, see above); the cache/ts wmt_* sums for the cavity runs include them and differ by up to 0.8 Sv in single classes. The class spacing of the FESOM std_dens array is uneven, with near-duplicate classes at 35.98, 36.595, 36.68, 36.75 and 36.875, which makes per-class formation rates noisy; derivatives are taken on a uniform 0.025 kg m-3 grid. The two densest classes (37.5 and 37.75) are 0.3 kg m-3 wide, F there is not meaningful and the plots stop at 37.2. The restoring term is not available in cache/pmean, its maps come from 30-yr means of the raw std_rest_flux files (cache/wmt/*_rest.nc) and are not included in fig13 and fig14, only in fig15. fw region means include runoff, so the shelf fw difference between the sets reflects the freshwater routing (coastal runoff versus basal melt) as much as sea ice. Ensemble spread is not available (single members), so differences of less than about 0.3 Sv in the dense formation between the two sets should not be over-interpreted; the pre-industrial decline of the cavity run (5.7 to 5.1 Sv) is of that order. The "dense water" threshold 37.0 is model relative; sigma2 above 37.0 occupies 0.2 to 0.5 percent of the volume south of 60S, sigma2 above 36.9 occupies 80 percent.

8.1.4 Suggested story lines

A surface-forcing explanation of the missing AABW: the model produces 4 to 5 Sv of dense shelf water by brine rejection, but only to sigma2 37.0 to 37.2, and this water is eroded within 1 to 2 years. Cavities increase the dense formation by 18 percent (Ross Sea +56 percent) mainly by moving the ice-sheet freshwater from the coast to the ice-shelf base, which keeps the surface shelf water saltier. Under warming the dense formation collapses between 2080 and 2150 depending on scenario, through freshening rather than reduced heat loss, and earlier with cavities. The SSS restoring active in both runs is an important methodological caveat.

Figures

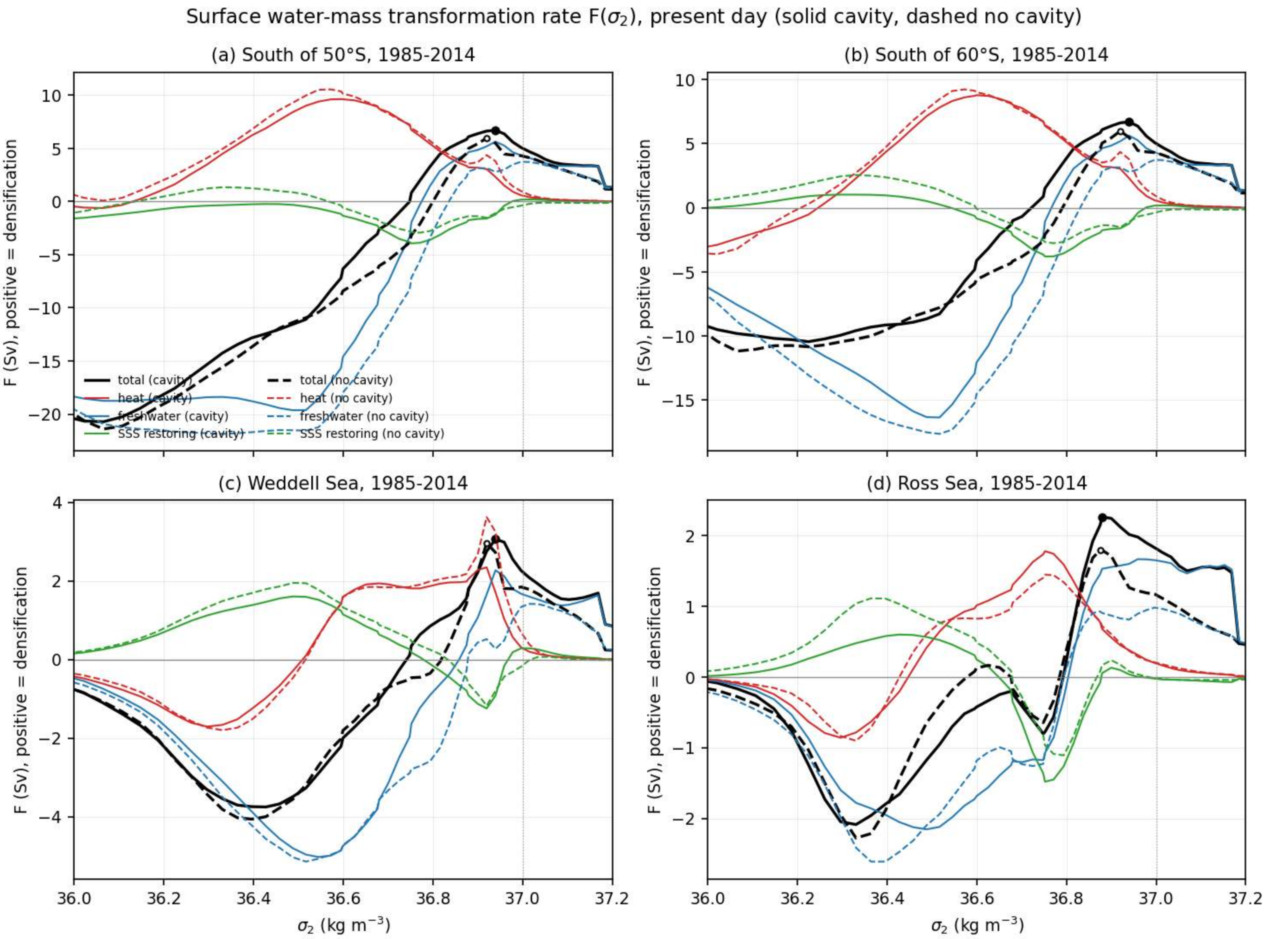

Figure 83: wmt__fig01__pd__transformation__components (wmt). Present-day (1985-2014) transformation rate F(sigma2) south of 50S (a), south of 60S (b), in the Weddell Sea box (c) and the Ross Sea box (d) for both sets, split into the heat (red), freshwater (blue) and SSS restoring (green) contributions and their sum (black). Dots mark the maximum of the total F above sigma2 36.5. South of 50S the total F peaks at 6.7 Sv at sigma2 36.94 (cavity) and 6.0 Sv at 36.92 (no cavity). In the dense classes (sigma2 above 36.9) the freshwater term dominates (brine release in coastal polynyas), the heat term is small (0.6 and 1.0 Sv at 37.0) because the surface is at the freezing point and ice covered. The restoring term is not negligible (up to 3.9 Sv in single classes near 36.8) and changes sign across the curve. The Ross Sea transformation is stronger with cavities (1.8 versus 1.2 Sv at 37.0), the Weddell Sea difference is smaller (2.3 versus 1.8 Sv). Curves for south of 50S and south of 60S coincide above 36.9 because all surface water denser than 36.9 lies south of 60S.

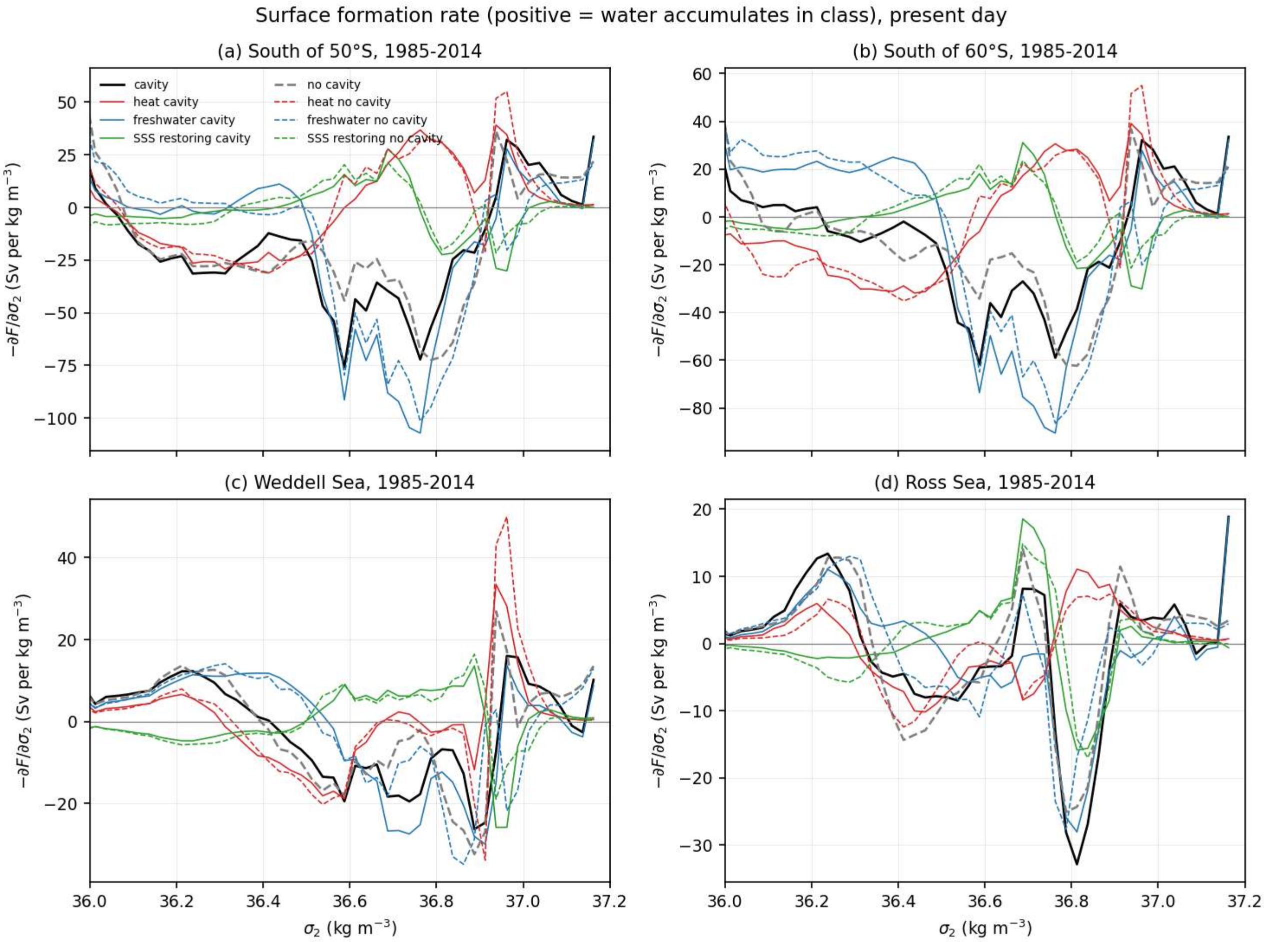

Figure 84: wmt__fig02__pd__formation__rate (wmt). Present-day formation rate, defined as minus the derivative of F with respect to sigma2 on a uniform 0.025 kg m-3 grid (Sv per kg m-3), same regions and components as fig01. Positive values mean water accumulates in that density class through surface forcing. Water is removed (negative) between 36.4 and 36.9 (net buoyancy gain by precipitation and sea-ice melt over the open ocean and gyres) and formed in the densest classes between 36.9 and 37.17 (brine rejection). The formation maximum south of 50S lies at 36.94 to 37.16 with 34 (cavity) and 36 (no cavity) Sv per kg m-3. Integrated over all classes denser than 37.0 the formation is 5.1 Sv (cavity) and 4.3 Sv (no cavity).

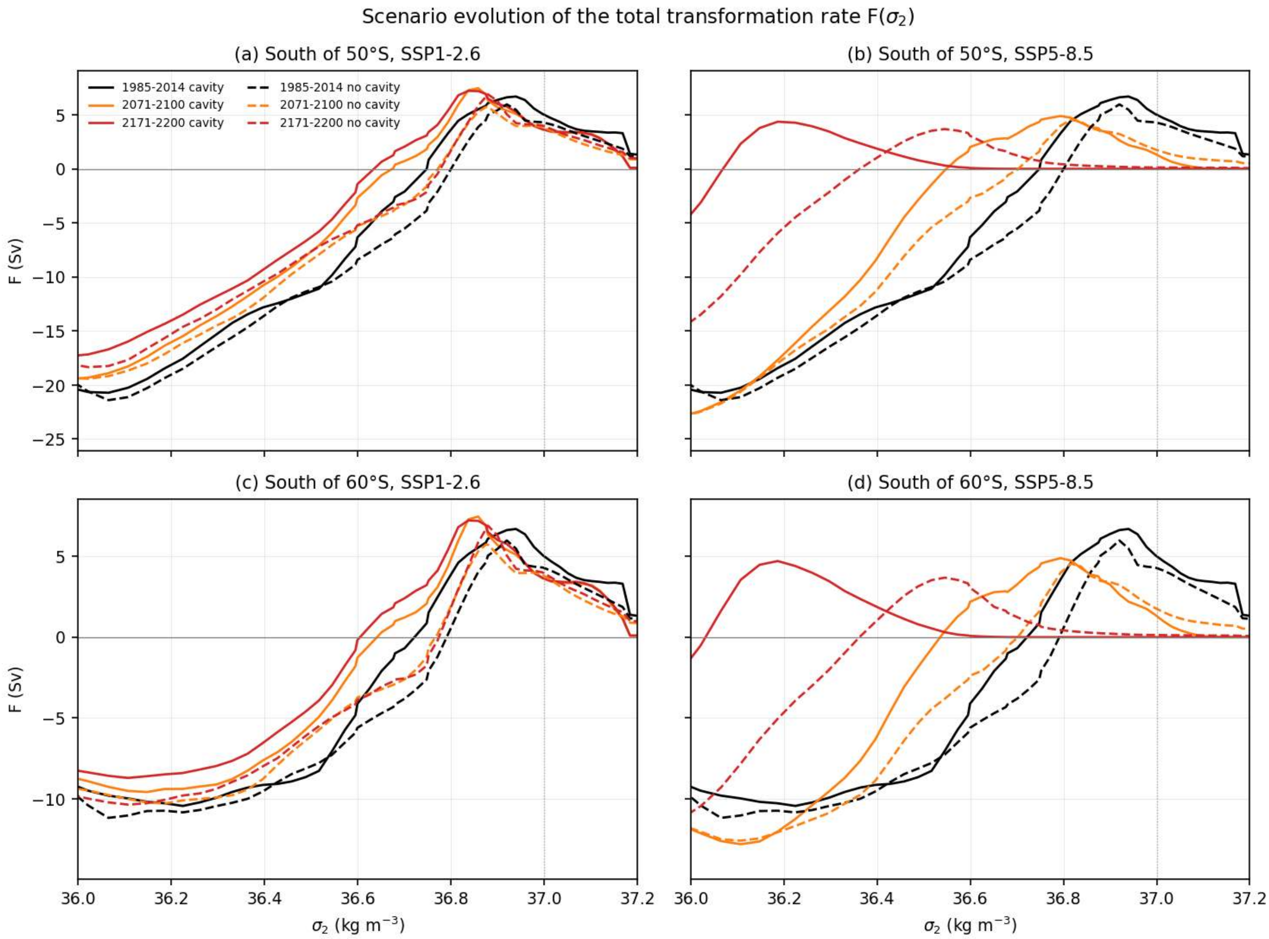

Figure 85: wmt_fig03_scenario_F_sigma (wmt). Total transformation rate F(sigma2) south of 50S (a, b) and south of 60S (c, d) for present day (black), 2071-2100 (orange) and 2171-2200 (red) under SSP1-2.6 (left) and SSP5-8.5 (right), both sets. Under SSP1-2.6 the curve keeps its shape and the peak shifts slightly to lighter water (36.85 to 36.88), with F at 37.0 falling from 5.1 to 3.7 Sv (cavity) and from 4.3 to 4.0 Sv (no cavity). Under SSP5-8.5 the whole curve migrates towards lighter classes, the peak is at 36.79 (cavity) and 36.82 (no cavity) by 2071-2100 and at 36.5 by 2171-2200, where F above 36.9 is zero (cavity) or 0.2 Sv (no cavity).

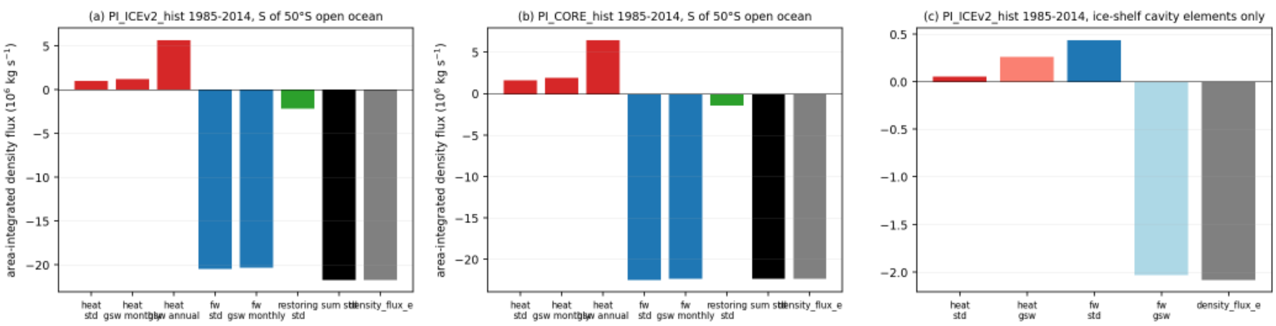

Figure 86: wmt_fig04_validation_density_flux (wmt). Validation of the diagnostic for 1985-2014. Panels a and b show the area-integrated surface density flux south of 50S (open ocean) for the two historical runs: the class-summed heat and freshwater terms from the FESOM diagnostic (std), independent estimates from the monthly fh, fw, SST and SSS climatologies with gsw alpha and beta (gsw monthly), the same estimate with annual means (gsw annual), the restoring term, the sum of the three std terms and the separately stored density_flux_e. Panel c shows the cavity elements of the cavity run only. The freshwater term agrees to 1 percent (-20.5 versus -20.4 and -22.5 versus -22.4 million kg/s), the heat term to 25 percent when the seasonal cycle is resolved (0.9 versus 1.2 and 1.6 versus 1.9 million kg/s) but is six times too large when annual means are used. The sum of the three std terms equals density_flux_e exactly (-21.7 million kg/s). Under the ice shelves the class-binned freshwater term has the wrong sign (+0.4 instead of -2.0 million kg/s), while density_flux_e (-2.1) agrees with the melt water input of 2360 Gt/yr. This is why cavity elements are excluded from F.

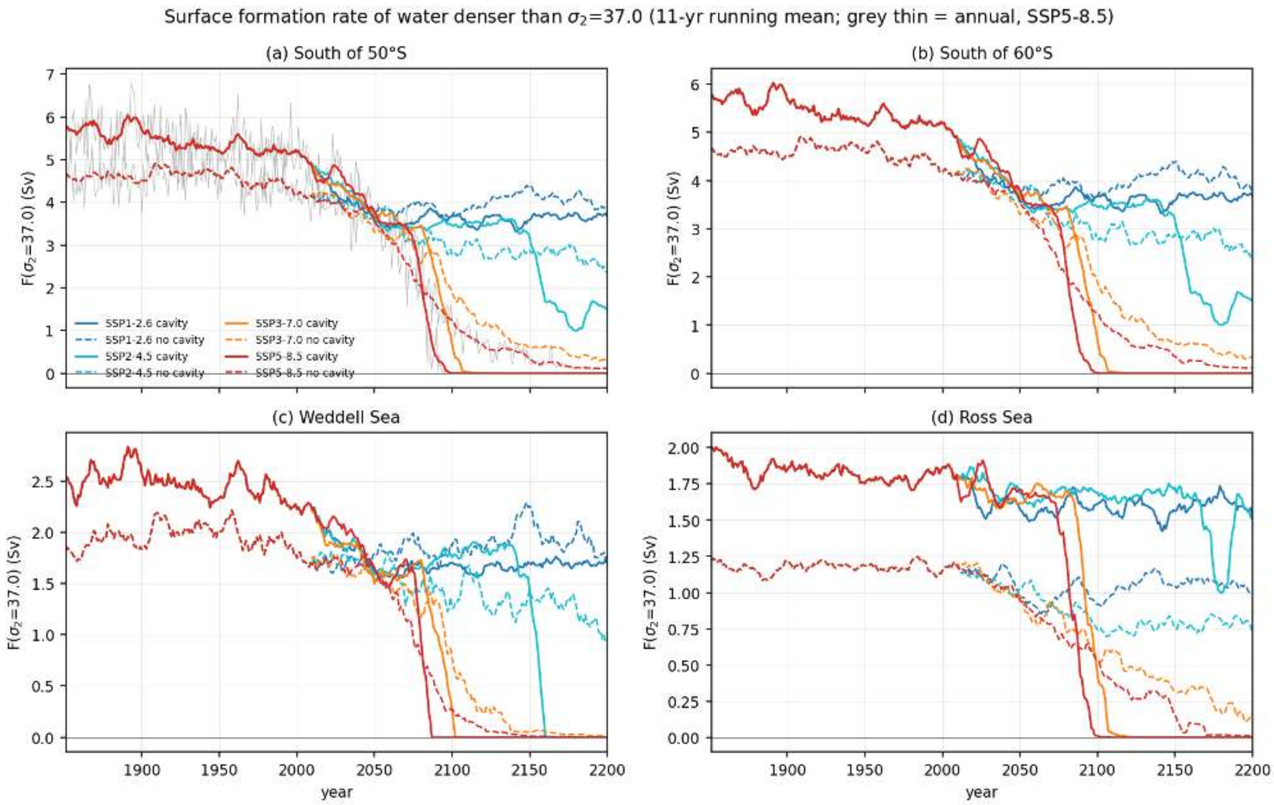

Figure 87: wmt_fig05_ts_dense_formation (wmt). Time series 1851-2200 of the surface formation rate of water denser than sigma2 37.0 (F at 37.0, 11-yr running mean; thin grey shows annual values of the cavity SSP5-8.5 chain) south of 50S (a), south of 60S (b), Weddell Sea (c) and Ross Sea (d) for the four scenarios and both sets. Present-day values are 5.1 (cavity) and 4.3 Sv (no cavity) south of 50S, 2.3 and 1.8 Sv in the Weddell Sea, 1.8 and 1.2 Sv in the Ross Sea. A slow decline from 5.7 Sv in 1851-1880 to 5.1 Sv at present occurs only in the cavity run. Under SSP5-8.5 the formation falls below 25 percent of the present value in 2084 (cavity) and 2103 (no cavity) and is zero (cavity, from about 2090) or 0.1 Sv (no cavity) in 2171-2200. Under SSP3-7.0 it vanishes in the cavity run around 2100 and drops to 0.4 Sv in the no-cavity run. Under SSP2-4.5 the cavity run loses the Weddell formation in 2157 while the no-cavity run keeps 1.2 Sv there. Under SSP1-2.6 formation stabilises at 3.7 (cavity) and 4.0 Sv (no cavity).

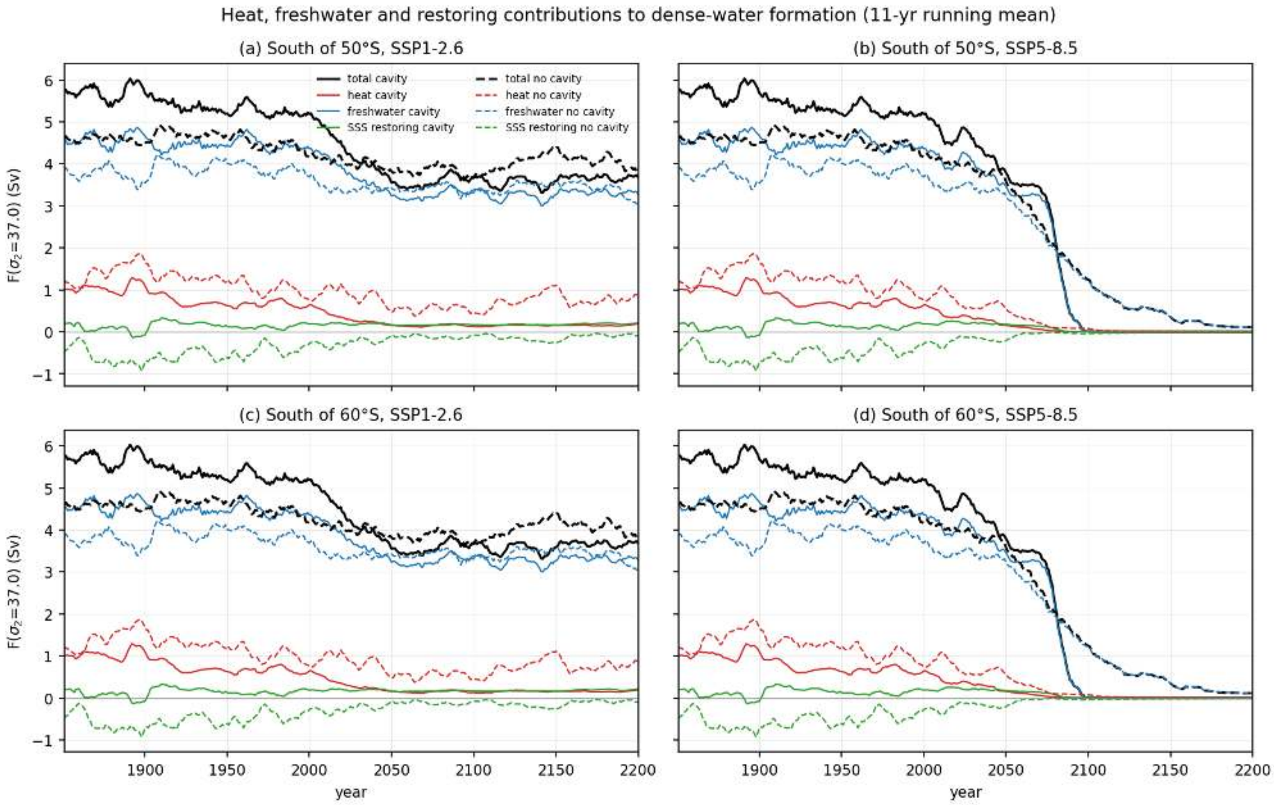

Figure 88: wmt_fig06_ts_components (wmt). Heat (red), freshwater (blue), restoring (green) and total (black) contributions to F at sigma2 37.0 south of 50S (a, b) and south of 60S (c, d) under SSP1-2.6 (left) and SSP5-8.5 (right), 11-yr running means, both sets. The freshwater term carries 85 percent (cavity, 4.3 of 5.1 Sv) and 87 percent (no cavity, 3.7 of 4.3 Sv) of the present-day dense-water formation. The heat term (0.6 and 1.0 Sv) disappears first under warming (0.02 and 0.1 Sv by 2071-2100 under SSP5-8.5) and the freshwater term follows (1.3 and 1.7 Sv by 2071-2100, zero by 2171-2200). The restoring contribution is +0.2 Sv (cavity) and -0.4 Sv (no cavity) at present and stays small in the dense classes.

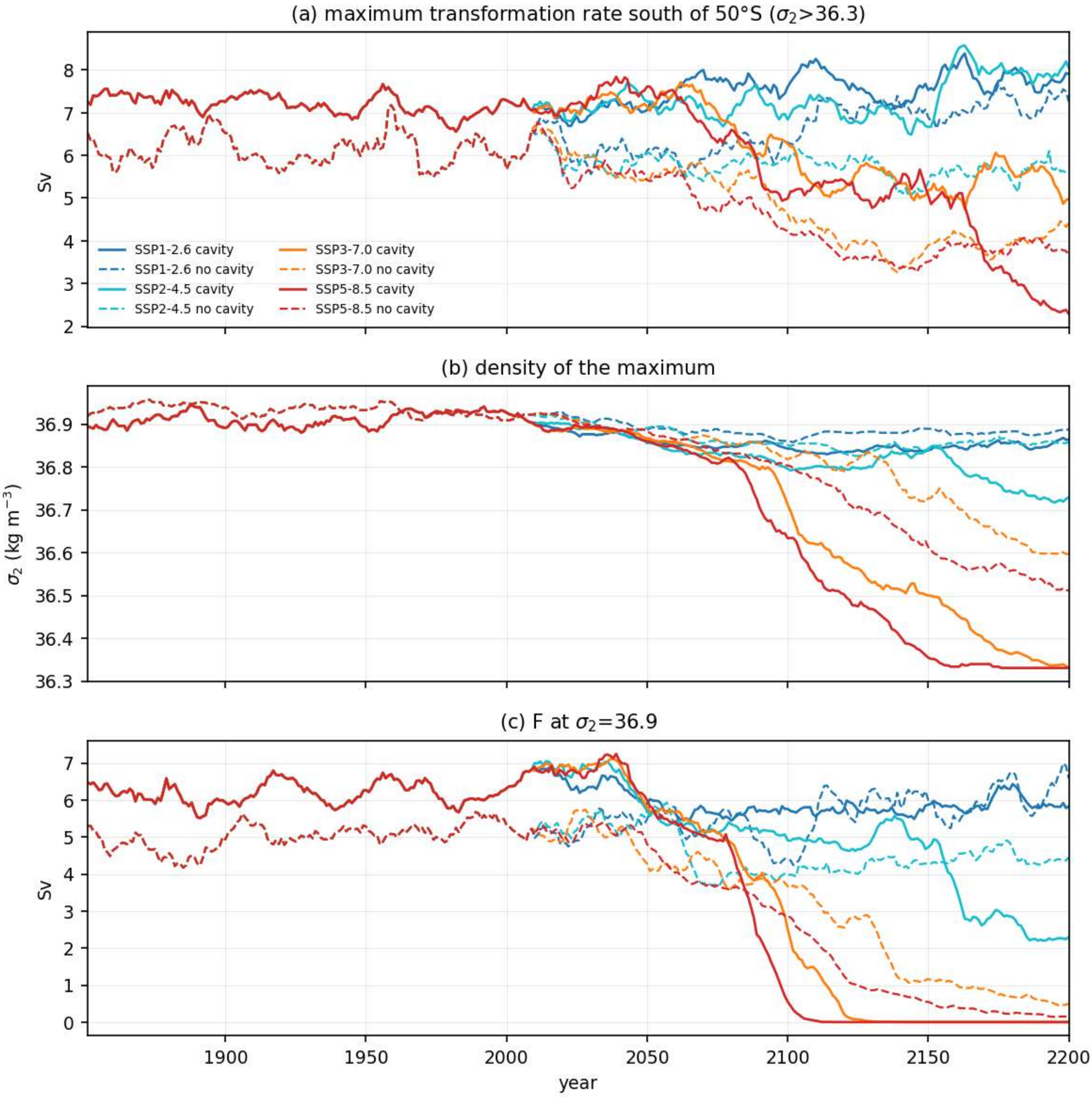

Figure 89: wmt__fig07__ts__peak__transformation (wmt). Time series of the maximum total transformation rate south of 50S above sigma2 36.3 (a), the density of the maximum (b) and F at sigma2 36.9 (c), 11-yr running means, all scenarios, both sets. The maximum stays at 5.7 to 8 Sv in all scenarios until 2100, so the Southern Ocean keeps densifying surface water, but the density at which this happens migrates from 36.92 (present) to 36.75 to 36.82 in 2071-2100 and to 36.33 to 36.54 in 2171-2200 under SSP5-8.5. F at 36.9 drops from 6.4 to 3.3 (cavity) and from 5.4 to 3.4 Sv (no cavity) by 2071-2100 under SSP5-8.5 and to zero and 0.2 Sv in 2171-2200.

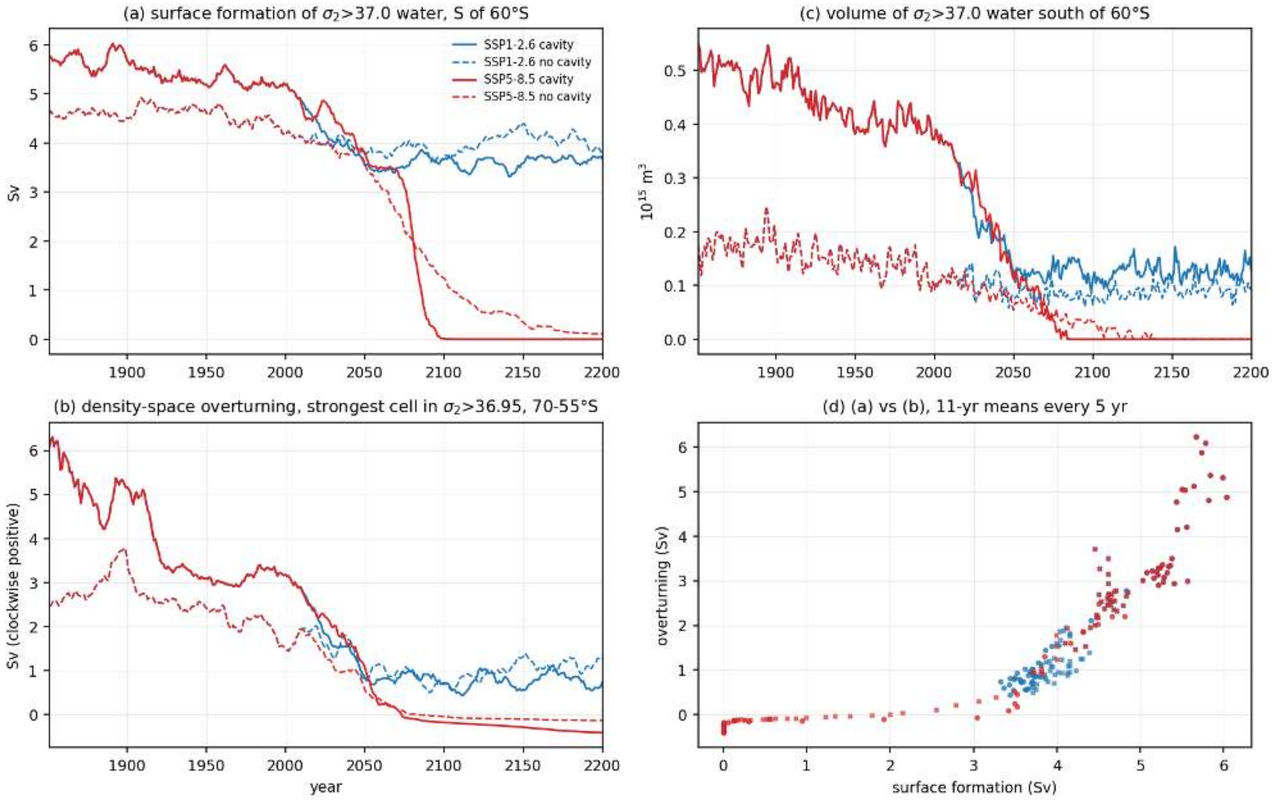

Figure 90: wmt_fig08_consistency_moc_census (wmt). Consistency of the surface formation with the interior. (a) surface formation of sigma2 above 37.0 water south of 60S, (b) strength of the densest overturning cell in density space (minimum of the global density-space stream function for sigma2 above 36.95 between 70S and 55S, sign reversed so that the clockwise AABW-type cell is positive), (c) volume of sigma2 above 37.0 water south of 60S from the census, (d) scatter of (a) against (b). SSP1-2.6 and SSP5-8.5 shown for both sets. The present-day dense cell is 3.1 Sv (cavity) and 1.8 Sv (no cavity), far smaller than the 5.1 and 4.3 Sv surface formation, and the dense volume is only 0.38 and 0.12 times 1e15 m3 (0.5 and 0.2 percent of the volume south of 60S), so the residence time of surface-formed dense water is 1 to 2 years. Most of the surface densification is eroded by mixing before it reaches the deep ocean. The 11-yr smoothed surface formation and dense cell are correlated at 0.86 to 0.96 along the scenario chains. Under SSP5-8.5 the dense cell and the dense volume vanish in 2071-2100 in both sets.

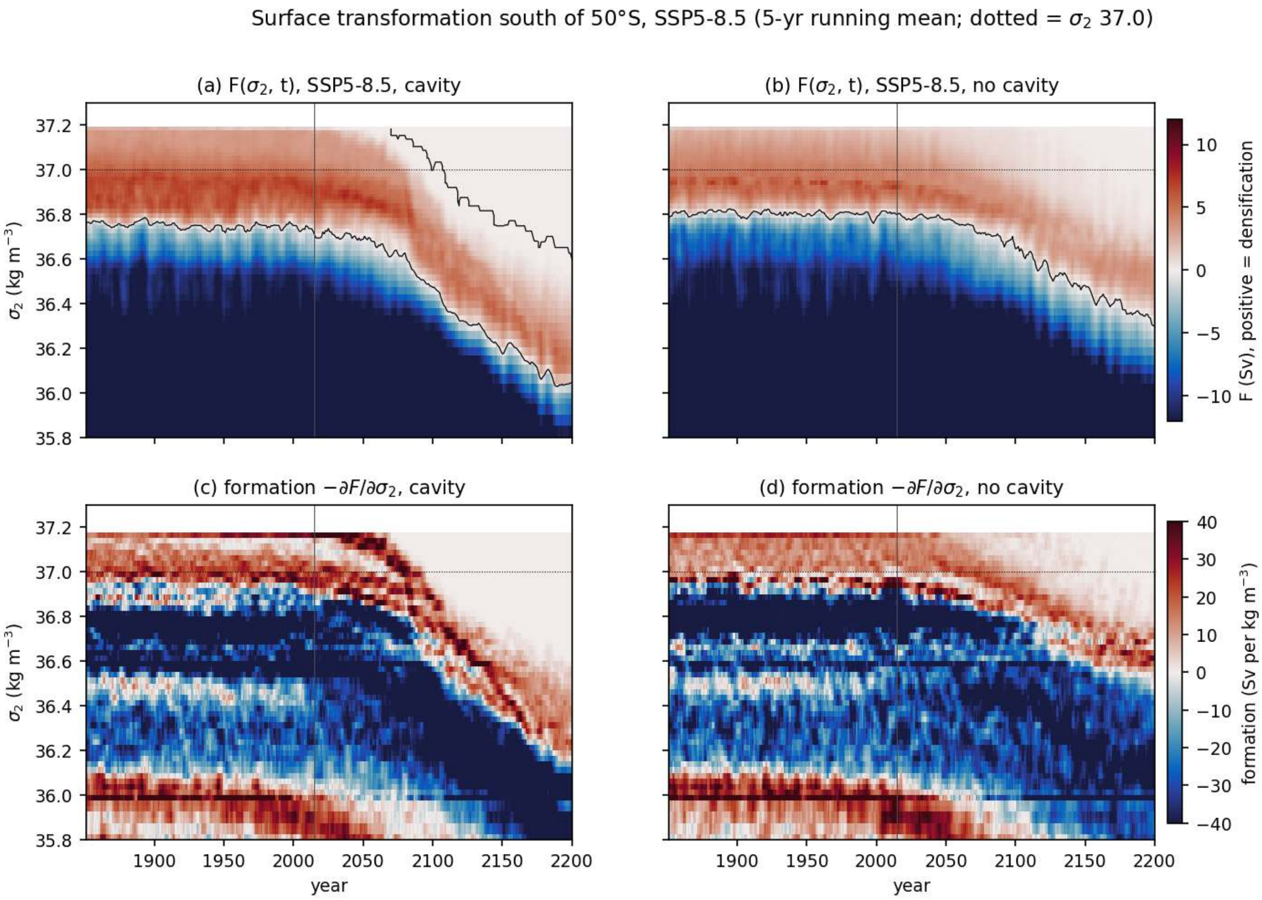

Figure 91: wmt_fig09_hovmoeller_ssp585 (wmt). Density-time Hovmoeller of the total transformation rate F(sigma2, t) south of 50S for the SSP5-8.5 chain (5-yr running mean), cavity run (a) and no-cavity run (b), and the corresponding formation rate minus dF/dsigma2 (c, d). Red is densification, blue is lightening, the black contour marks F = 0, the dotted line sigma2 37.0 and the vertical line the 2014/2015 transition. The zero crossing, which separates the lightening of subpolar water from the densification of coastal water, moves from sigma2 36.7 at present to 36.0 by 2200 in both sets, and the densest transformed class moves from 37.18 to about 36.6. The shift starts around 2040 and is similar in both sets, the cavity run being slightly faster in the densest classes after 2080.

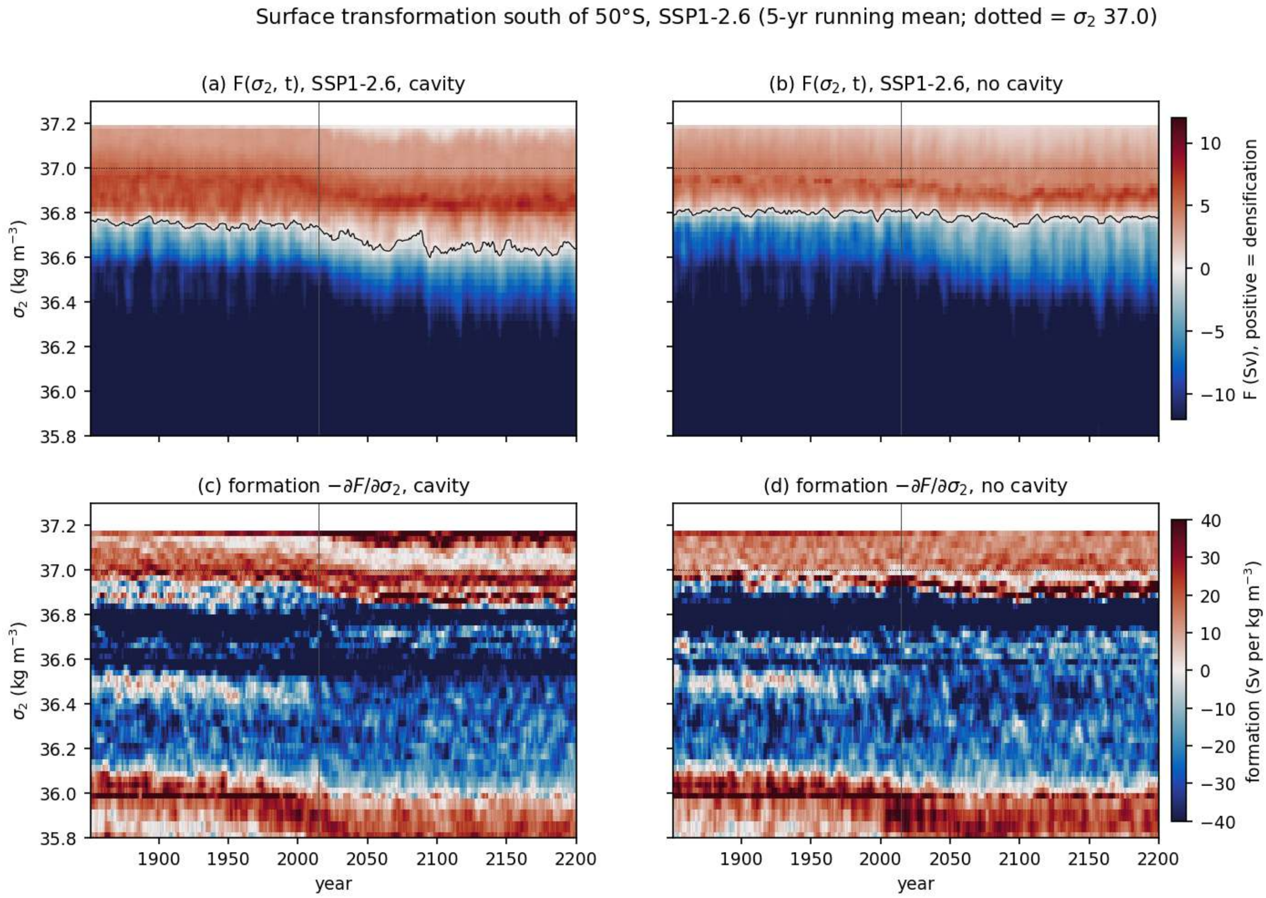

Figure 92: wmt_fig10_hovmoeller_ssp126 (wmt). As fig09 for SSP1-2.6. The transformation pattern shifts by less than 0.1 kg m-3 after 2050 and remains stationary until 2200 in both sets. The densest transformed class stays at 37.18 throughout.

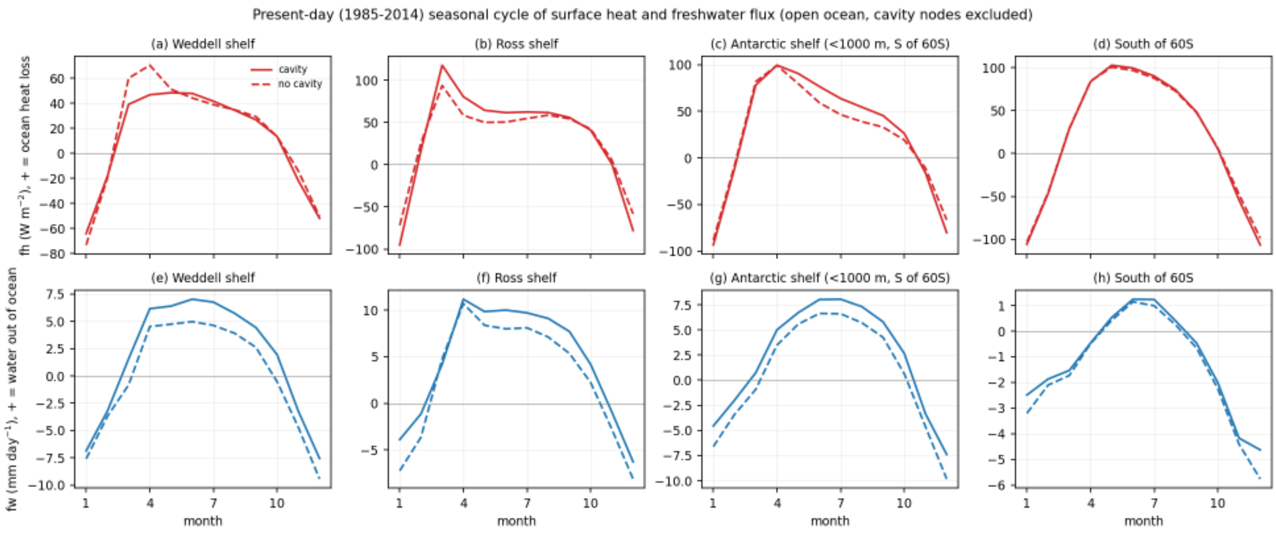

Figure 93: wmt_fig11_flux_seasonal_cycle (wmt). Present-day (1985-2014) seasonal cycles of the region-mean surface heat flux fh (top, W m-2, positive is ocean heat loss) and freshwater flux fw (bottom, mm/day, positive is water leaving the ocean, that is the sum of evaporation minus precipitation, sea-ice freezing minus melting and minus runoff) for the Weddell shelf, Ross shelf, the whole Antarctic shelf (depth below 1000 m, south of 60S) and the region south of 60S, both sets, cavity nodes excluded. Winter (JJA) heat loss is 41 and 39 W m-2 on the Weddell shelf, 62 and 55 on the Ross shelf, 65 and 48 on the whole shelf and 88 and 86 south of 60S (cavity, no cavity). Winter fw is 6.5 and 4.5 mm/day on the Weddell shelf, 9.6 and 7.7 on the Ross shelf, meaning that net brine release in the cavity run is larger. A large part of this difference is not sea ice: in the no-cavity run 1.1 to 1.8 mm/day of runoff enters at the coast (ice-sheet discharge routed by the HD model) and is included in fw, in the cavity run this water enters as melt under the ice shelves (2360 Gt/yr) at the excluded cavity nodes and runoff south of 60S is zero. Sea-ice brine and precipitation minus evaporation cannot be separated in the fw output.

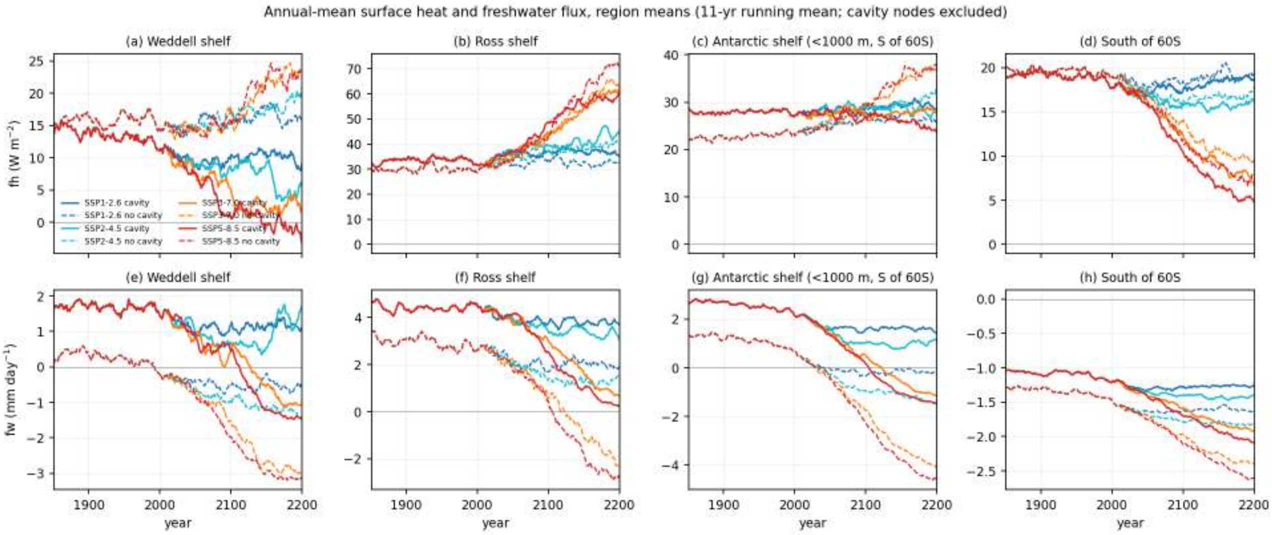

Figure 94: wmt_fig12_flux_timeseries (wmt). Annual-mean fh (top) and fw (bottom) region means 1851-2200 (11-yr running mean) for the same four regions, all scenarios, both sets, cavity nodes excluded. Under SSP5-8.5 the heat loss south of 60S falls from 18 to 5 (cavity) and from 19 to 7 W m-2 (no cavity) by 2171-2200, while on the Ross and Antarctic shelves the heat loss increases (Ross shelf 32 to 58 and 30 to 70 W m-2) as the winter ice cover retreats and exposes warmer water. fw becomes more negative everywhere (shelf mean 2.3 to -1.4 and 0.6 to -4.5 mm/day) because of more precipitation and less sea-ice formation, and it is this freshening, not the heat flux change, that removes the dense-water formation. Under SSP1-2.6 both fluxes are stable after 2050.

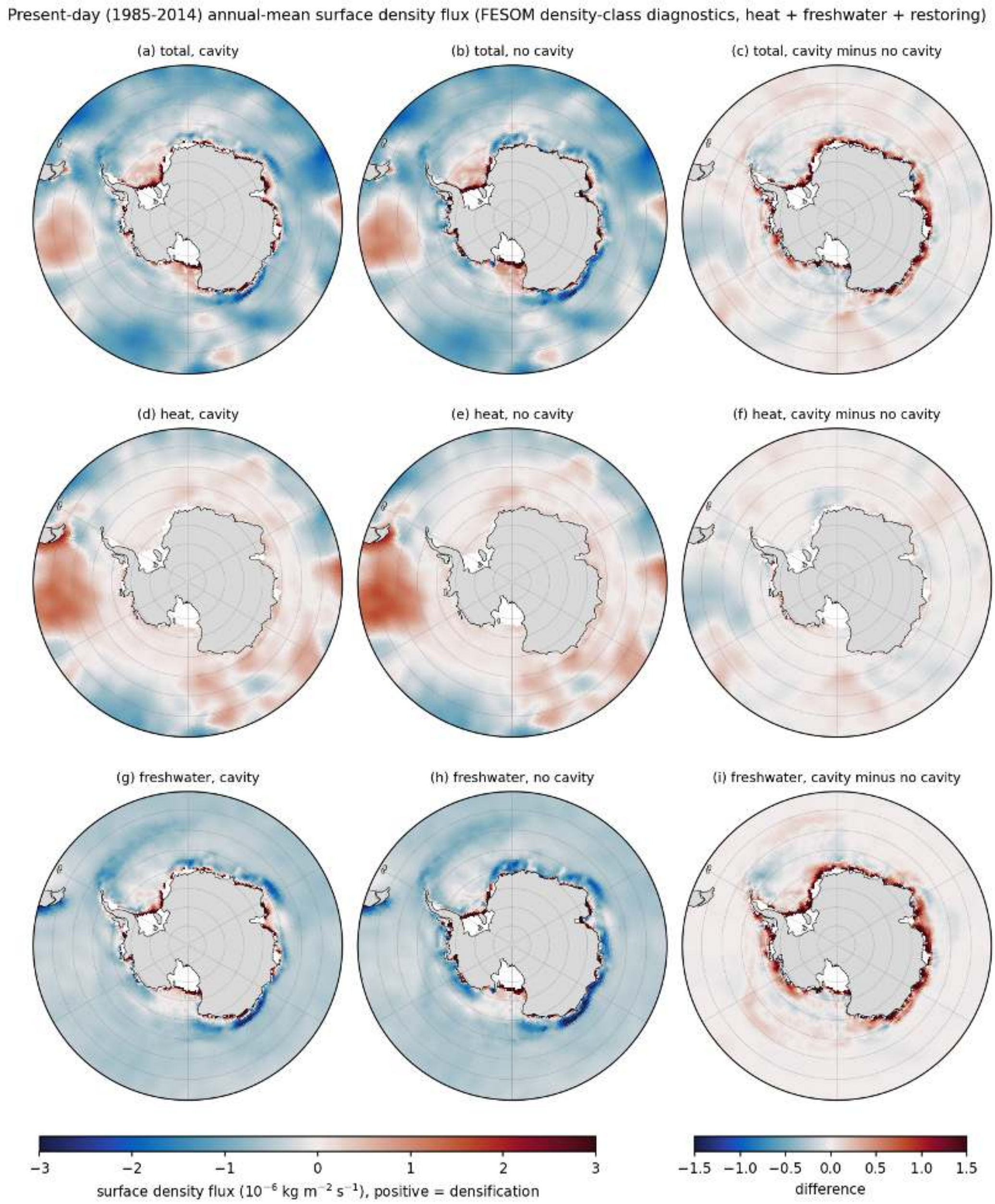

Figure 95: wmt_fig13_map_density_flux_pd (wmt). Present-day annual-mean surface density flux (1e-6 kg m-2 s-1, positive is densification) south of 50S from the FESOM class diagnostics summed over classes (heat plus freshwater, restoring not included in the maps), for the cavity run (left), the no-cavity run (middle) and the difference (right) for the total (a-c), the heat part (d-f) and the freshwater part (g-i). Densification is confined to a narrow coastal band (coastal polynyas and the shelves, shelf mean +0.82 and +0.46 units) and to the Drake Passage/Agulhas sector of the ACC (heat loss), while the gyres and the open Southern Ocean are lightened (south of 50S mean -0.48 and -0.49, freshwater part -0.45 and -0.49). The cavity run densifies the coastal band more strongly (difference up to +1.5 at the coast) because it has no coastal runoff; away from the coast the two sets differ by less than 0.2 units.

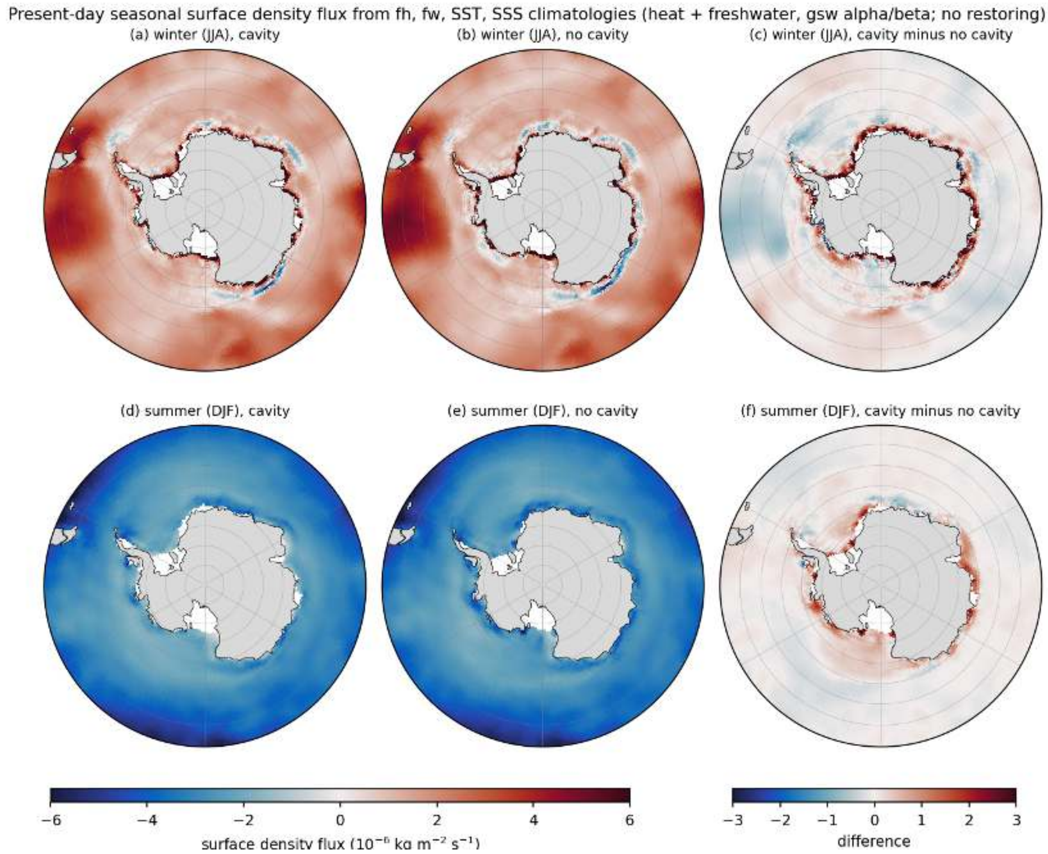

Figure 96: wmt_fig14_map_density_flux_season (wmt). Present-day winter (JJA, top) and summer (DJF, bottom) surface density flux from the monthly fh, fw, SST and SSS climatologies with gsw alpha and beta (heat plus freshwater, no restoring), cavity run (left), no-cavity run (middle), difference (right). In winter the mean density flux south of 50S is +1.7 units (heat part 1.8, freshwater -0.03), on the Antarctic shelf +3.0 (cavity) and +2.4 (no cavity) of which 2.5 and 2.0 come from brine release. In summer the whole region gains buoyancy (-3.1 south of 50S, -2.2 and -2.7 on the shelf). The annual gsw-based means (-0.42 and -0.45 south of 50S) agree with the class-diagnostic maps of fig13 to within 0.06 units.

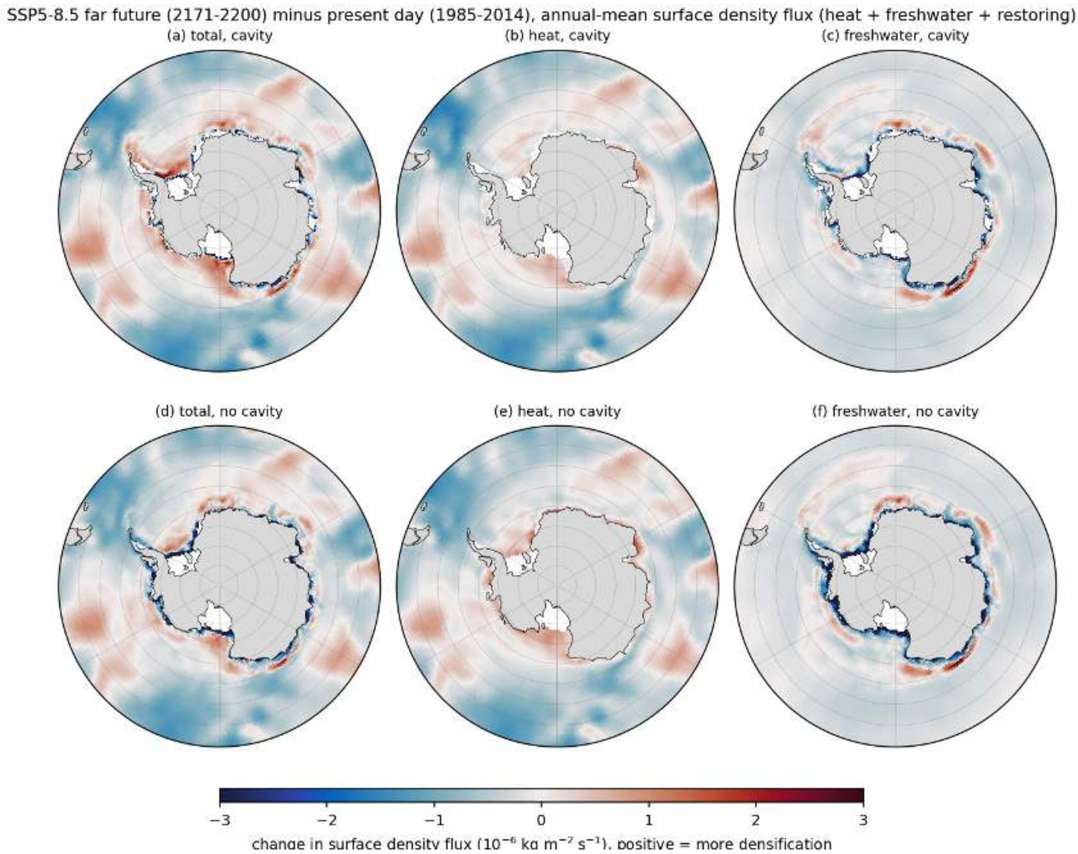

Figure 97: wmt__fig15__map__density__flux__far585__minus__pd (wmt). Change of the annual-mean surface density flux between 2171-2200 under SSP5-8.5 and 1985-2014 (heat plus freshwater plus restoring), for the cavity run (top) and the no-cavity run (bottom): total (a, d), heat part (b, e), freshwater part (c, f). The coastal densification band weakens (shelf-mean freshwater change -1.05 and -1.48 units), the heat part increases the densification on the shelves (where the ice cover retreats) and decreases it north of 60S. The mean change south of 50S is -0.20 (cavity) and -0.28 units (no cavity), about half of the present-day mean. Both sets show the same pattern, the no-cavity run reacts more strongly on the shelf because its coastal runoff increases with the warmer climate.

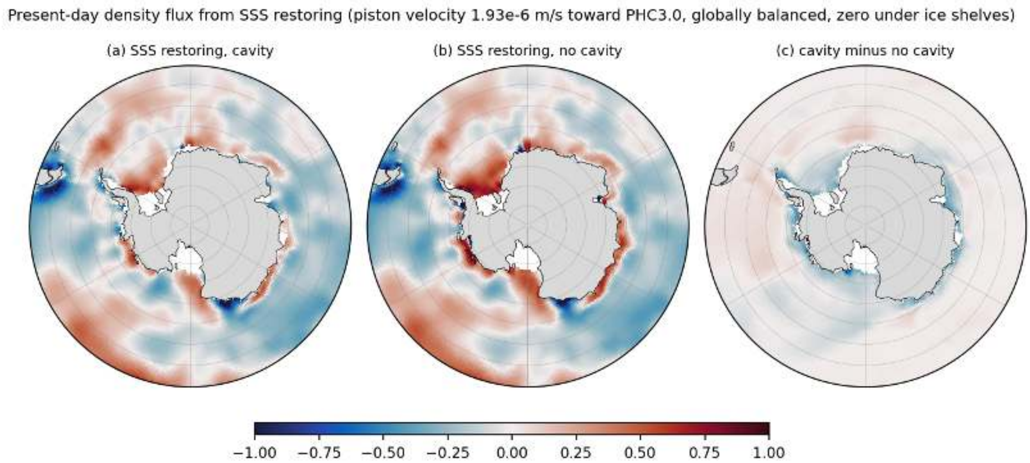

Figure 98: wmt__fig16__map__restoring__pd (wmt). Present-day density flux due to the sea-surface salinity restoring towards PHC3.0 (piston velocity 1.929e-6 m/s, global mean removed, zero under ice shelves) for the cavity run (a), the no-cavity run (b) and the difference (c), 1e-6 kg m-2 s-1, positive where salt is added. The restoring adds salt (densifies) in the Weddell Sea (box mean +0.12 and +0.14), on the Antarctic shelf (+0.08 and +0.22) and in the Ross Sea of the no-cavity run (+0.10), that is where the model is too fresh, and removes salt over the ACC. The magnitude is comparable to the annual-mean total flux of fig13, so the restoring is a first-order term in the surface buoyancy budget of the shelves and reduces the freshwater anomalies that distinguish the two sets.

9 Circulation, stratification and hydrographic sections

Findings

9.1 circ theme findings: circulation, stratification, sections (AWIESM-2.1 cavity vs no cavity)

Conventions: cavity = PI_ICEv2 set, no cavity = PI_CORE set; pd = 1985-2014, far = 2171-2200. Gyre strength = minimum barotropic streamfunction inside the Weddell (60W-30E, 75S-55S) and Ross (160E-130W, 75S-60S) boxes with psi = 0 at the Antarctic coast (it therefore includes the Antarctic Slope Current). Numbers are in cache/circ/{gyre,psi,section,strat,cdw}_stats.json.

1. ACC offset between the sets. Drake Passage transport is 93 Sv in the cavity run and 79 Sv in the no-cavity run for the whole historical period (1851-2014 means 93.1 and 78.1 Sv), and the zonal-mean psi at 30S is 113 versus 99 Sv. The 14 Sv offset exists from 1851, so it is a property of the two spin-ups (different mesh and bathymetry under the ice shelves, different deep density structure) rather than a transient response. The psi difference map (circ_psi_maps c) shows the extra transport spread over the whole ACC belt with maxima (+20-25 Sv) in the Indian and Pacific sectors.

2. Gyres. Weddell gyre 108 Sv (cavity) and 114 Sv (no cavity) at pd, Ross gyre 21 and 27 Sv; the no-cavity run has slightly stronger subpolar gyres and a clearly stronger Antarctic Slope Current (bottom speed over the 1000-3000 m slope band 0.032 versus 0.024 m/s, westward component -0.019 versus -0.012 m/s). The Weddell gyre values are large compared with observations (about 50-60 Sv) because the index is referenced to the coast and includes the slope current and the model's broad cyclonic circulation around the whole continent (psi reaches -40 Sv almost everywhere along the coast).

3. Scenario response of the circulation. Under SSP5-8.5 (2171-2200 minus pd) Drake transport rises by +9 Sv (cavity) and +19 Sv (no cavity), the ACC core between 45S and 55S intensifies by 20-40 Sv in psi while the net transport at 30S rises by only 6 and 15 Sv, i.e. the ACC sharpens and shifts. The Ross gyre strengthens from 22 to 37 Sv (cavity) and 28 to 33 Sv (no cavity) with a monotonic scenario ordering (SSP1-2.6 26/25, SSP2-4.5 32/32, SSP3-7.0 36/34 Sv); the Weddell gyre from 108 to 127 (cavity) and 114 to 119 Sv (no cavity). Under SSP1-2.6 Drake transport falls by 4 Sv in the cavity run and rises by 5 Sv in the no-cavity run; the two sets converge (89.5 versus 84 Sv) because the no-cavity run loses its Maud Rise convection. Gyre trends over 2015-2200 are about -10 Sv per century for both gyres in the cavity run under SSP5-8.5.

4. Bottom circulation and dense-water pathways. The bottom velocity maps show the slope current and gyre rims as the only fast bottom flows (0.05-0.3 m/s); there is no coherent down-slope plume at the Filchner sill, the western Ross Sea or the Adelie sill in either set. This matches the sections: the densest shelf water (sigma2 37.0 at 30W, 37.03 at 180E) is at best as dense as the 36.9-36.95 water filling the basins below 1000 m, so shelf water cannot sink to the abyss and the deep Weddell/Ross bottom stays at 1.75-2.0 degC. A paper figure that illustrates the missing AABW plume could combine circ_sec_weddell30W rows 1-2 (no cold bottom layer, sigma2 37.0 only on the shelf) with circ_bottom_velocity a-b (no slope plume).

5. Cavity effects on hydrography at pd. Cavity minus no cavity: deep Weddell at 0E and 30W 0.2-0.25 degC cooler below 1500 m and 0.02-0.03 psu fresher; zonal-mean Southern Ocean below 2000 m 0.1-0.3 degC cooler; upper 500 m south of 65S up to 0.5 degC cooler and 0.1 psu fresher; Ross shelf bottom 0.13 degC cooler and 0.03 psu fresher; Amundsen slope Tmax 0.5 degC warmer (2.43 versus 1.91 degC). T at 400 m on the shelves is 0.0 degC (cavity) versus 0.26 degC (no cavity) circumpolar, the cavity run being cooler in every sector except the Weddell interior (+0.2-0.5 degC at the CDW core in the eastern Weddell). Caveat: the regional shelf boxes in DML, Amery, Adelie compare different domains (327e3 km2 of cavity area is open ocean in PI_CTRL), which explains part of the coastal cooling/freshening ring in circ_cdw_tmax c and circ_N2_maps c.

6. Stratification and convection control. The stratification index sigma2(200-1000) minus sigma2(0-200) is 0.119 kg/m3 (cavity) versus 0.107 (no cavity) at Maud Rise and N2(100-300 m) 5.3e-6 versus 4.7e-6 s-2, i.e. the no-cavity run is 10 percent less stratified exactly where it convects every winter; on the Ross shelf the cavity run is the less stratified one (0.10 versus 0.15 kg/m3; N2 5.1e-6 versus 8.1e-6 s-2), consistent with its warmer Ross winter SST and larger heat loss reported by the atmos theme. Under SSP5-8.5 the index south of 60S rises fourfold (0.23 to 0.55 kg/m3 cavity, 0.21 to 0.51 no cavity) and N2(100-300 m) by 60-70 percent, which suppresses open-ocean convection in both sets; under SSP1-2.6 the increase is only 0.03-0.04 kg/m3 (Maud Rise 0.12 to 0.16 cavity, 0.11 to 0.13 no cavity), so intermittent convection can persist.

7. Shelf warming by CDW intrusion (SSP5-8.5, 2171-2200 minus pd, shelves 400-1000 m bottom, S of 60S, cavity nodes excluded). T at 400 m: Weddell +1.5/+1.9 degC (cavity/no cavity), DML +1.7/+2.4, Prydz +1.9/+2.1, Wilkes-Adelie +0.9/+1.0, Ross +2.0/+1.9, Amundsen +1.4/+1.4, Bellingshausen +1.3/+1.3; circumpolar +1.5/+1.7 to absolute values of 1.49/1.93 degC. Section shelf-bottom warming: 30W +2.4/+2.7, 180E +3.3/+2.9, 70E +1.6/+2.1, 110W +1.2/+1.4 degC, and shelf-bottom freshening of 0.3-0.5 psu (cavity) versus 0.1-0.2 psu (no cavity). By 2071-2100 (end) shelf T400 has risen by only 0.2-0.3 degC circumpolar (Amundsen +0.6/+0.3), so most of the shelf warming happens after 2100 when the winter halocline erodes. Under SSP1-2.6 shelf T400 changes stay within +-0.3 degC.

8. Zonal-mean change. SSP5-8.5 far minus pd: upper 1000 m south of 50S +2 to +4 degC and -0.2 to -0.4 psu, deep ocean +0.3 to +0.6 degC; sigma2 -0.4 kg/m3 above 500 m. The cavity run freshens more over the shelf/slope south of 70S (about -0.4 psu below 500 m versus -0.1 to -0.2) because ice-shelf melt increases. SSP1-2.6: upper 1000 m +0.2 to +0.5 degC, deep +0.1 to +0.2 degC.

Caveats and data issues. a) cache/ts wg_psi/rg_psi/acc_psi are not gyre strengths: diag.barotropic_streamfunction returns psi with the ACC negative, so acc_psi (nanmax) equals minus the Weddell gyre minimum and wg_psi/rg_psi equal minus the ACC-side psi maximum in each box. Use cache/circ/_gyre.nc instead (analyses/circ/gyre_ts.py). drake in cache/ts is fine. b) Nearest-neighbour regridding of depth-integrated transport produces strong meridional striping in psi; the maps and indices here use node-averaged transports with linear triangle interpolation plus a 3-point zonal smoothing (circ_lib.psi_from_geo). Gyre minima are still sensitive to small-scale noise at the +-5 Sv level. c) The gyre boxes and the "relative to coast" definition include the slope current; a closed-contour definition would give smaller numbers (roughly 60-70 Sv for the Weddell gyre interior judging from the maps). d) The two sets are single members; the historical difference maps mix the cavity effect with internal variability (30-year means), especially for psi (+-5 Sv) and upper-ocean T/S. e) Sections use nearest-node sampling (max 80 km), so the shelf part of each section represents a handful of nodes; the vertical axis is square-root stretched. f) N2, Kv were stored on interfaces (48 levels); maps use the 100-300 m interface mean of annual means, so winter mixed-layer N2 minima are not resolved.

Suggested story lines. (i) Same atmosphere, different ocean: the cavity mesh gives a 14 Sv stronger ACC and a weaker slope current from the spin-up onward, and both sets lack a dense shelf-to-abyss pathway, which is the primary reason for the CDW-like abyss. (ii) Where each set convects is set by a 10 percent stratification difference (Maud Rise for no cavity, Ross shelf for cavity), and a fourfold stratification increase under SSP5-8.5 shuts both off while the Ross gyre spins up by 50 percent. (iii) CDW reaches every shelf after 2100 under SSP5-8.5 (+1.5 to +2.5 degC at 400 m) and the cavity run stays cooler and fresher on the shelves because melt water is explicitly released at depth.

Figures

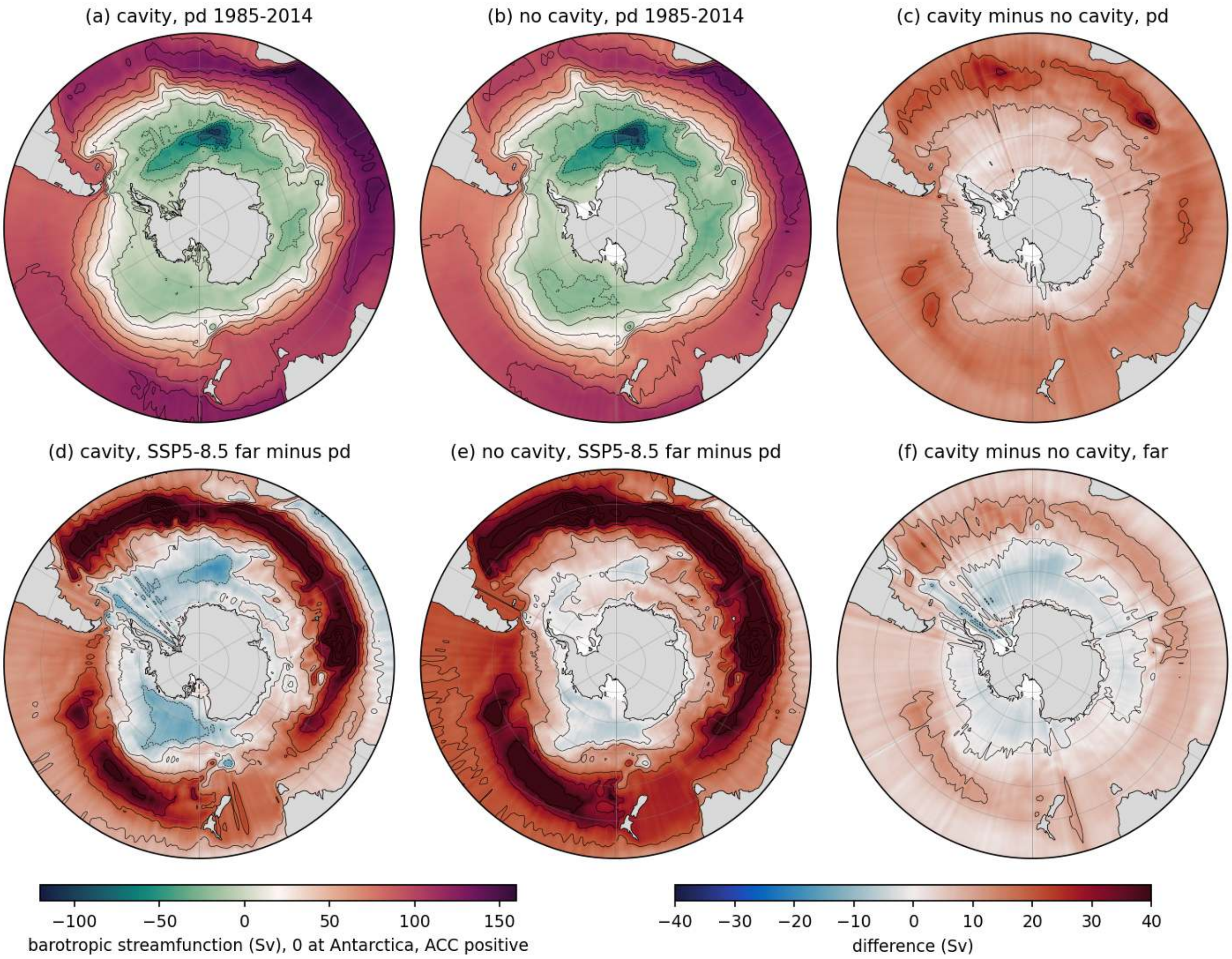

Figure 99: circ_psi_maps (circ). Barotropic streamfunction (Sv) south of 30S. (a) cavity pd, (b) no cavity pd, (c) cavity minus no cavity at pd, (d) cavity SSP5-8.5 far minus pd, (e) no cavity SSP5-8.5 far minus pd, (f) cavity minus no cavity in the far SSP5-8.5 period. Contours every 20 Sv (a, b) and 10 Sv (c-f). Depth-integrated transports from the rotated u_geo, v_geo were averaged to nodes, linearly interpolated to a 0.5 degree grid and integrated northward; a 3-point zonal running mean is applied. Take-away: the Weddell gyre minimum relative to the coast is 108 Sv (cavity) and 114 Sv (no cavity), the Ross gyre 21 and 27 Sv, and the total eastward transport at 30S is 113 versus 99 Sv, so the cavity run has a 13-14 Sv stronger ACC but slightly weaker gyres. Under SSP5-8.5 both sets intensify the ACC core by 20-40 Sv between 45S and 55S while the net transport at 30S only rises by 6 Sv (cavity) and 15 Sv (no cavity), i.e. the ACC shifts and sharpens; the Weddell and Ross gyres deepen by 19 and 15 Sv (cavity) and 6 and 5 Sv (no cavity).

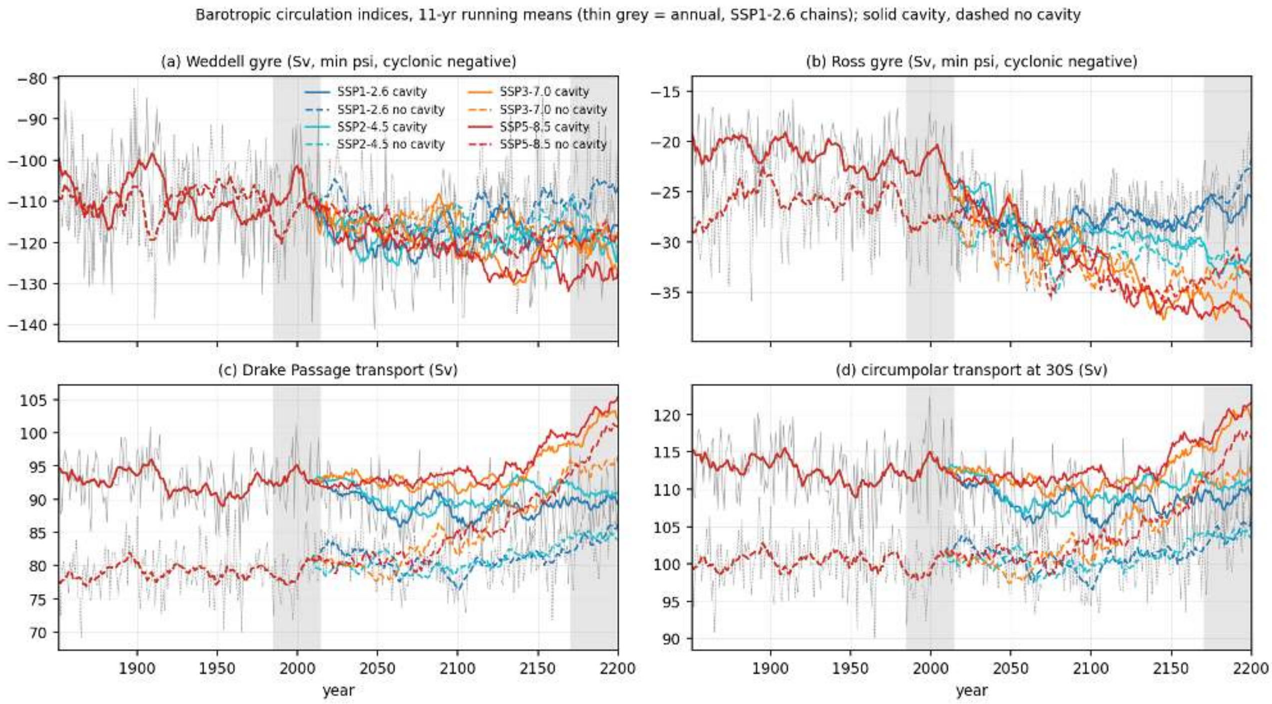

Figure 100: circ_gyre_ts (circ). Yearly indices 1851-2200 from raw u, v ('analyses/circ/gyre_ts.py'), 11-year running means, solid = cavity, dashed = no cavity, scenario colours; thin grey = annual values of the SSP1-2.6 chains; shaded = pd and far windows. (a) Weddell gyre strength (minimum psi in 60W-30E, 75S-55S), (b) Ross gyre (160E-130W, 75S-60S), (c) Drake Passage transport at 65W, (d) zonal-mean psi at 30S. Take-away: Drake transport is 93 Sv (cavity) versus 79 Sv (no cavity) throughout the historical period, a constant 14 Sv offset inherited from the spin-ups. Under SSP5-8.5 it rises to 103 and 98 Sv by 2171-2200 (+9 and +19 Sv), under SSP1-2.6 it falls by 4 Sv (cavity) and rises by 5 Sv (no cavity). The Ross gyre strengthens monotonically from 22 to 37 Sv (cavity) and 28 to 33 Sv (no cavity) under SSP5-8.5 with a clear scenario ordering; the Weddell gyre strengthens from 108 to 127 Sv (cavity) and 114 to 119 Sv (no cavity) with large interannual variability (+-15 Sv).

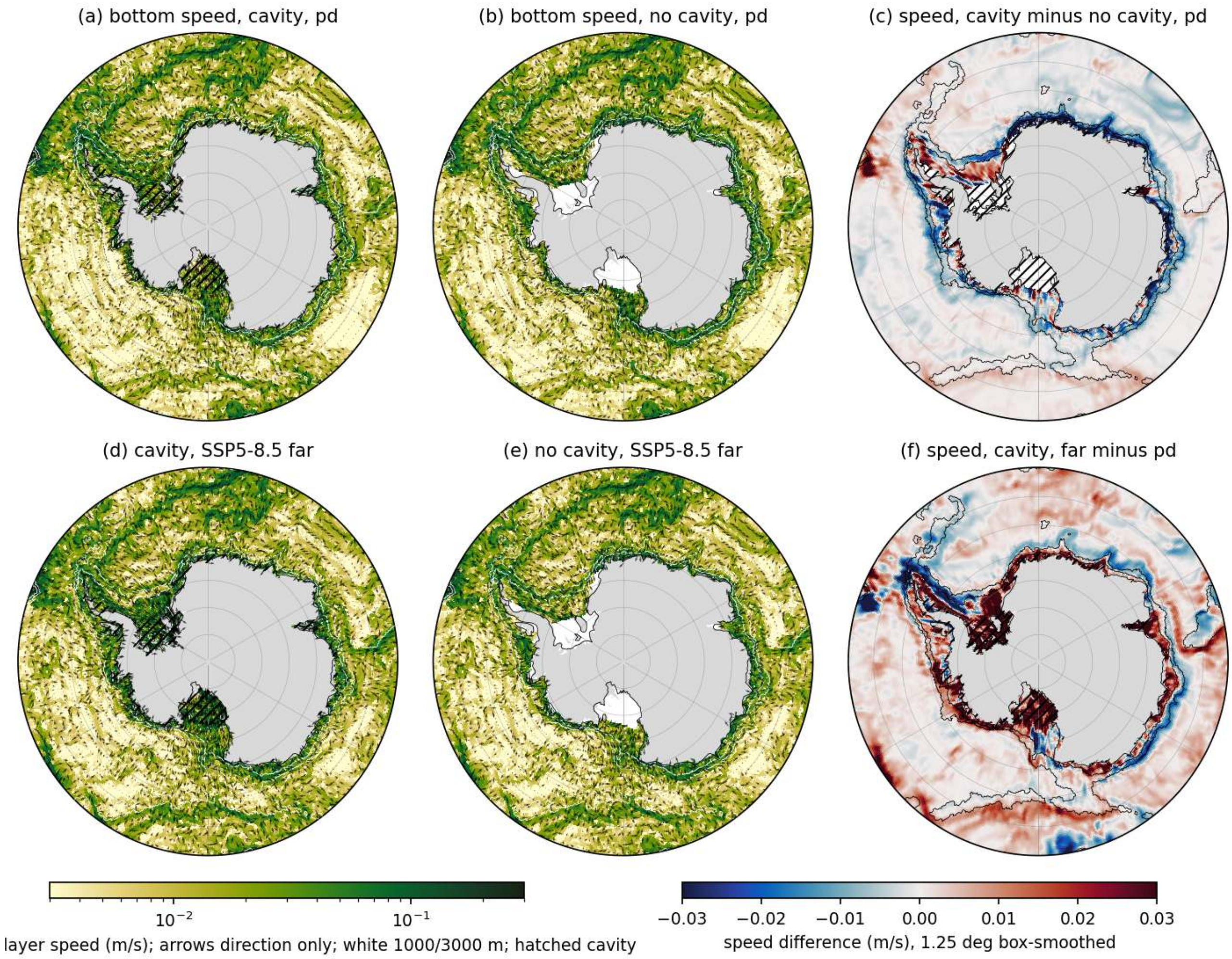

Figure 101: circ__bottom__velocity (circ). Bottom-layer (deepest wet layer of each element) speed (log colour) with direction arrows south of 55S; white contours 1000 and 3000 m isobaths, hatched = ice-shelf cavities. (a) cavity pd, (b) no cavity pd, (c) speed difference cavity minus no cavity (1.25 degree box smoothed), (d) cavity SSP5-8.5 far, (e) no cavity SSP5-8.5 far, (f) cavity far minus pd. Take-away: the strongest bottom flow (0.05-0.3 m/s) is the westward Antarctic Slope Current over the 1000-3000 m band and the cyclonic rims of the Weddell and Ross gyres; there is no distinct down-slope dense plume signature at the Filchner, Ross or Adelie sills. Mean bottom speed over the circumpolar slope band (1000-3000 m, S of 60S) is 0.024 m/s (cavity) versus 0.032 m/s (no cavity), the westward component -0.012 versus -0.019 m/s, so the slope current is about 35 percent weaker in the cavity run (the shelf geometry differs, see caveats). Under SSP5-8.5 the slope current weakens further in the no-cavity run (-0.015 m/s) and is unchanged in the cavity run, while under-ice-shelf currents in the FRIS and Ross cavities accelerate by 0.01-0.03 m/s.

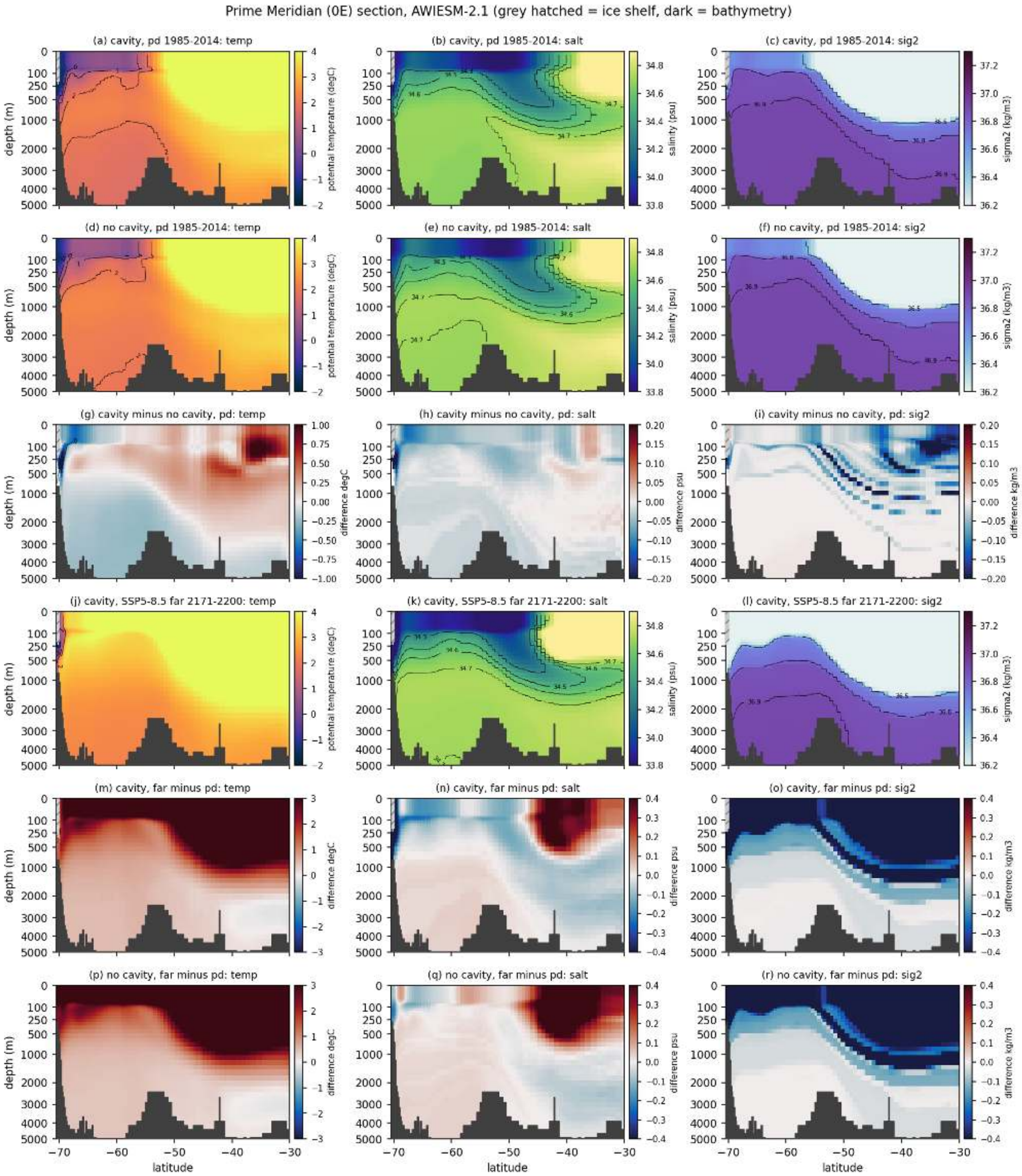

Figure 102: circ_sec_greenwich (circ). Prime Meridian section 70S-30S of potential temperature, salinity and sigma2. Rows: cavity pd, no cavity pd, cavity minus no cavity at pd, cavity SSP5-8.5 far, cavity far minus pd, no cavity far minus pd. Depth axis is square-root stretched; dark = bathymetry, grey hatched = ice shelf (Fimbul, cavity set only). Contours: T -1, 0, 1, 2 degC; S 34.3 to 34.7; sigma2 36.5 to 37.1. Take-away: neither set has a cold dense bottom layer; the deepest Weddell water at 0E is 2.2-2.4 degC and sigma2 36.9, the 37.0 surface exists only at the shelf (<300 m) in the cavity run. The cavity run is 0.2-0.5 degC cooler and 0.03 psu fresher in the deep Weddell (below 1500 m) and has a thicker cold (<0 degC) surface layer near the coast. By 2171-2200 under SSP5-8.5 the whole column south of 60S warms by 0.5-1.0 degC below 1000 m and by more than 3 degC in the upper 500 m; the shelf bottom freshens by 0.5 (cavity) and 0.1 psu (no cavity) and the upper ocean loses more than 0.4 kg/m3 of density, i.e. stratification strengthens strongly.

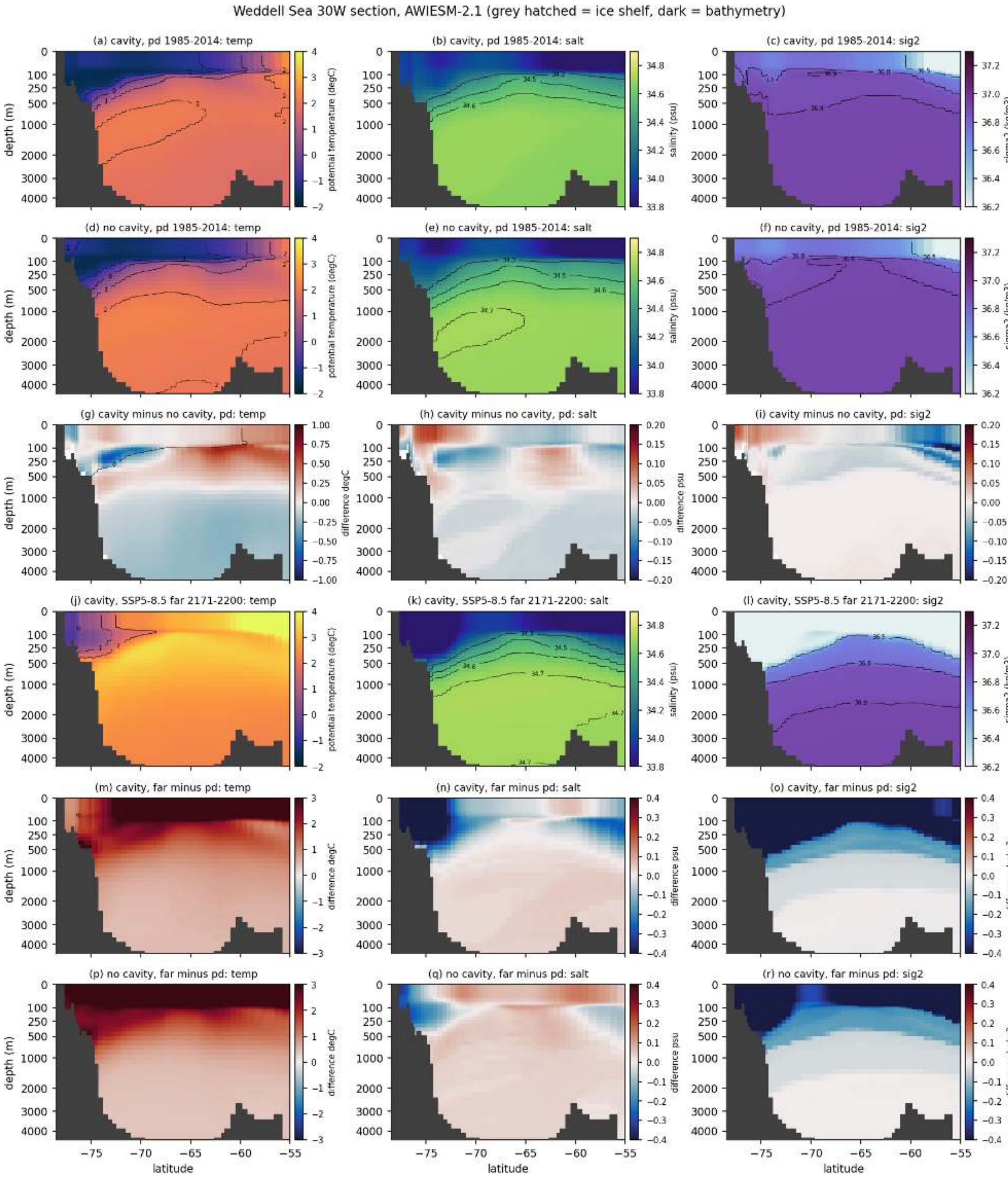

Figure 103: circ_sec_weddell30W (circ). As above for 30W, 78.5S-55S across the southern Weddell Sea (Filchner/Ronne shelf to the gyre centre). Take-away: shelf bottom water at pd is -1.1 degC, 34.16 psu (cavity) and -1.0 degC, 34.18 psu (no cavity), about 0.3 psu too fresh for HSSW, and the deep basin bottom is 1.75 and 1.98 degC. The densest shelf water (sigma2 36.9) is lighter than the 36.9 water filling the basin below 1000 m, so shelf water cannot sink to the bottom. In 2171-2200 the shelf bottom warms to 1.3 degC (cavity) and 1.7 degC (no cavity) and the shelf freshens by 0.45 and 0.16 psu.

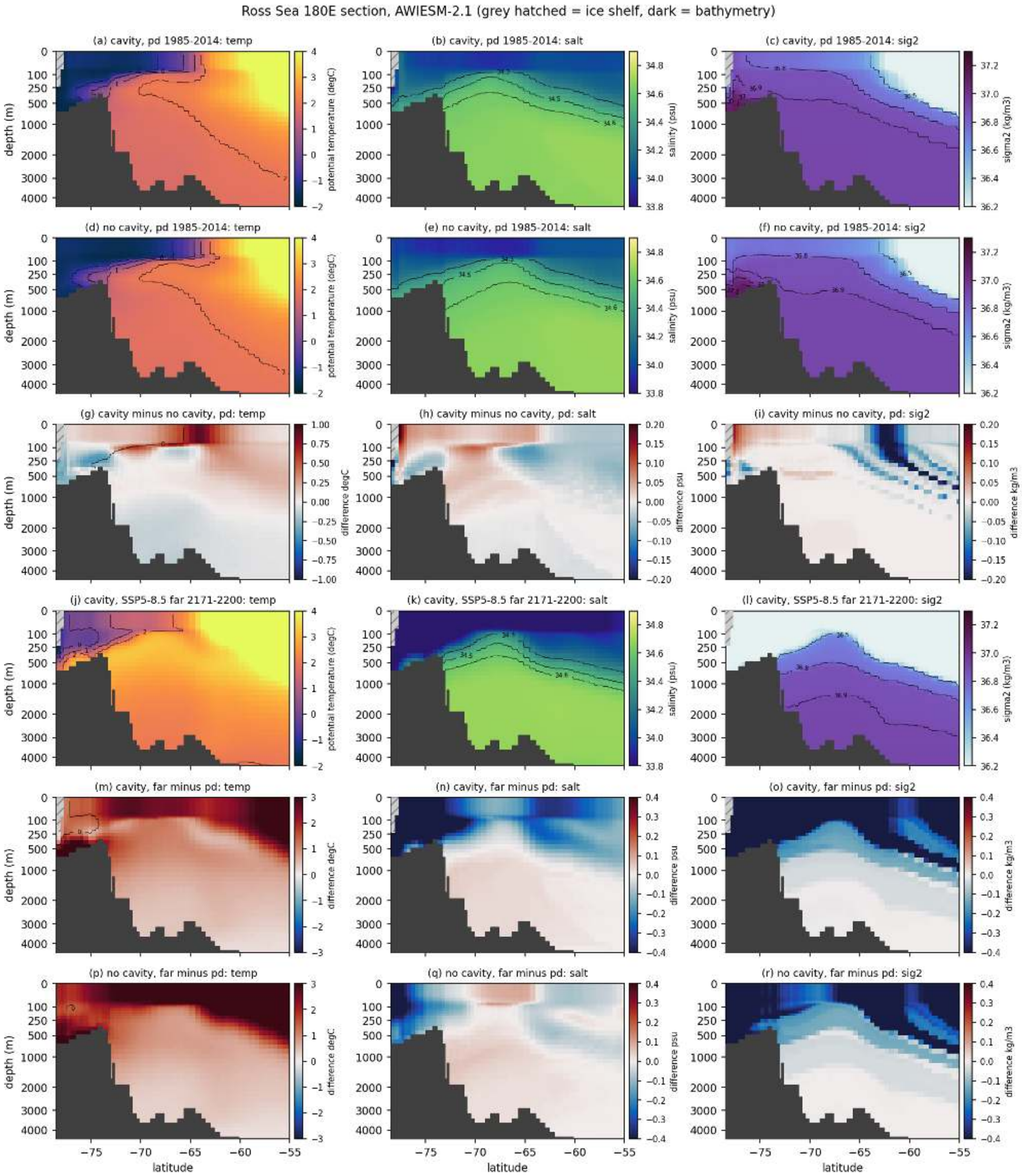

Figure 104: circ_sec_ross180 (circ). As above for 180E, 78.5S-55S across the Ross Sea shelf, slope and gyre. Take-away: the Ross shelf holds the densest model shelf water (bottom sigma2 37.03, S 34.41-34.44, T -0.8 to -0.6 degC) but it remains confined to the shelf. Under SSP5-8.5 the Ross shelf bottom warms to 2.5 (cavity) and 2.2 degC (no cavity) and freshens by 0.3 and 0.14 psu, the largest shelf change of all sections, and the shelf bottom density drops to 36.2 (cavity) and 36.7 (no cavity).

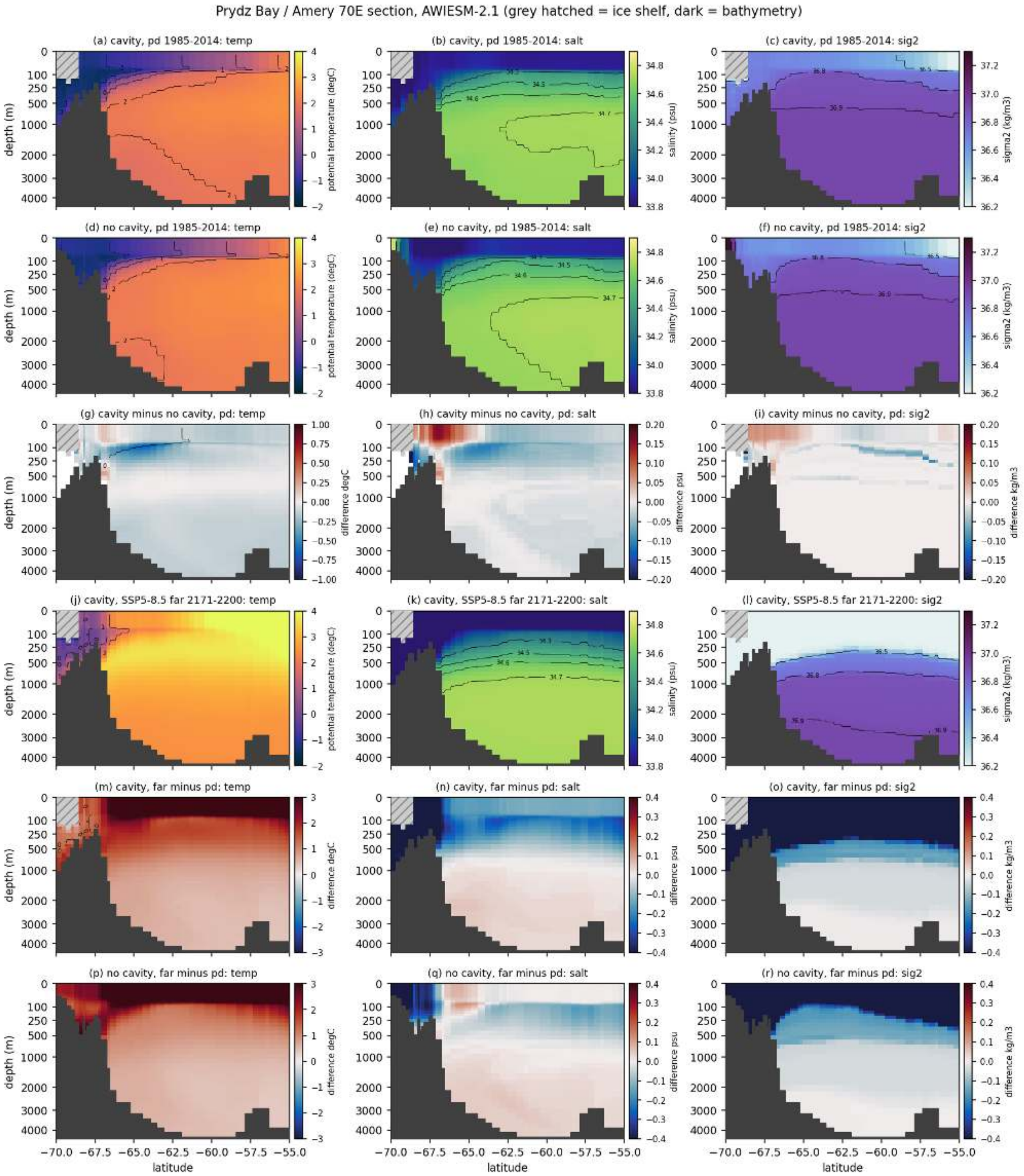

Figure 105: circ_sec_prydz70E (circ). As above for 70E, 70S-55S (Prydz Bay/Amery shelf; note that the Amery cavity and parts of the shelf are open ocean in the no-cavity mesh). Take-away: shelf bottom T -0.6/-0.8 degC and S 34.03/34.21 at pd (cavity/no cavity); CDW (T>1.5) sits on the slope at 300-1000 m in both sets. In 2171-2200 the shelf bottom warms to 1.0 and 1.3 degC and freshens by 0.5 and 0.4 psu.

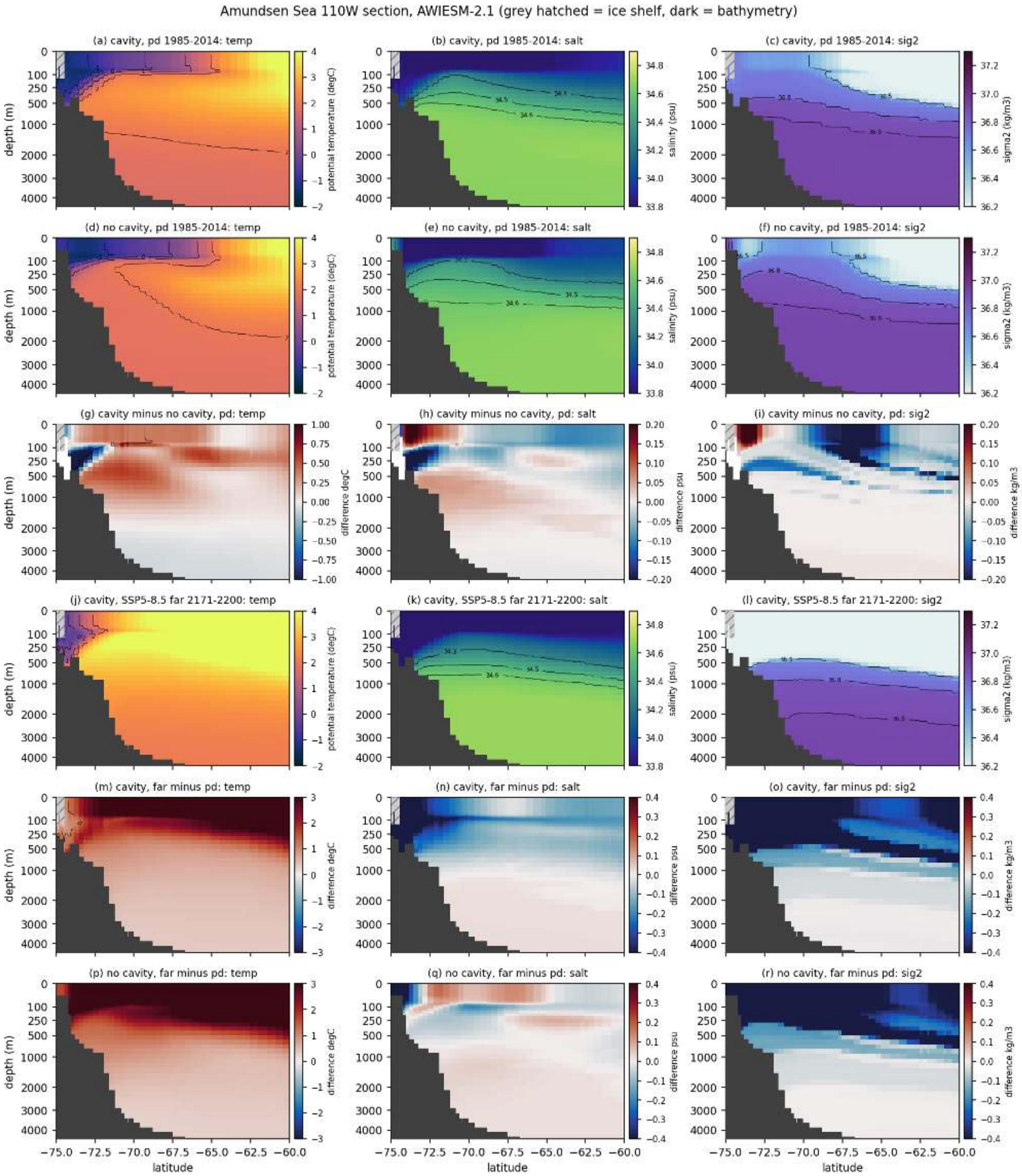

Figure 106: circ_sec_amundsen110W (circ). As above for 110W, 75S-60S (Amundsen Sea shelf and slope). Take-away: this is the warmest shelf at pd (bottom 0.9-1.0 degC, CDW present at 400 m), consistent with observations; the cavity run has a 0.5 degC warmer slope Tmax (2.43 versus 1.91 degC below 200 m). By 2171-2200 the shelf bottom reaches 2.2 (cavity) and 2.4 degC (no cavity) and the sub-200 m Tmax reaches 3.9 degC.

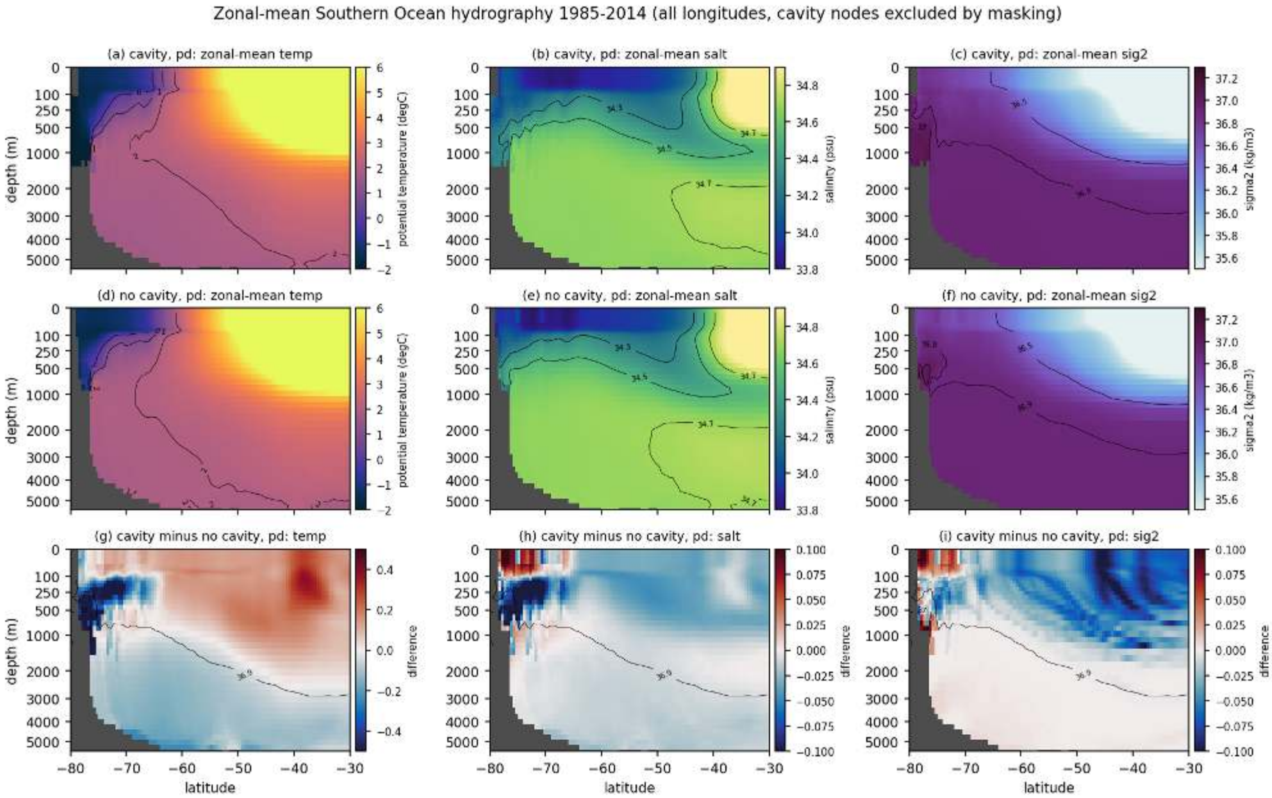

Figure 107: circ_zonal_pd (circ). Zonal-mean (all longitudes, 0.5 degree grid, cavity nodes excluded) potential temperature, salinity and sigma2 for 80S-30S at pd: (a-c) cavity, (d-f) no cavity, (g-i) cavity minus no cavity. Depth axis square-root stretched. Take-away: the zonal-mean Southern Ocean below 2000 m is 0.1-0.3 degC cooler and about 0.02 psu fresher in the cavity run; south of 65S the upper 500 m is up to 0.5 degC cooler and 0.1 psu fresher, with the largest differences on the shelf and slope (ice-shelf melt water). The sigma2 = 37.0 surface outcrops only at the shelf south of 75S in both sets and no water denser than 37.1 exists in the zonal mean.

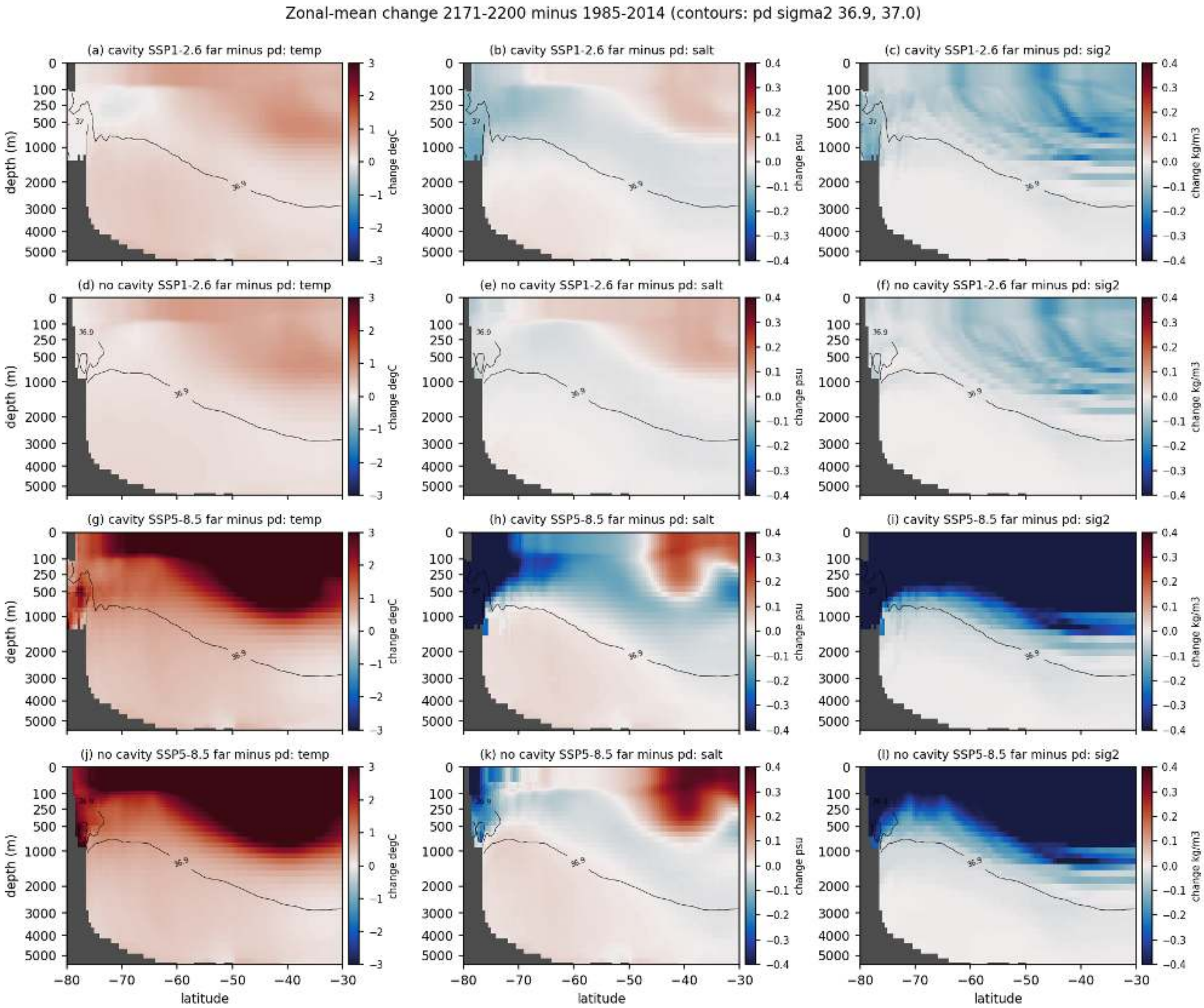

Figure 108: circ__zonal__change (circ). Zonal-mean change 2171-2200 minus 1985-2014 for SSP1-2.6 (a-f) and SSP5-8.5 (g-l), cavity (rows 1, 3) and no cavity (rows 2, 4); contours = pd sigma2 36.9 and 37.0. Take-away: under SSP5-8.5 the upper 1000 m south of 50S warms by 2-4 degC and freshens by 0.2-0.4 psu, the deep ocean warms by 0.3-0.6 degC; sigma2 decreases by more than 0.4 kg/m3 above 500 m and by 0.1-0.2 down to 1500 m. The cavity run freshens more on the shelf and slope south of 70S (-0.4 psu below 500 m) because of increased ice-shelf melt. Under SSP1-2.6 the changes are 0.2-0.5 degC warming of the upper 1000 m and a surface freshening of 0.05-0.1 psu, with the deep ocean warming by 0.1-0.2 degC.

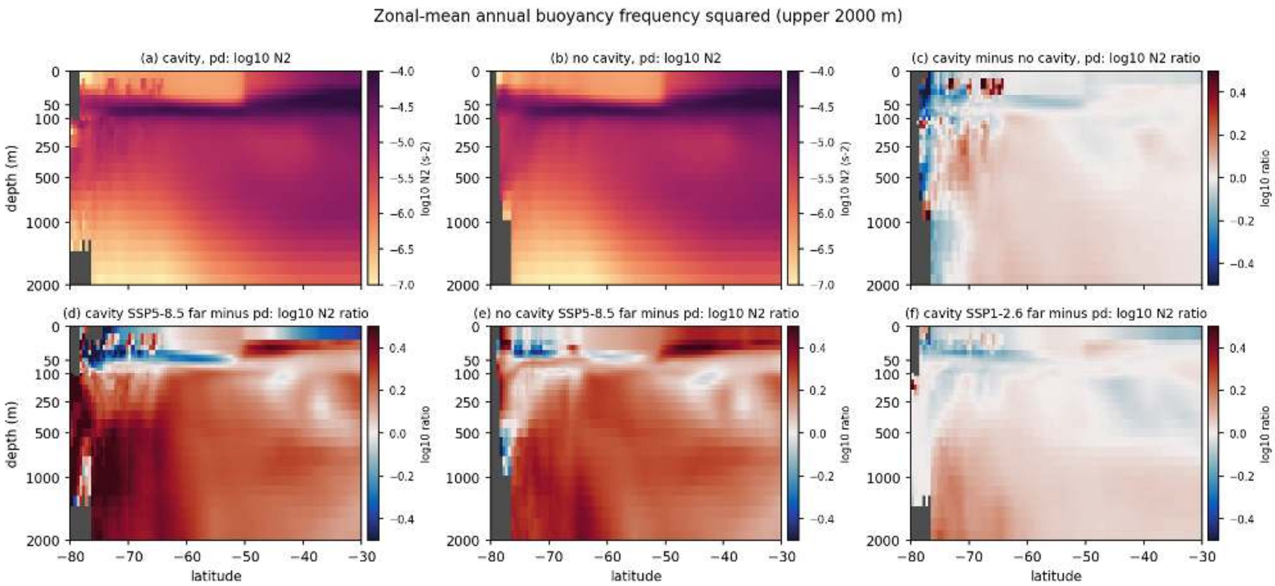

Figure 109: circ_N2_zonal (circ). Zonal-mean annual buoyancy frequency squared (log10 N2, interfaces) in the upper 2000 m. (a) cavity pd, (b) no cavity pd, (c) log10 ratio cavity/no cavity, (d, e) log10 ratio far SSP5-8.5 over pd, (f) far SSP1-2.6 over pd. Take-away: the pycnocline (N2 > 1e-5 s-2) sits at 50-150 m south of 60S and deepens to 500-1000 m north of 50S. The cavity run is slightly more stratified in the upper 300 m of the 70S-60S band (ratio 1.1-1.3); under SSP5-8.5 N2 in the upper 300 m south of 55S increases by a factor of 1.6-2.0 in both sets.

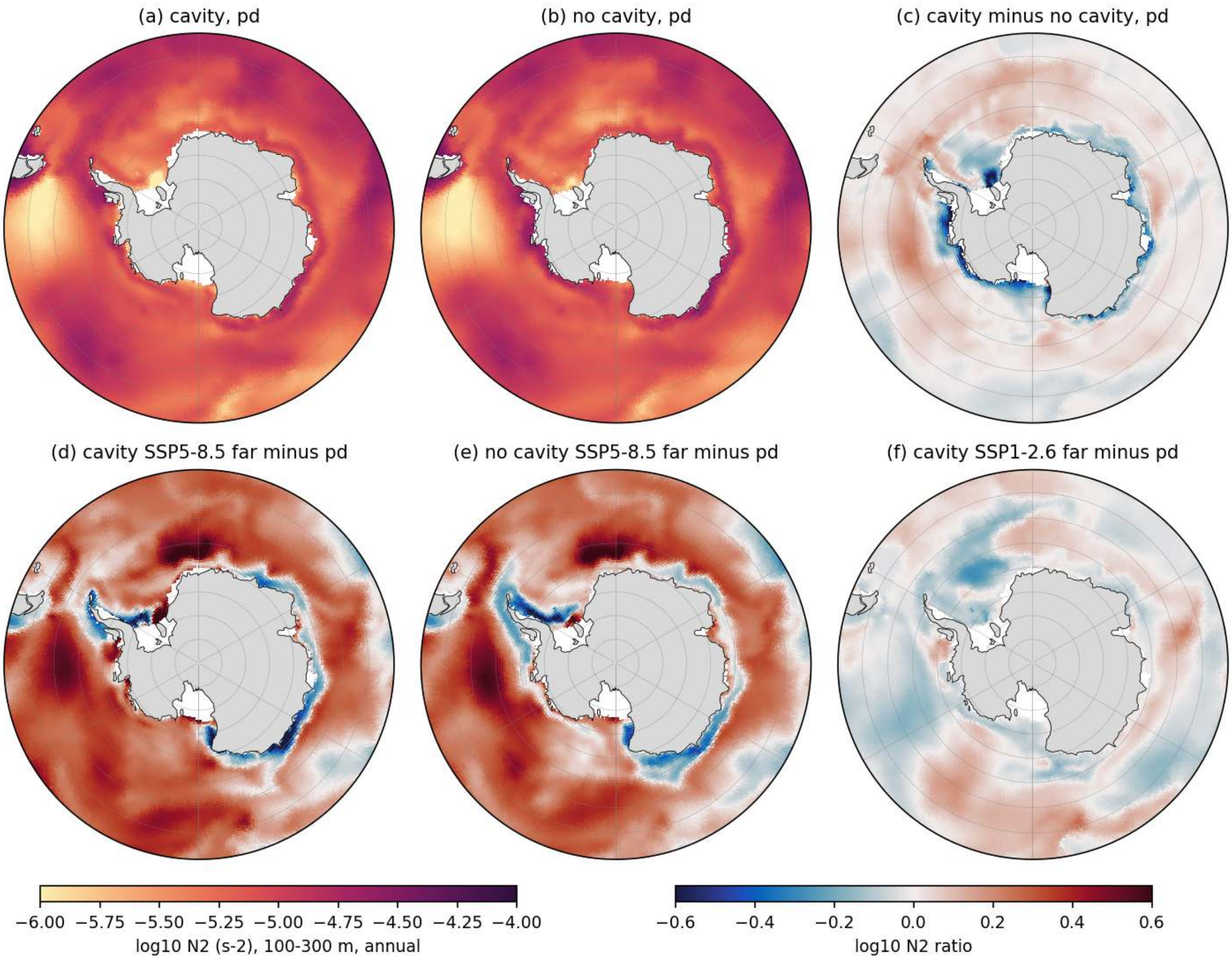

Figure 110: circ__N2__maps (circ). Maps south of 50S of log10 N2 averaged over 100-300 m (annual), same panel layout as above (cavity nodes excluded). Take-away: at pd the weakest 100-300 m stratification (N2 about 5e-6 s-2) is found over the Ross shelf in the cavity run (5.1e-6 versus 8.1e-6 in the no-cavity run) and near Maud Rise in the no-cavity run (4.7e-6 versus 5.3e-6), matching where each set convects. Under SSP5-8.5 N2 south of 60S increases by 60-70 percent (ssa mean 7.7e-6 to 1.28e-5 cavity, 7.9e-6 to 1.18e-5 no cavity), except on the East Antarctic and Ross shelves where it decreases (blue rings) as the winter halocline erodes.

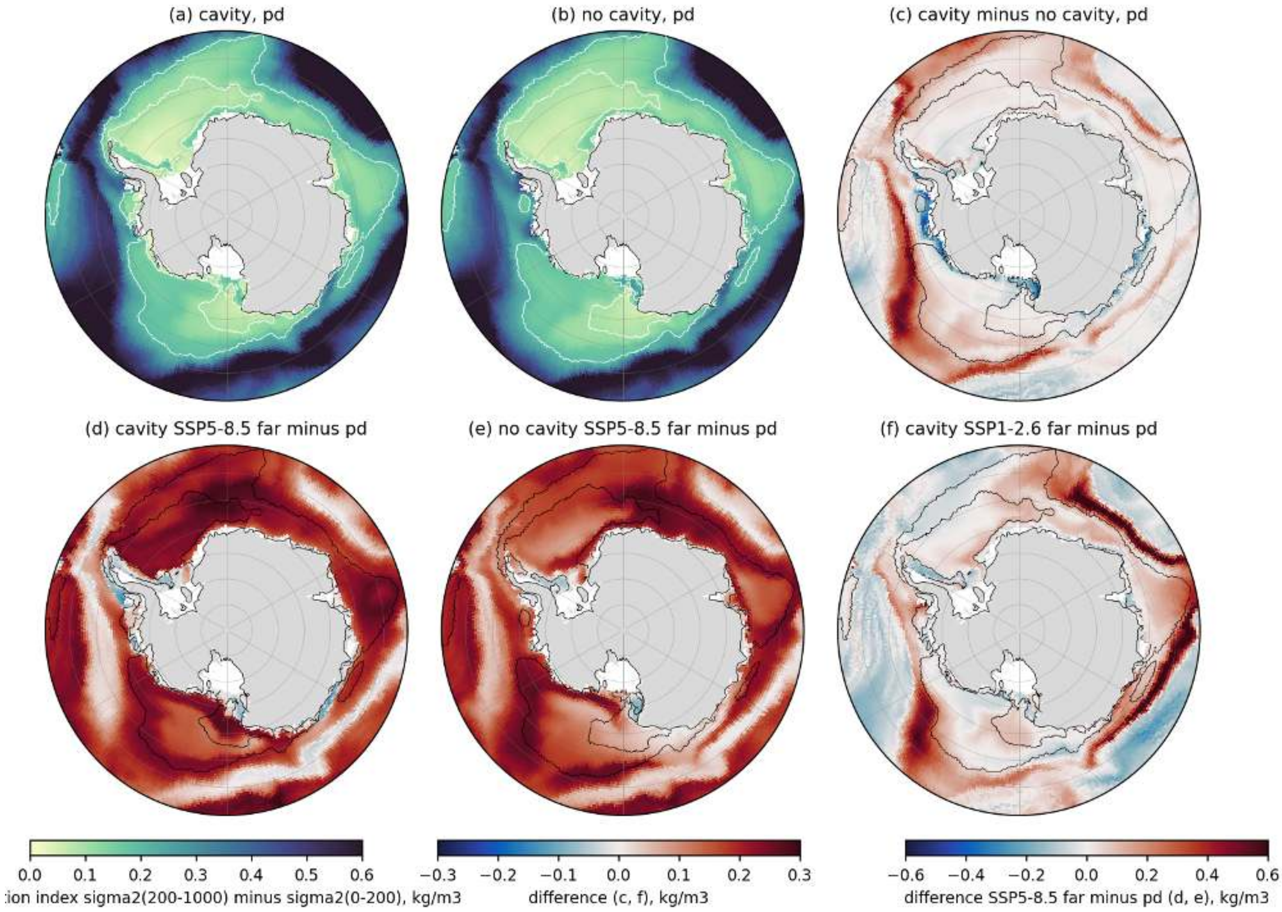

Figure 111: circ_strat_maps (circ). Upper-ocean stratification index sigma2(200-1000 m) minus sigma2(0-200 m) (kg/m3) south of 55S, cavity nodes excluded. (a) cavity pd, (b) no cavity pd, (c) cavity minus no cavity, (d, e) SSP5-8.5 far minus pd for cavity and no cavity (scale +-0.6), (f) cavity SSP1-2.6 far minus pd. White/black contours 0.1 and 0.2 kg/m3 of the reference. Take-away: the weakly stratified (<0.15 kg/m3) deep Weddell and Ross gyre interiors are the convection-prone areas; at Maud Rise the index is 0.119 (cavity) versus 0.107 kg/m3 (no cavity), consistent with the no-cavity run convecting every winter and the cavity run only intermittently; on the Ross shelf it is 0.10 versus 0.15. Under SSP5-8.5 the index rises to 0.55 (cavity) and 0.51 kg/m3 (no cavity) south of 60S, a fourfold increase that shuts down open-ocean convection; under SSP1-2.6 it rises only by 0.03-0.04 kg/m3.

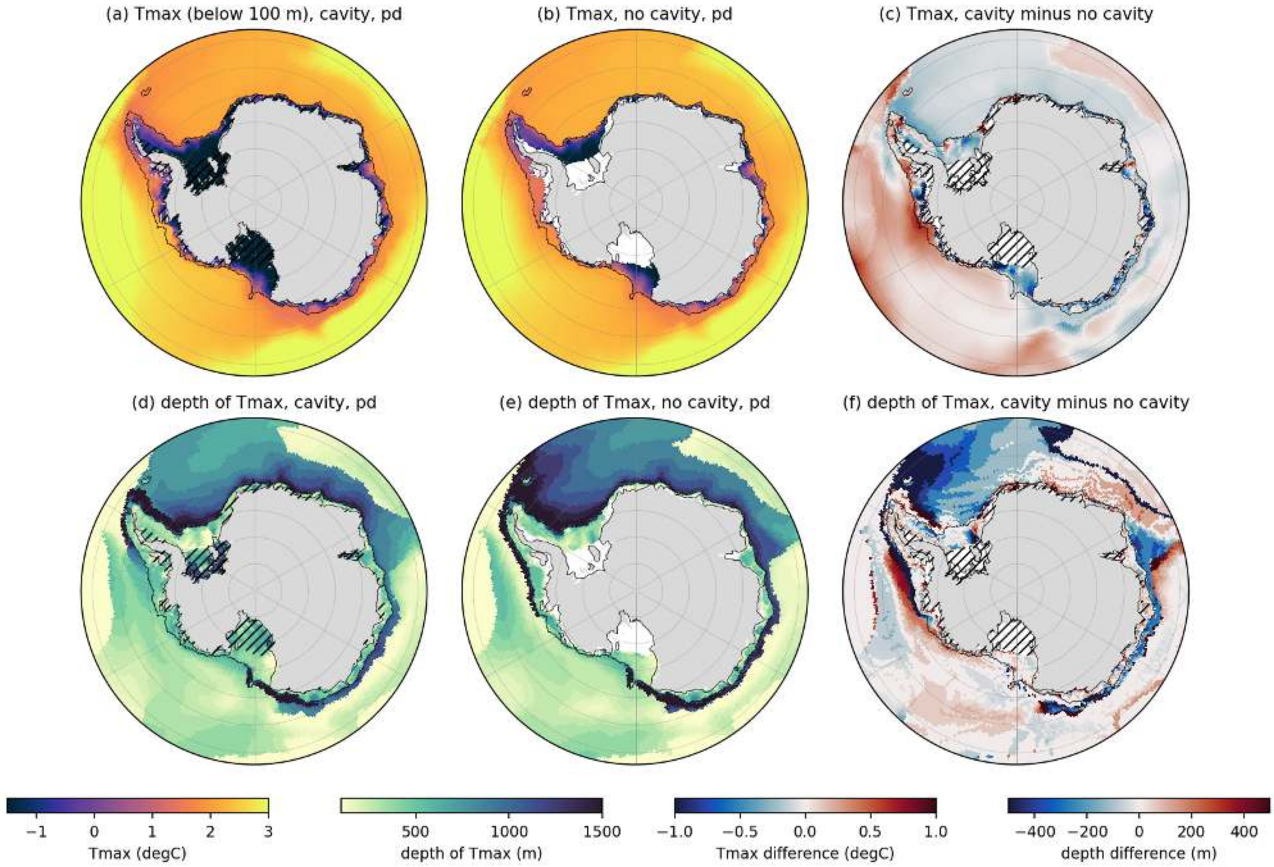

Figure 112: circ_cdw_tmax (circ). Subsurface temperature maximum below 100 m (Tmax) and its depth, south of 58S at pd: (a, d) cavity, (b, e) no cavity, (c, f) cavity minus no cavity. Black contour 1000 m isobath, hatched = cavities. Take-away: Tmax on the shelves (bottom <1000 m, S of 60S, cavity nodes excluded) is 0.12 degC (cavity) and 0.23 degC (no cavity) circumpolar mean, 1.4 degC in the Amundsen and -0.5 degC in the Weddell and Ross seas; the cavity run is 0.2-0.6 degC cooler at the CDW core on the East Antarctic and Ross slopes and 0.2-0.5 degC warmer in the eastern Weddell interior. The depth of Tmax is 350-570 m on the shelves and deeper than 1000 m in the gyre interiors.

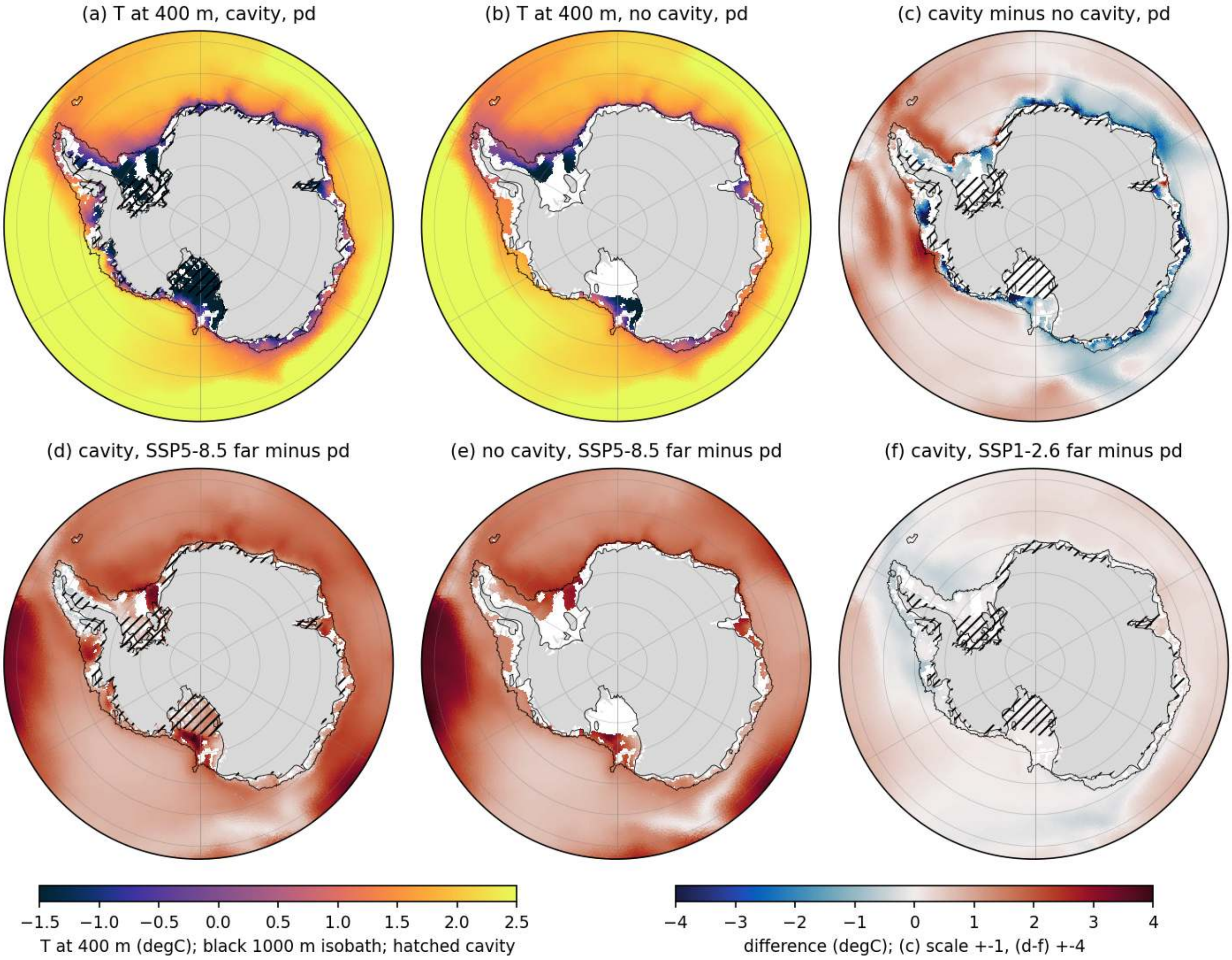

Figure 113: circ__cdw__t400 (circ). Temperature at 400 m (where the bottom is deeper than 400 m), (a) cavity pd, (b) no cavity pd, (c) cavity minus no cavity (scale +-1 degC), (d) cavity SSP5-8.5 far minus pd, (e) no cavity SSP5-8.5 far minus pd, (f) cavity SSP1-2.6 far minus pd (scale +-4 degC). Take-away: at pd the shelf (400-1000 m bottom) mean T400 is 0.0 degC (cavity) and 0.26 degC (no cavity), with the warmest shelves in the Amundsen (1.1/1.8 degC) and Bellingshausen (0.7/1.2) seas and the coldest in the Weddell (-0.6/-0.5) and Ross (-0.7/-0.4) seas. Under SSP5-8.5 every shelf sector warms by 1.5-2.5 degC by 2171-2200 (circumpolar shelf mean 1.49 cavity, 1.93 no cavity; Weddell 0.9/1.4, Ross 1.2/1.5, Amundsen 2.6/3.2), so CDW-type water reaches all shelves; the cavity run stays 0.3-0.9 degC cooler on the shelves. Under SSP1-2.6 the shelf changes are within +-0.3 degC.

10 Meridional overturning circulation

Findings

10.1 MOC theme: findings

All numbers come from cache/ts/<EXP>_moc.nc via analyses/moc/compute_indices.py (indices in cache/moc/moc_indic
scenario table in cache/moc/moc_changes.csv). Periods: early 1851-1880, pd 1985-2014, end 2071-2100, far 2171-2200. Lower-cell indices are negative; "weaker" means less negative. Percent changes are changes in magnitude relative to pd. Cavity = PI_ICEv2 set, no cavity = PI_CORE set.

10.1.1 Diagnostic conventions (checked on the data)

Depth-space moc_z is the Eulerian overturning from the resolved vertical velocity only (no bolus contribution), cumulated from the north. NADW cell positive, abyssal cell negative. South of 38S the Eulerian Deacon cell (about 43 Sv) reaches the sea floor, so the depth-space abyssal cell is only defined north of about 38S and the requested index "at 60S below 2000 m" is zero in every run. Density-space moc_dens gives the northward transport of all water lighter than sigma2 and therefore uses the same sign convention (dense AABW cell negative). The dense lower cell lives in a very narrow band of model classes, sigma2 36.85 to 36.92, which is the CDW-like density of the model abyss (overview theme: the observed AABW class sigma2 > 37.16 is essentially empty). All "AABW cell" statements are therefore about the model's densest overturning, not about water in the observed AABW class. The Atlantic sector mask is open zonally south of 35S, so Atlantic curves are only interpreted north of 35S.

10.1.2 Findings

1. Present-day lower cell is weak in depth space and confined to the abyss north of 38S. The global abyssal cell minimum (all latitudes, below 2000 m) is -10.4 Sv (cavity) and -9.5 Sv (no cavity), located near 20S-10S at 3000-3500 m. At 30S below 2500 m it is -5.7 Sv (cavity) and -5.2 Sv (no cavity), and at 30N below 2500 m -1.2 and -0.4 Sv. Interannual standard deviation of the global abyssal minimum is about 2 Sv, so the cavity minus no-cavity difference of 0.5-0.9 Sv is only marginally outside noise but is consistent in all periods and in all three depth-space indices. Observational and inverse estimates of AABW formation or abyssal overturning are of order 10-20 Sv (for example Orsi et al. 1999, Lumpkin and Speer 2007, Talley 2013), so the model abyssal cell is at the low end even though the streamfunction is measured in a much lighter class than observed AABW. This comparison is indicative only because the observed numbers refer to a density class the model does not produce.

2. In density space the present-day lower cell is stronger and the cavity effect is reversed. The minimum south of 50S over classes sigma2 >= 36.6 is -20.9 Sv (cavity) and -30.2 Sv (no cavity), located at 55S-62S in the 36.88-36.92 classes. At 60S it is -15.1 and -27.8 Sv, at 30S -7.5 and -6.7 Sv. So the no-cavity run overturns about 9 Sv more dense water around Antarctica at pd but exports no more of it northward (the 30S values are within 1 Sv, the cavity run even slightly stronger). This extra no-cavity overturning is local recirculation in the Weddell sector linked to the Maud Rise convection (finding 7). It was not present in the early period (early values -24.3 Sv cavity versus -17.3 Sv no cavity), the no-cavity lower cell strengthened through the historical period while the cavity one weakened (see finding 7).

3. AMOC is similar in both sets and much stronger than the lower cell. AMOC maximum at 26N (Atlantic sector, below 300 m) is 15.4 Sv (cavity) and 16.2 Sv (no cavity) at about 1040 m, 10.1 Sv at 45N in both. The Atlantic abyssal cell is nearly absent, -1.0/-0.8 Sv at 30S and -3.9/-3.7 Sv at the equator. In density space the AMOC at 26N is 14.7/15.1 Sv. The cavity run has a 0.9 Sv weaker AMOC at 26N over the full historical period (early 16.8 versus 18.5 Sv), which is larger than the 30-yr noise (standard deviation 1.3 Sv annually, about 0.5 Sv for 30-yr means) and may reflect the slightly stronger abyssal inflow and fresher Southern Ocean surface in the cavity run.

4. Scenario response of the lower cell is strongly index dependent. Under SSP5-8.5 the cavity run loses 25 percent of the global abyssal cell minimum by 2171-2200 (-10.4 to -7.8 Sv), 77 percent of the lower cell at 30S (-5.7 to -1.3 Sv), 69 percent of the density-space lower cell south of 50S (-20.9 to -6.6 Sv) and 96 percent at 30S (-7.5 to -0.3 Sv). SSP3-7.0 is nearly as strong (28, 71, 66 and 91 percent). SSP2-4.5 gives 17, 48, 25 and 62 percent, SSP1-2.6 8, 26, 20 and 34 percent. The no-cavity run responds with the same ordering (SSP3-7.0: 31, 57, 66, 78 percent; SSP2-4.5: 8, 23, 45, 34 percent; SSP1-2.6 shows no weakening of the depth-space cells by 2171-2200 and 39 percent in the density-space cell south of 50S). Under SSP5-8.5 the no-cavity run loses 32 percent of the global abyssal minimum (-9.5 to -6.5 Sv), 57 percent of the lower cell at 30S (-5.2 to -2.3 Sv), 75 percent of the density-space cell south of 50S (-30.2

to -7.7 Sv) and 77 percent at 30S (-6.6 to -1.5 Sv), so by 2171-2200 the two sets converge to the same weak state (abyssal minimum -7.8 versus -6.5 Sv, 30S lower cell -1.3 versus -2.3 Sv, density cell south of 50S -6.6 versus -7.7 Sv). The global abyssal minimum is the most resilient index because it sits far from the Southern Ocean (10S-20S, 3500 m) and responds with a lag; it never drops below half of its pd value in any run.

5. Timing of the 50 percent loss (11-yr running mean, sustained for 5 years). Cavity SSP5-8.5: density-space lower cell at 30S in 2102, AMOC at 26N in 2107, depth-space lower cell at 30S in 2137, density-space lower cell south of 50S in 2145. Cavity SSP3-7.0: 2104, 2118, 2151, 2144. Cavity SSP2-4.5: only the depth-space 30S cell (2180) and the density 30S cell (2152) cross the threshold. Cavity SSP1-2.6: no index crosses. No cavity SSP3-7.0: 2129 (density 30S), 2118 (AMOC), 2174 (depth 30S), 2036 (density south of 50S, but this reflects the collapse of the anomalously strong pd state rather than a response beyond the early-period level). No cavity SSP5-8.5: 2118 (density 30S), 2110 (AMOC), 2175 (depth 30S), 2034 (density south of 50S). The density-space lower cell south of 50S in the no-cavity run drops below half its pd value already 2036-2055 in all scenarios because its pd value is inflated by the Maud Rise convection which stops after 2020-2030.

6. SSP1-2.6 shows partial recovery and no return to pd. The cavity density-space lower cell south of 50S weakens to -14.4 Sv (end) and recovers to -16.7 Sv (far), that is 80 percent of pd. The no-cavity one goes from -30.2 to -17.0 (end) and -18.6 Sv (far), so the two sets converge at 16-19 Sv by 2200. The depth-space abyssal cell does not weaken at all under SSP1-2.6 (cavity -10.4 to -9.6 Sv, no cavity -9.5 to -10.0 Sv), the lower cell at 30S loses 26 percent in the cavity run and nothing in the no-cavity run. AMOC at 26N drops to 11.5/12.5 Sv by 2071-2100 and recovers to 13.1/14.5 Sv by 2171-2200.

7. Drivers of lower-cell variability differ between the sets, and open-ocean convection is not a good predictor. The no-cavity run convects every September near Maud Rise (MLD > 2000 m over 0.10-0.16e6 km2), the cavity run in very few years (0.00-0.02e6 km2). Yet the density-space lower cell of the no-cavity run was weaker than the cavity one until about 1950 (-17 versus -24 Sv in 1851-1880). On annual time scales the correlation between the September convection area and any lower-cell index is below 0.15 in both sets (moc_lagcorr_hist a, g). On centennial scales the no-cavity lower cell strengthened from -17 (early) to -30 Sv (pd) while its convection area grew from 0.10 to 0.16e6 km2 and its September sea-ice extent fell from 18.3 to 14.7e6 km2. In the cavity run the lower cell weakened from -27 Sv (1951-1980) to -21 Sv (pd) while total ice-shelf melt rose from 4250 to 4670 Gt/yr and September sea ice fell from 15.0 to 13.5e6 km2; on decadal scales the cavity lower cell correlates with sea-ice extent (r = -0.67) and with melt (r = +0.67). The shelf-deep bottom density contrast is tiny (0.01-0.04 kg m-3) and explains little (annual r 0.2-0.35 cavity, negative no cavity). The depth-space abyssal cell shows no significant relation to any driver on annual scales. So in this model the abyssal overturning is not set by winter convection area, and the density-space lower cell around Antarctica responds to the freshwater and sea-ice state (melt in the cavity run, polynya convection in the no-cavity run) rather than to shelf water density.

8. The Deacon cell and the upper density-space cell are nearly scenario independent. The Eulerian Deacon cell maximum (43 Sv at pd) changes by less than 5 percent in all scenarios (up to +4.5 Sv at 2071-2100 under SSP3-7.0 and SSP5-8.5, consistent with the stronger and poleward shifted westerlies reported by the atmos theme). The upper density-space cell south of 45S (18-19 Sv) changes by less than 10 percent.

9. Atlantic abyssal inflow strengthens as the AMOC weakens. At 30S below 2500 m the Atlantic abyssal cell doubles, from -1.0 to -2.0 Sv (cavity SSP5-8.5, far) and from -0.8 to -1.8 Sv (no cavity SSP3-7.0, far), while the AMOC at 26N falls from 15-16 to 4-6 Sv. This is the classic replacement of NADW by southern-sourced water below 2500 m, which happens even though the Southern Ocean lower cell itself weakens.

10.1.3 Caveats and suspicious points

The depth-space abyssal cell from Eulerian w alone is masked by the Deacon cell south of 38S, so no depth-space index at 60S can be formed; the density-space diagnostic is the only one that sees the cell around Antarctica. The density-space cell lives in 2-3 classes of width 0.02 kg m-3, so the per-latitude minimum jumps between classes as the deep ocean warms, producing diagonal stripes in the latitude-time plots (moc_hovmoeller, lower rows). A depth-mapped version of the density MOC (item 5 of the brief) was not produced because std_dens_Z in the pmean files is zero or near zero for most elements and std_dens_H is zero everywhere, so the class-depth mapping could not be verified. The first year of the scenario chain is noisy (annual index minima vary by 2-4 Sv from year to year). The historical sets are single members, so cavity minus no-cavity differences below about 1 Sv (depth space) or 5 Sv (density space south of 50S) in 30-yr means should be treated as within internal variability. The no-cavity pd density-space lower cell (-30 Sv) is an anomalous state reached only after 1950 and coincides with the Maud Rise convection maximum; its scenario percent changes therefore overstate the warming response.

10.1.4 Story lines for a paper

Cavities make the present-day abyssal cell 0.5-0.9 Sv stronger in depth space and the AMOC about 1 Sv weaker, but they do not change the basic weak-AABW state of the model. The large density-space difference around Antarctica (-21 versus -30 Sv) is a regional recirculation tied to the Maud Rise polynya of the no-cavity run, not an export difference, and it disappears within two decades of the scenarios. The warming response is scenario dependent and index dependent: the export of dense water at 30S and the dense cell around Antarctica lose two thirds or more under SSP3-7.0 and SSP5-8.5 by 2200, while the abyssal cell in the tropics loses only a quarter, and SSP1-2.6 stabilises all indices at 70-100 percent of pd. Convection area is a poor predictor of overturning in this model, which is a useful warning for diagnostics that use polynya area as an AABW proxy.

Figures

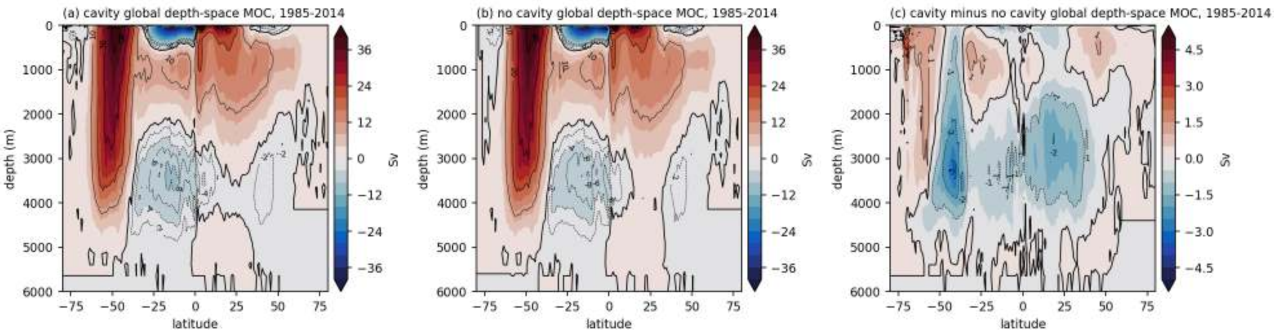

Figure 114: moc_z_global_pd (moc). Global depth-space meridional overturning streamfunction (Sv) averaged over 1985-2014 for (a) the cavity run PI_ICEv2_hist, (b) the no-cavity run PI_CORE_hist and (c) cavity minus no cavity. Black contours every 2 Sv in the lower cell and every 10 Sv in the upper cells, zero line thick. The Eulerian Deacon cell (about 43 Sv, 60S-40S) dominates the Southern Ocean and reaches the sea floor, so the abyssal cell is only visible north of about 38S. The abyssal cell minimum is -10.4 Sv (cavity) and -9.5 Sv (no cavity) near 20S-10S at 3000-3500 m. Its value at 30S below 2500 m is -5.7 Sv (cavity) and -5.2 Sv (no cavity). The cavity run has a 0.5 to 1 Sv stronger abyssal cell between 40S and 40N and a slightly weaker upper Deacon limb.

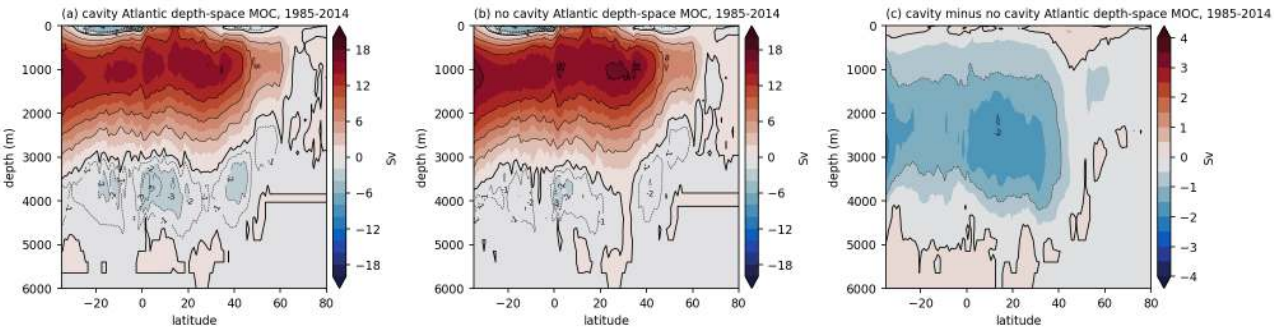

Figure 115: moc_z_atl_pd (moc). As moc_z_global_pd but for the Atlantic sector north of 35S. The AMOC maximum at 26N is 15.4 Sv (cavity) and 16.2 Sv (no cavity), both at about 1040 m, and 10.1 Sv at 45N in both runs. The Atlantic abyssal cell is very weak, -1.0 Sv (cavity) and -0.8 Sv (no cavity) at 30S below 2500 m and -3.9/-3.7 Sv at the equator. The cavity run has a slightly weaker NADW cell (by 0.9 Sv at 26N) and a slightly stronger abyssal inflow.

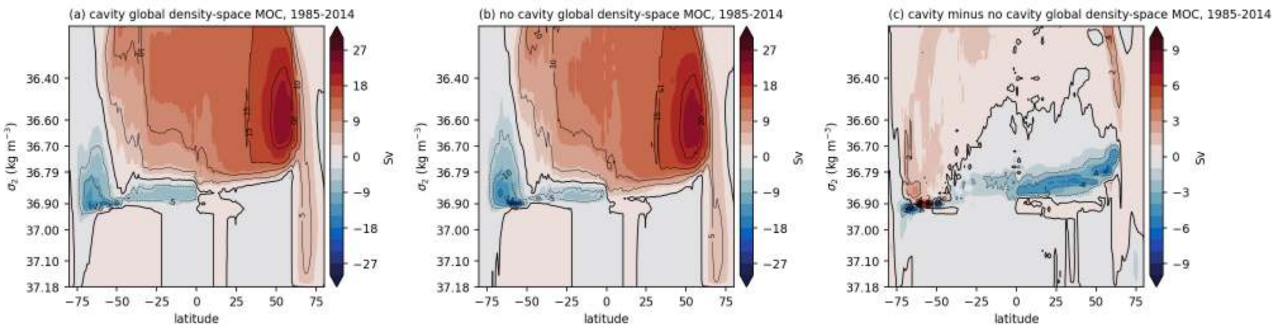

Figure 116: moc_dens_global_pd (moc). Global density-space overturning streamfunction (Sv) for 1985-2014 on a stretched sigma2 axis (one FESOM density bin per tick interval, 36.0 to 37.18 shown, dummy bins omitted) for (a) cavity, (b) no cavity and (c) the difference. The dense lower cell sits in a very narrow class band, sigma2 36.85 to 36.92, and extends from 75S to about 20S. Its minimum south of 50S is -20.9 Sv (cavity) and -30.2 Sv (no cavity) near 55S-62S, that is the no-cavity run overturns 9 Sv more dense water at present day. At 30S the lower cell is -7.5 Sv (cavity) and -6.7 Sv (no cavity). The upper (NADW and Deacon) cell reaches 18-19 Sv in both runs.

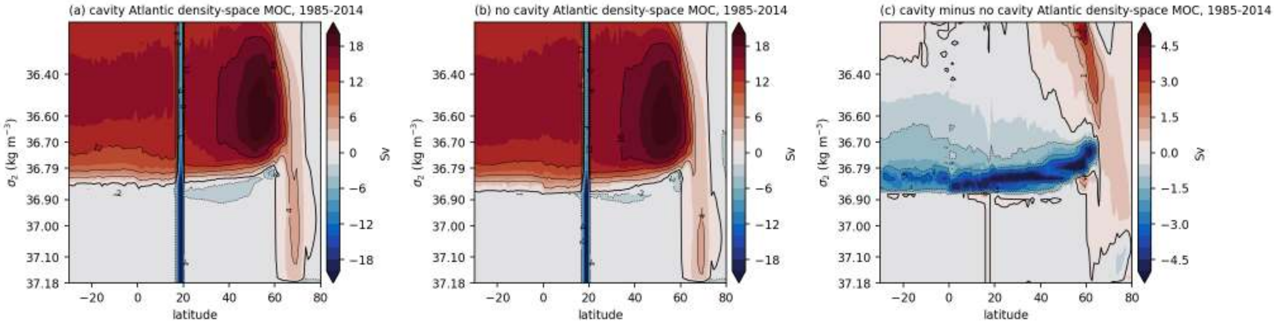

Figure 117: moc_dens_atl_pd (moc). As moc_dens_global_pd for the Atlantic sector north of 30S. The density-space AMOC at 26N is 14.7 Sv (cavity) and 15.1 Sv (no cavity) in classes lighter than sigma2 36.8. The dense return flow below the NADW class is under 2 Sv. Differences between the two sets are below 1 Sv except at the class boundaries.

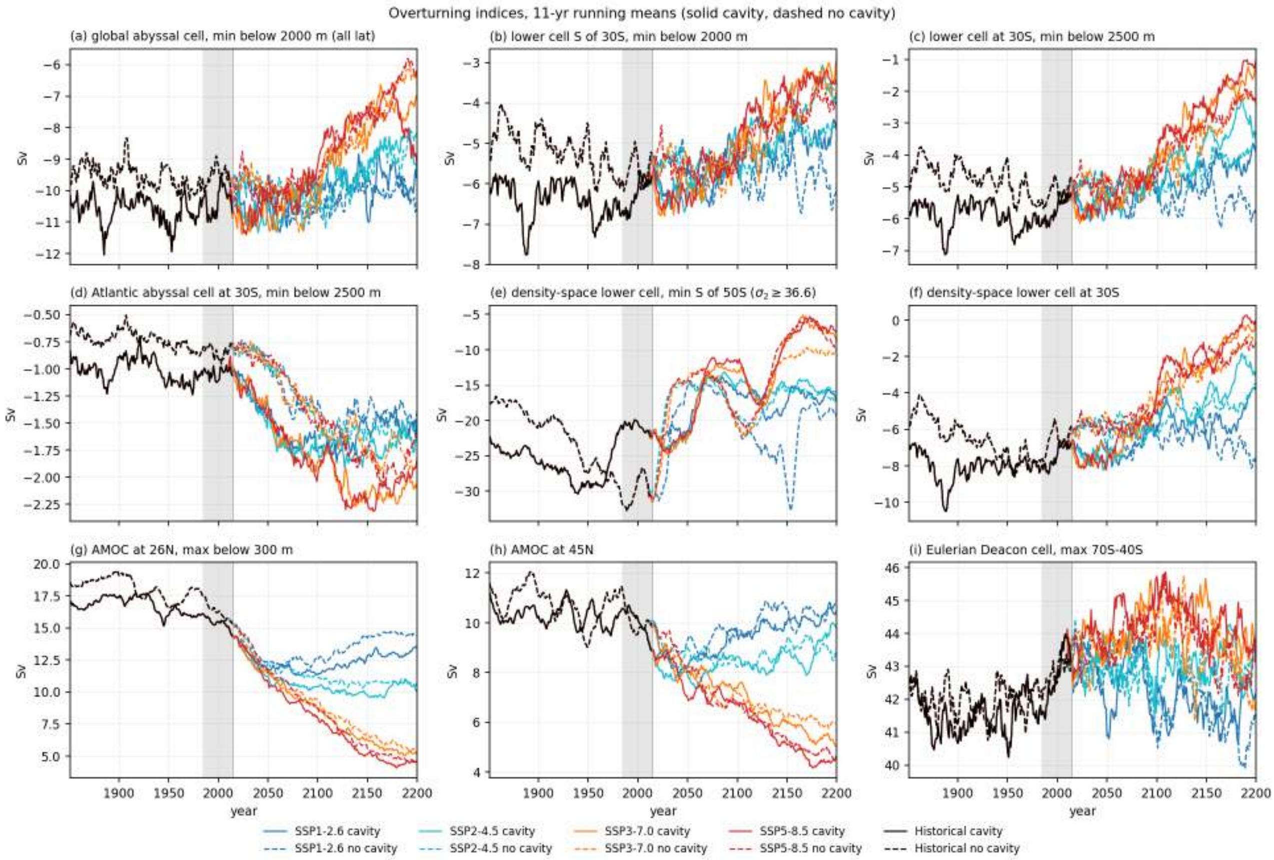

Figure 118: moc_indices_timeseries (moc). Eleven-year running means of overturning indices, 1851-2200, for all four SSP chains and both sets (solid cavity, dashed no cavity, black for the shared historical part, grey band pd). (a) global abyssal cell, minimum below 2000 m over all latitudes; (b) lower cell south of 30S, minimum below 2000 m; (c) lower cell at 30S, minimum below 2500 m; (d) Atlantic abyssal cell at 30S; (e) density-space lower cell, minimum south of 50S over sigma2 >= 36.6; (f) density-space lower cell at 30S; (g) AMOC at 26N; (h) AMOC at 45N; (i) Eulerian Deacon cell maximum. The abyssal cell weakens only 25-32 percent by 2171-2200 under SSP5-8.5 and SSP3-7.0, whereas the lower cell at 30S loses 71-77 percent (cavity) and 57 percent (no cavity) and the density-space lower cell south of 50S loses 66-75 percent in both sets. The AMOC at 26N drops from 15.4 to 4.4 Sv (cavity) and from 16.2 to 4.8 Sv (no cavity) under SSP5-8.5 and to 13-14.5 Sv under SSP1-2.6. The Deacon cell changes by less than 5 percent.

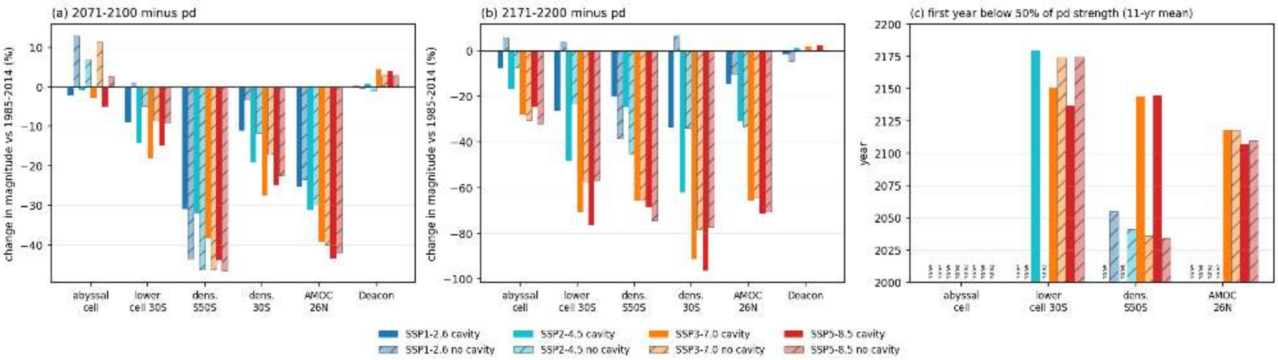

Figure 119: moc_scenario_summary (moc). Summary of the scenario response of the main indices. (a) and (b) change in magnitude of the period mean for 2071-2100 and 2171-2200 relative to 1985-2014 in percent, per scenario (colours) and set (plain cavity, hatched no cavity). (c) first year in which the 11-yr running mean drops below half of the pd strength for at least 5 years; "none" means that this never happens before 2200. Under SSP5-8.5 the cavity run crosses the 50 percent threshold in 2102 (density lower cell at 30S), 2107 (AMOC 26N), 2137 (depth lower cell at 30S) and 2145 (density lower cell south of 50S), the no-cavity run in 2118, 2110, 2175 and 2034 (the last because its pd value is inflated by the Maud Rise convection). The global abyssal cell never drops below half of its pd value in any scenario.

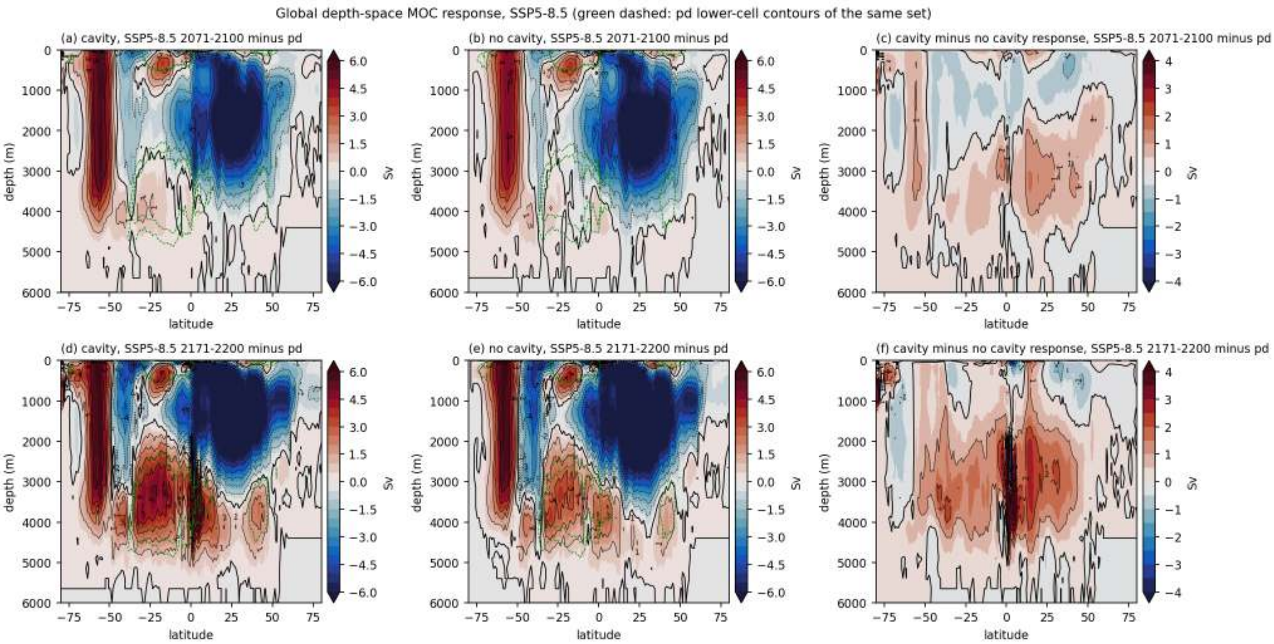

Figure 120: moc_z_response_ssp585 (moc). Change of the global depth-space streamfunction (Sv) under SSP5-8.5 for 2071-2100 minus pd (a-c) and 2171-2200 minus pd (d-f), cavity (a, d), no cavity (b, e) and cavity minus no cavity response (c, f). Green dashed lines show the -2 and -4 Sv pd contours of the lower cell of the same set. The NADW cell weakens by more than 6 Sv throughout the Atlantic depth range, the Deacon cell strengthens by 2-4 Sv in its core, and the abyssal cell weakens (positive anomaly) by 3-5 Sv between 40S and 10N below 2500 m by 2171-2200. The two sets respond alike. The cavity run loses 1-3 Sv more of the abyssal cell between 40S and 40N by 2171-2200 (panel f) because its pd cell was stronger, so both sets end at the same weak state.

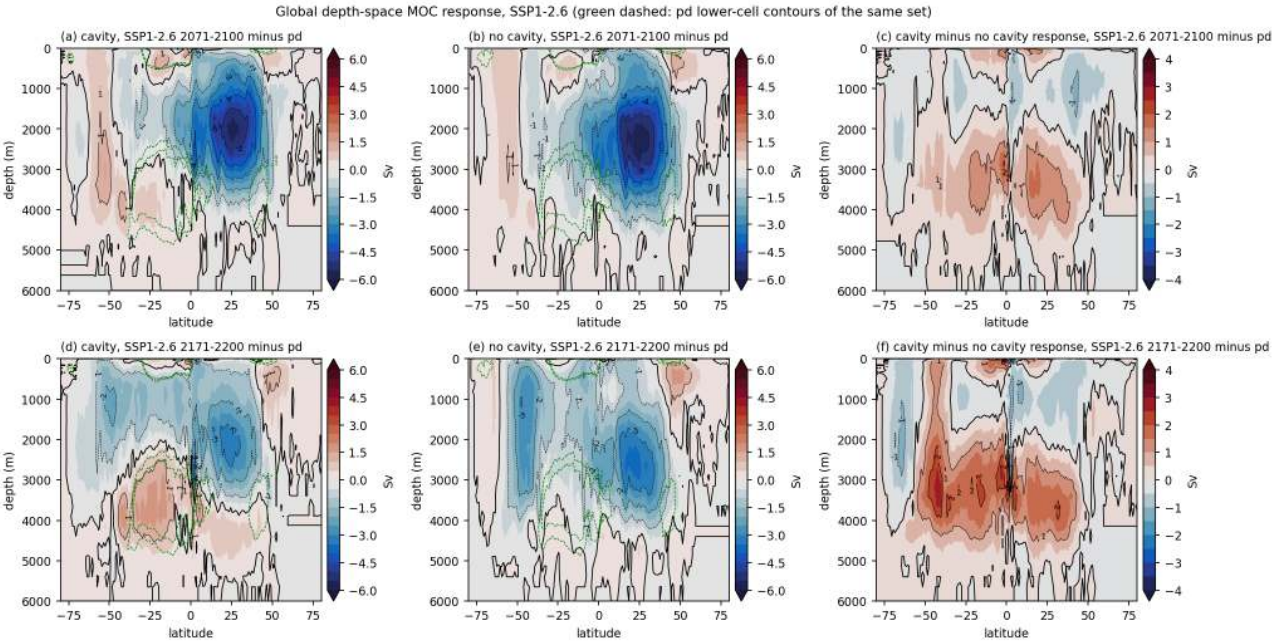

Figure 121: moc_z_response_ssp126 (moc). As moc_z_response_ssp585 for SSP1-2.6. The NADW cell recovers partly after 2100 and the abyssal cell change stays within 1-2 Sv. In the no-cavity run the abyssal cell is slightly stronger than at pd by 2171-2200.

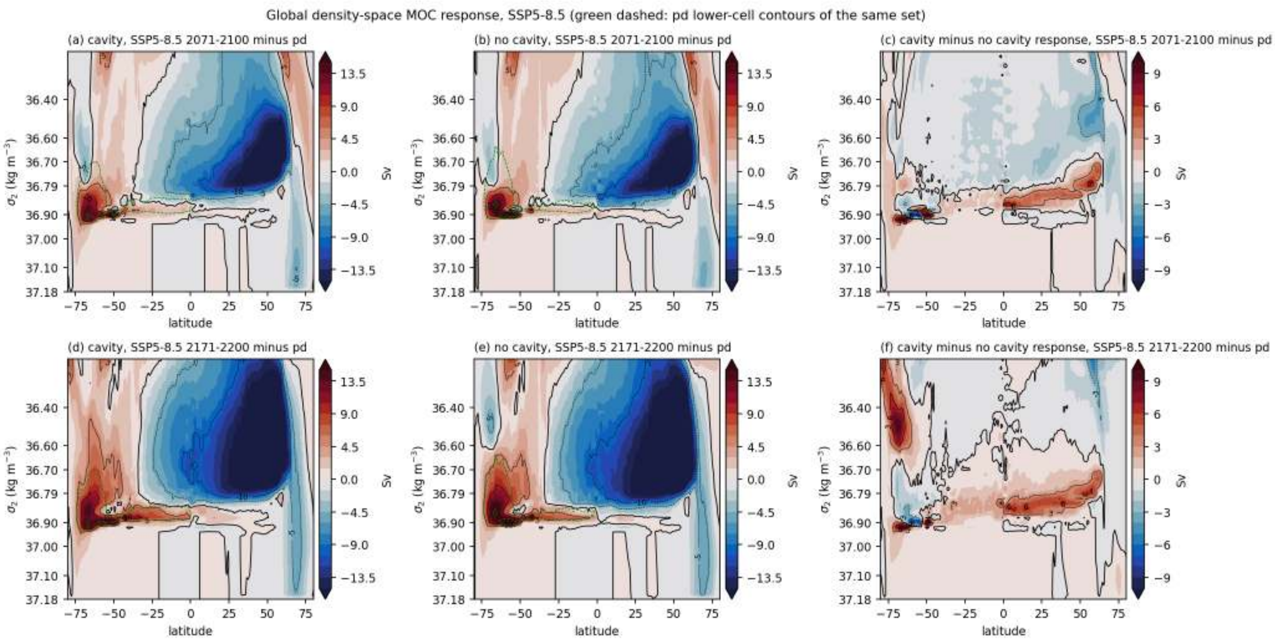

Figure 122: moc_dens_response_ssp585 (moc). Change of the global density-space streamfunction (Sv) under SSP5-8.5, layout as moc_z_response_ssp585, green dashed lines are the pd -5 and -15 Sv contours of the lower cell. The dense lower cell south of 50S (sigma2 36.85-36.92) weakens by 10-15 Sv by 2071-2100 and by more by 2171-2200, and the NADW cell in the 36.6-36.8 classes weakens by more than 10 Sv. Part of the dipole pattern at the lower-cell classes reflects a shift of the cell to lighter classes as the deep ocean warms, not only a weakening.

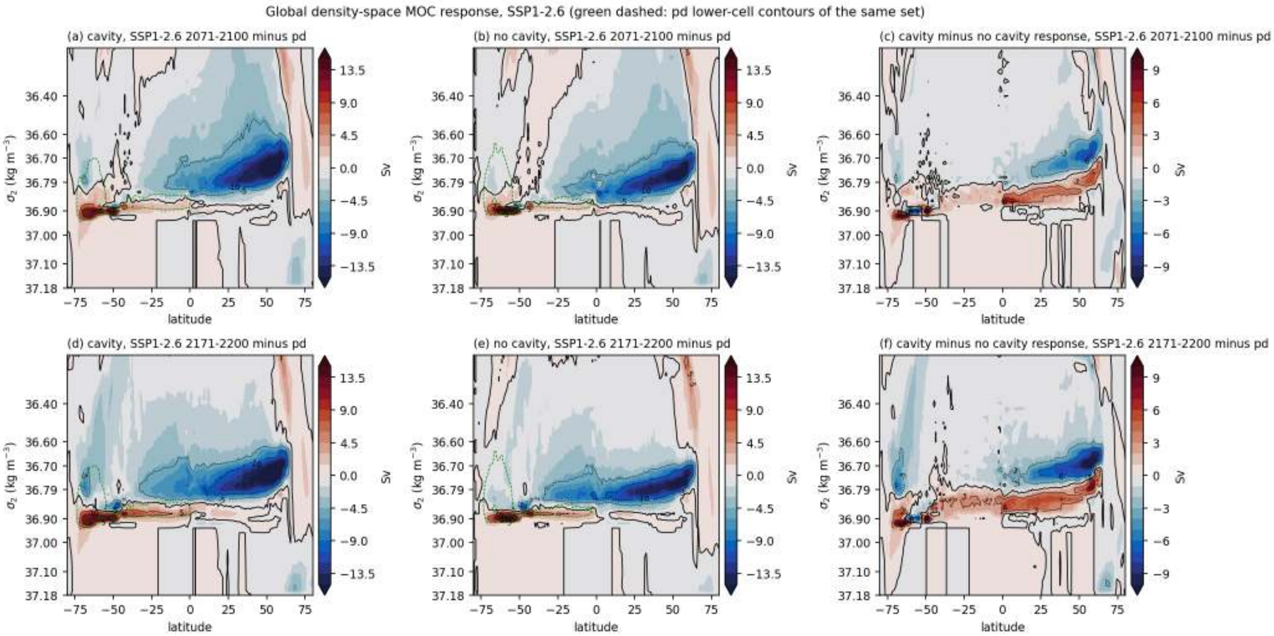

Figure 123: moc_dens_response_ssp126 (moc). As moc_dens_response_ssp585 for SSP1-2.6. The dense lower cell south of 50S still weakens by 5-15 Sv in both sets by 2071-2100 and only partly recovers by 2171-2200. The NADW-class weakening is 10 Sv at 2071-2100 and smaller afterwards.

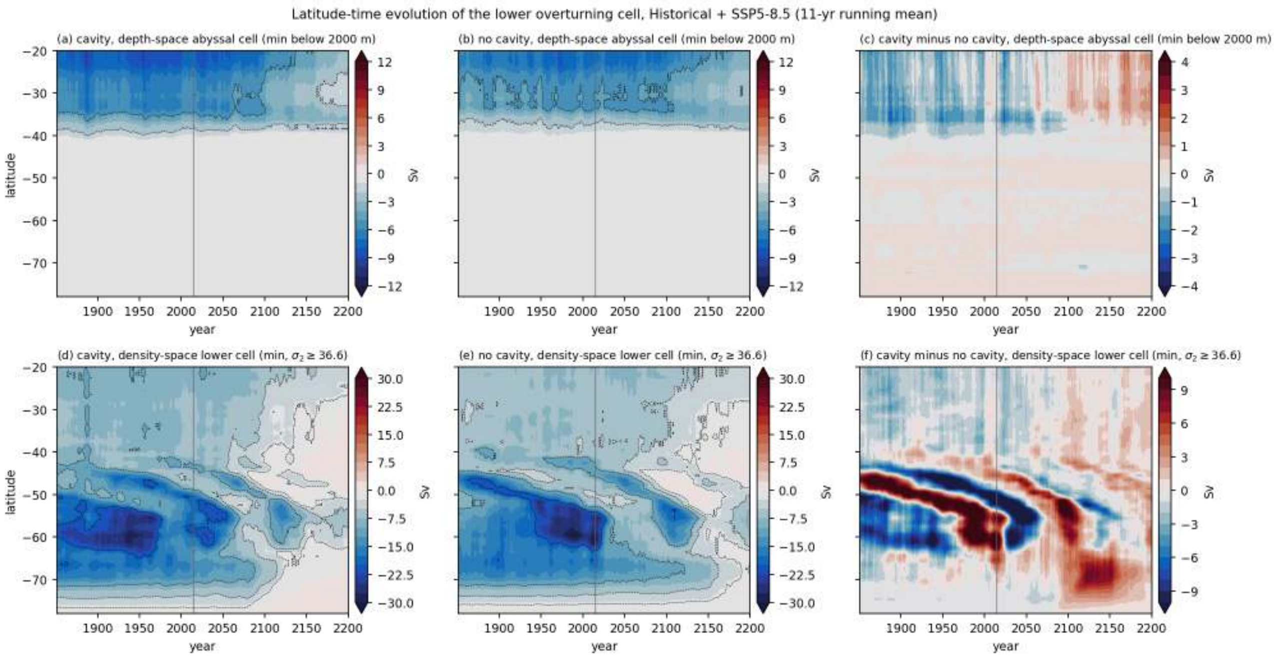

Figure 124: moc_hovmoeller_ssp585 (moc). Latitude-time evolution (11-yr running means) of the lower overturning cell for the historical plus SSP5-8.5 chain. Top row, depth-space abyssal cell strength per latitude (minimum below 2000 m); bottom row, density-space lower cell per latitude (minimum over sigma2 >= 36.6); cavity (a, d), no cavity (b, e) and difference (c, f). In depth space the abyssal cell exists only north of 38S (south of that the Eulerian Deacon cell dominates). In density space the cell is strongest at 55S-65S. Diagonal stripes in the density-space panels come from the slow drift of the lower-cell core to lighter classes with the coarse class resolution, so the difference panel (f) should be read with care. The density-space cell south of 55S collapses within two decades after 2020 in the no-cavity run (from -30 to -15 Sv) and more gradually in the cavity run, and both sets reach 5-8 Sv by 2170. The depth-space cell north of 38S weakens after 2100.

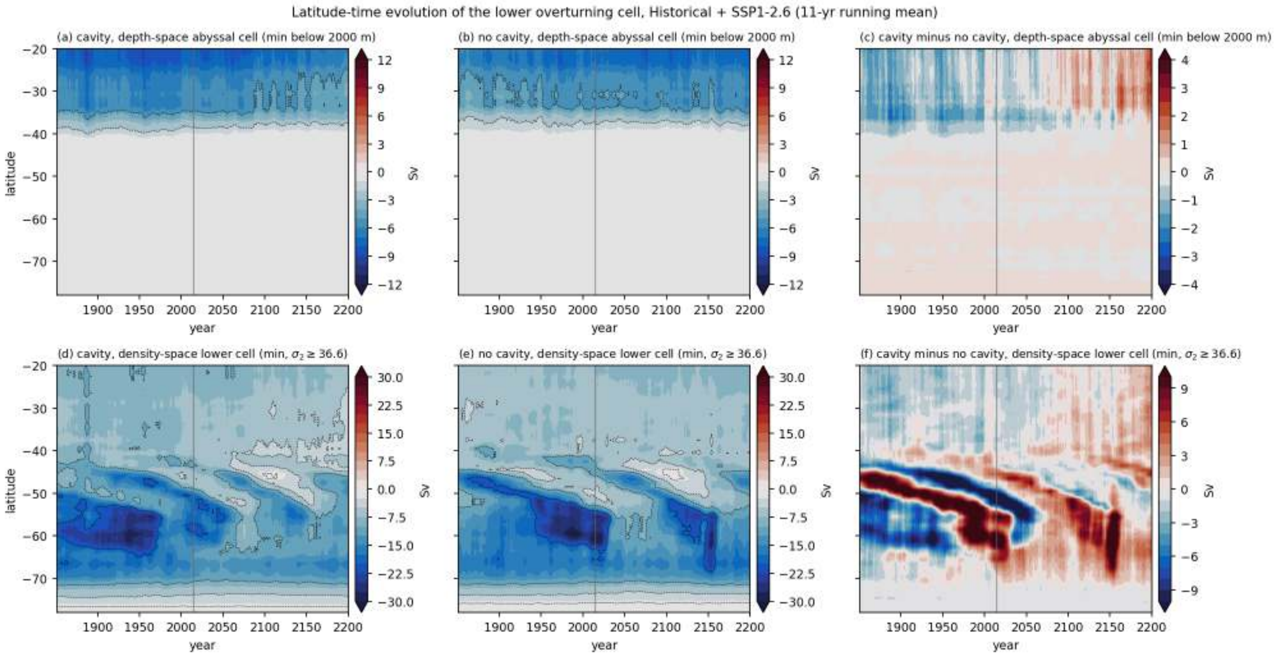

Figure 125: moc_hovmoeller_ssp126 (moc). As moc_hovmoeller_ssp585 for SSP1-2.6. The no-cavity run keeps a 15-30 Sv density-space lower cell south of 55S until 2200 with a renewed strengthening after 2100, the cavity run stays weaker (10-18 Sv) after 2050.

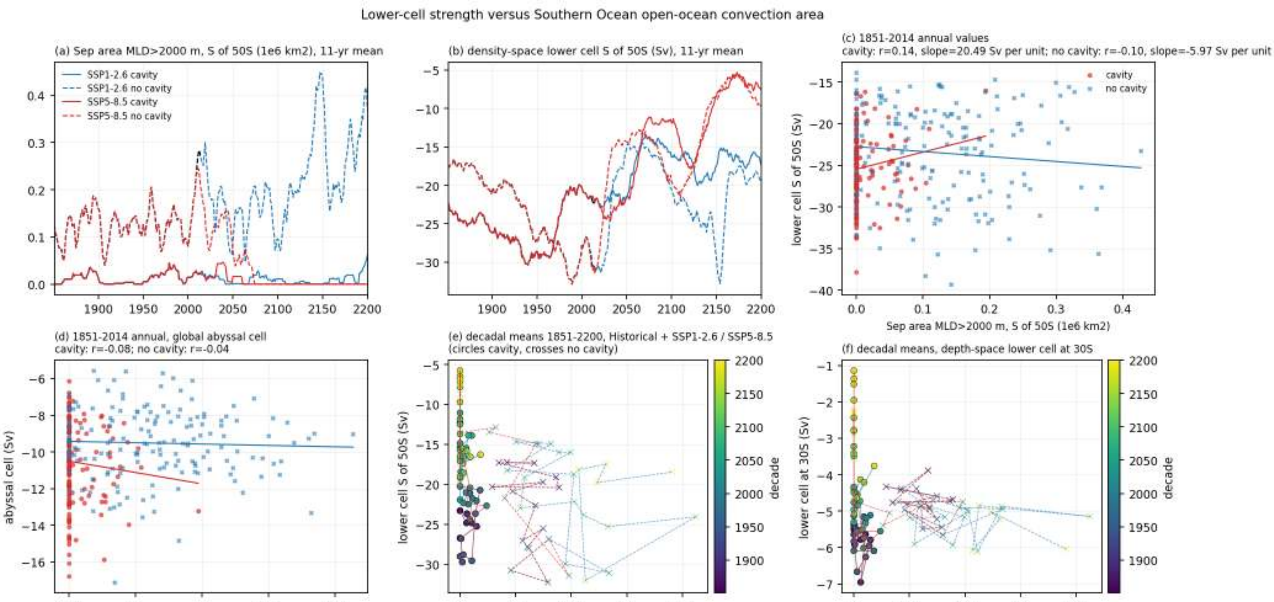

Figure 126: moc_vs_convection (moc). Relationship between lower-cell strength and Southern Ocean open-ocean convection. (a) September area with mixed layer deeper than 2000 m south of 50S (1e6 km2) and (b) the density-space lower cell south of 50S (Sv), 11-yr running means, historical plus SSP1-2.6 and SSP5-8.5 chains. (c) and (d) annual values 1851-2014, lower cell and global abyssal cell versus convection area, with linear fits and correlations for each set. (e) and (f) decadal means 1851-2200 (colour is the decade, circles cavity, crosses no cavity) for the density-space lower cell and the depth-space lower cell at 30S. The no-cavity run convects every winter near Maud Rise (0.10-0.16e6 km2) while the cavity run does so only in a few years, yet the interannual correlation between convection area and lower-cell strength is weak ($|r| < 0.15$ for both sets). On centennial scales the no-cavity lower cell strengthens from -17 Sv (1851-1880) to -30 Sv (1985-2014) as its convection area grows from 0.10 to 0.16e6 km2, whereas the cavity lower cell weakens from -27 Sv (1951-1980) to -21 Sv (pd) without convection.

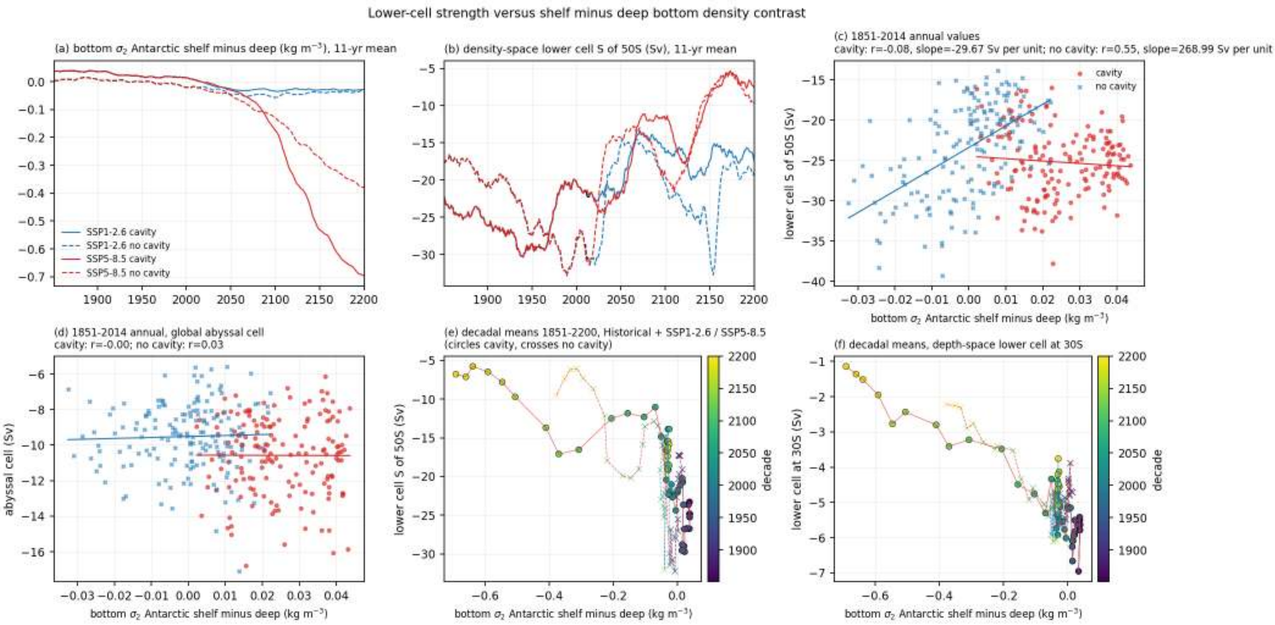

Figure 127: moc_vs_density_contrast (moc). As moc_vs_convection but with the bottom sigma2 contrast between the Antarctic shelf and the deep Southern Ocean (ant_shelf minus ant_deep from the ts cache, cavity nodes excluded) as the driver. The shelf is denser than the deep ocean by only 0.04 (cavity, early) and 0.01 (no cavity, early) kg m-3 and the contrast vanishes or reverses during the historical period. The annual correlations with the lower cell are 0.2-0.35 in the cavity run and weakly negative in the no-cavity run.

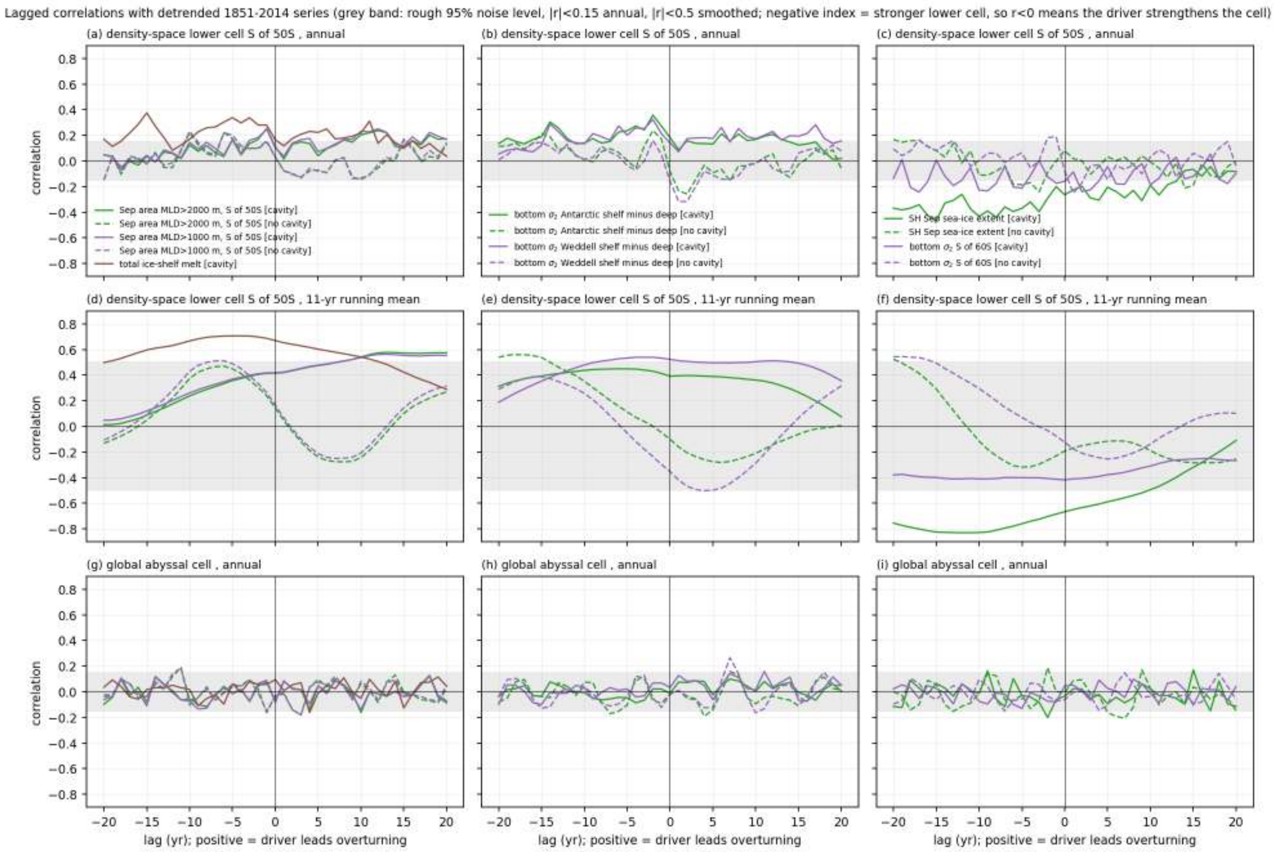

Figure 128: moc_lagcorr_hist (moc). Lagged correlations between detrended historical (1851-2014) driver series and the density-space lower cell south of 50S (a-c annual, d-f 11-yr running means) and the global abyssal cell (g-i annual). Positive lag means the driver leads. Drivers are the September convection area (MLD $>$ 2000 m and $>$ 1000 m), the total ice-shelf melt (cavity only), the shelf minus deep bottom density contrast (all Antarctica and Weddell), the September SH sea-ice extent and the bottom sigma2 south of 60S. Since the lower-cell index is negative, $r < 0$ means that a larger driver value goes with a stronger cell. On annual scales no driver explains more than 15 percent of the variance in either set. On decadal scales the cavity lower cell co-varies with sea-ice extent ($r = -0.67$ at lag 0, more ice with a stronger cell) and with ice-shelf melt ($r = +0.67$, more melt with a weaker cell) but these series have only about 15 degrees of freedom and share the historical trend structure. No driver shows a consistent decadal relation in the no-cavity run.

11 AABW volume census and deep-ocean change

Findings

11.1 Findings, theme census (AABW volume census, water-mass classes, deep-ocean change)

Scripts: analyses/census/fine_census.py (SLURM, 0.005 kg m-3 sigma2 census per year from raw temp/salt/density_dMOC, output cache/census/*fcens.nc), plot_census.py, plot_deep.py, plot_maps.py, clib.py. Numbers are in analyses/census/stats*{census,deep,maps}.json.

11.1.1 Definitions and data caveats

1. Density variable. The model output density_dMOC is written as float32 of the full density (about 1036.9 kg m-3) and is therefore quantised in steps of 1.2e-4 kg m-3 (only 424 distinct values in the deep Southern Ocean). Its abyssal distribution is also much narrower (10th to 90th percentile 36.909-36.921 below 2000 m south of 30S) than TEOS-10 sigma2 computed from the same T and S (36.896-36.970). Together with the near-homogeneity of the abyss this makes any fixed threshold on density_dMOC produce sawtooth jumps of 30 percent from year to year. The census therefore uses TEOS-10 sigma2 from gsw (same as the overview theme). Both versions are stored in cache/census (vol/T/S/Z model density, vol_g/T_g/S_g/Z_g gsw). The coarse cache census cens_vol (89 FESOM std_dens bins) puts 40 percent of the global volume into the single bin 36.900-36.920 and cannot resolve the abyssal classes.

2. Model AABW class. The observed AABW class (sigma2 > 37.16) is empty in the model (overview theme), so the class is defined model-relative as TEOS-10 sigma2 >= 36.92. This is the lower edge of the abyssal mode at present day and holds 24 (cavity) and 23 (no cavity) percent of the global volume, 34 and 31 percent south of 30S, 70 and 66 percent south of 60S, with mean properties 1.88-1.99 degC and 34.67-34.69 psu at 3200 m. Sensitivity thresholds 36.90, 36.93, 36.94, 36.96 were tested. The densest 10 percent of the volume south of 30S lies above 36.945 (cavity) and 36.935 (no cavity) at pd. Water above 36.96 is absent south of 30S at pd (0.15 and 0.05 percent) and disappears before 1950. The shelf (ant_shelf) is the only region with denser classes (10 percent above 37.075 cavity, 37.01 no cavity).

3. Drift. Both control states drift. The abyss below 2000 m warms by 0.081 (cavity) and 0.077 K per century (no cavity) globally and 0.094 and 0.069 K per century south of 30S over 1851-2014 with no sign of slowing, the global deep OHC grows by 540 ZJ between 1851 and 2100 identically in all four scenarios, and the AABW class volume shrinks by 25 (cavity) and 27 percent (no cavity) between 1851-1880 and 1985-2014. Every deep quantity below must be read against this drift. Drift-corrected numbers (historical trend times period separation) are given where relevant (stats dTdc_*).

11.1.2 Findings

4. Present-day AABW class volume. 315 (cavity) and 297 (no cavity) x 1e15 m3 globally, 142 and 129 south of 30S, 49 and 46 south of 60S, 14.3 and 13.6 in the Weddell Sea, 6.3 and 5.7 in the Ross Sea, 36 and 32 in the Atlantic, 40 and 34 in the Indian and 66 and 63 in the Pacific sector south of 30S. The cavity run has 6 (global) to 10 percent (south of 30S) more dense water, and its densest classes are 0.1-0.15 degC colder and 0.01-0.02 kg m-3 denser. For the densest 10 percent south of 30S the difference is 36.945 versus 36.935. The no-cavity run with its Maud Rise convection does NOT produce more dense-class volume, the convection ventilates a warm abyss and the cavity run's colder slope and bottom water (Weddell 0.25 K, deep Southern Ocean 0.2 K) wins in the census.

5. Fate of the class. The class volume south of 30S falls by 49 percent (cavity) and 58-61 percent (no cavity) between pd and 2071-2100 and by 87-94 percent by 2171-2200 in every scenario. The Atlantic and Indian sectors and the Weddell Sea lose the class completely between 2100 and 2150, the Pacific sector and the Ross Sea keep 1.3-3 x 1e15 m3 (Ross) and 8-16 x 1e15 m3 (Pacific) in 2171-2200. Exponential decay times after 2015 are 60-80 years south of 30S, 14-30 years in the Weddell Sea, 100-240 years in the Ross Sea. The e-folding and percent numbers are the same within 10 percent for SSP1-2.6 and SSP5-8.5, so the loss is the continuation of the spin-up drift plus a threshold-crossing effect of the homogeneous abyssal mode, not a scenario response. The same holds for the density of the densest 10 percent of the Southern Ocean, which falls from 36.965 (cavity) and 36.955 (no cavity) in 1851 to 36.905 in 2200 at 0.02 kg m-3 per century in all scenarios.

6. Cavities and the decline rate. The no-cavity class empties earlier (10-25 years in the Atlantic and Indian sectors and the Weddell Sea, decline rates 2015-2100 of 7.7-8.7 percent per decade south of 30S/60S against

6.2-6.7 with cavities, 11.5-12.3 against 6.8-7.7 in the Weddell Sea and Atlantic sector), because it starts lighter. Expressed as a lag, the no-cavity abyss is 30-50 years ahead of the cavity abyss on the same drift curve. The offset between the two sets closes after 2100 in all scenarios. Cavities therefore change the volume of dense water by 6-10 percent and delay its loss by a few decades but do not change the character of the decline.

7. Deep-ocean heat uptake. Below 2000 m the global change is drift dominated (end minus pd 0.07-0.08 K, far minus pd 0.14-0.19 K, drift-corrected far 0.0-0.04 K). South of 30S below 2000 m the cavity run warms more in every scenario, 0.13-0.14 K (end) and 0.23-0.34 K (far) against 0.09-0.12 and 0.15-0.29 K without cavities, drift-corrected far 0.06 (SSP1-2.6) to 0.17 K (SSP5-8.5) with cavities against 0.03 to 0.16 K without. In OHC the Southern Ocean below 2000 m gains 415 (cavity) and 341 ZJ (no cavity) by 2171-2200 in SSP5-8.5 and 320 and 221 ZJ in SSP1-2.6, i.e. 70-100 ZJ more with cavities, while the 0-1000 m layer gains 90 ZJ less with cavities in SSP5-8.5 (1329 versus 1418 ZJ) consistent with the weaker Southern Ocean surface warming of the cavity set found by the atmos theme. Part of the deep surplus is drift (the cavity run drifts 0.025 K per century faster below 2000 m south of 30S) and part is a real response (the Indian and Pacific sectors at 3000 m warm 0.05-0.1 K more by 2171-2200). The Atlantic below 2000 m cools after 2100 in all scenarios (far minus pd drift-corrected -0.05 to -0.08 K) following the AMOC weakening, slightly more so with cavities. At 1000-2000 m the scenario response dominates everywhere (global far 0.32 K SSP1-2.6 to 0.77 K SSP5-8.5, Atlantic 1.0 K) and the sets differ by less than 0.03 K.

8. Depth structure. Warming above 0.5 K reaches 1000 m globally by 2100 and 2000 m by 2200 in SSP5-8.5 and stays within 1000 m in SSP1-2.6. The cavity minus no-cavity anomaly difference is negative (up to 0.6 K in SSP5-8.5, 0.2-0.4 K in SSP1-2.6) in the upper 500 m south of 30S after 2050-2100 and positive (0.05-0.1 K) below 2000 m, i.e. the cavity set stores relatively more of its Southern Ocean heat gain at depth.

9. Maps. Temperature change 2171-2200 minus pd in SSP5-8.5 at 3000 m is 0.16 (cavity) and 0.15 K (no cavity) globally and 0.32 and 0.27 K south of 30S, with warming of 0.3-0.8 K in the Southern Ocean and Indo-Pacific and cooling of up to 0.5 K in the Atlantic. Bottom warming is 0.67 and 0.65 K globally and 0.62 and 0.61 K south of 60S. The cavity run warms more at 3000 m south of 40S but less on the Antarctic shelves and in the Weddell bottom layer and cools more in the deep North Atlantic. SSP2-4.5 gives almost the same 3000 m change (0.16 and 0.15 K) and half the bottom change, again indicating that the 3000 m signal is mostly drift.

10. Class properties. The class mean temperature south of 30S rises from 1.81 (1851) to 1.88 degC (pd) in the cavity run while the no-cavity class sits at 1.95-1.99 degC, both reach 2.0 degC by 2100 and 2.01-2.03 by 2200 in SSP5-8.5. Salinity stays within 34.67-34.69. The class mean depth deepens from 3200 m to 3600 m (end) and 3900-4500 m (far) because the class retreats into the deepest basins.

11.1.3 Caveats and suspicious points

11. Single members, and the scenario chains share the same 2015 state per set, so scenario to scenario differences in the deep ocean are well defined but cavity minus no-cavity differences in the upper ocean include internal variability (visible as stripes in the Hovmoeller difference panels).

12. The hard upper edge of the abyssal density distribution and its drift mean that percent-change and e-folding numbers depend on the threshold and on the period. Use the density of the densest 10 percent (figure aabw_class_properties_timeseries d, h) as the robust lightening index.

13. Layer volumes and OHC exclude cavity nodes (patched cache), the census includes them (by design). Basin sector regions are full basins extended to Antarctica (atl_sector etc.) in the OHC and Hovmoeller figures and basins south of 30S (atl_so30 etc.) in the census figures.

14. The 3000 m and bottom maps regrid each mesh by nearest neighbour to 1 degree before differencing, so the difference panels show mesh artefacts on steep slopes and on the shelf where the cavity mesh has ice shelf instead of open ocean.

11.1.4 Suggested story lines

15. The model has no AABW in the observational sense, its abyss is a single CDW-like mode that drifts lighter at 0.02 kg m-3 per century and warms at 0.08 K per century. Ice-shelf cavities make this mode 6-10 percent larger and 0.01-0.02 kg m-3 denser and delay its disappearance by 30-50 years but do not alter the drift. The paper should present the census as a diagnosis of the abyssal state and its sensitivity to cavities, and keep scenario statements for the upper 2000 m.

16. Cavities shift Southern Ocean heat uptake downward in the warming scenarios, 70-100 ZJ more below 2000 m and 90 ZJ less above 1000 m south of 30S by 2200 in SSP5-8.5, in line with the weaker surface warming of the cavity set. Whether this is cavity physics or the drift difference needs the difference of the drift-corrected series, which is 0.0-0.03 K and therefore marginal.

Figures

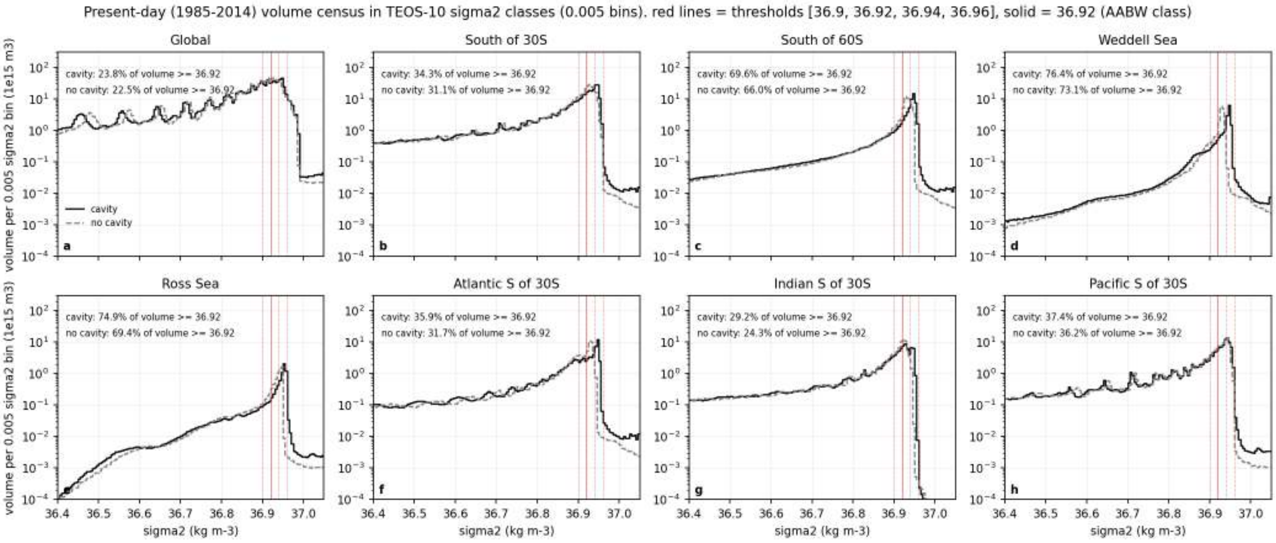

Figure 129: census_pd_distribution (census). Present-day (1985-2014) volume per 0.005 kg m-3 sigma2 class (log scale) for the global ocean, south of 30S, south of 60S, the Weddell Sea, the Ross Sea and the Atlantic, Indian and Pacific sectors south of 30S, cavity (black solid) and no cavity (grey dashed). Red vertical lines mark the thresholds 36.90, 36.92 (solid, the AABW class used in this theme), 36.94 and 36.96. The abyss of both runs is one nearly homogeneous mode with a sharp upper edge. In the cavity run the mode peaks at 36.94-36.95 and ends at 36.955, in the no-cavity run it peaks at 36.92-36.93 and ends at 36.945. The class sigma2 >= 36.92 holds 24 (cavity) and 23 (no cavity) percent of the global volume, 34 and 31 percent south of 30S, 70 and 66 percent south of 60S. The densest 10 percent of the volume south of 30S lies above 36.945 (cavity) and 36.935 (no cavity). Essentially no water exists above 36.96 south of 30S (0.15 and 0.05 percent).

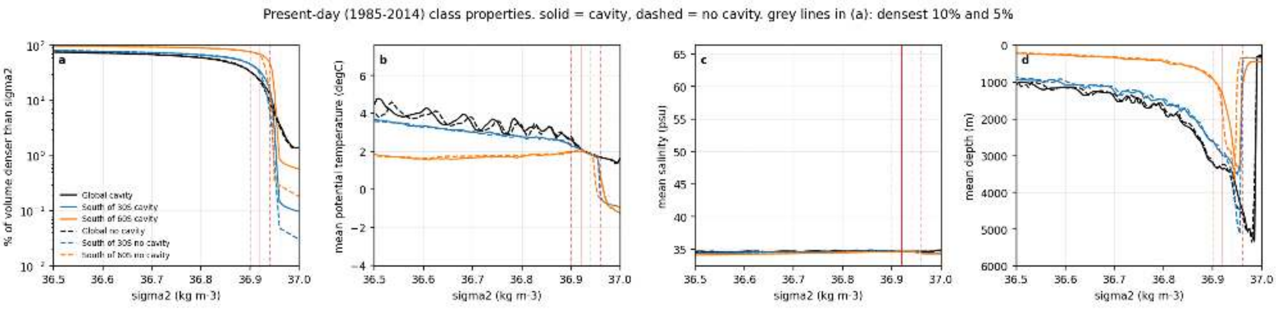

Figure 130: census_class_properties (census). Present-day class properties as a function of sigma2 for the global ocean (black), south of 30S (blue) and south of 60S (orange), cavity solid and no cavity dashed. (a) Percentage of the volume denser than a given sigma2 (log scale, grey lines at 10 and 5 percent). (b) Volume-weighted mean potential temperature, (c) salinity and (d) mean depth of each 0.005 class. The abyssal mode (36.90-36.96) has 1.8-2.0 degC and 34.67-34.69 psu at 3000-4500 m in both runs, i.e. the model AABW class is CDW-like. The cavity run has its densest classes 0.1-0.15 degC colder than the no-cavity run, which is why its mode is 0.01-0.02 kg m-3 denser.

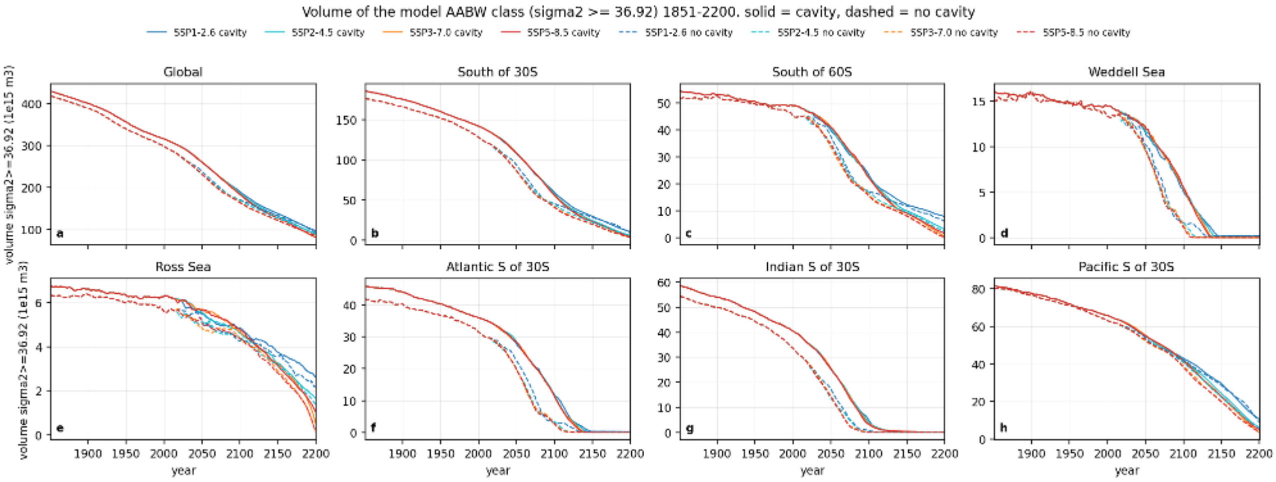

Figure 131: aabw_volume_timeseries (census). Volume of the AABW class (sigma2 >= 36.92) 1851-2200 for the global ocean, south of 30S, south of 60S, Weddell Sea, Ross Sea and the three sectors south of 30S, all four scenario chains and both sets. Present-day volumes are 315 (cavity) and 297 (no cavity) x 1e15 m3 globally and 142 and 129 x 1e15 m3 south of 30S. The class shrinks by 25 (cavity) and 27 (no cavity) percent already between early and pd, and by 49 (cavity) and 58-61 (no cavity) percent between pd and end south of 30S. The decline is almost identical in all four scenarios (far minus pd south of 30S is -89 to -93 percent cavity, -87 to -94 percent no cavity), so it is a drift of the abyss, not a scenario response. The no-cavity class empties 10-25 years earlier in the Atlantic and Indian sectors and in the Weddell Sea, the Pacific sector and the Ross Sea keep dense water longest.

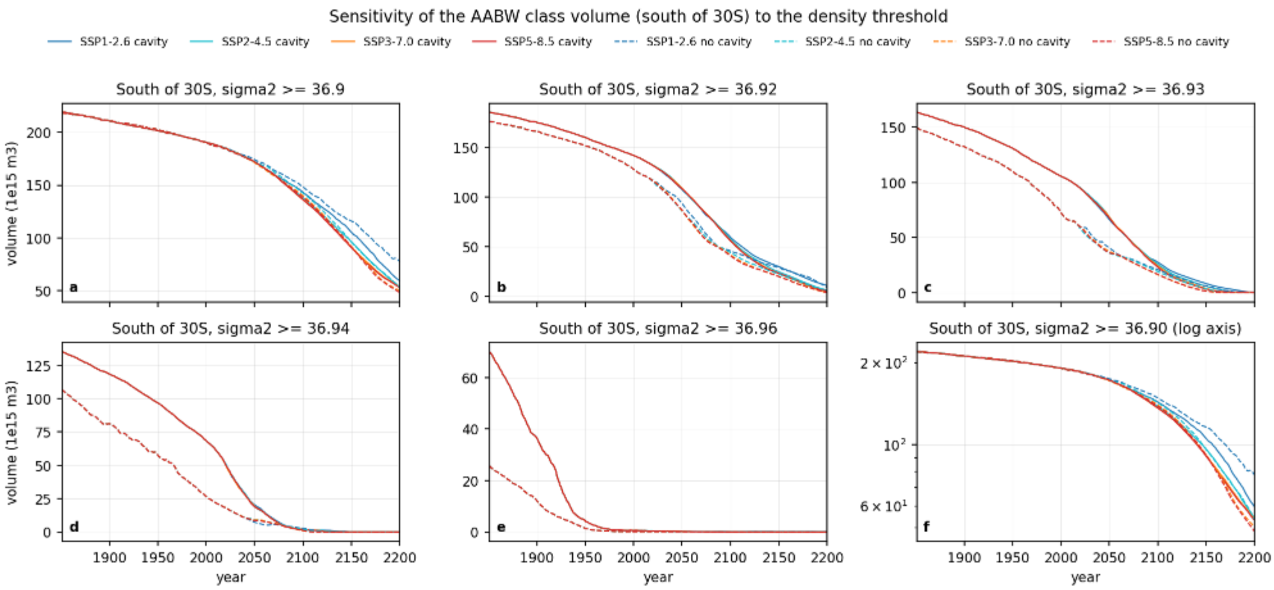

Figure 132: aabw_volume_threshold_sensitivity (census). As the previous figure for the volume south of 30S with thresholds 36.90, 36.92, 36.93, 36.94 and 36.96 (panel f repeats 36.90 on a log axis). The timing of the collapse moves with the threshold because the abyssal mode drifts to lower density at about 0.02 kg m-3 per century and crosses each threshold in turn, but the statements above (scenario independence, earlier loss without cavities, larger dense volume with cavities) hold for all thresholds. Water above 36.96 disappears before 1950 in both sets.

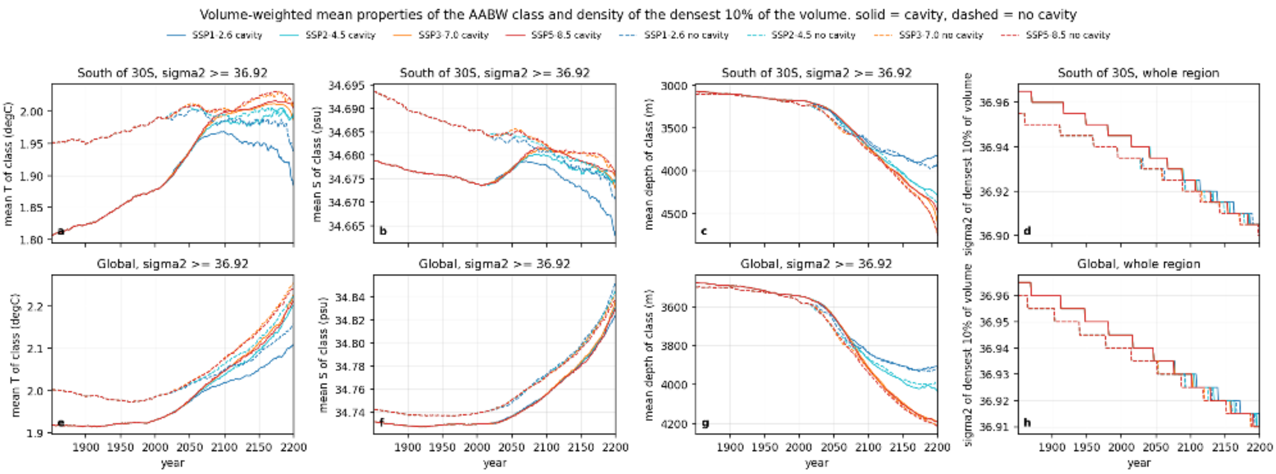

Figure 133: aabw_class_properties_timeseries (census). Volume-weighted mean potential temperature (a, e), salinity (b, f) and mean depth (c, g) of the AABW class (sigma2 >= 36.92) south of 30S (top) and globally (bottom), and (d, h) the sigma2 below which the densest 10 percent of the regional volume lies (0.005 resolution). The class south of 30S is 1.81 degC in 1851 and 1.88 degC at pd in the cavity run but 1.95-1.99 degC in the no-cavity run, it sinks from 3200 m to 3600 m (end) and 3900-4500 m (far) as the lighter upper part is lost. The densest 10 percent of the Southern Ocean volume lightens from 36.965 (cavity) and 36.955 (no cavity) in 1851 to 36.905 in 2200, by 0.06 kg m-3 in 350 years, at almost the same rate in all scenarios, and the 0.01 kg m-3 offset between the sets closes after about 2100.

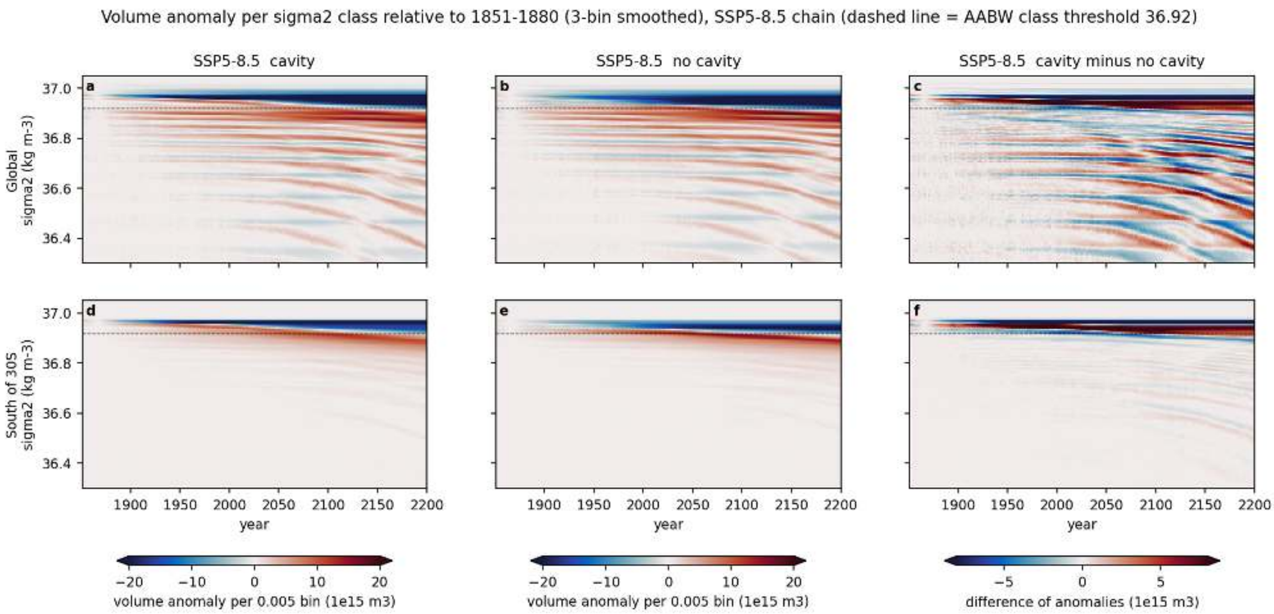

Figure 134: hov_density_ssp585 (census). Hovmoeller diagram of the volume anomaly per 0.005 sigma2 class relative to 1851-1880 (3-bin running mean in density) for the SSP5-8.5 chain, global (top) and south of 30S (bottom), cavity (left), no cavity (middle) and the difference of the anomalies (right). The dashed line is the 36.92 threshold. The abyssal mode loses volume above 36.93 and gains below it from 1851 on, the band of gain migrates steadily to lower density (36.90-36.92 by 2100, 36.85-36.90 by 2200), and after 2050 volume is lost across the whole column above 36.7 as the upper ocean warms. The difference panel shows the cavity run keeping more volume in the 36.93-36.96 classes until about 2100.

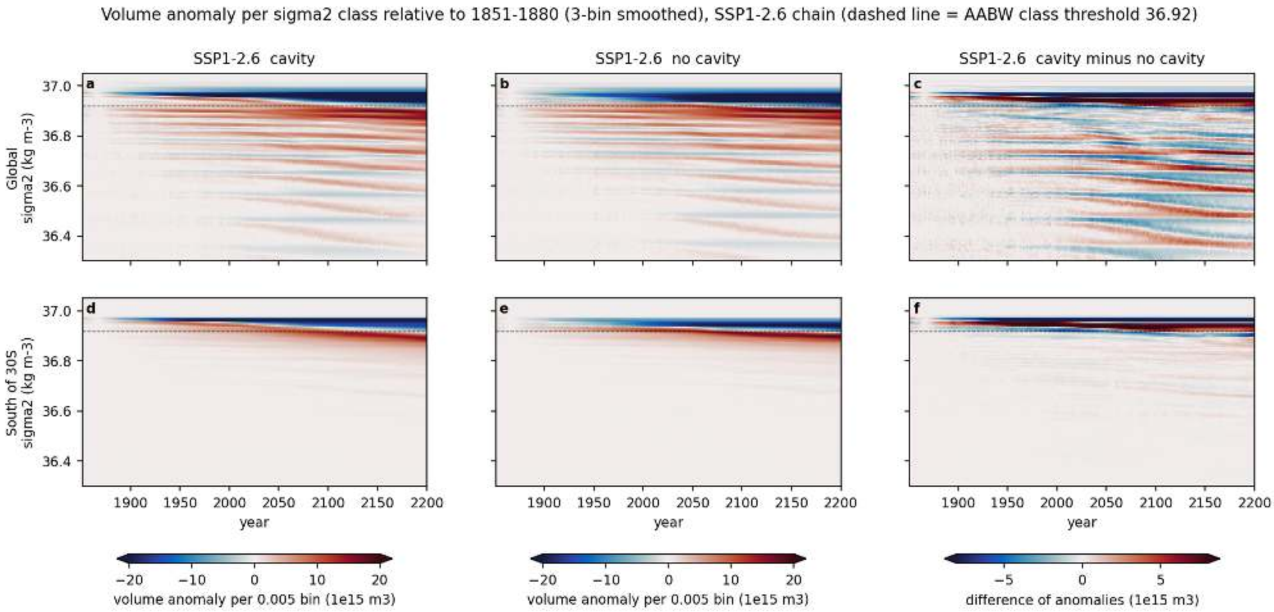

Figure 135: hov_density_ssp126 (census). As the previous figure for the SSP1-2.6 chain. The deep part of the diagram (sigma2 > 36.85) is nearly indistinguishable from SSP5-8.5, the scenario signal is confined to classes lighter than 36.8.

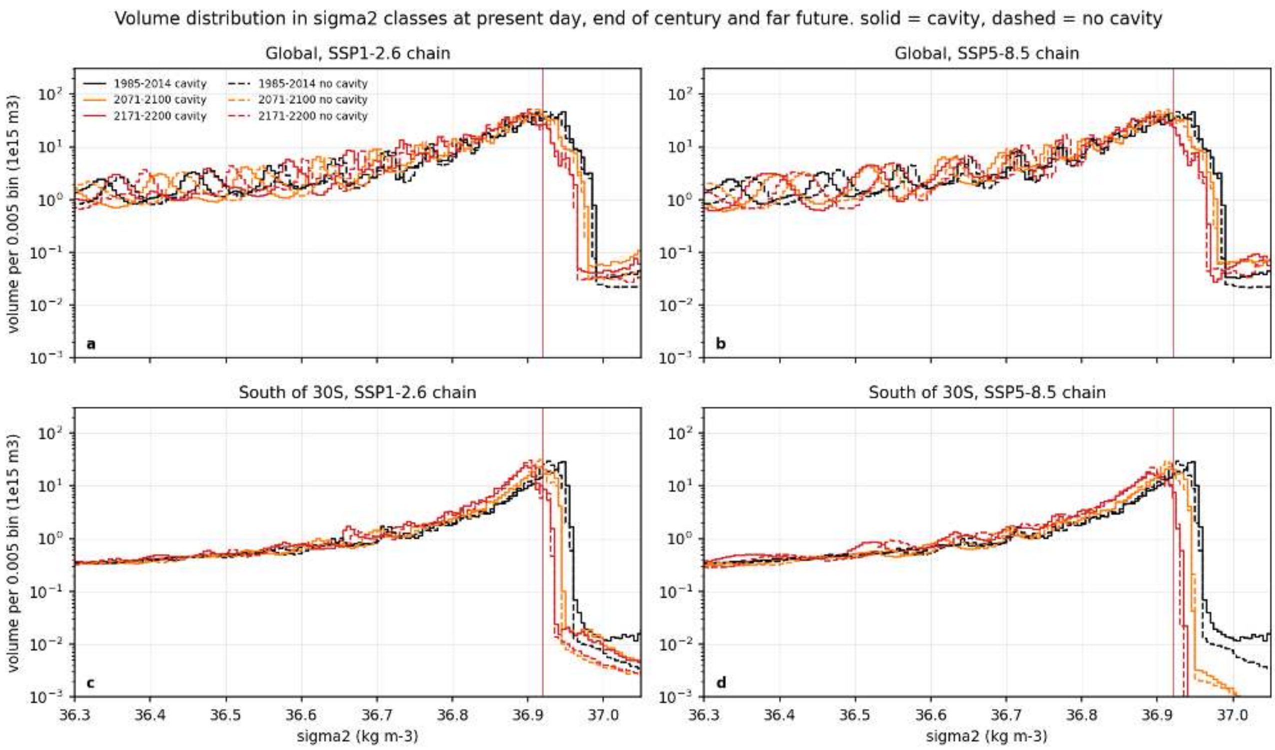

Figure 136: census_distribution_scenarios (census). Volume per 0.005 sigma2 class at pd (black), end (orange) and far (red) for the SSP1-2.6 (left) and SSP5-8.5 (right) chains, global (top) and south of 30S (bottom), cavity solid and no cavity dashed. The abyssal mode shifts from 36.94-36.95 (pd) to 36.90-36.92 (end) and 36.87-36.90 (far) in both sets and both scenarios. In SSP5-8.5 the classes between 36.3 and 36.8 lose up to an order of magnitude of volume by 2171-2200, in SSP1-2.6 they are nearly unchanged.

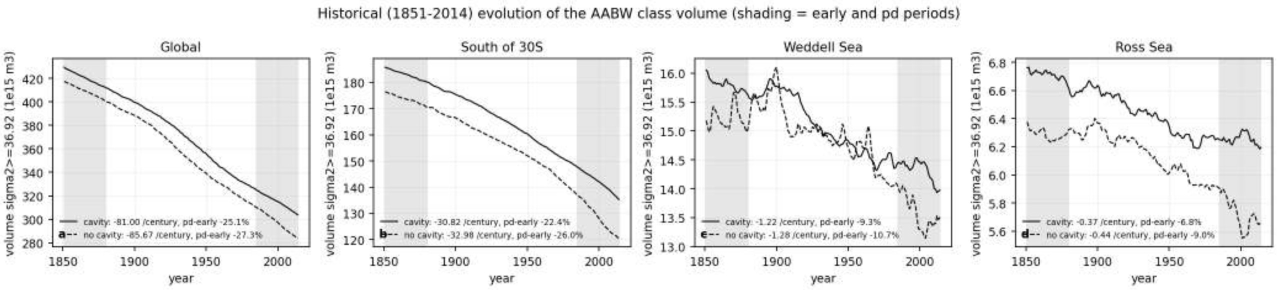

Figure 137: hist_drift_aabw_volume (census). Historical (1851-2014) evolution of the AABW class volume (sigma2 >= 36.92) for the global ocean, south of 30S, the Weddell Sea and the Ross Sea, cavity solid and no cavity dashed, with linear trends and the pd minus early change. The class loses 81 (cavity) and 86 (no cavity) x 1e15 m3 per century globally and 31 and 33 x 1e15 m3 per century south of 30S, i.e. 22-27 percent between early and pd. The Weddell and Ross Sea volumes fall by 9-11 and 7-9 percent. This is a spin-up drift of the abyss common to both sets.

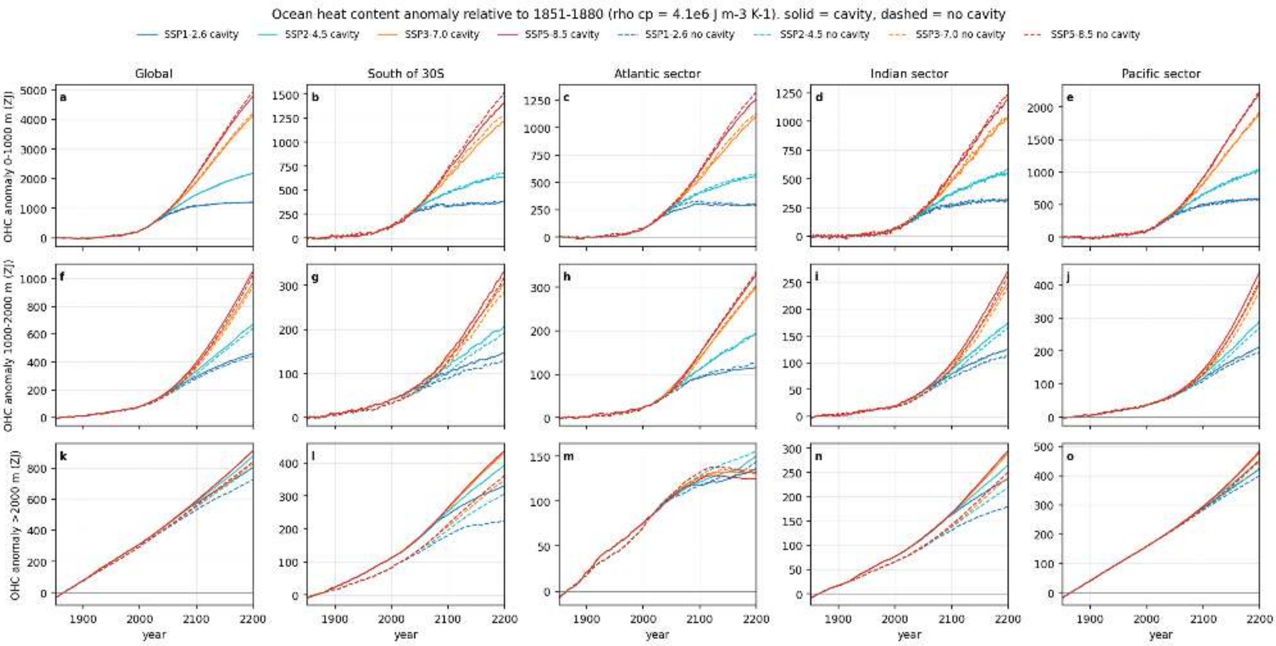

Figure 138: ohc_timeseries (census). Ocean heat content anomaly relative to 1851-1880 (ZJ, rho cp = 4.1e6 J m-3 K-1) for 0-1000 m (top), 1000-2000 m (middle) and below 2000 m (bottom), for the global ocean, south of 30S and the Atlantic, Indian and Pacific sectors (full basins including their Southern Ocean part), all scenario chains and both sets. Cavity nodes are excluded from the layer means. Global 0-1000 m uptake by 2171-2200 is 1190-1200 ZJ (SSP1-2.6) and 4440 (cavity) to 4560 (no cavity) ZJ (SSP5-8.5). Below 2000 m the curves of all scenarios lie on top of each other until about 2100 (global 540 ZJ by end, 770-860 ZJ by far), which identifies a scenario-independent drift of 0.08 K per century. South of 30S below 2000 m the cavity run gains 30-70 ZJ more than the no-cavity run in all scenarios.

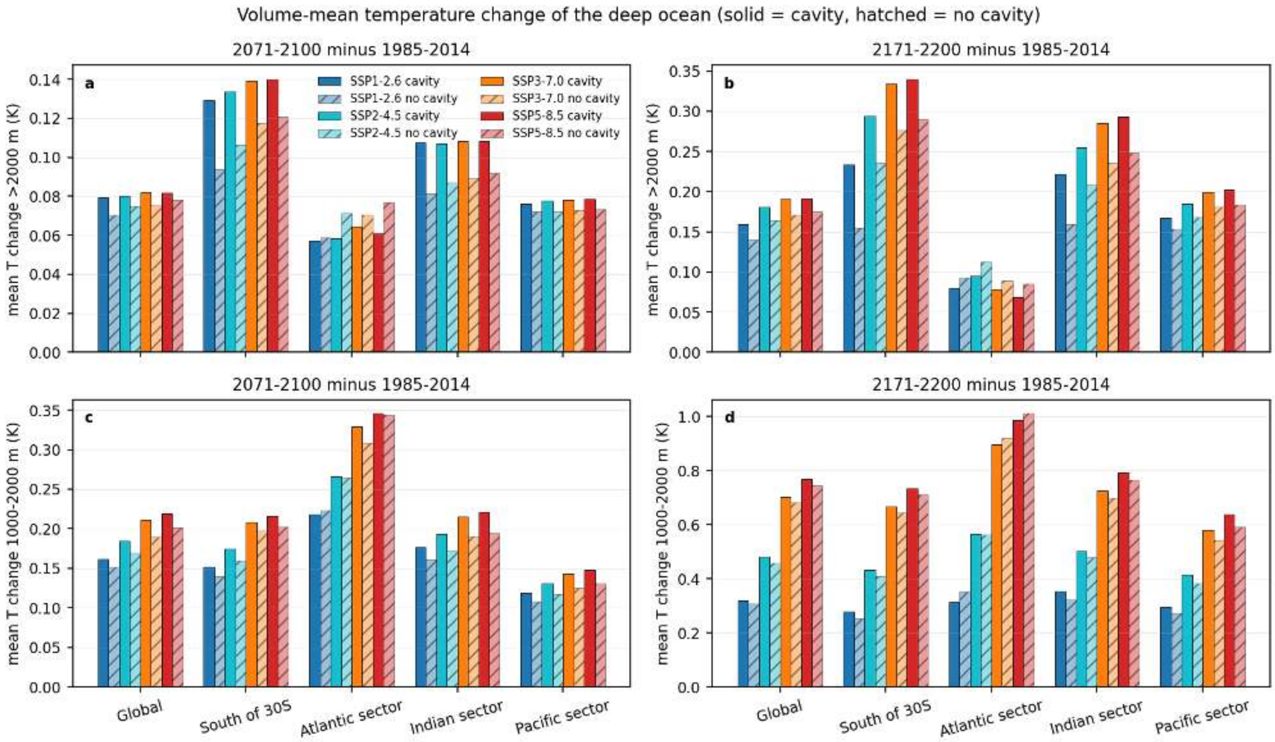

Figure 139: deep_warming_bars (census). Volume-mean temperature change below 2000 m (top) and 1000-2000 m (bottom) between pd and end (left) and pd and far (right) per region, scenario and set (solid bars cavity, hatched bars no cavity). Values are not drift corrected. Below 2000 m the global change is 0.07-0.08 K (end) and 0.14-0.19 K (far) with almost no scenario dependence, south of 30S it is 0.13-0.14 K (cavity) versus 0.09-0.12 K (no cavity) at end and 0.23-0.34 K versus 0.15-0.29 K at far. After subtracting the historical drift the far deep warming south of 30S is 0.06 (SSP1-2.6) to 0.17 K (SSP5-8.5) and the global one 0.0-0.04 K, while the Atlantic below 2000 m cools by 0.05-0.08 K (AMOC weakening). At 1000-2000 m the scenario response dominates (global far 0.32 K SSP1-2.6, 0.77 K SSP5-8.5; Atlantic up to 1.0 K).

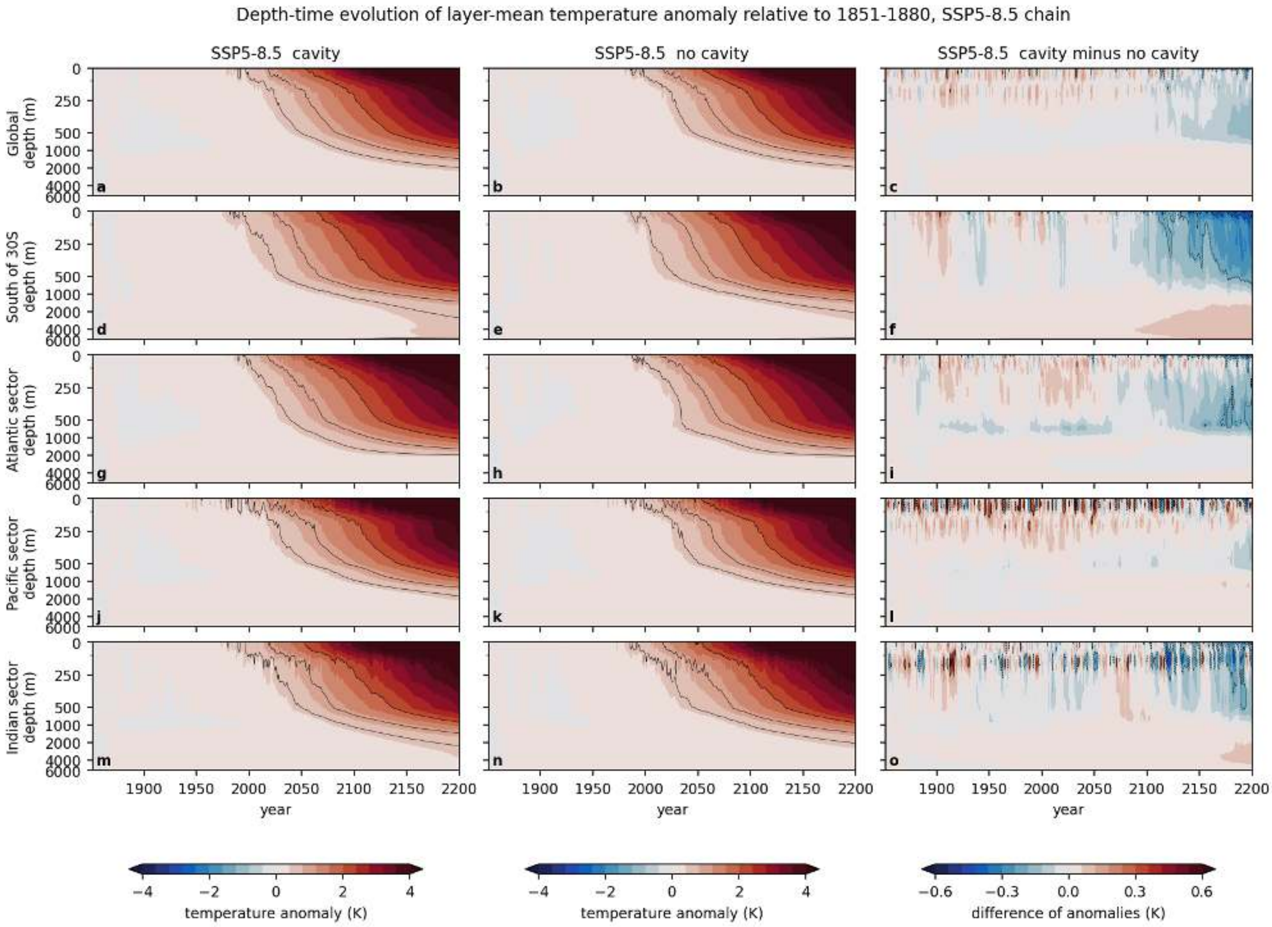

Figure 140: hov_depth_temp_ssp585 (census). Depth-time Hovmoeller of the layer-mean temperature anomaly relative to 1851-1880 for the SSP5-8.5 chain, global, south of 30S, Atlantic, Pacific and Indian sectors (rows), cavity (left), no cavity (middle) and cavity minus no cavity (right, note the smaller colour range). The depth axis is stretched above 500 m. Warming of more than 0.5 K reaches 1000 m globally by 2100 and 2000 m by 2200, the abyss warms by 0.1-0.3 K through drift. The cavity run warms less in the upper 500 m south of 30S after 2100 (by up to 0.6 K, the sea-ice and surface feedback discussed by the atmos theme) but slightly more below 2000 m (0.05-0.1 K).

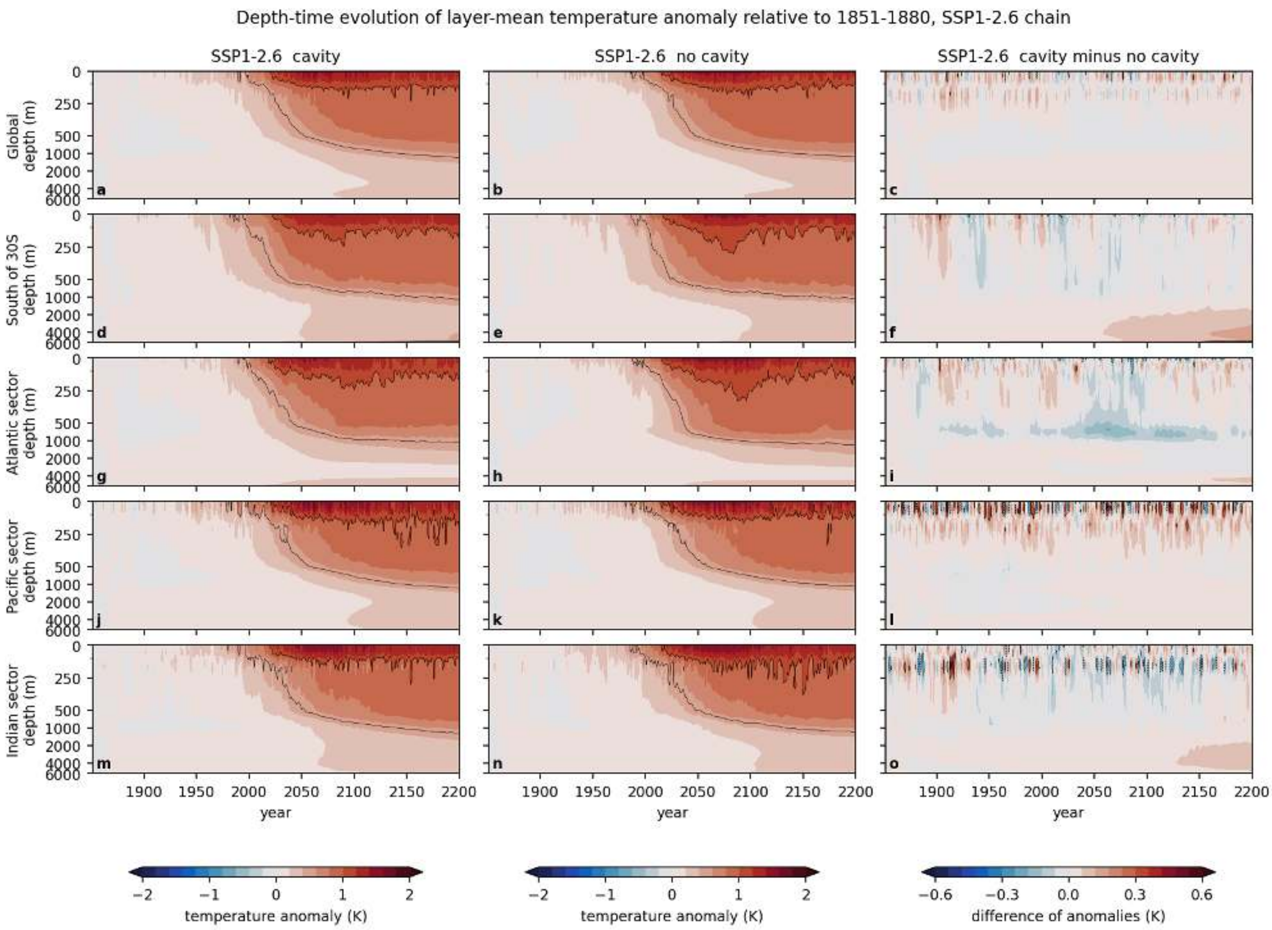

Figure 141: hov_depth_temp_ssp126 (census). As the previous figure for the SSP1-2.6 chain. Warming above 0.5 K stays within the upper 1000 m, the deep signal is the same drift as in SSP5-8.5. The cavity minus no-cavity difference in the upper 500 m south of 30S is 0.2-0.4 K cooler in the cavity run after 2050 and the difference below 2000 m is 0.05 K warmer.

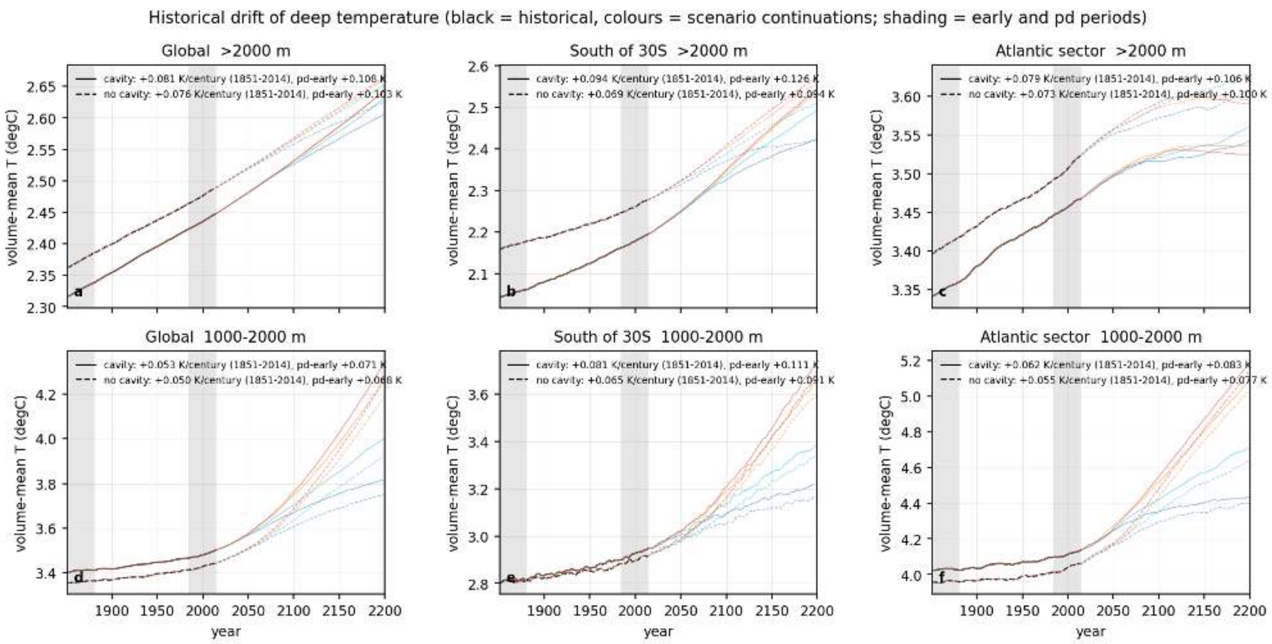

Figure 142: hist_drift_deepT (census). Volume-mean temperature below 2000 m (top) and at 1000-2000 m (bottom) for the global ocean, south of 30S and the Atlantic sector, 1851-2200, black for the historical runs and faded colours for the scenario continuations, with the 1851-2014 linear trend and the pd minus early change in the legend. Both control states drift warm below 2000 m at 0.08 (cavity) and 0.077 K per century (no cavity) globally and 0.094 and 0.069 K per century south of 30S, with no sign of slowing down over 1851-2014. The cavity run is 0.04 K (global) and 0.12 K (south of 30S) colder below 2000 m in 1851 and converges towards the no-cavity run during the 21st century.

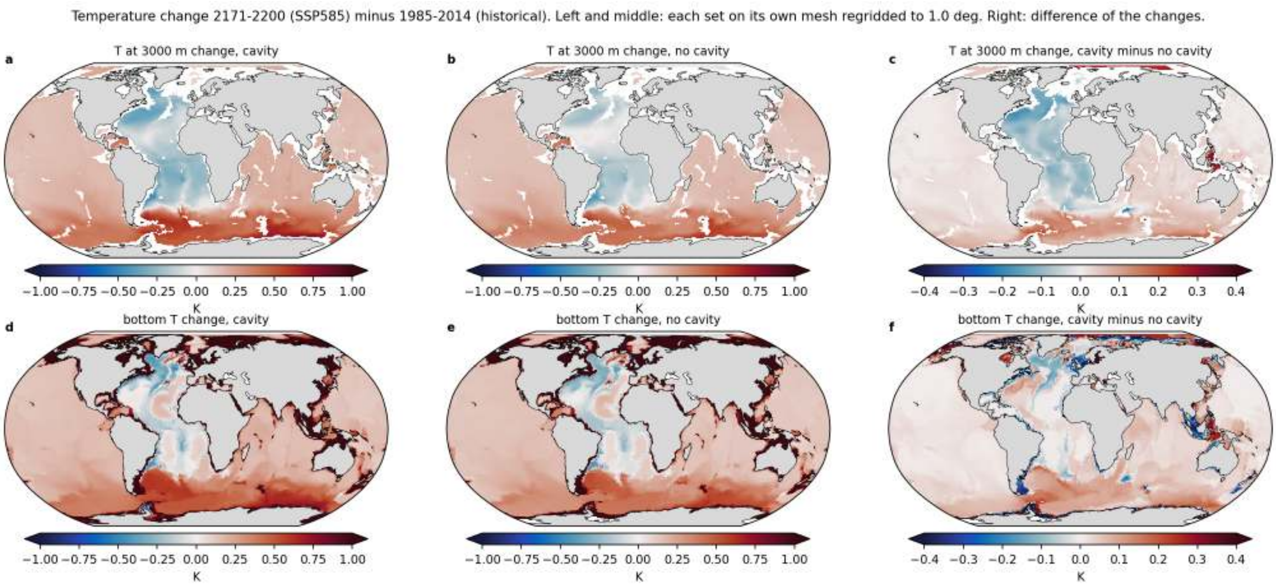

Figure 143: map_deepT_change_ssp585 (census). Temperature change 2171-2200 (SSP5-8.5) minus 1985-2014 at 3000 m (top) and at the bottom (bottom row) for the cavity run (left), the no-cavity run (middle), both regridded to 1 degree, and the difference of the two changes (right). The Southern Ocean and Indo-Pacific warm by 0.3-0.8 K at 3000 m while the Atlantic cools by up to 0.5 K (AMOC weakening). Area means at 3000 m are 0.16 (cavity) and 0.15 K (no cavity) globally and 0.32 and 0.27 K south of 30S. Bottom warming is 0.67 and 0.65 K globally (shelves included) and 0.62 and 0.61 K south of 60S. The cavity run warms 0.05-0.1 K more at 3000 m south of 40S but less on the Antarctic shelves and in the Weddell Sea bottom layer, and cools more in the deep North Atlantic.

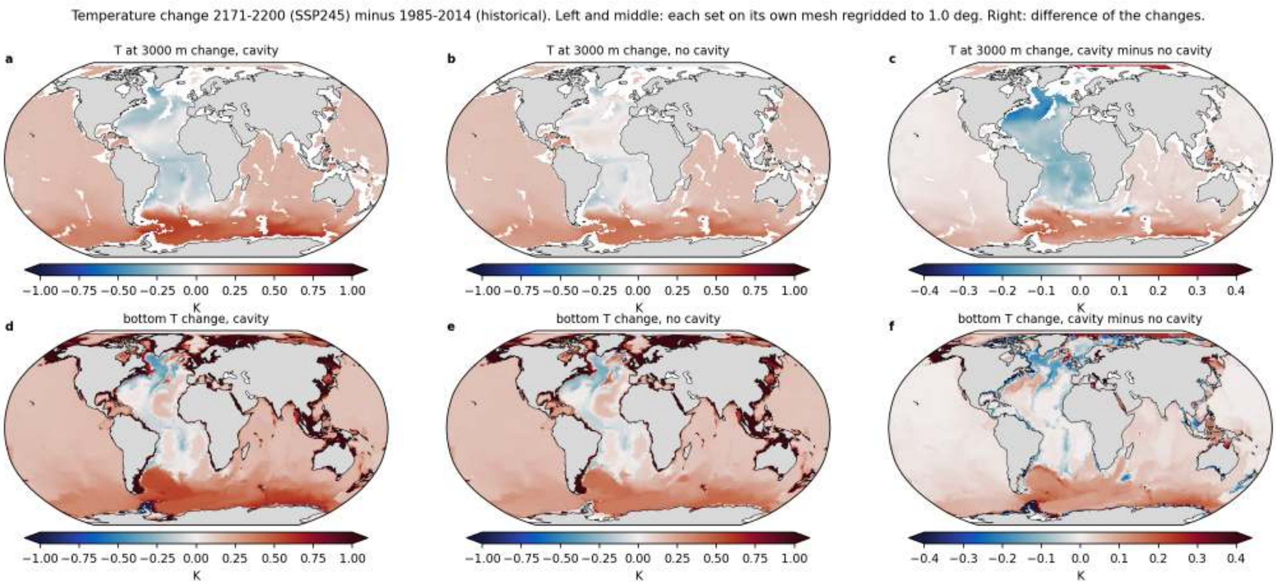

Figure 144: map__deepT__change__ssp245 (census). As the previous figure for SSP2-4.5. The pattern is the same with about half the amplitude at the bottom (global 0.33 and 0.31 K, south of 30S 0.35 and 0.29 K) and nearly the same amplitude at 3000 m (0.16 and 0.15 K globally), which confirms that the 3000 m change is largely drift.

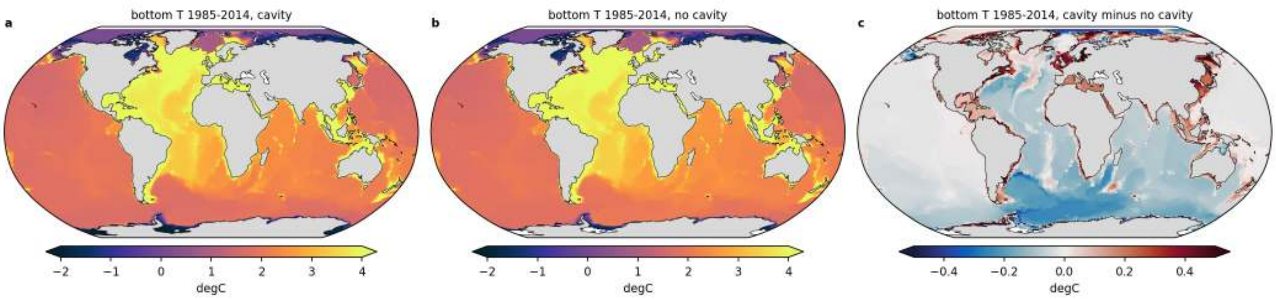

Figure 145: map__bottomT__pd (census). Present-day (1985-2014) bottom potential temperature for the cavity run (a), the no-cavity run (b) and the difference (c, regridded to 1 degree). The abyss is 1.5-2.5 degC almost everywhere south of 30S. The cavity run is 0.1-0.3 K colder in the deep Weddell and Ross Seas and along the Antarctic slope, and warmer only on parts of the shelf where the cavity mesh replaces open ocean by ice shelf.

12 Abyssal signal propagation, trends and emergence

Findings

12.1 Propagation theme: findings (abyssal signal propagation, trends, time of emergence)

Scripts: `analyses/propagation/prop_compute.py` (SLURM, per set), `prop_thickness.py`, `prop_plots.py`; derived data in `cache/propagation/prop_{cav,nocav}.nc`, `thick_{cav,nocav}.nc`; numbers in `analyses/propagation/s`
All area means exclude cavity nodes; "deep" means bottom depth > 3000 m (sector Hovmoellers, emergence) or > 3500 m (regional boxes, pathways). "Drift-corrected" = per-node linear 1851-1950 trend extrapolated and removed.

12.1.1 A. Data and diagnostic caveats (read first)

1. Deep-ocean drift dominates the raw abyssal trends. The area-mean bottom temperature of the deep (>3000 m) sea floor rises by 0.089 (cavity) and 0.083 degC per century (no cavity) already in 1851-1950, when forcing is negligible, and the global deep mean keeps rising at the same 0.08 to 0.10 degC per century through 2200 in every scenario (fig01, fig14). Relative to 1851-1880 the global deep mean anomaly is 0.20 degC in 2071-2100 and 0.29 degC in 2171-2200 under SSP5-8.5, but after drift correction it is 0.005 and 0.011 degC, i.e. the forced global deep mean change is essentially zero because Southern Ocean warming and Atlantic cooling cancel. The drift is uniform in space (histogram in fig09l, 0.085 degC per century median, interquartile range 0.07 to 0.10) which makes a per-node linear correction defensible, but the correction is linear and cannot capture a decelerating spin-up drift; the residual over 250 years of extrapolation is uncertain by a few hundredths of a degree. The 1000-yr spin-ups were clearly not equilibrated in the abyss. The drift south of 60S is 0.09 (cavity) versus 0.03 degC per century (no cavity) in 1851-1950, so the two spin-ups also differ in how far from equilibrium the Southern Ocean is. Recommendation: any abyssal trend or budget in this project should be quoted drift-corrected or with the 1851-1950 trend given alongside.

2. density_dMOC is not usable for fine density thresholds. The model output density_dMOC takes only ~285 distinct bottom values on 64000 deep nodes and is nearly constant (36.911 to 36.919) below 700 m in a deep Weddell profile where gsw sigma2 from the same T and S varies from 36.92 to 36.95; deep bottom values cluster at 36.918, 37.255 and 38.05. The census theme threshold "density_dMOC >= 36.90" therefore selects a coarse class that covers 40 percent of the ocean volume (5.3 x 10ˆ17 m3) and a 2700 m thick layer south of 60S at pd, and it cannot discriminate the cavity from the no-cavity abyss. All sigma2 quantities in this theme are recomputed with gsw (`diag.sigma2`, labelled sig2g). The census theme should be told.

3. The model abyss has no dense southern end member. gsw bottom sigma2 of nodes deeper than 3000 m lies between 36.94 and 36.98 from 70S to 60N at pd, with the deep Southern Ocean (36.94 to 36.95) marginally lighter than the North Pacific and North Atlantic abyss (36.97 to 36.98) (fig13). Any "AABW layer" defined by a density threshold therefore appears in the North Pacific and Arctic as well (fig12). This is consistent with the overview finding of a CDW-like abyss and an abyssal cell of only -7 Sv.

4. Emergence with a 2 sigma criterion is hypersensitive in the abyss. The interannual std of the detrended 1851-1950 bottom temperature is 0.003 degC (median of deep nodes, 10-90 percent range 0.0006 to 0.017 degC), so a 0.01 degC deviation is "significant". Without drift correction 80 to 85 percent of the deep sea floor has emerged by 1900 (fig09i). Results are given for the 2 sigma criterion (as requested) and for an absolute 0.05 degC (0.005 psu) criterion that represents a physically detectable change; the latter should be used for interpretation.

12.1.2 B. Trends (fig01 to fig05)

5. Historical 1951-2014. Deep sea floor area-mean trend 0.090 (cavity) and 0.085 degC per century (no cavity), i.e. identical to the drift; south of 60S 0.14 and 0.13 degC per century, so a forced component of about 0.05 to 0.1 degC per century is already present in the Southern Ocean by 1951-2014. Bottom salinity south of 60S changes by less than 0.01 psu per century.

6. SSP5-8.5 2015-2100. Deep sea floor south of 60S warms at 0.28 (cavity) and 0.22 degC per century (no cavity); per sector south of 60S: Atlantic 0.30/0.25, Pacific 0.21/0.16, Indian 0.38/0.32 degC per century. The Atlantic abyss north of 30S cools at -0.05 to -0.2 degC per century (area-mean Atlantic-sector trend 0.05 degC per century, with 40 percent of the deep Atlantic area showing a significant negative trend

against the drift). In 2101-2200 the Atlantic-sector deep mean trend is 0.03 (cavity) and -0.003 degC per century (no cavity), 55 to 57 percent of the area cooling significantly, while the Southern Ocean south of 60S still warms at 0.23/0.19 degC per century. Bottom salinity south of 60S increases by 0.020/0.023 psu per century (2015-2100), the deep Atlantic freshens by -0.011/-0.009 psu per century (2015-2100) and -0.017/-0.020 (2101-2200). gsw bottom sigma2 decreases everywhere in the Southern Ocean (-0.03 to -0.06 kg/m3 per century) and in the Atlantic abyss (freshening wins over cooling).

7. SSP1-2.6 2015-2100 gives 80 to 90 percent of the SSP5-8.5 Southern Ocean warming rate (0.26/0.18 degC per century south of 60S) and the same Atlantic cooling pattern. In 2101-2200 the no-cavity SSP1-2.6 abyss south of 60S cools slightly again (-0.008 degC per century, Atlantic sector) while the cavity run keeps warming at 0.15 degC per century.

8. Cavity effect on trends. The cavity run warms faster than the no-cavity run in the deep Southern Ocean in every period (difference 0.05 to 0.08 degC per century south of 60S in 2015-2100 and 2101-2200, and 0.06 degC per century already in the drift period), and its abyssal Indian sector warms by 0.02 to 0.04 degC per century more. Part of this is the larger disequilibrium of the cavity spin-up (item 1), part is the different Southern Ocean state (cavity run starts 0.25 degC colder in the deep Weddell and has no regular Maud Rise convection). Differences outside the Southern Ocean are below 0.02 degC per century.

12.1.3 C. Propagation pattern (fig06, fig07, fig11, fig14)

9. No equatorward-propagating abyssal warming front exists. After drift correction the Southern Ocean bottom warming (0.3 to 0.6 degC by 2200 under SSP5-8.5) is confined to south of 40S in all sectors. The 50 percent arrival year of the band-mean warming is 2071 to 2095 for every band from 74S to 40S in both sets (fig06), and along the western Atlantic pathway the four warming boxes (65S to 38S) reach 50 percent between 2068 and 2092 with no systematic increase with distance (fitted slope -4 to +1 yr per 1000 km, within the uncertainty of a 0 slope). The warming is therefore a quasi-simultaneous adjustment of the whole Southern Ocean abyss (reduced dense-water supply and isopycnal heave) rather than an advected anomaly from the Weddell Sea. Propagation speed cannot be estimated meaningfully; the detectable signal arrives at 40S at the same time as at 65S.

10. The Atlantic abyss north of 30S cools, and the cooling spreads from the north. Drift-corrected bottom temperature in the Brazil basin, equatorial Atlantic and NW Atlantic falls by 0.17 to 0.6 degC by 2200 (SSP5-8.5, both sets; largest at 40N). The 50 percent arrival year is earliest at 30 to 40N (2086 to 2097) and latest at 20 to 28S (2116 to 2134), a southward delay of roughly 30 to 40 years over 7000 km (3 to 5 yr per 1000 km). This is the signature of NADW weakening (less warm, salty NADW reaching the abyss; the bottom salinity freshens at the same time) and not of an AABW change. The boundary between warming and cooling sits at 30 to 35S in both sets and does not move with time.

11. Pacific and Indian abyss. Only the Ross Sea box (0.2 degC by 2200) and the band 55 to 65S warm detectably; the abyssal Pacific north of 52S changes by less than 0.03 degC by 2200 in both sets and both scenarios (no 0.05 degC emergence anywhere north of 45S, fig08). The Indian abyss north of 40S changes by less than 0.1 degC. Abyssal warming rates in the Pacific boxes (0.08 to 0.10 degC per century) equal the drift in all scenarios.

12. Scenario independence of the Southern Ocean abyssal warming. Deep Weddell bottom temperature (cavity) rises from 1.76 degC (pd) to 2.21, 2.25, 2.25 and 2.26 degC in 2171-2200 under SSP1-2.6, 2-4.5, 3-7.0 and 5-8.5; the no-cavity run from 2.00 to 2.21, 2.38, 2.40 and 2.41 degC. Argentine basin (cavity) 1.78 to 2.17/2.31/2.35/2.37 degC. The 2015-2100 trends (0.29 to 0.30 degC per century cavity Weddell) are the same in all scenarios; scenarios only separate after 2100, and then by at most 0.2 degC. The abyssal Southern Ocean response is therefore a committed response to the historical forcing (and possibly to the ongoing drift) rather than a response to 21st-century emissions. Under SSP1-2.6 the no-cavity Weddell warming stops around 2140 and reverses (-0.01 degC per century in 2101-2200) when deep convection resumes; the cavity run under SSP1-2.6 flattens after 2150 but does not cool.

13. Cavity effect on propagation. The cavity run has the same arrival years (differences below 10 to 15 years, fig08c,f and fig11e,f), the same warming-cooling boundary and a 0.05 to 0.15 degC larger warming in the Indian-sector abyss by 2200. South of 65S the cavity run warms slightly less (it is colder to begin with). The propagation pattern is therefore insensitive to the cavities; the cavity effect is on the present-day state (deep Weddell 0.25 degC colder, denser abyss, 1200 versus 175 m of dense layer south of 60S) and on rates, not on timing.

12.1.4 D. Time of emergence (fig08 to fig10)

14. With the 2 sigma criterion (drift-corrected) 44 (cavity) and 53 percent (no cavity) of the deep sea floor area have emerged by 2014 and 92/97 percent (SSP5-8.5) or 84/93 percent (SSP1-2.6) by 2200. Median emergence is 1950 to 1975 in the Southern Ocean and in the Atlantic (cooling), 2000 to 2120 in the Pacific. These numbers mainly reflect the tiny abyssal noise and the drift correction uncertainty.

15. With the 0.05 degC criterion the emerged deep area is 10 percent (both sets) in 2014, 26 percent in 2050, 36/41 percent (cavity/no cavity) in 2100 and 47/50 percent in 2200 under SSP5-8.5; under SSP1-2.6 35/37 percent in 2100 and 42/48 percent in 2200. Median emergence year by latitude (SSP5-8.5): 2005 to 2030 at 55 to 70S (warming), 2045 to 2070 at 30 to 50S, 2045 to 2085 in the Atlantic between 20S and 40N (cooling), never in the Pacific and Indian abyss north of 40S. Bottom salinity with a 0.005 psu criterion emerges in the Southern Ocean (salinification) between 2000 and 2060 and in the Atlantic and Arctic abyss (freshening) between 2040 and 2120.

16. Cavity versus no cavity emergence differs by less than 20 years over most of the deep ocean for both criteria; the no-cavity run emerges 5 to 10 years earlier in the Southern Ocean with the 2 sigma criterion (its 1851-1950 drift south of 60S is smaller, so less of the signal is removed), and the cavity run emerges earlier in the Indian sector abyss.

12.1.5 E. Dense-layer thickness (fig12, fig12b, fig13)

17. Using gsw sigma2 >= 36.945 (densest 10 percent of the cavity-run pd volume), the dense layer is 1211 m thick on average south of 60S in the cavity run but only 175 m in the no-cavity run at pd (global means 395 versus 283 m; dense volume 1.32 versus 0.94 x 10ˆ17 m3). The cavity run thus has a denser Southern Ocean abyss (0.25 degC colder Weddell bottom water, fig14), which is the clearest cavity signal in this theme. By 2171-2200 the dense layer has vanished from the Southern Ocean in both sets and both scenarios (0 to 14 m south of 60S) and the global dense volume is 0.30 to 0.34 x 10ˆ17 m3 (remaining in the North Pacific, Arctic and Mediterranean). The thickness numbers depend strongly on the threshold because bottom sigma2 is so uniform (a 0.01 kg/m3 shift changes pd thickness by several hundred metres); the disappearance by 2200 and the cavity minus no-cavity contrast at pd are robust.

12.1.6 F. Story lines

18. Under warming the model abyss does not show the classical AABW-shutdown signature of an abyssal warming that spreads from the Weddell and Ross seas into the basins. Instead the Southern Ocean abyss warms quasi-simultaneously south of 40S (committed, scenario-independent), the Atlantic abyss north of 30S cools and freshens from the north (NADW weakening), and the Pacific abyss does nothing. The reason is the model state: there is no dense AABW to shut down (bottom sigma2 is uniform and the deep Southern Ocean is the lightest abyss), so the 21st-century response is an adjustment of a CDW-filled Southern Ocean rather than the loss of a bottom-water pathway.

19. Ice-shelf cavities make the present-day Southern Ocean abyss colder and denser (dense-layer thickness 1200 versus 175 m south of 60S, deep Weddell 0.25 degC colder) and make the deep Southern Ocean warm 20 to 30 percent faster in the scenarios, but they do not change where or when the abyssal signals emerge. The comparison is complicated by the larger spin-up disequilibrium of the cavity run south of 60S (0.09 versus 0.03 degC per century drift).

20. Suspicious or open points. (i) The abyssal drift of 0.085 degC per century in both sets should be confirmed against the PI spin-up tails (PI_ICEv2_prod, PI_CTRL_prod) if they can be retrieved; the drift correction here assumes linearity. (ii) density_dMOC quantisation (item 2) affects the census theme. (iii) The no-cavity SSP1-2.6 cooling after 2140 coincides with renewed deep convection and deserves a look in the convection theme. (iv) The Indian-sector cavity minus no-cavity warming difference (0.1 degC by 2200) has no obvious mechanism and may be drift-related.

Figures

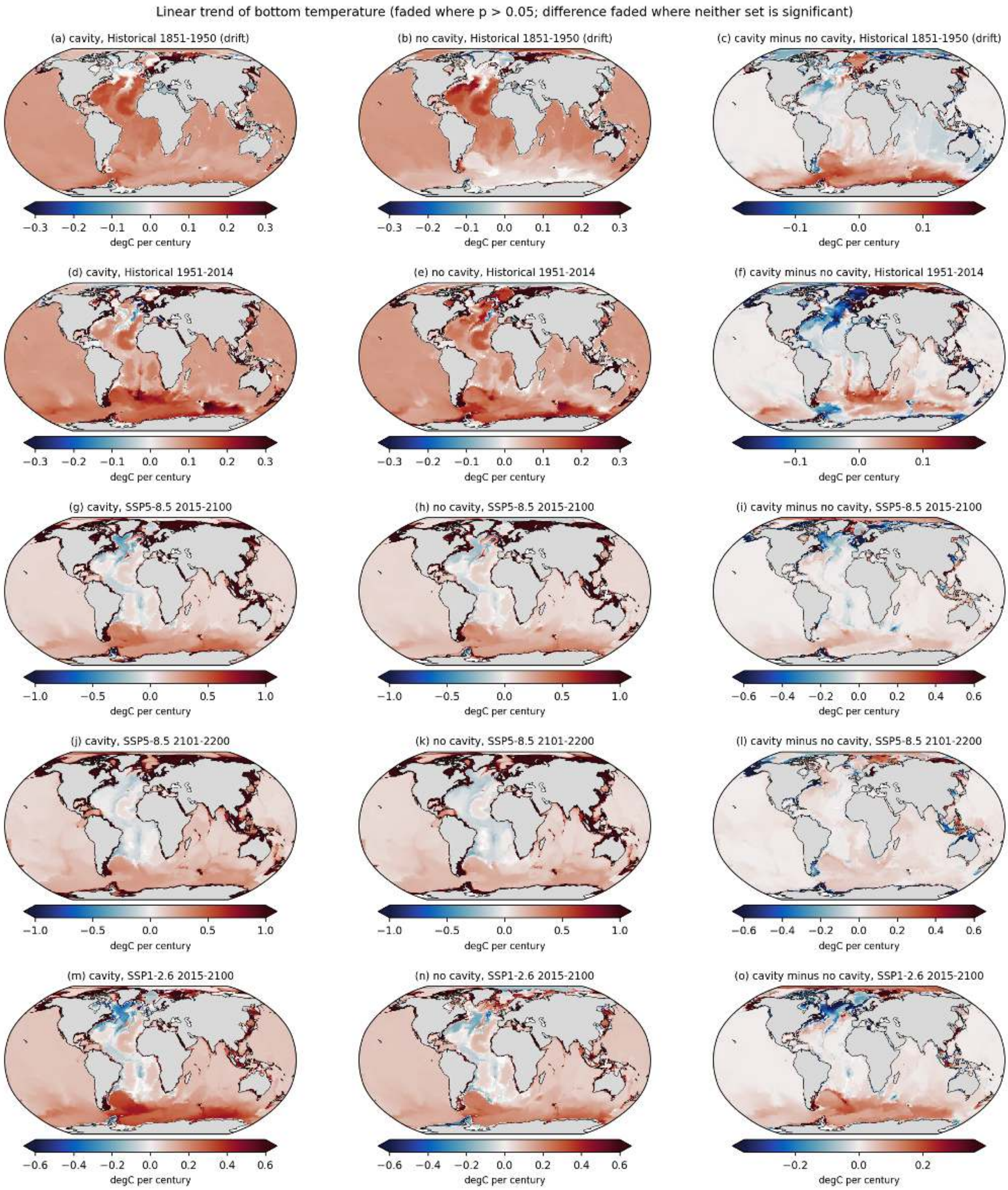

Figure 146: fig01__trend__btemp__global (propagation). Linear trends of bottom temperature (degC per century) on a Robinson projection. Rows are the periods 1851-1950 (historical drift), 1951-2014, SSP5-8.5 2015-2100, SSP5-8.5 2101-2200 and SSP1-2.6 2015-2100. Columns are cavity, no cavity and cavity minus no cavity. Colours are faded where the trend is not significant ($p > 0.05$, two-sided t test on the OLS slope); the difference is faded where neither set is significant. Take-away. The deep ocean warms by 0.08 to 0.09 degC per century almost everywhere already in 1851-1950, which is spin-up drift. The forced signal in 2015-2100 is a Southern Ocean warming of 0.2 to 0.4 degC per century south of 40S (strongest in the Indian sector) and an abyssal cooling of the Atlantic north of 30S of -0.05 to -0.2 degC per century that reverses the drift. The cavity run warms 0.05 degC per century faster in the Southern Ocean than the no-cavity run.

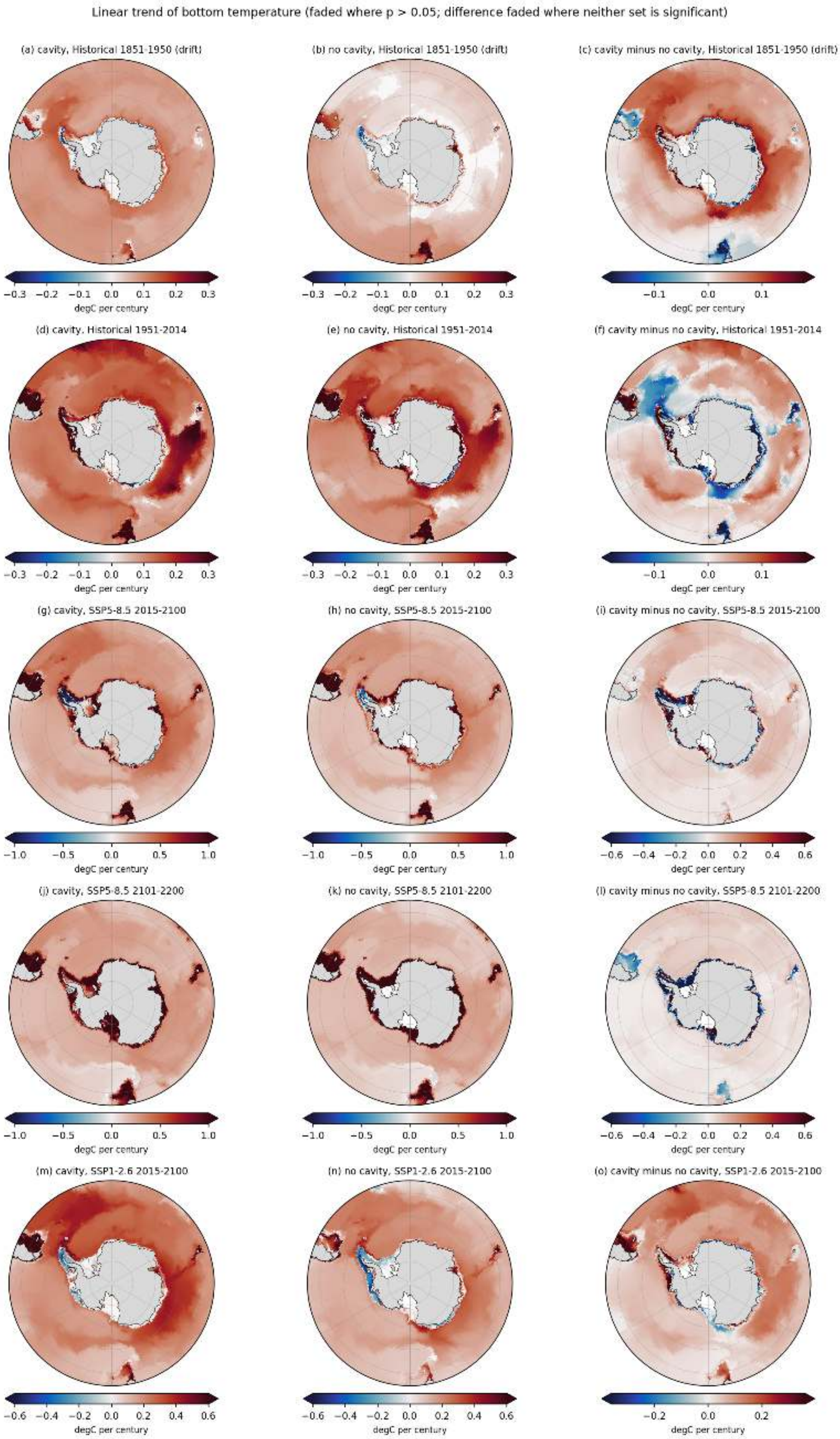

Figure 147: fig02__trend__btemp__south (propagation). Same as fig01 for the south polar view (south of 45S, 0.5 deg grid). Take-away. The 1851-1950 drift south of 60S is 0.09 degC per century in the cavity run but only 0.03 in the no-cavity run, so the cavity run is further from equilibrium in the Southern Ocean. In 2015-2100 (SSP5-8.5) the deep Weddell and Indian sectors warm by 0.3 to 0.5 degC per century in both sets, the Ross sector by 0.2 degC per century. Shelf and slope bottom waters warm by 0.5 to more than 1 degC per century. The cavity minus no-cavity difference is positive over most of the deep Southern Ocean (0.05 to 0.1 degC per century) and negative along the coast where the cavity mesh has a different geometry.

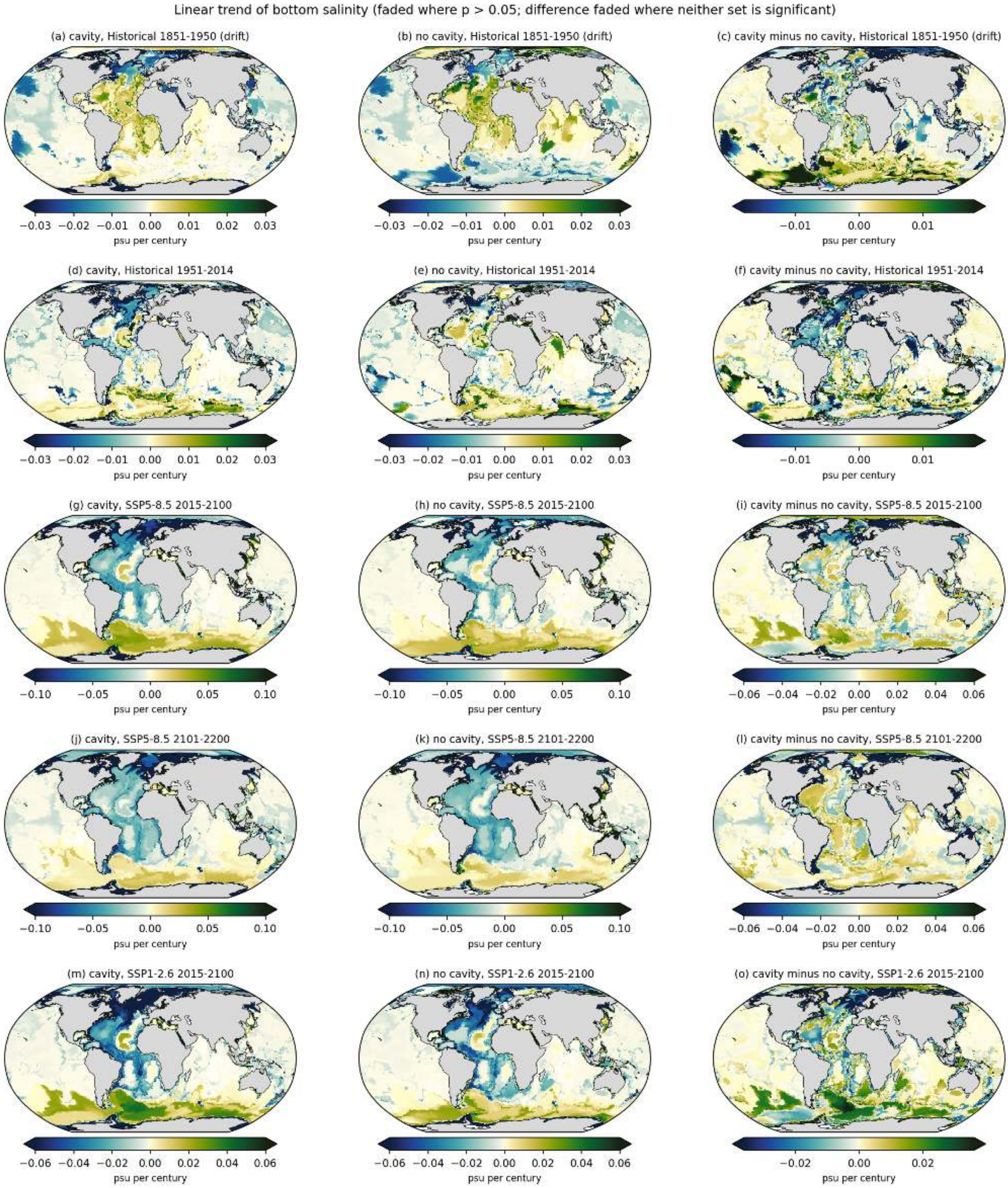

Figure 148: fig03_trend_bsalt_global (propagation). Same layout as fig01 for bottom salinity (psu per century). Take-away. The Southern Ocean bottom salinifies by 0.02 to 0.03 psu per century under SSP5-8.5 2015-2100 (shelf waters and the deep Weddell both), while the Atlantic abyss north of 30S freshens by 0.01 to 0.05 psu per century. The combination of Atlantic abyssal cooling and freshening is the signature of a weakening and lightening NADW source, not of AABW. The cavity minus no-cavity difference is small (below 0.01 psu per century) over the open ocean.

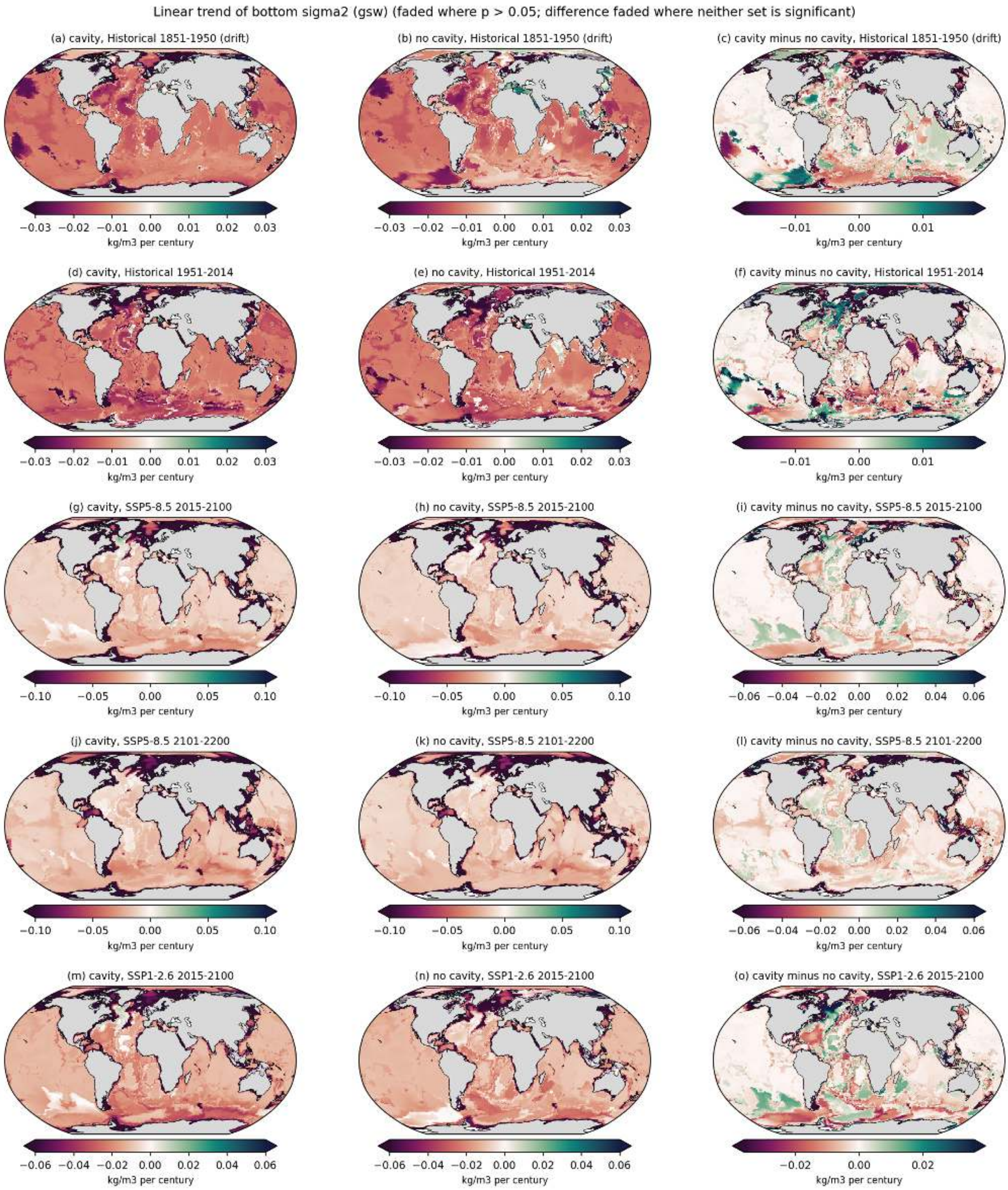

Figure 149: fig04__trend__bsig2__global (propagation). Same layout as fig01 for bottom sigma2 computed with gsw from bottom T and S (kg/m3 per century). Take-away. Bottom density decreases everywhere except in parts of the North Atlantic; the Southern Ocean bottom lightens by 0.03 to 0.06 kg/m3 per century in 2015-2100 under SSP5-8.5 because warming dominates over salinification. The abyssal Atlantic lightens despite cooling because freshening dominates.

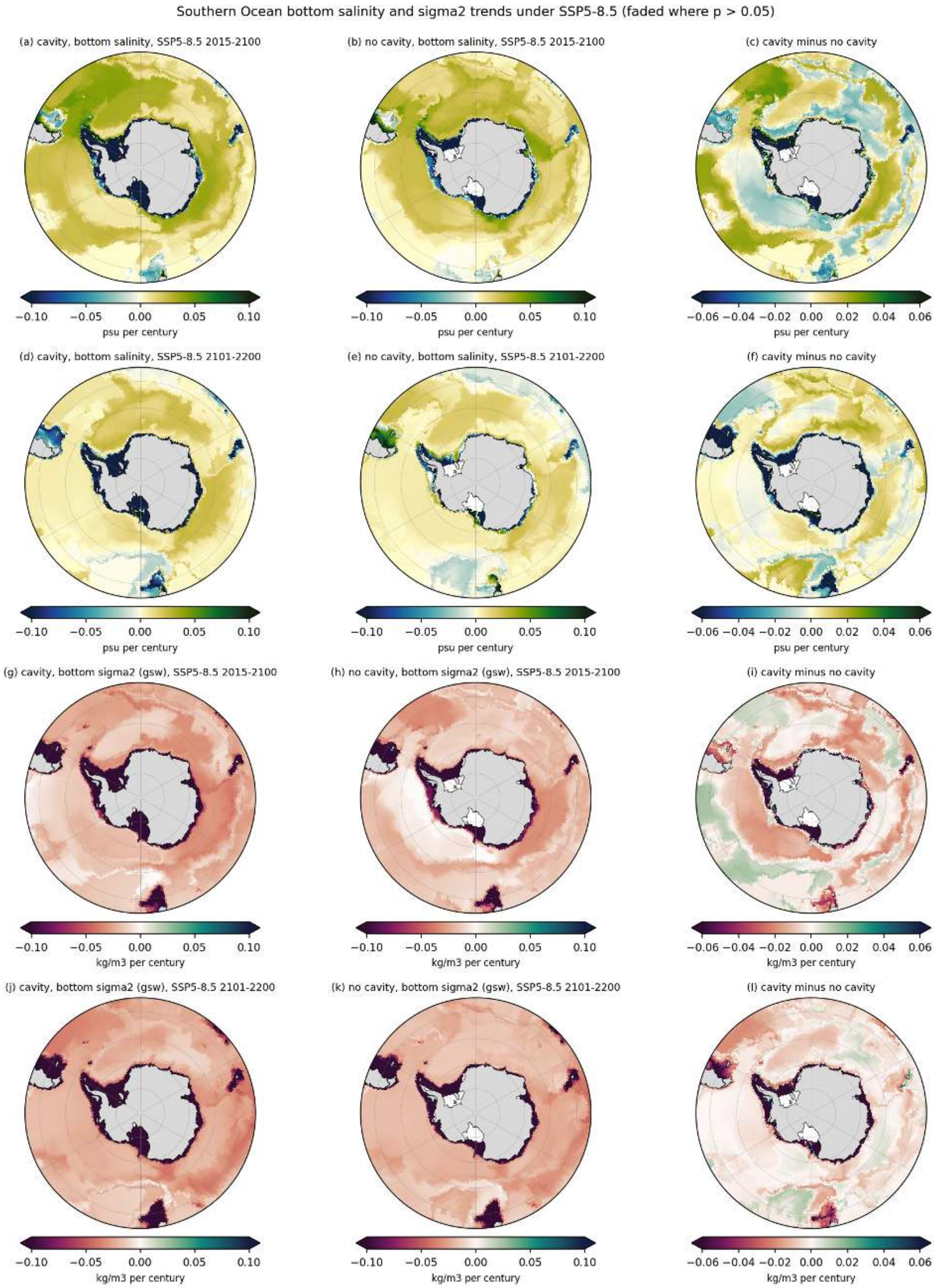

Figure 150: fig05__trend__salt__sig2__south (propagation). South polar view of the bottom salinity (rows 1 and 2) and gsw bottom sigma2 (rows 3 and 4) trends for SSP5-8.5 2015-2100 and 2101-2200, cavity, no cavity and difference. Take-away. The deep Weddell and Ross bottom waters become saltier by 0.02 to 0.04 psu per century and lighter by 0.04 to 0.08 kg/m3 per century. The cavity run shows a stronger coastal salinification in 2015-2100 in the Weddell and Amundsen sectors, consistent with CDW intrusion replacing shelf water, and a weaker one around Dronning Maud Land where PI_CTRL has open ocean and PI_ICEv2 has cavities.

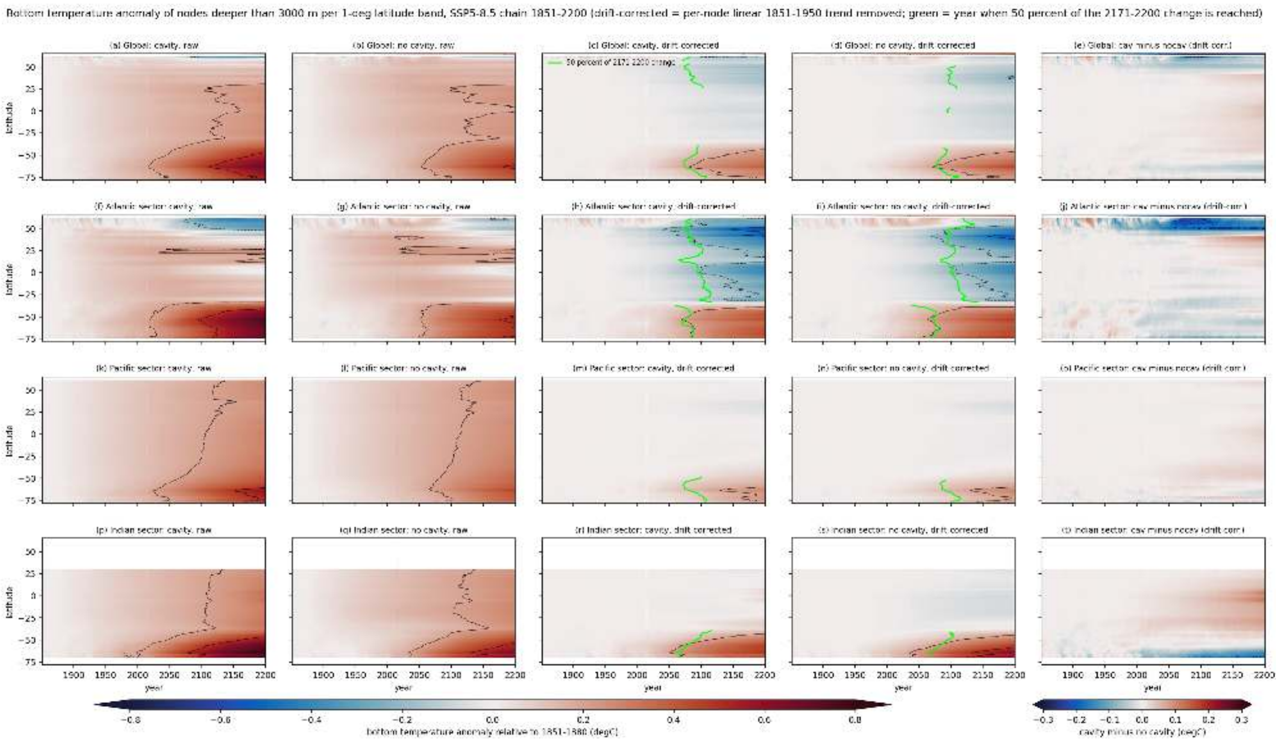

Figure 151: fig06_hovmoeller_btemp_ssp585 (propagation). Hovmoeller diagrams of the bottom temperature anomaly (relative to 1851-1880) of non-cavity nodes deeper than 3000 m, area-weighted per 1 deg latitude band, 1851-2200 for the SSP5-8.5 chain. Rows are global, Atlantic sector, Pacific sector and Indian sector. Columns are cavity raw, no cavity raw, cavity drift-corrected, no cavity drift-corrected and the drift-corrected cavity minus no-cavity difference. Black contours at plus and minus 0.2, 0.5 and 1 degC. The green line marks the year in which each band reaches 50 percent of its 2171-2200 anomaly (only bands with a final anomaly above 0.1 degC). Take-away. The raw panels are dominated by a uniform drift warming (0.3 degC by 2200). After drift correction the forced signal is a warming confined to south of 40S (0.3 to 0.6 degC by 2200) and a cooling of the Atlantic abyss north of 30S (-0.2 to -0.5 degC), while the Pacific and Indian abyss north of 40S change by less than 0.1 degC. The 50 percent arrival year is 2070 to 2095 at all latitudes south of 40S in both sets, so the warming appears almost simultaneously over the whole Southern Ocean rather than spreading from the Weddell Sea northward. The cavity run is 0.05 to 0.15 degC warmer than the no-cavity run in the Indian sector abyss by 2200 and slightly cooler south of 65S.

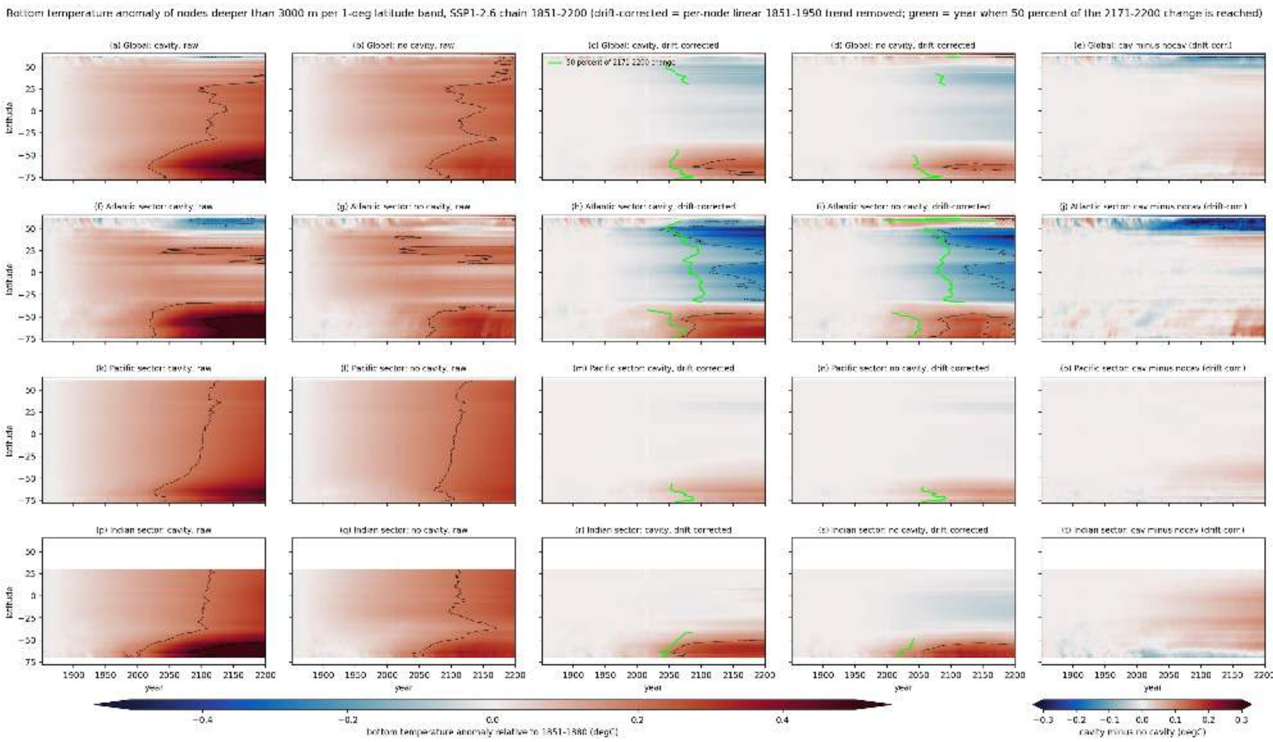

Figure 152: fig07__hovmoeller__btemp__ssp126 (propagation). As fig06 for the SSP1-2.6 chain. Take-away. The drift-corrected Southern Ocean warming reaches 0.2 to 0.35 degC by 2200 and the Atlantic abyssal cooling -0.15 to -0.4 degC, about two thirds of the SSP5-8.5 response. In the no-cavity SSP1-2.6 run the deep Weddell warming stops around 2140 and the abyss south of 60S cools again by 2200 (trend -0.01 degC per century in 2101-2200), which coincides with a return of open-ocean deep convection in that run.

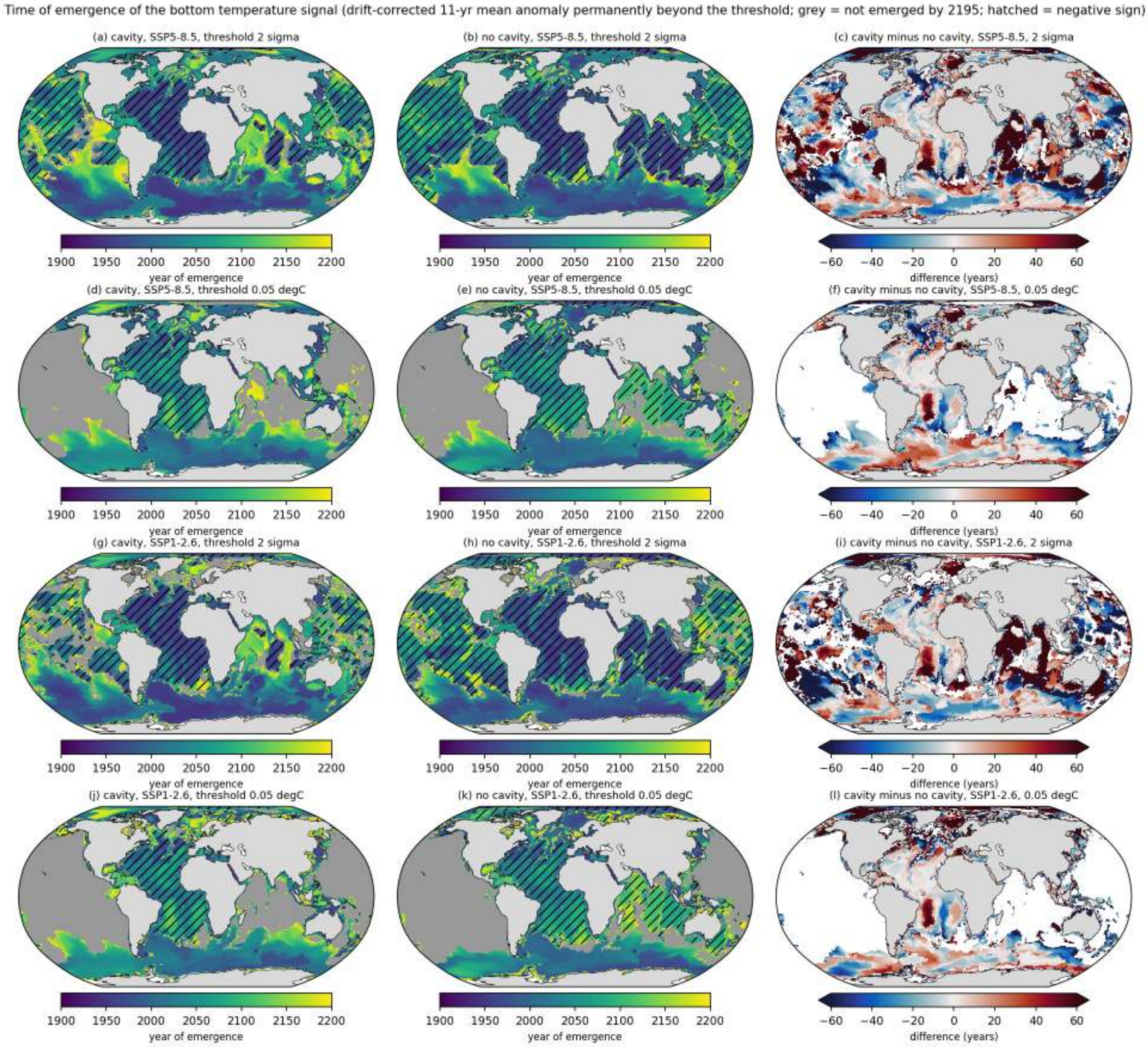

Figure 153: fig08_toe_btemp_maps (propagation). Time of emergence of the bottom temperature signal, defined as the first year from which the 11-yr running mean of the drift-corrected anomaly stays permanently beyond a threshold until 2195. Rows are SSP5-8.5 with a 2 sigma threshold (sigma = std of the detrended 1851-1950 annual values per node), SSP5-8.5 with an absolute threshold of 0.05 degC, and the same two for SSP1-2.6. Columns are cavity, no cavity and cavity minus no cavity (years). Grey means not emerged by 2195 and hatching marks emergence with a negative sign (cooling). Take-away. Abyssal interannual noise is tiny (median sigma 0.003 degC below 3000 m), so with the 2 sigma criterion 44 to 53 percent of the deep sea floor has emerged by 2014 and 92 to 97 percent by 2200. With the 0.05 degC criterion the Southern Ocean south of 45S emerges between 2005 and 2060, the Atlantic abyss (cooling) between 2020 and 2080, and the Pacific and Indian abyss north of 40S never emerge. The cavity and no-cavity emergence years differ by less than 20 years over most of the deep ocean.

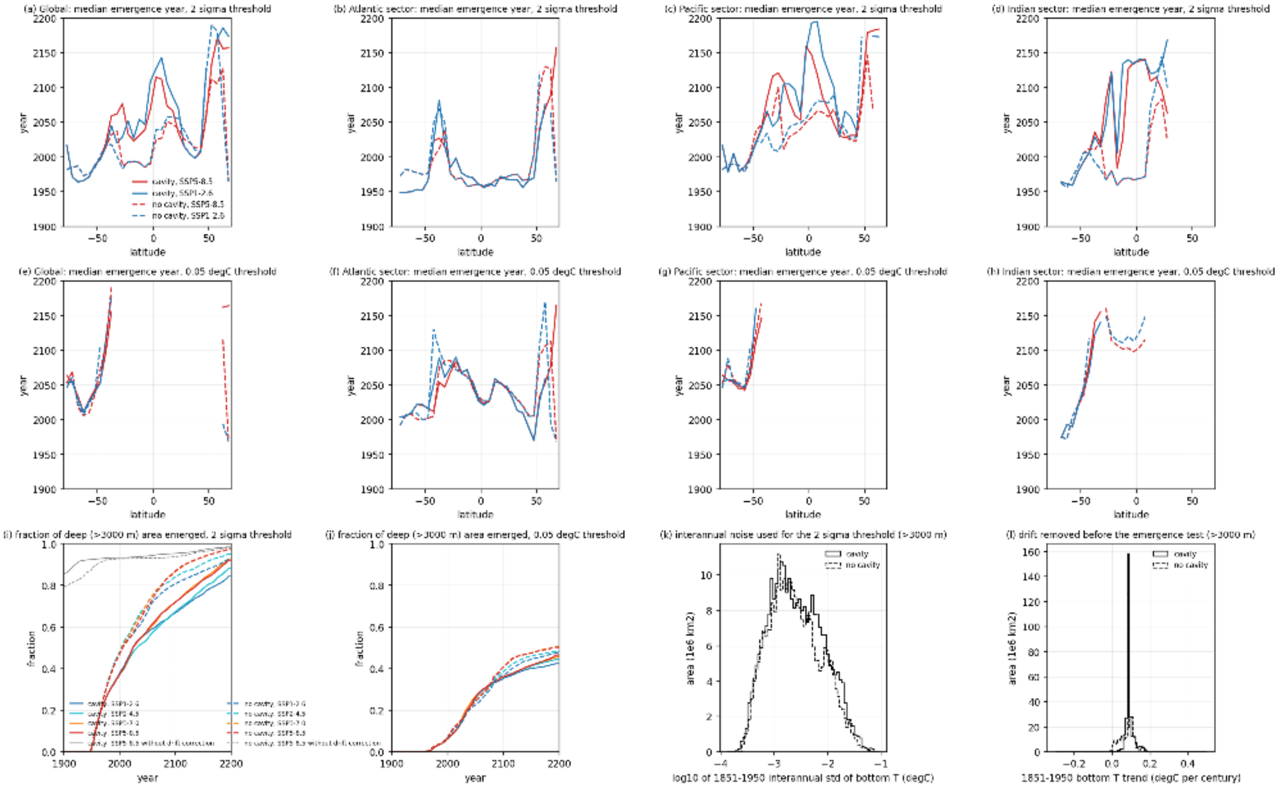

Figure 154: fig09_toe_latitude_summary (propagation). Panels (a) to (d) give the area-weighted median emergence year of bottom temperature (2 sigma threshold) per 5 deg latitude band for nodes deeper than 3000 m, globally and per sector, for SSP5-8.5 (red) and SSP1-2.6 (blue), cavity (solid) and no cavity (dashed). Panels (e) to (h) repeat this for the 0.05 degC threshold. Panel (i) shows the fraction of deep (>3000 m) sea floor area emerged as a function of time for all four scenarios and both sets with the 2 sigma threshold, plus the SSP5-8.5 result without drift correction in grey. Panel (j) is the same for the 0.05 degC threshold. Panels (k) and (l) are area-weighted histograms of the interannual noise and of the removed 1851-1950 drift for the deep nodes. Takeaway. Without drift correction 80 to 85 percent of the deep sea floor has "emerged" by 1900, which shows that the raw emergence is drift, not forcing. With the 0.05 degC threshold the emerged fraction is 10 percent in 2014, 36 to 41 percent in 2100 and 42 to 50 percent in 2200, with little dependence on scenario (the emerged area in SSP1-2.6 is only 3 to 5 percent smaller than in SSP5-8.5). Median emergence (0.05 degC) is 2005 to 2030 at 55 to 70S and 2045 to 2085 in the Atlantic between 30S and 40N.

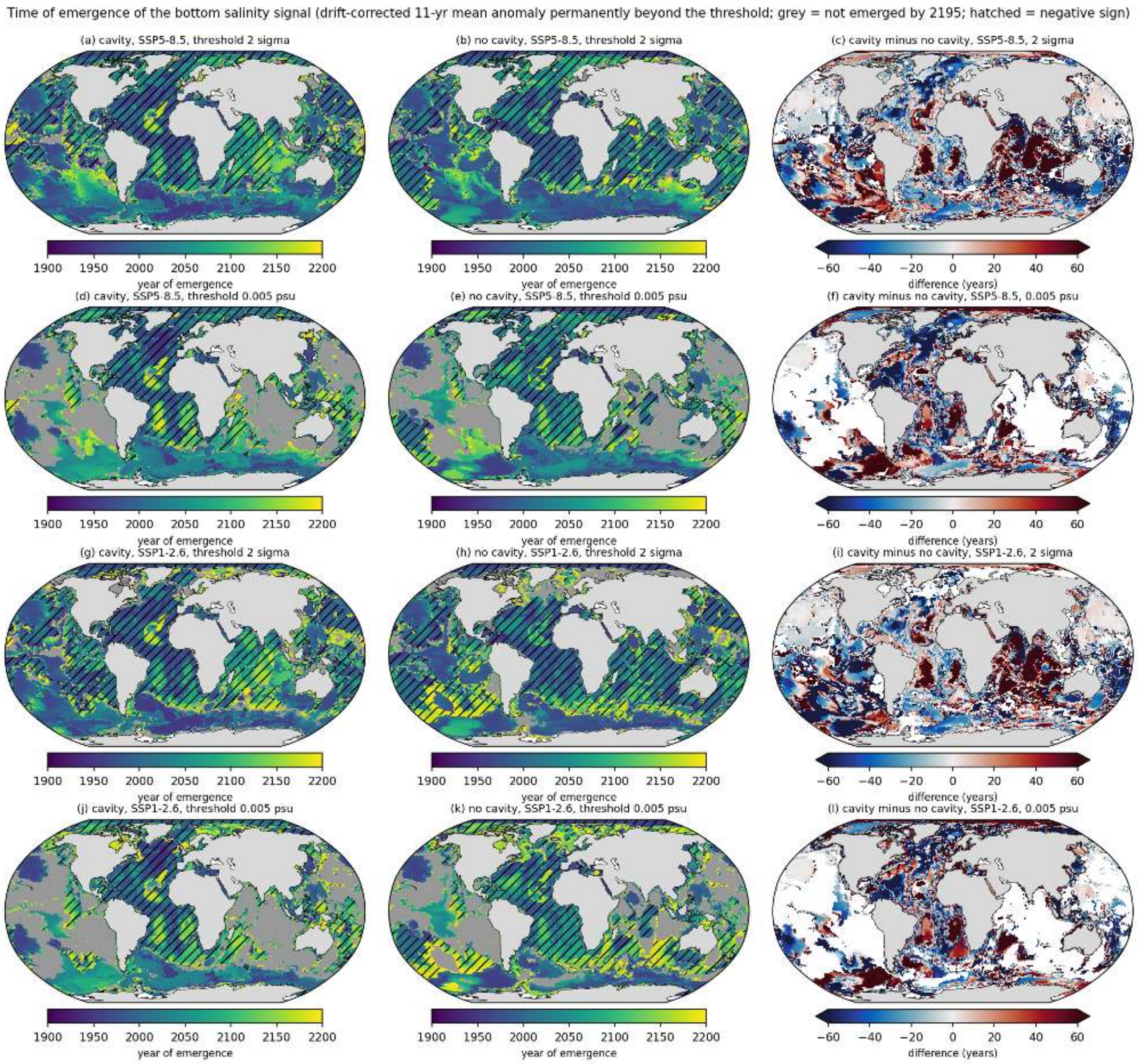

Figure 155: fig10_toe_bsalt_maps (propagation). As fig08 for bottom salinity, with 2 sigma and 0.005 psu thresholds. Take-away. Bottom salinity emerges with a positive sign (salinification) in the Southern Ocean and with a negative sign (freshening) in the Atlantic and Arctic abyss. With the 0.005 psu threshold the Southern Ocean south of 50S emerges between 2000 and 2060, the North Atlantic abyss between 2040 and 2120, and the abyssal Pacific does not emerge by 2195.

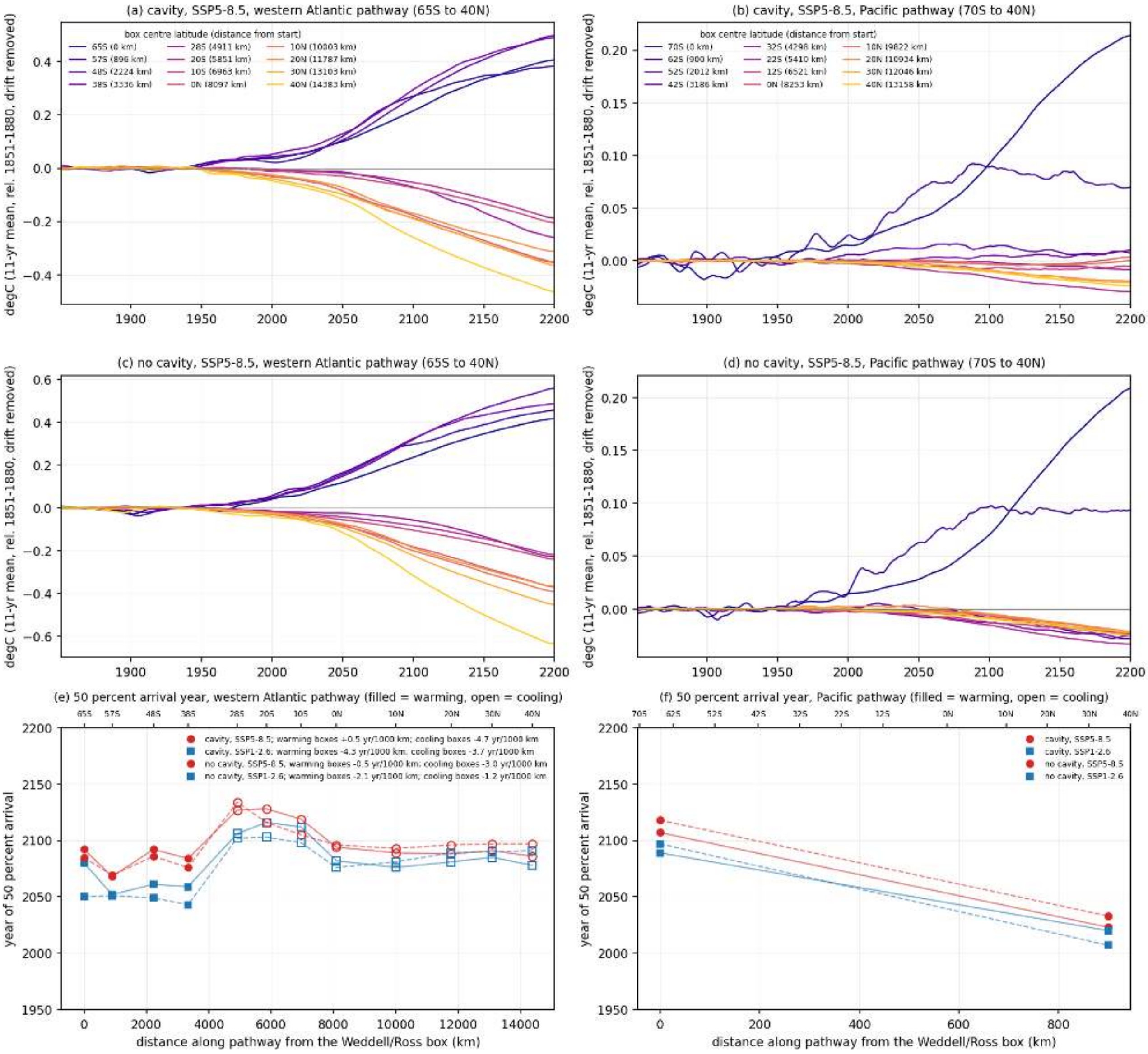

Figure 156: fig11_propagation_lag (propagation). Panels (a) to (d) show the 11-yr running mean drift-corrected bottom temperature anomaly (SSP5-8.5) in 10 deg by 8 deg boxes deeper than 3500 m along the western Atlantic pathway (65S, 30W to 40N, 55W; 12 boxes, 14400 km) and along a Pacific pathway (70S, 178W to 40N, 170W; 12 boxes, 13200 km), colour coded by distance from the first box. Panels (e) and (f) show for each box the year in which the anomaly reaches 50 percent of its 2171-2200 value (boxes with a final anomaly smaller than 0.05 degC are omitted), filled markers for warming and open markers for cooling boxes, SSP5-8.5 in red and SSP1-2.6 in blue, cavity solid and no cavity dashed. The legend gives the least-squares slope of arrival year versus distance in years per 1000 km for warming and for cooling boxes separately. Take-away. Along the Atlantic pathway the four southern boxes (65S to 38S) warm by 0.37 to 0.53 degC by 2200 with 50 percent arrival between 2068 and 2092 and no systematic delay with distance (slope between -4 and +1 yr per 1000 km, i.e. faster than can be resolved, consistent with a basin-wide adjustment rather than an advective front). The eight northern boxes (28S to 40N) cool by 0.17 to 0.6 degC with 50 percent arrival between 2086 and 2134, the cooling arriving first in the north (40N, 2086 to 2097) and last at 20 to 28S (2116 to 2134), so the cooling spreads southward from the North Atlantic at roughly 3 to 5 yr per 1000 km. In the Pacific only the Ross Sea box (70S) and the 62S box change by more than 0.05 degC (0.2 and 0.07 to 0.09 degC); the abyssal Pacific north of 52S changes by less than 0.03 degC by 2200 in both sets. The lag of maximum correlation (stored in stats_propagation.json) is 0 to 6 years within the warming domain and ill defined (100 to 150 years) between the warming and cooling domains.

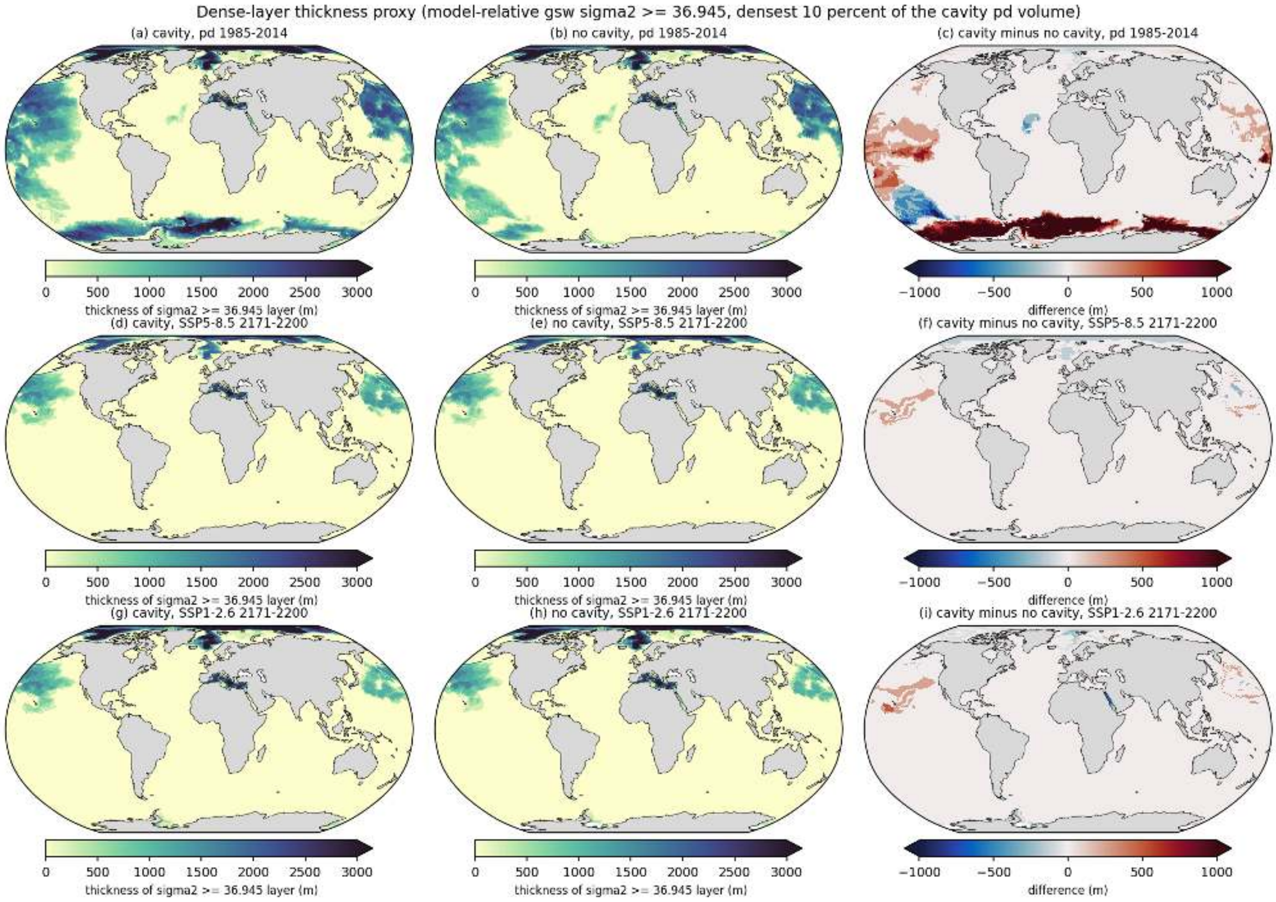

Figure 157: fig12_aabw_thickness_maps (propagation). Thickness (m) of the water column with gsw sigma2 >= 36.945, computed from the period-mean temperature and salinity ('cache/pmean') with the wet mask applied. The threshold is the volume-weighted 90th percentile of the cavity-run 1985-2014 sigma2 (densest 10 percent of the ocean volume), a model-relative definition of the densest class. Rows are pd 1985-2014, SSP5-8.5 2171-2200 and SSP1-2.6 2171-2200; columns cavity, no cavity, cavity minus no cavity. Take-away. At present day the densest class fills the abyssal Southern Ocean south of 55S (1200 m mean thickness south of 60S in the cavity run, 175 m in the no-cavity run, where only the Weddell abyss is dense enough) but it is also present in the North Pacific, the Arctic and the Mediterranean, because the model abyss has a nearly uniform sigma2 of 36.94 to 36.98 from 70S to 60N (fig13) and the North Pacific is marginally denser than the Weddell abyss. By 2171-2200 the dense class has disappeared completely from the Southern Ocean in both scenarios and both sets (mean thickness south of 60S 0 to 14 m); globally the dense volume drops from 1.32 (cavity) and 0.94 (no cavity) to 0.30 to 0.34 x 10^17 m3.

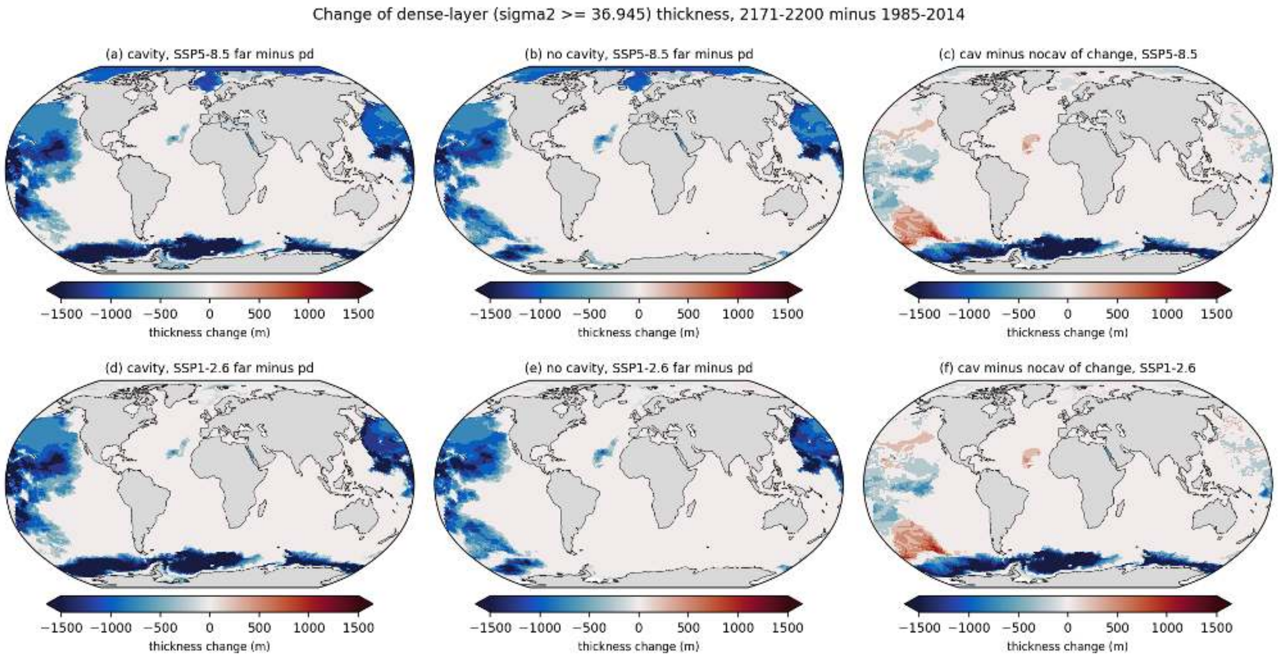

Figure 158: fig12b_aabw_thickness_change (propagation). Change of the thickness of the sigma2 >= 36.945 layer, 2171-2200 minus 1985-2014, for SSP5-8.5 (top) and SSP1-2.6 (bottom), cavity, no cavity and the difference of the two changes. Take-away. The Southern Ocean loses the whole dense layer (-1500 to -3000 m in the Weddell-Enderby basin of the cavity run), the North Pacific loses 500 to 1500 m. The loss is larger in the cavity run because it started with a thicker dense layer (the cavity minus no-cavity difference of the change is -1000 to -2000 m south of 60S). Thresholds 0.01 kg/m3 lower or higher shift the pd thickness by several hundred metres, so the absolute numbers are definition dependent while the disappearance by 2200 is robust.

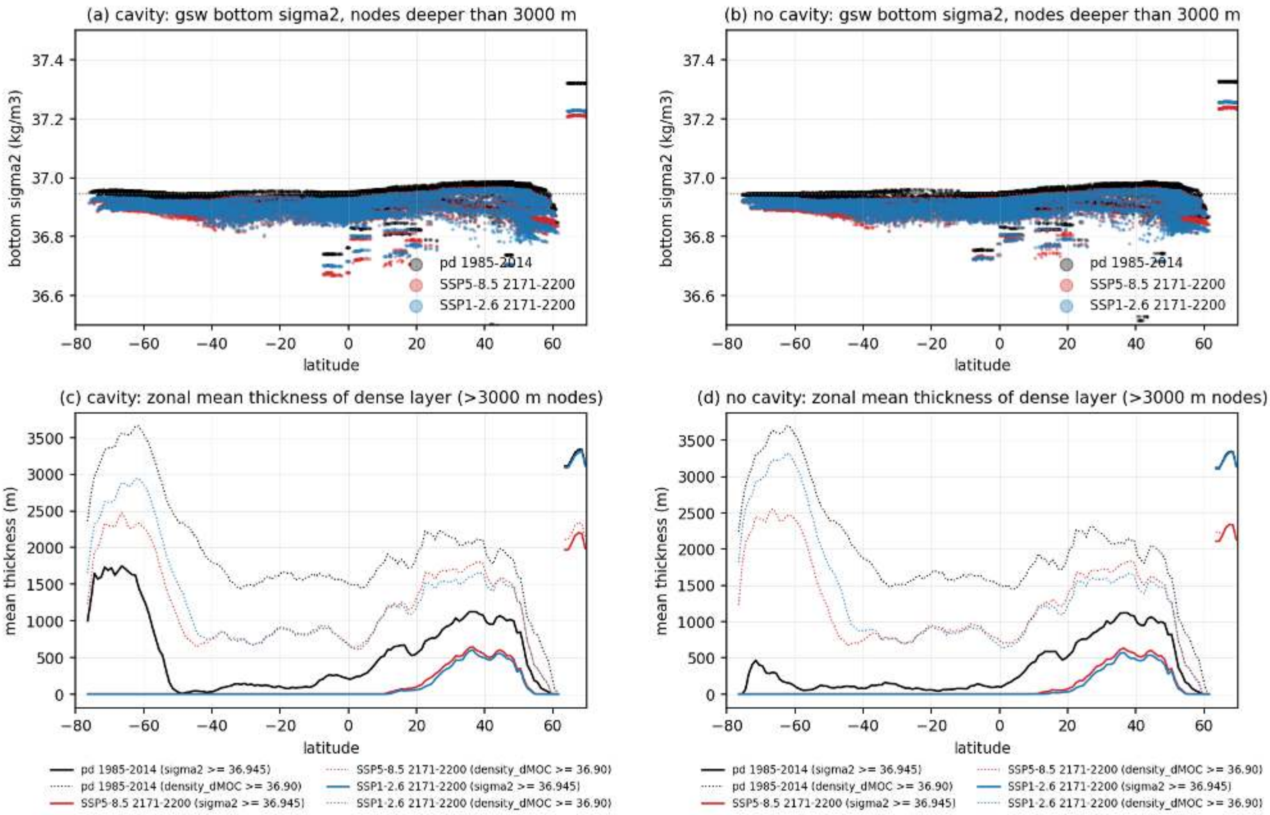

Figure 159: fig13_bottom_sigma2_latitude (propagation). Panels (a) and (b) give the gsw bottom sigma2 of a random subset of nodes deeper than 3000 m versus latitude for pd (black), SSP5-8.5 2171-2200 (red) and SSP1-2.6 2171-2200 (blue), cavity and no cavity; the dotted line is the 36.945 threshold. Panels (c) and (d) give the zonal mean thickness of the dense layer for both definitions (solid gsw sigma2 >= 36.945, dotted density_dMOC >= 36.90 as used by the census theme). Take-away. The abyss has a remarkably uniform bottom sigma2 of 36.94 to 36.98 at present day with the densest values in the North Pacific and North Atlantic (36.97 to 36.98) and the lightest in the deep Southern Ocean (36.94 to 36.95); there is no dense southern end member. By 2200 bottom sigma2 drops to 36.85 to 36.92 everywhere. The density_dMOC layer definition gives thicknesses two to three times larger and does not reproduce the cavity minus no-cavity contrast, because density_dMOC is nearly constant (36.91 to 36.92) below 700 m.

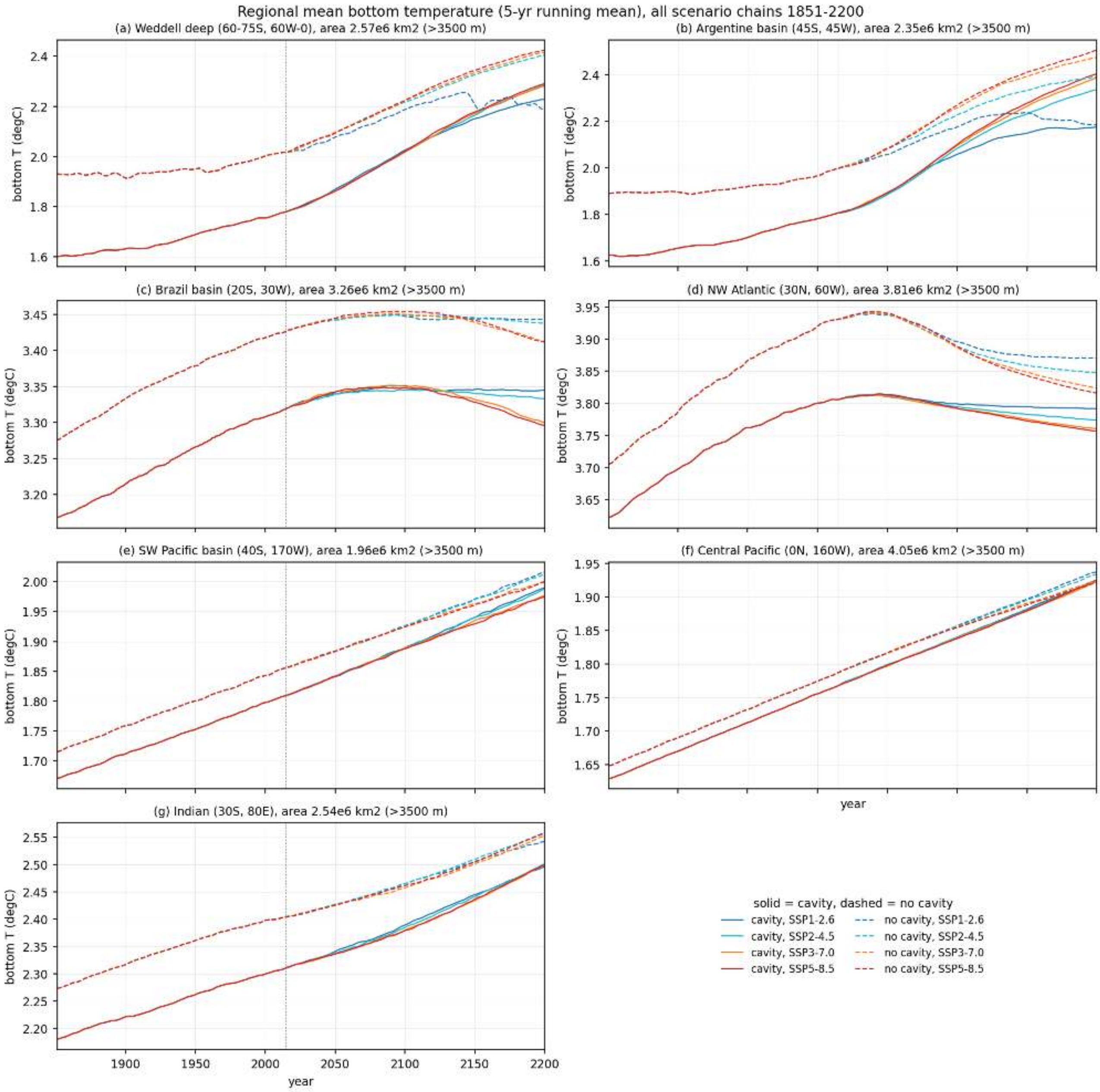

Figure 160: fig14_regional_btemp_timeseries (propagation). Area-weighted bottom temperature (5-yr running mean) of seven regional boxes restricted to depths below 3500 m for all four scenario chains 1851-2200, cavity solid and no cavity dashed, scenario colours. Boxes are the deep Weddell Sea (60-75S, 60W-0E), the Argentine basin (38-52S, 58-35W), the Brazil basin (12-28S, 38-20W), the NW Atlantic (24-36N, 75-45W), the SW Pacific basin (34-46S, 180-160W), the central Pacific (7S-7N, 172-148W) and the Indian Ocean (23-37S, 70-92E). Take-away. Deep Weddell bottom water is 1.76 degC (cavity) and 2.00 degC (no cavity) at present day and warms to 2.26 and 2.41 degC by 2171-2200 under SSP5-8.5 (0.29 and 0.25 degC per century in 2015-2100). The Southern Ocean boxes warm almost identically in all four scenarios; the spread between SSP1-2.6 and SSP5-8.5 at 2200 is only 0.05 degC in the Weddell and 0.2 degC in the Argentine basin. The Brazil basin and NW Atlantic warm until 2050-2090 (drift) and then cool under the stronger scenarios (-0.05 degC per century in 2101-2200 for SSP5-8.5). The Pacific boxes follow the drift line (0.08 to 0.09 degC per century) with no visible scenario response.